PD4A4-D00

BF4 WITH ATS2600
CX9 + GRACE + AST2600 BOARD

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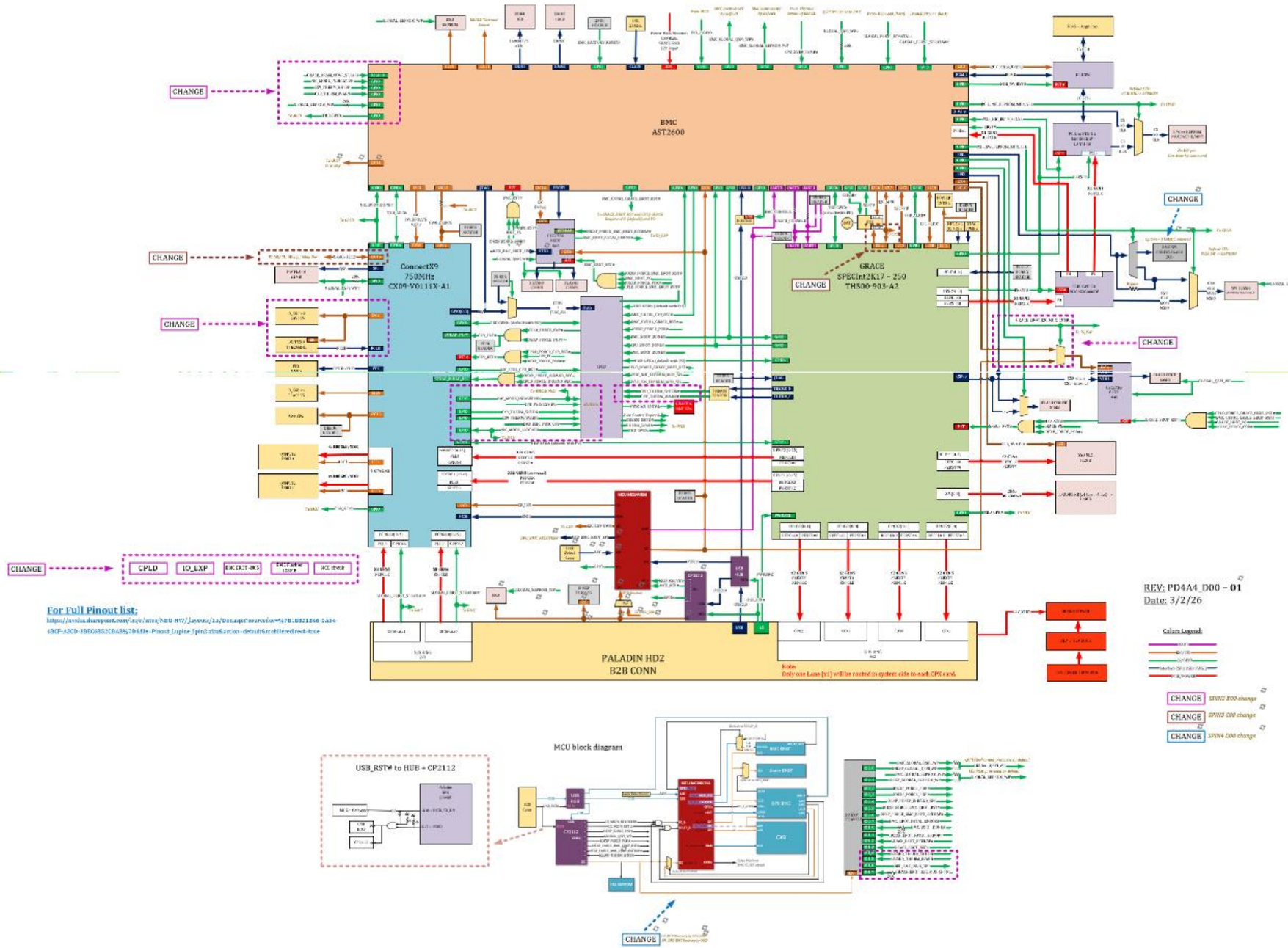
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
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PAGE DETAIL	Block Diagram Drawing

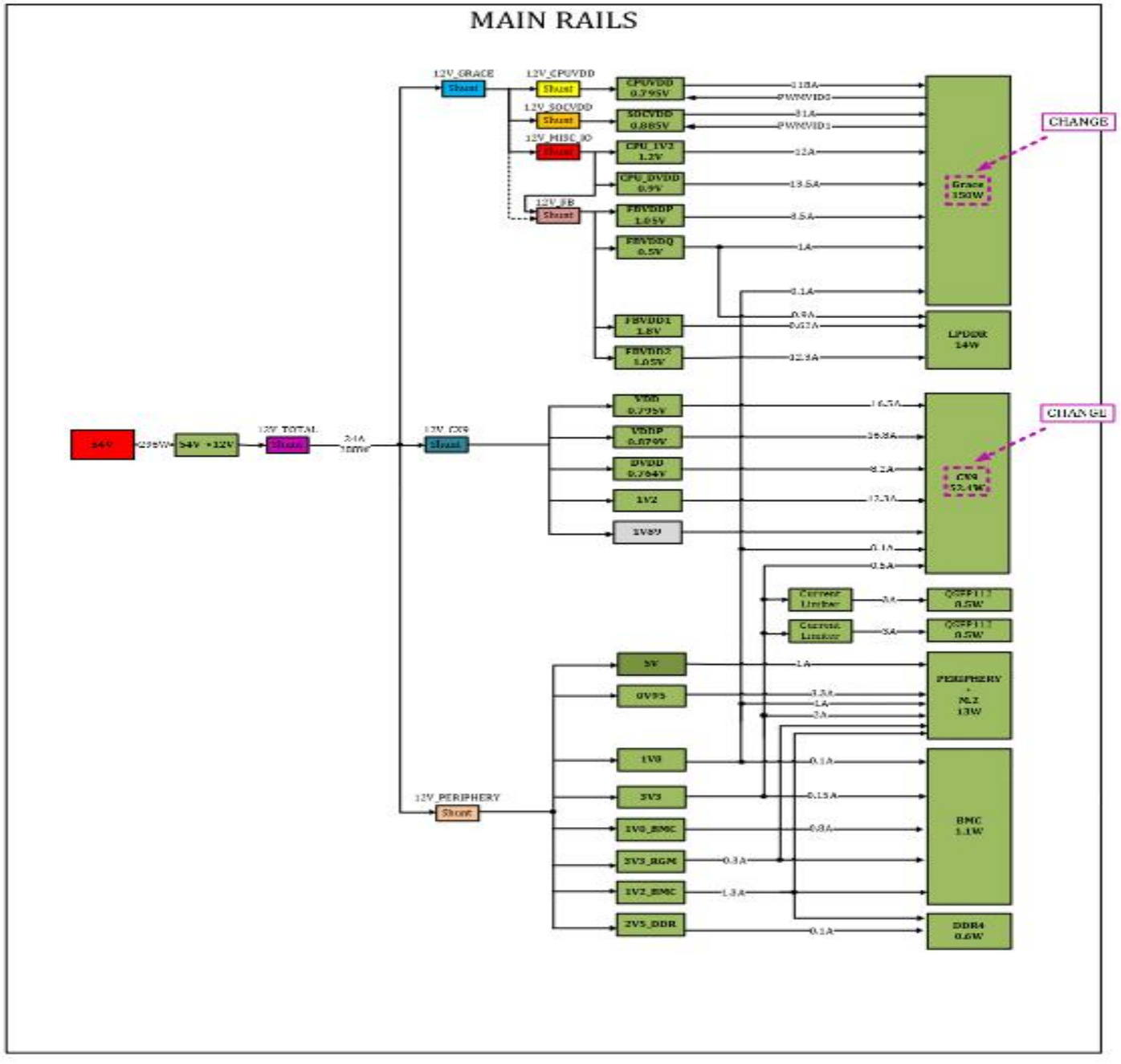
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MAIN RAILS



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Power Block

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Strap Table

Strap Description	Assigned GPIOs		Value Description
Flash Present	strap0	GP[0]	0 - Live Fish Mode, No FW
			1 - FW present
Bypass Core PLL (Debug)	strap1	GP[1]	0 - PLL Bypass
			1 - No PLL Bypass
Primary Pcore	strap2, strap3	GP[18],GP[17]	0,0 - Primary Pcore is 0
			0,1 - Primary Pcore is 1
			1,0 - Primary is Pcore 2
			1,1 - Reserved
Socket Direct/Multi Host	strap4	GP[21]	0 - Socket Direct Mode
			1 - Multi-Host Mode
2PCores_Active	strap5	GP[22]	0 - 1 PCORE Active
			1 - 2 PCOREs Active
Number Of Hosts	strap6, strap7	GP[27], GP[26]	0,0 - 1x16 PCIe Socket per PEX/PCORE
			0,1 - 2x8 PCIe Socket per PEX/PCORE
			1,0 - 4x4 PCIe Socket per PEX/PCORE
			1,1 - 8x2 PCIe Socket per PEX/PCORE
GP[27] is the most significant bit. {GP[27],GP[26]}			
Pcore0 Lane Reversal	strap8	GP[28]	0 - No lane reversal for Primary PEX PCIe[15:0]
			1 - x16 Lane reversal for Primary PEX PCIe[15:0]
Core OSC Frequency	strap9, strap10	GP[37], GP[36]	1,1 - 156.25 MHz
			1,0 - Reserved
			0,1 - Reserved
			0,0 - Reserved
Tester_Master_Active_n	strap11	GP[38]	0 - Enable tester mode
			1 - Disable tester mode
Boot_Debug_n	strap12	GP[39]	0 - Pause YU boot and enable boot debug
			1 - No pause YU boot and enable boot debug
Enable_inband_recovery	strap13	GP[45]	0 - Prevent PCIe operation in LF mode
			1 - No prevent PCIe operation in LF mode

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Power Sequence Diagram

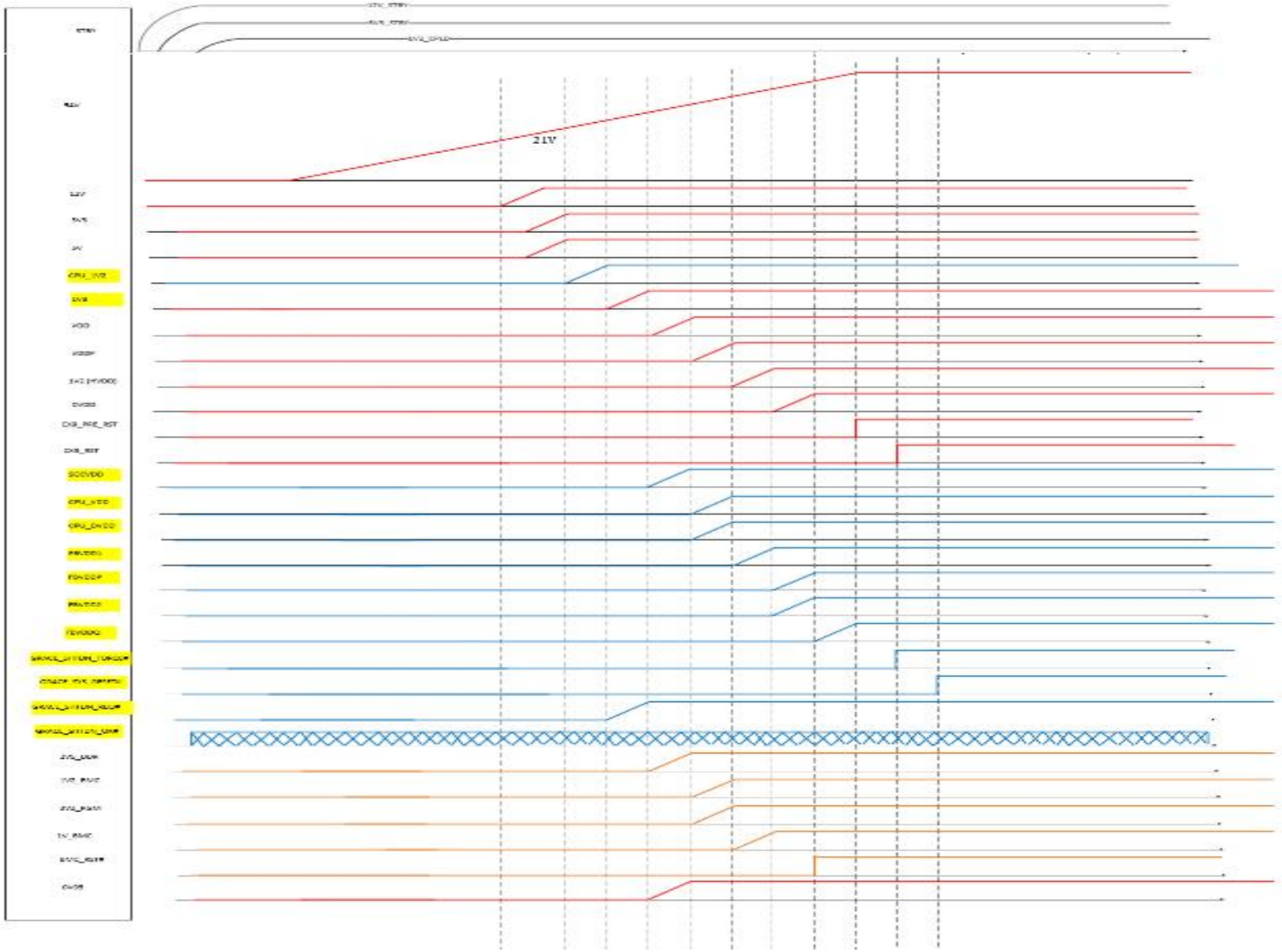
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Power Up Sequence



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PAGE DETAIL	Power Up Diagram

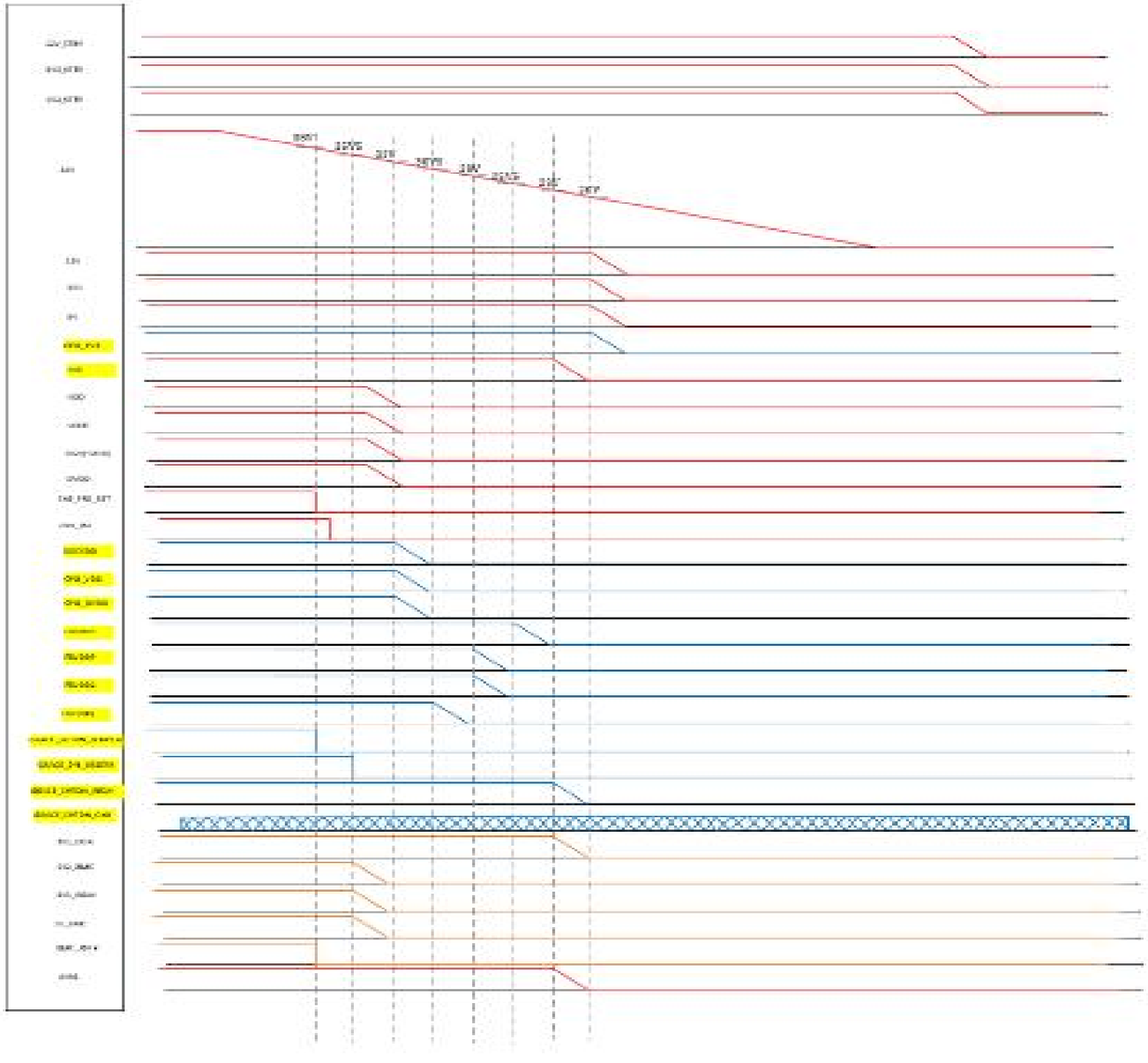
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Power down Sequence



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SPXO Oscillator

3V3_RCLK_R — LB13 600ohm — 3V3_RCLK — R656 0.249ohm 0402 1% COMMON — 3V3

C1713 22uF 6.3V 20% X6S 0603 COMMON
C3602 22uF 6.3V 20% X6S 0603 COMMON
C3338 22uF 6.3V 20% X6S 0603 COMMON
C1714 22uF 6.3V 20% X6S 0603 COMMON

0.03A

GND

C3601 0.1uF 50V

R156 4.7k 1% 0402 NA
R2671 4.7k 1% 0402 NA
Y1 156.25MHz X1G0061111000515
R2670 0ohm 0.05 ohm 0402
R157 0ohm 0.05 ohm 0402

LVDS_SMD_B NA
3V3_RCLK_R_EN
S_SPXO_NC
LVDS_RCLK_OSC_P
LVDS_RCLK_OSC_N
VCC OUT OE OUT* NC GND

PCB CO-LAYOUT
C1742 10nF
C3601 10nF
C3602 10nF
C1743 10nF
PCB CO-LAYOUT

LVDS_RCLK_TCXO_P
LVDS_RCLK_TCXO_N

R1157 10k 1% 0402 COMMON
R1147 10k 1% 0402 COMMON
TP98

22> IN S_CX_RST_1V8 — L23 RST# 1.8V
S_CCLK — U14 CCLK 1.8V
S_CCLK_SELECT — V16 CCLK_SELECT 1.8V
HCSL_RCLK_P — A9 PL_PLL_REFCLKP
HCSL_RCLK_N — A7 PL_PLL_REFCLKN

R551 49.9ohm
R552 49.9ohm
S_RCLK_PU
1V2
R549 1k 0402 1% COMMON
R550 499ohm 0402 1% COMMON
C1147 2.2uF 0402 COMMON 20%
C3650 2.2uF 0402 COMMON 20%

The purpose of the circuit is to generate
the CLK signal with 0.4V Common Mode DC
according the requirement of the CX-9
HSCL level differential clock input
This DC level circuit is fed by HVDD to
to eliminate ESD risk to the input clock
buffer before the HVDD 1.2V is up

I2C 3-pin Con

GND
26> 24< 8< 108> — OUT / BI — I2C_SCL0 I2C_SDA0 — J51 3_HOLES_FOOTPRINT_ONLY NA 2.54MM N/A NON-PHYSICAL NA
8<> 24<> 26<> 108<>

ConnectX-9

U1 CX09-V0111X-A1 BGA COMMON
1/15 Control

Internal PU/PD - 20k ohm

		STRAP0_FNP#/CPLD_TDI/SER_DO/MDIO7	3.3V		GPIO[0]	R23
		STRAP1_CPLL_BYP#/CPLD_TCK/SER_CLK/MDC7	3.3V		GPIO[1]	R25
		CPLD_TMS/SER_CAP/MDIO8	3.3V		GPIO[2]	T23
		CPLD_TDO/SER_DI/MDC8	3.3V	PU	GPIO[3]	T25
		NCSI_ARB_IN	3.3V	PU	GPIO[4]	C25
		NCSI_ARB_OUT	3.3V	PU	GPIO[5]	L25
		NCSI_REF_CLK	3.3V	PU	GPIO[6]	G23
		NCSI_BMC_TXD0/IST_DONE	3.3V	PU	GPIO[7]	J23
		NCSI_BMC_TXD1	3.3V	PU	GPIO[8]	G25
		NCSI_BMC_TX_EN	3.3V	PU	GPIO[9]	J25
		NCSI_BMC_RXD0	3.3V	PU	GPIO[10]	C23 SE_NCSI_RX_D0_CX
		NCSI_BMC_RXD1	3.3V	PU	GPIO[11]	A25 SE_NCSI_RX_D1_CX
		NCSI_BMC_CRS_DV	3.3V	PD	GPIO[12]	A23 SE_NCSI_CRS_DV_CX
		WAKE0#	3.3V	PU	GPIO[13]	G21
		SPI_CS1/OVT_SHTDN	1.8V	PD	GPIO[14]	V48
		I2C_SDA0/MDIO0	3.3V	PU	GPIO[15]	V56
		I2C_SCL0/MDC0	3.3V	PU	GPIO[16]	T64
		STRAP2_PRIMARY_PCORE0/I2C_SDA7/SPI_MASTER_MOSI	3.3V		GPIO[17]	T60
		STRAP3_PRIMARY_PCORE1/I2C_SCL7/SPI_MASTER_CLK	3.3V		GPIO[18]	P64
		I2C_SDA8/SPI_MASTER_MISO/BOOT_DEBUG	3.3V	PU	GPIO[19]	V52
		I2C_SCL8/SPI_MASTER_CS	3.3V	PU	GPIO[20]	V60
		STRAP4_MULTI_HOST/I2C_SDA9/SLAVE_MOSI/PWM0	3.3V	PU	GPIO[21]	E21
		STRAP5_2PCORES_ACTIVE/I2C_SCL9/SLAVE_MISO/TACH0	3.3V	PD	GPIO[22]	C21
		SPI_CLK	1.8V		GPIO[23]	A13 SE_SPI_CLK_CX
		SPI_CS0#	1.8V	PU	GPIO[24]	C13 SE_SPI_CS0_L_CX
		PRE_RESET_N	3.3V	PU	GPIO[25]	A19
		STRAP6_PCI_PART0/SPI_IO0	1.8V		GPIO[26]	C15 SE_SPI_IO0_STRP_PCI_PART0_CX
		STRAP7_PCI_PART1/SPI_IO1	1.8V		GPIO[27]	C17 SE_SPI_IO1_STRP_PCI_PART1_CX
		STRAP8_PCT_REVERSAL/SPI_IO2	1.8V		GPIO[28]	A15 SE_SPI_IO2_STRP_PCORE0_RVRSL_CX
		SPI_IO3	1.8V		GPIO[29]	A17 SE_SPI_IO3_CX
		I3C_SDA1/MDIO1	3.3V		GPIO[30]	V44
		I3C_SCL1/MDC1	3.3V		GPIO[31]	T37
		I3C_SDA2/MDIO2	3.3V		GPIO[32]	T52 I2C_SDA2_CX
		I3C_SCL2/MDC2	3.3V		GPIO[33]	V40 I2C_SCL2_CX
		I3C_SDA3/MDIO3	3.3V		GPIO[34]	T56
		I3C_SCL3/MDC3	3.3V		GPIO[35]	V37
		STRAP9_OSC_FREQ0/I3C_SDA4/MDIO4	3.3V		GPIO[36]	V29 I2C_SDA4_STRP_CORE_FREQ0_CX
		STRAP10_OSC_FREQ1/I3C_SCL4/MDC4	3.3V		GPIO[37]	T33 I2C_SCL4_STRP_CORE_FREQ1_CX
		STRAP11_TEST_MASTER_ACTIVE_N/I3C_SDA5/MDIO5	3.3V		GPIO[38]	T48
		STRAP12_BOOT_DEBUG_N/I3C_SCL5/MDC5	3.3V		GPIO[39]	T44
		I3C_SDA6/MDIO6/PWM1	3.3V		GPIO[40]	T29 I3C_SDA6_CX
		I3C_SCL6/MDC6/TACH1	3.3V		GPIO[41]	T40 I3C_SCL6_CX
		I2C_SDA10/SPI_SLAVE_CLK/PWM2	3.3V	PU	GPIO[42]	P33 I2C_SDA10_CX
		I2C_SCL10/SPI_SLAVE_CS/TACH2	3.3V	PU	GPIO[43]	P56 I2C_SCL10_CX
		PERST0#	3.3V		GPIO[44]	AH21 S_CX9_PERST_CX_L
		STRAP13_ENABLE_INBAND_RECOVERY/OVT_WRNG	3.3V	PU	GPIO[45]	R17 S_ENABLE_INBAND_RECOVERY
		I2C_SDA13/MDIO11/PWM_VID2	3.3V		GPIO[46]	R21 I2C_SDA13_CX
		I2C_SCL13/MDC11/PWM_VID3	3.3V		GPIO[47]	T19 I2C_SCL13_CX
		PPSIN	3.3V	PD	GPIO[48]	F19
		STRAP14_SECONDERY_PCORE/PPS_OUT	3.3V	PU	GPIO[49]	D19
		I2C_SDA11/MDIO9/PWM_VID0/PVS_MODE0	3.3V		GPIO[50]	T12
		I2C_SCL11/MDC9/PWM_VID1/PVS_MODE1	3.3V		GPIO[51]	P48
		I2C_SDA12/7/MDIO10/PERST1#	3.3V		GPIO[52]	P40 S_CX9_PERST_BIF_CX_L
		I2C_SCL12/MDC10/PERST2#	3.3V		GPIO[53]	V33
		PERST3#	3.3V		GPIO[54]	BB21

FUNCMODE 1.8V
JTAG_TCK 1.8V
JTAG_TDI 1.8V
JTAG_TDO 1.8V
JTAG_TMS 1.8V
JTAG_TRST# 1.8V

L_FUNCMODE V64
JTAG_TCK AM64
JTAG_TDI AH64
JTAG_TDO AM66
JTAG_TMS AT64
JTAG_TRST_L AH66

1V8 — R1587 10k 1% 0402 NA — R214 10k 1% 0402 COMMON

Should be placed
Phisically on the traces between
the ConnectX-9 and the SPI Flash

23> IN S_FNP_L — R2330 4.7k 0402 1% COMMON — S_FNP_R_L — R1030 4.7k 0402 1% NA — R1213 45.3k 1% 0402 COMMON

J10 2.92MM 2.54MM NORM NA

P-Up - No Core PLL Bypass
NA- Primary PCORE is PCORE0
NA- Primary PCORE is PCORE1
NA - Multi Host mode
NA - 2 PCORE Active
P-Up - 156.25MHz Core OSC FREQ0
P-Up - 156.25MHz Core OSC FREQ1
P-Up - Tester master active
P-Up - Boot debug
P-Up - Enable Inband recovery

1V8 — 3V3

R1041 4.7k — R1338 4.7k — R1339 4.7k — R3043 4.7k — R1013 4.7k — R1051 1k — R1052 100k — R1053 1.5k — R1080 1.5k — R1082 1.5k — R2331 4.7k — R2332 4.7k — R2333 4.7k — R1008 45.3k — R1024 1.5k — R1025 1.5k — R1265 4.7k — R1446 4.7k — R2310 49.9k

PD-Primary PCORE is PCORE0
PD-Primary PCORE is PCORE1
PD - SD
NA - 2 PCORE Active
L GP26 PULL RES
L GP27 PULL RES
L GP28 PULL RES
NA - 156.25MHz Core OSC FREQ0
NA - 156.25MHz Core OSC FREQ1
NA - Tester master active
NA - Boot debug
NA - Disable Inband recovery
PD - Secondary Pcore

R1107 4.7k — R1108 4.7k — R1109 4.7k — R1110 4.7k — R2209 45.3k — R30 45.3k — R1602 45.3k — R2383 45.3k — R3107 4.7k — R3108 4.7k — R3109 4.7k — R3110 4.7k — R3111 4.7k — R2332 1k

1V8 — R1021 4.7k 0402 1% NA — R1020 4.7k 0402 1% COMMON — R1022 4.7k 0402 1% COMMON — R1035 4.7k 0402 1% COMMON — R1034 4.7k 0402 1% NA — R1036 4.7k 0402 1% NA

GPIO[26] P. UP - 2 hosts per Pcore
GPIO[27] P. Down
GPIO[28] P. Down: No PCIe Lane Reversal for PEX0

GND

3V3 — C4150 0.1uF 16V X7R — GND
S_CX9_OVT_SHTDN — U268 SN74LV1T08DCKR — S_3V3_CX9_OVERT OUT 27< 39<

SPI Flash Memory

First Source: ISS - 170-0683-000

Second Source: Macronix - 170-0684-500

U9 IS25WP512MG-WHILE-TR_512Mb TBGA24_B COMMON

24> IN	SE_SPI_CS0_L	C2	CS*		
8> IN	SE_SPI_CLK	B2	SCK	VDD	B4
8<> BI	SE_SPI_IO0_STRP_PCI_PART0	D3	DO0		
8<> BI	SE_SPI_IO1_STRP_PCI_PART1	D2	DO1		
8<> BI	SE_SPI_IO2_STRP_PRIMARY_RVRSL	C4	DO2		
8<> BI	SE_SPI_IO3	D4	DQ3/HOLD*/RST*		
	SNN_FLASH_RST	A4	RST*/NC	VSS	B3
23> IN	S_SPI_WP_L	C5	WP*		
	SNN_FLASH_A2	A2	NC	NC	D1 SNN_FLASH_D1
	SNN_FLASH_A3	A3	NC	NC	D5 SNN_FLASH_D5
	SNN_FLASH_A5	A5	NC	NC	E1 SNN_FLASH_E1
	SNN_FLASH_B1	B1	NC	NC	E2 SNN_FLASH_E2
	SNN_FLASH_B5	B5	NC	NC	E3 SNN_FLASH_E3
	SNN_FLASH_C1	C1	NC	NC	E4 SNN_FLASH_E4
	SNN_FLASH_C3	C3	NC	NC	E5 SNN_FLASH_E5

1V8 — C3605 0.1uF 50V — GND

DC-TCXO Oscillator

Address 0x5A
SiTime
1 0 1 A1 0 A0 0

RAKON Address 0x50

3V3_RCLK_R — C1746 0.1uF 50V — GND

3V3 — R122 1.5k 0402 NA — R123 1.5k 0402 NA

Y14 SIT5977A-FQ20002 5MD10_D COMMON

9	VDD	CLKP	7 LVDS_RCLK_TCXO_P
1	NC / OE	CLKN	6 LVDS_RCLK_TCXO_N
2	SCLK		
10	SDA		
3	A1 / NC	NC	8 SNN_TCXO_NF
5	A0 / NC	GND	4

3V3_RCLK_R — R1206 4.7k 0402 COMMON — R1208 4.7k 0402 COMMON — R1207 4.7k 0402 NA — R1209 4.7k 0402 NA — GND

4.7k COMMON R1156 0402 — S_RCLK_TCXO_OE
I2C_SCL4_DC_TCXO
I2C_SDA4_DC_TCXO
S_TCXO_ADD_A1
S_TCXO_ADD_A0

I2C_SCL4_DC_TCXO — 0ohm R124 — I2C_SCL10_CX — 0ohm R3007 — I2C_SCL4_STRP_CORE_FREQ1
I2C_SDA4_DC_TCXO — 0ohm R3008 — I2C_SDA4_STRP_CORE_FREQ0 — 0ohm R30 — I2C_SDA10_CX

ConnectX-9 signal routing

		0ohm R3009	SE_NCSI_ARB_OUT	
			SE_NCSI_ARB_IN	IN 19>
			SE_NCSI_CLK	IN 19>
			SE_NCSI_TX_D0	IN 19>
			SE_NCSI_TX_D1	IN 19>
			SE_NCSI_TX_EN	IN 19>
		0ohm R3109	SE_NCSI_RX_D0	OUT 19<
		0ohm R3110	SE_NCSI_RX_D1	OUT 19<
		0ohm R3111	SE_NCSI_CRS_DV	OUT 19<
			S_INTERRUPT_GPIO_L	IN 24>
			S_CX9_OVT_SHTDN	OUT
			I2C_SDA0	BI 8<> 24<> 26<> 108<>
			I2C_SCL0	IN 8> 24< 26> 108>
			S_STRP_PRIMARY_PCORE0	
			S_STRP_PRIMARY_PCORE1	
			S_CX9_BOOT_DONE_L	
			S_INTERRUPT_GPIO_EXP2_L	OUT 28< 40<
			L_STRAP_SD_MH	IN 24>
			S_STRAP_2_PCORE_ACTIVE	
		0ohm R1459	SE_SPI_CLK	
		0ohm R1460	SE_SPI_CS0_L_CX_R	OUT 8<
			S_PRE_RESET_N	OUT 24<
		0ohm R2330	SE_SPI_IO0_STRP_PCI_PART0	IN 22>
		0ohm R2331	SE_SPI_IO1_STRP_PCI_PART1	BI 8<>
		0ohm R1464	SE_SPI_IO2_STRP_PRIMARY_RVRSL	BI 8<>
		0ohm R1465	SE_SPI_IO3	BI 8<>
			I2C_SDA1	BI 9<>
			I2C_SCL1	OUT 9<
		0ohm R2288	I2C_SDA2	BI 20<>
		0ohm R2289	I2C_SCL2	IN 20>
			I2C_SDA3	IN 9<>
			I2C_SCL3	BI 9<
		0ohm R1525	I2C_SDA4_STRP_CORE_FREQ0	OUT 9<
		0ohm R1526	I2C_SCL4_STRP_CORE_FREQ1	BI 24<>
			I2C_SDA5_STRP_TEST_MAS_ACT	OUT 24> 24<
			I2C_SCL5_STRP_BOOT_DBUG_N	OUT 24<>
		0ohm R1529	I3C_BMC_CX_SDA	OUT 24<
		0ohm R1530	I3C_BMC_CX_SCL	BI 26<>
		0ohm R3005	I2C_BMC_GRACE_MNG_SDA	IN 26>
		0ohm R3006	I2C_BMC_GRACE_MNG_SCL	BI 26< 64<>
		R2332 200ohm	S_CX9_PERST_PRI_L	IN 26> 64>
		0ohm R1359	S_ENABLE_INBAND_RECOVERY_L	IN 28< 112>
		0ohm R1531	I2C_SDA13	IN 23> 15<> 18<> 24<> 77<>
		0ohm R1532	I2C_SCL13	BI 15<> 18<> 17<> 18<> 19>
				OUT 23< 15< 17< 18< 19> 77<
			SE_PPS_IN	IN 24>
			SE_PPS_OUT	OUT 24<
			S_PORT0_PGOOD	IN 9>
			S_PORT1_PGOOD	IN 9>
		R2333 200ohm	S_CX9_PERST_BIF_L	IN 28< 112>
			S_PCORE1_PERST_L	IN 73>
			S_PCORE2_PERST_L	IN 73>

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	ConnectX-9 Control

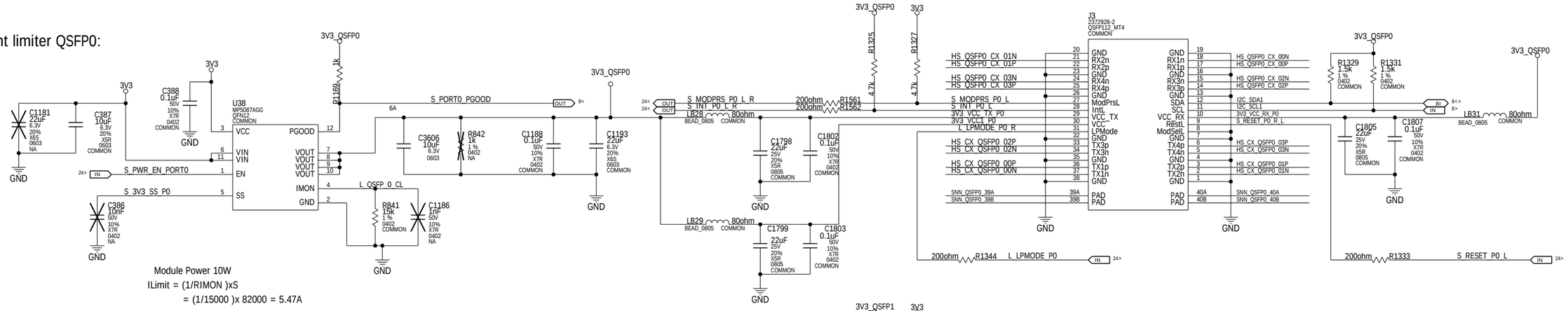
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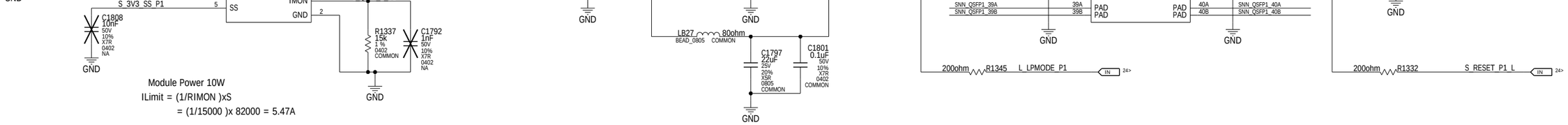
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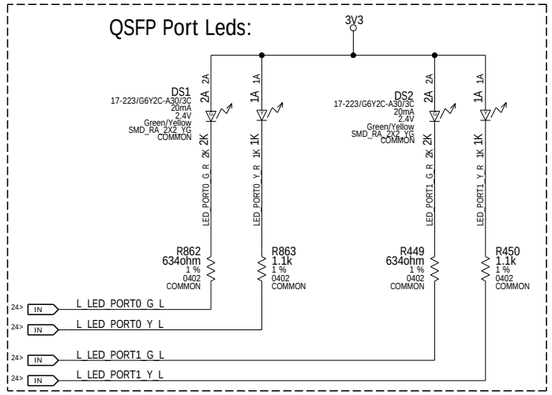
Current limiter QSFP0:



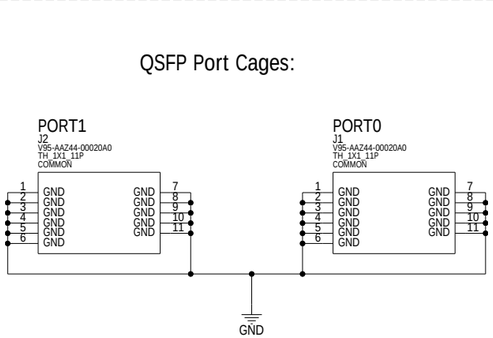
Current limiter QSFP1:



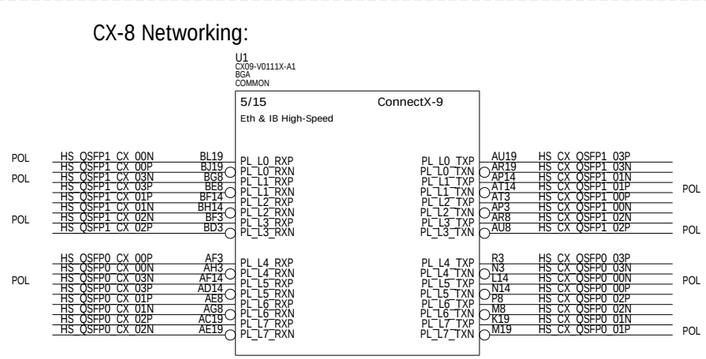
QSFP Port Leds:



QSFP Port Cages:



CX-8 Networking:



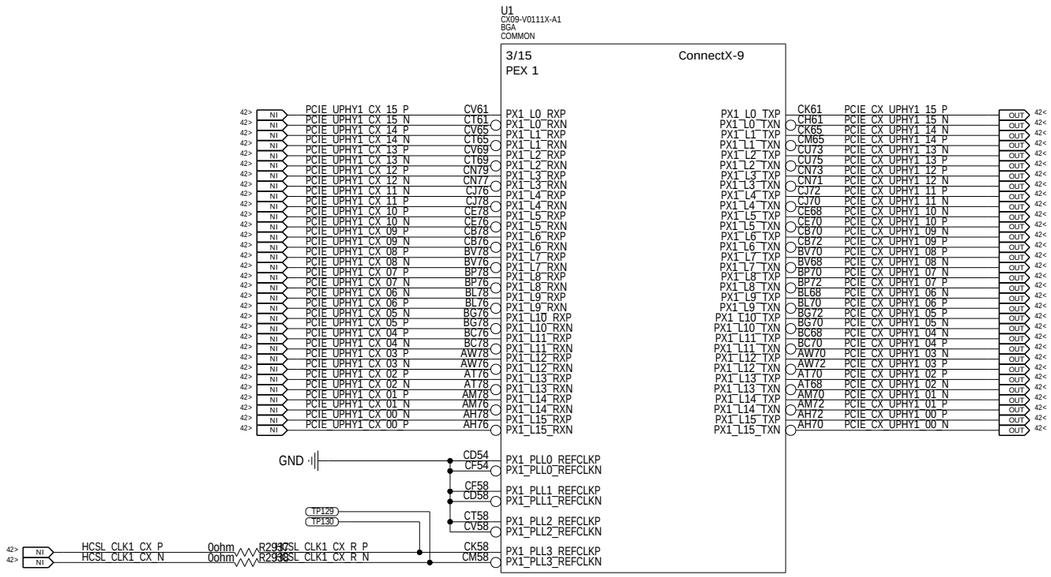
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Network Connectivity

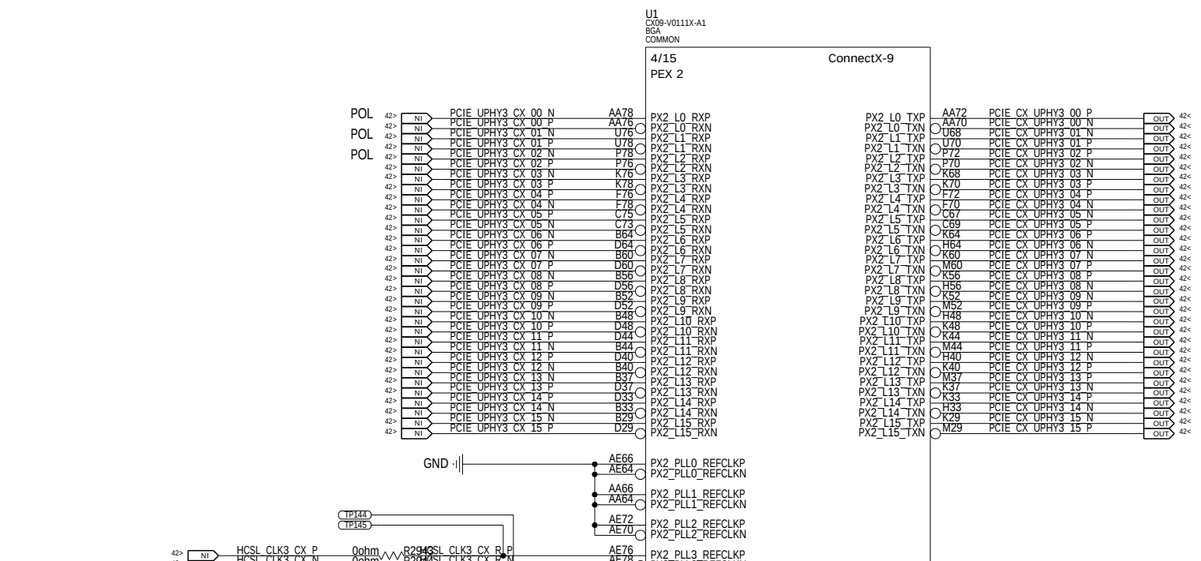
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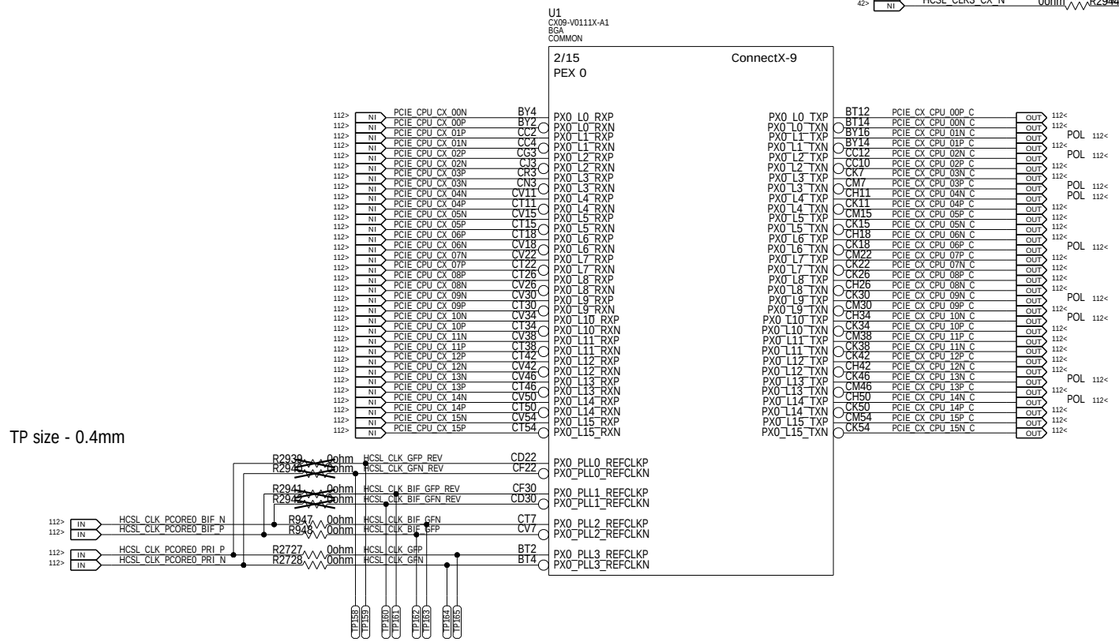
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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	ConnectX-9 PCIe

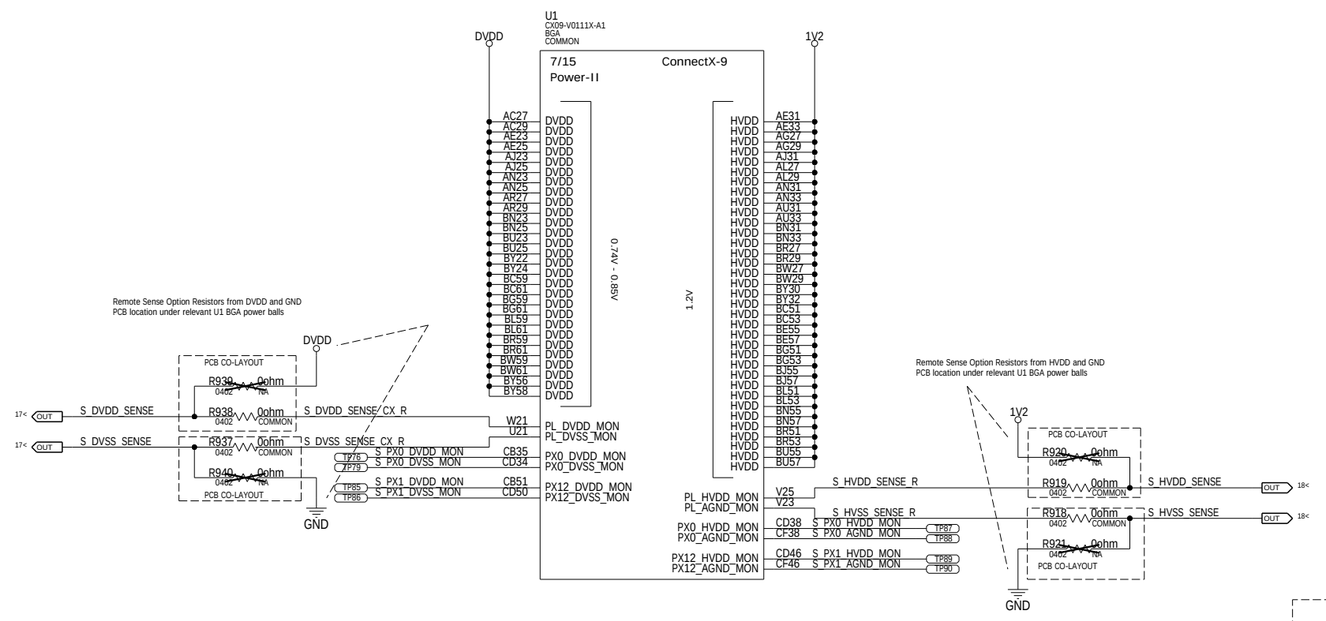
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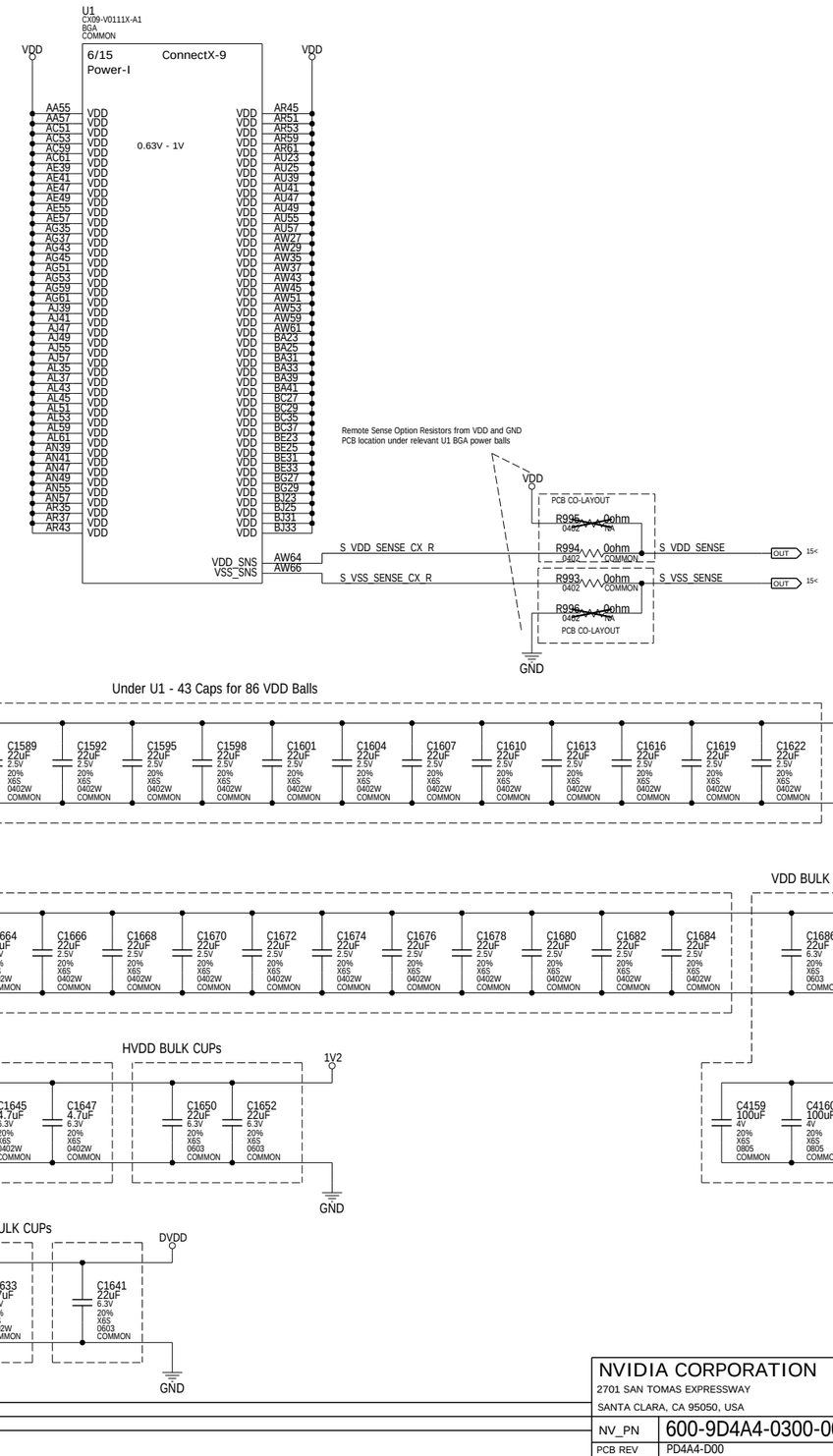
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ConnectX-9: Power #1





Under U1 - 43 Caps for 86 VDD Balls

C1747 C1589 C1592 C1595 C1598 C1601 C1604 C1607 C1610 C1613 C1616 C1619 C1622
22uF 2.5V 20% X6S 0402W COMMON

Under U1 - 43 Caps for 86 VDD Balls

C1625 C1628 C1631 C1634 C1636 C1639 C1642 C1644 C1646 C1648 C1649 C1651 C1653 C1654 C1655 C1656 C1658 C1660 C1662 C1664 C1666 C1668 C1670 C1672 C1674 C1676 C1678 C1680 C1682 C1684
22uF 2.5V 20% X6S 0402W COMMON

VDD BULK CUPs

C1686 C1688
22uF 6.3V 20% X6S 0603 COMMON

C4159 C4160 C4161
100uF 4V 20% X6S 0805 COMMON

Under U1 - 17 Caps for 34 HVDD Balls

C1602 C1605 C1608 C1611 C1614 C1617 C1620 C1623 C1626 C1629 C1632 C1635 C1637 C1640 C1643 C1645 C1647
4.7uF 6.3V 20% X6S 0402W COMMON

HVDD BULK CUPs

C1650 C1652
22uF 6.3V 20% X6S 0603 COMMON

Under U1 - 15 Caps for 28 DVDD Balls

C1591 C1594 C1597 C1600 C1603 C1606 C1609 C1612 C1615 C1618 C1621 C1624 C1627 C1630 C1633
4.7uF 6.3V 20% X6S 0402W COMMON

DVDD BULK CUPs

C1641
22uF 6.3V 20% X6S 0603 COMMON

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	ConnectX-9 Power

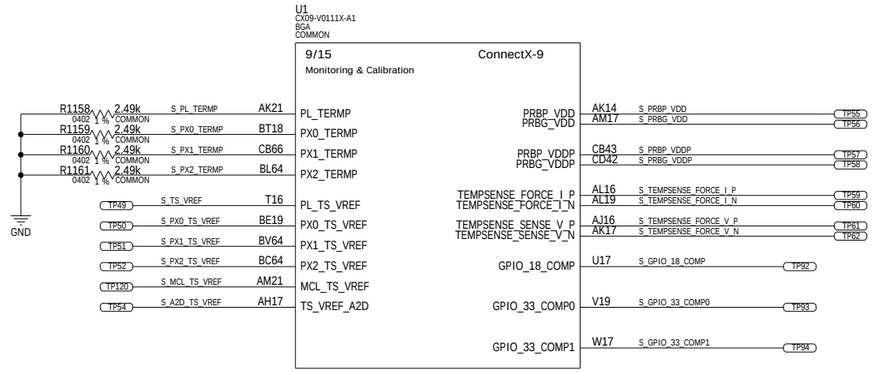
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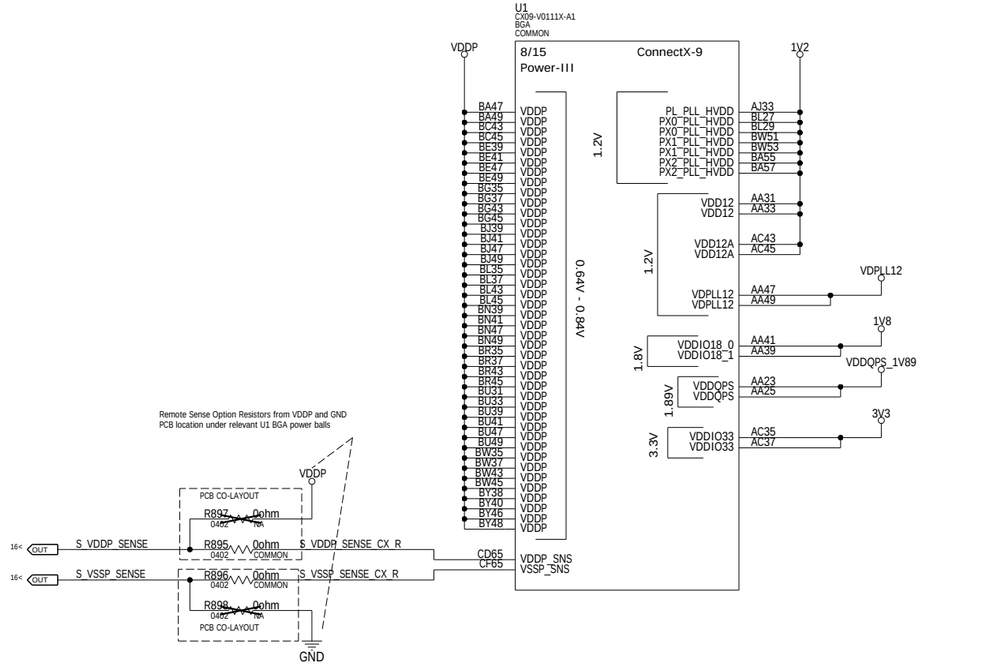
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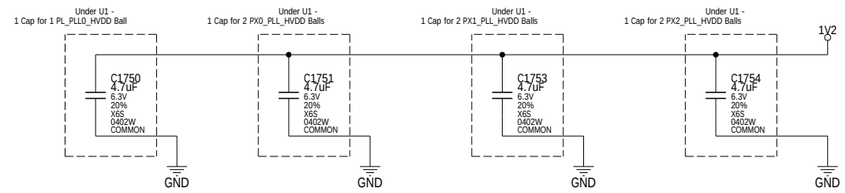
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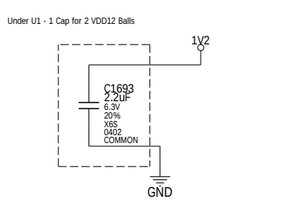
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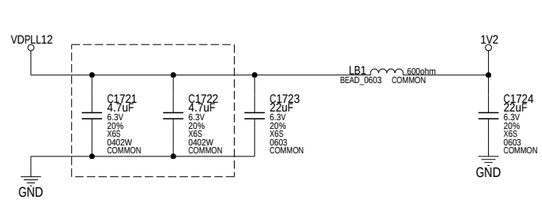
ConnectX-9: Power #2

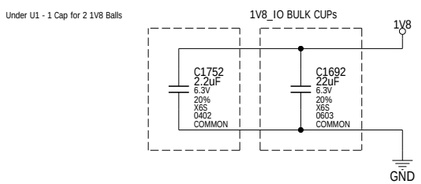


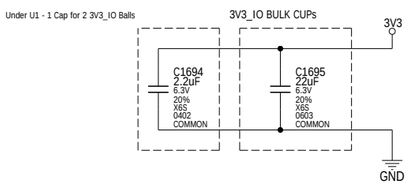


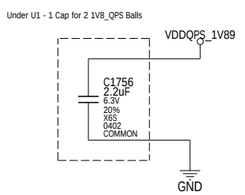


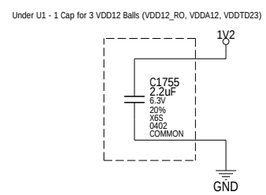


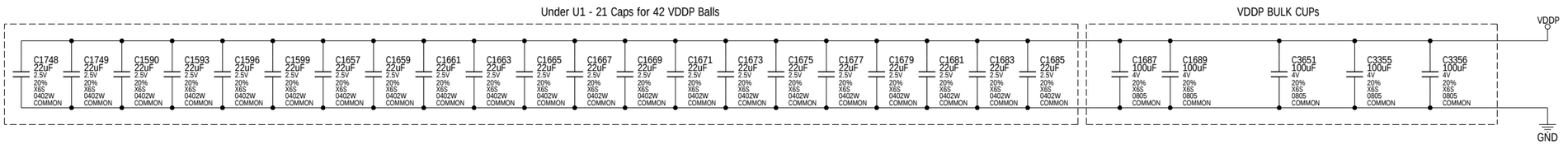












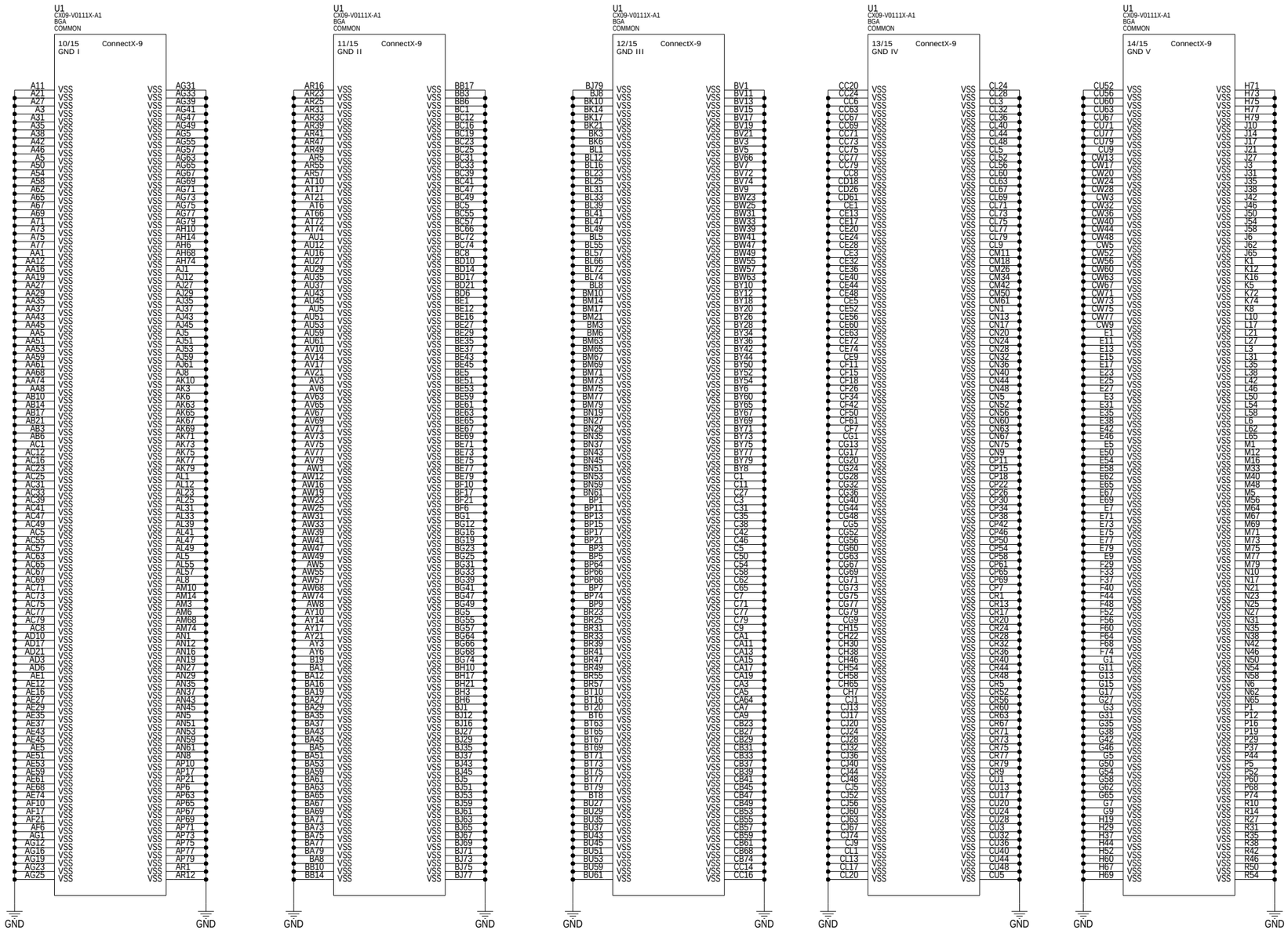
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	ConnectX-9 Power and Monitoring

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	ConnectX-9 GND

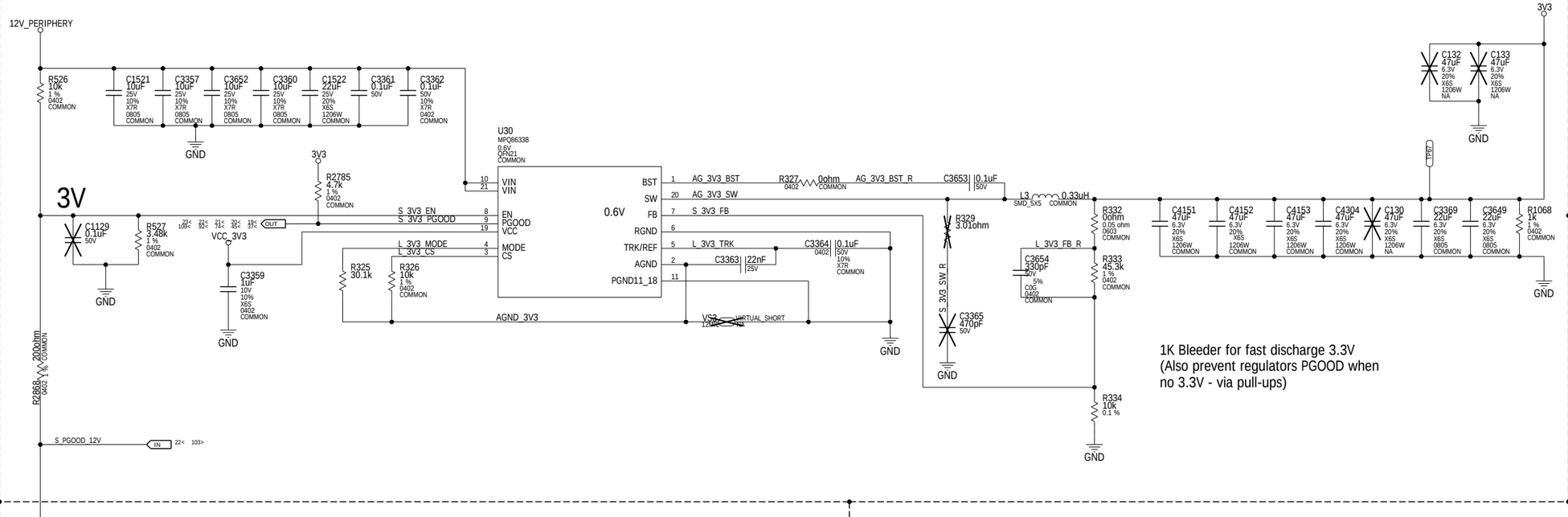
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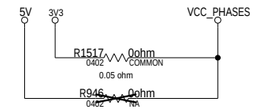
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Power: 3.3V

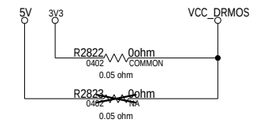


5V OR 3V3 VCC_PHASES



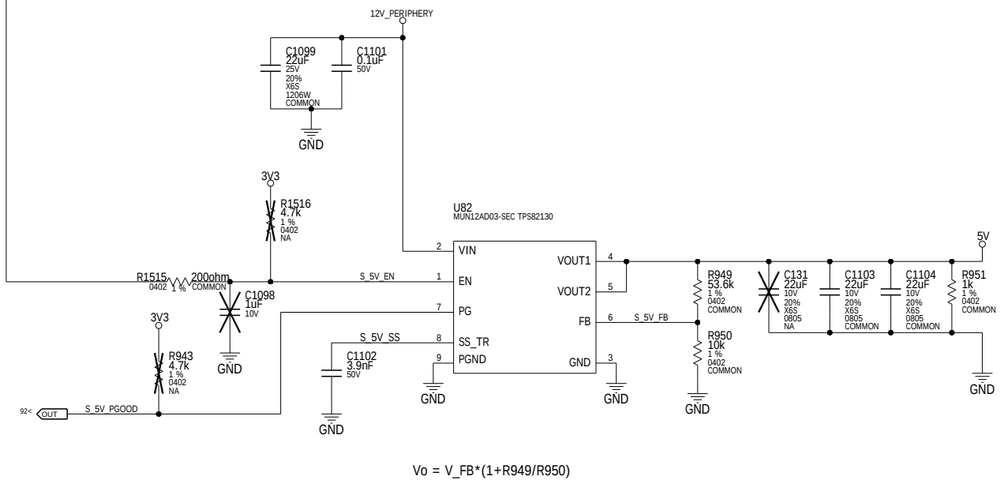
MPS - 3V3
Infinion - 5V

5V OR 3V3 VCC_DRMOS



MPS - 3V3
Infinion - 5V

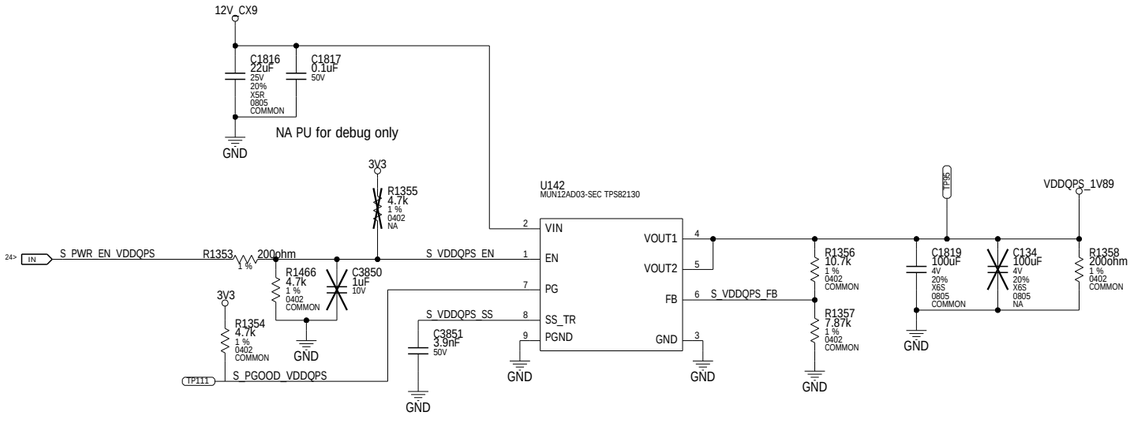
Power: 5V



Vo = V_FB*(1+R949/R950)

Power: VDDQPS

Power: 1.89V QPS



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Power 3V3, 5V, VDDQPS

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Address 0x6B

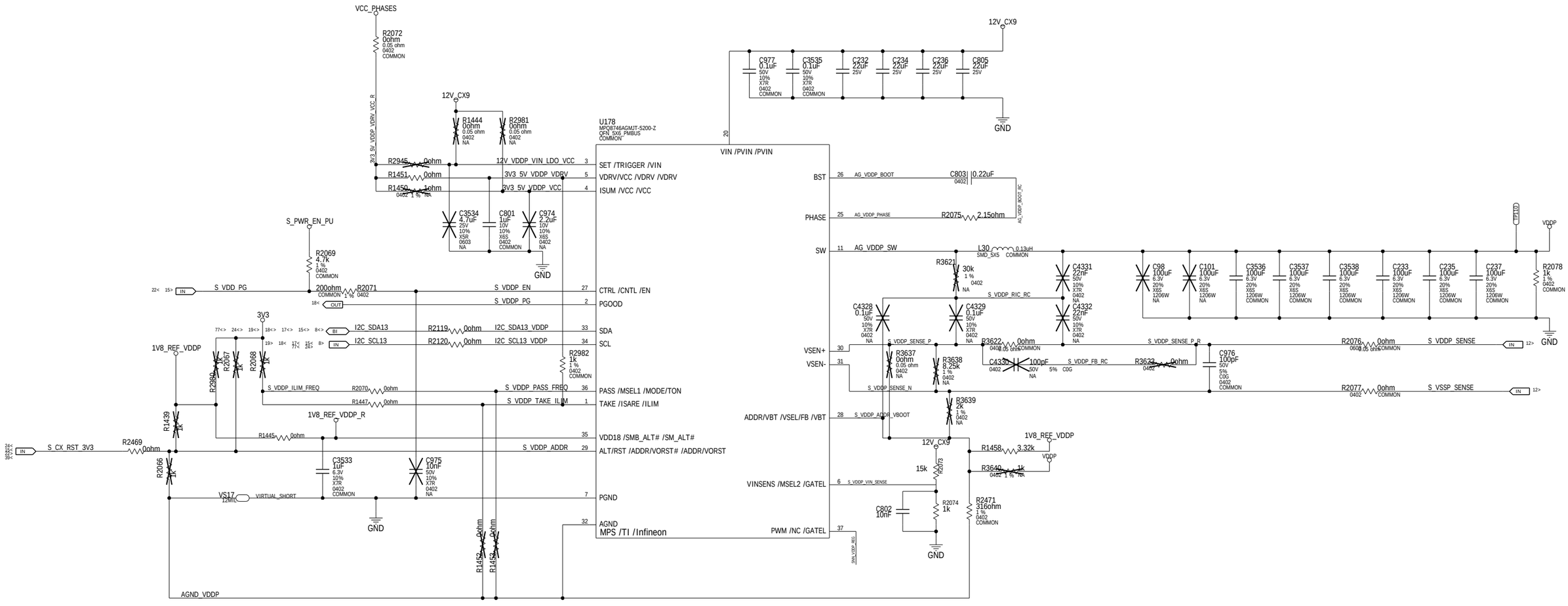
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Power VDD

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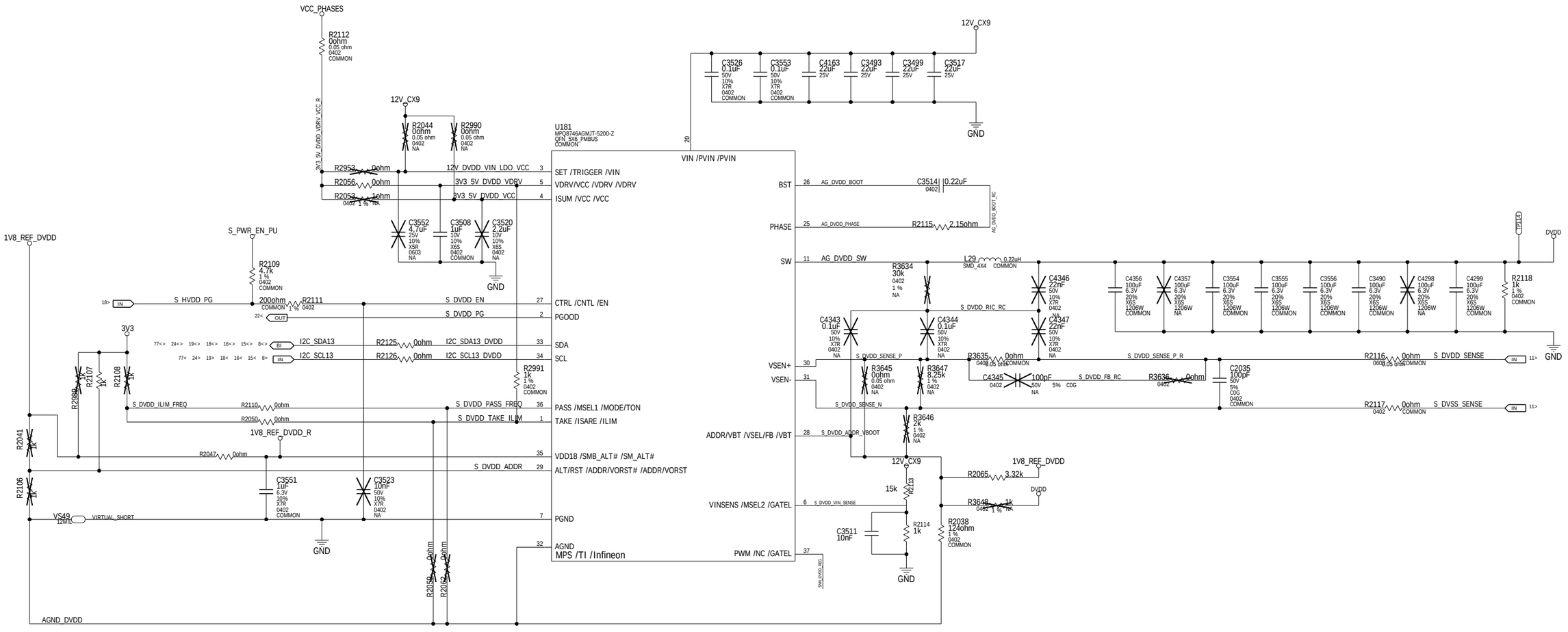
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Power VDDP

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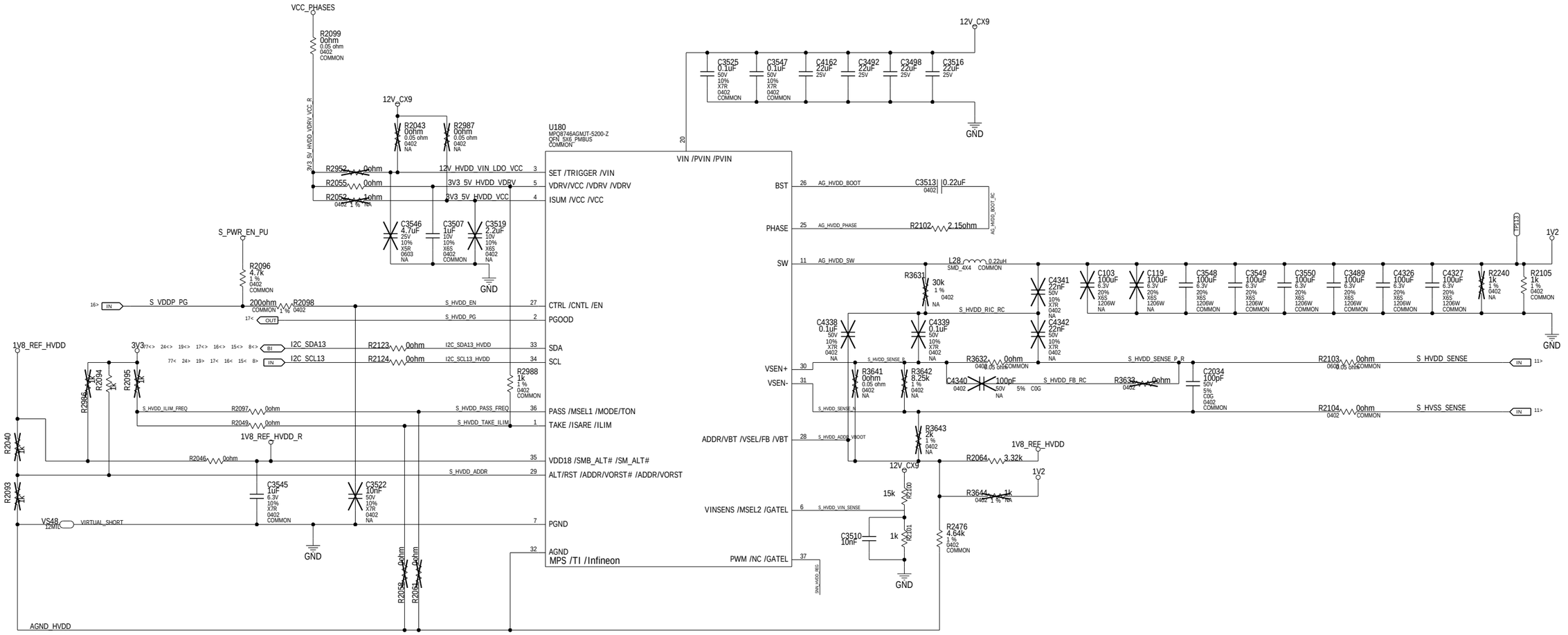
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Power DVDD

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Power HVDD

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NCSI MUX

BMC

CX9

CPLD

SMBUS GPIO

GRACE EROT

BMC OR MCU

USB ISOLATION

USB_HUB

BMC

SMBUS 3-pin Con

PMBUS 3-pin Con

PCB CO-LAYOUT

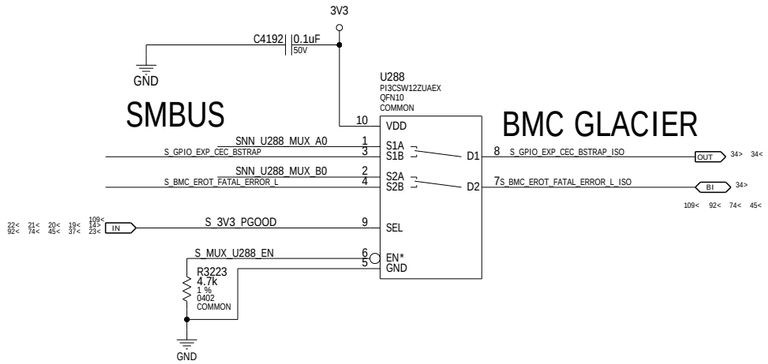
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Isolators

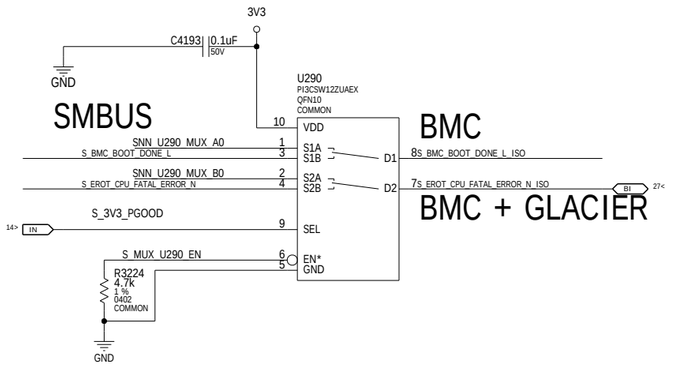
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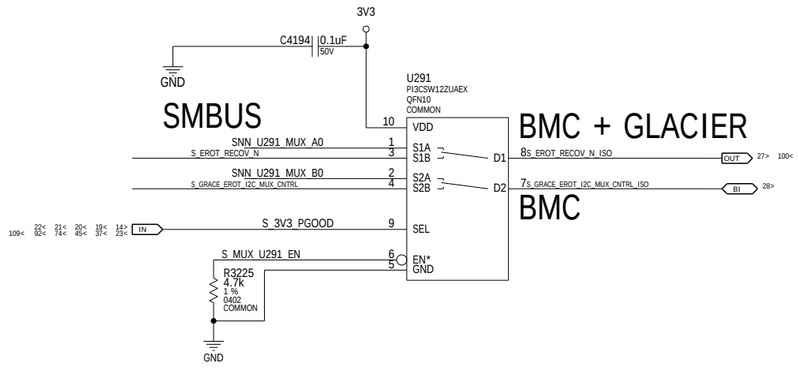
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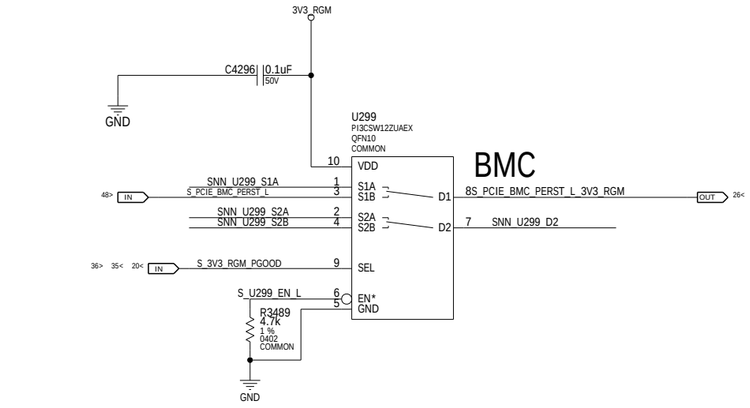
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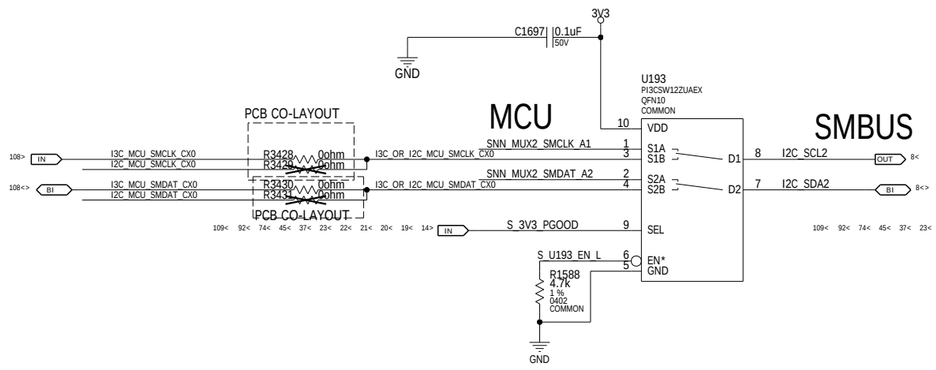
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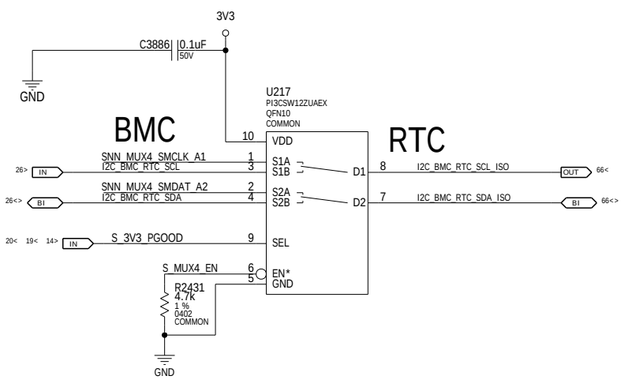


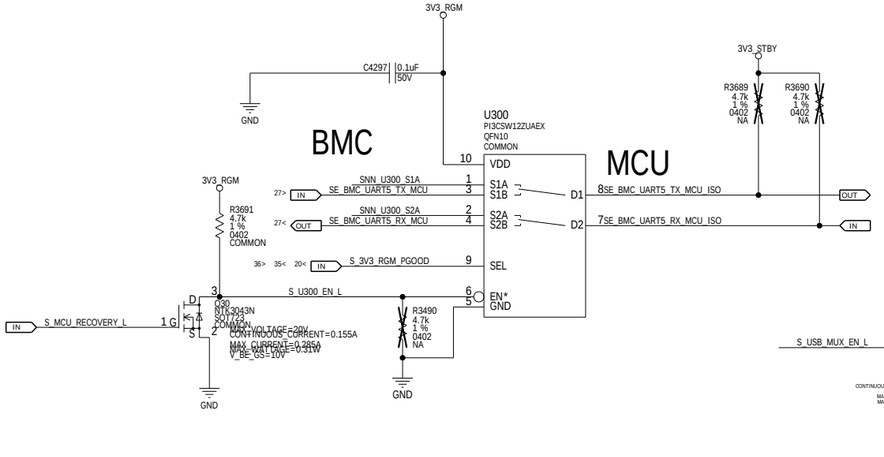


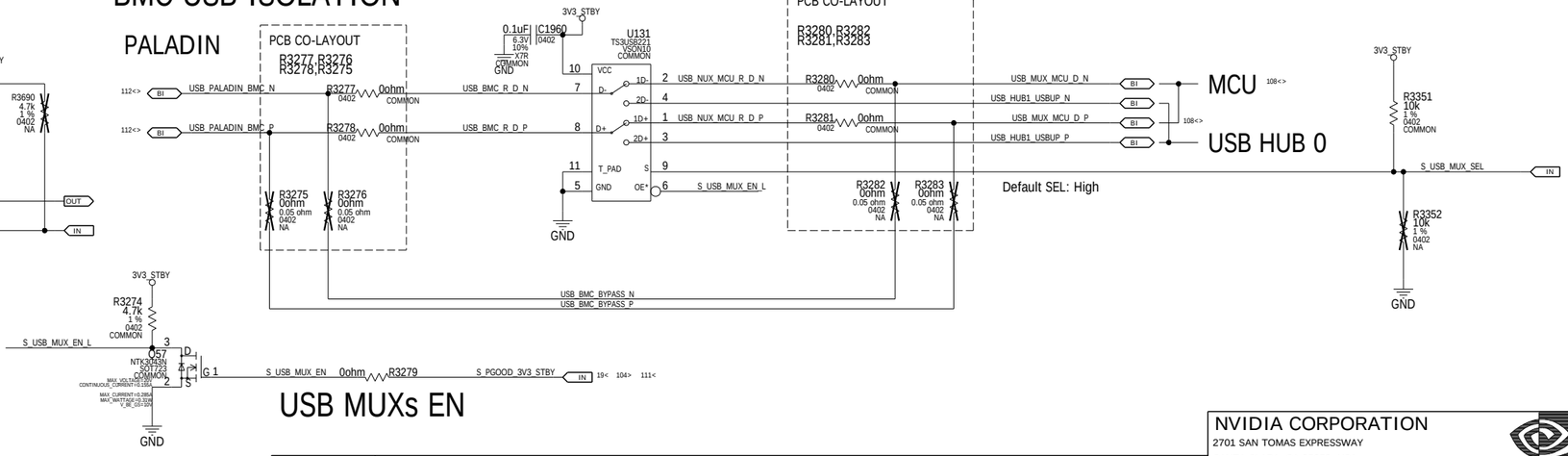












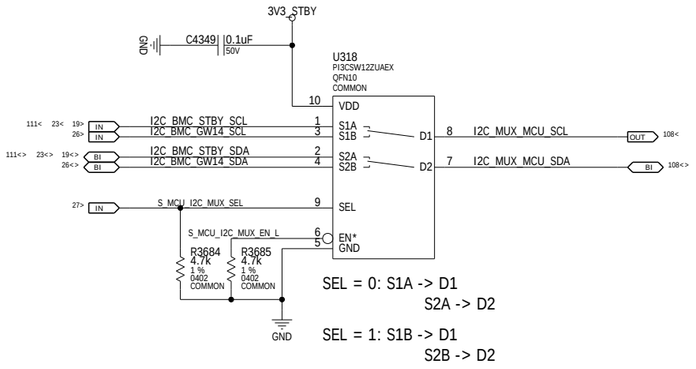
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Isolators #2

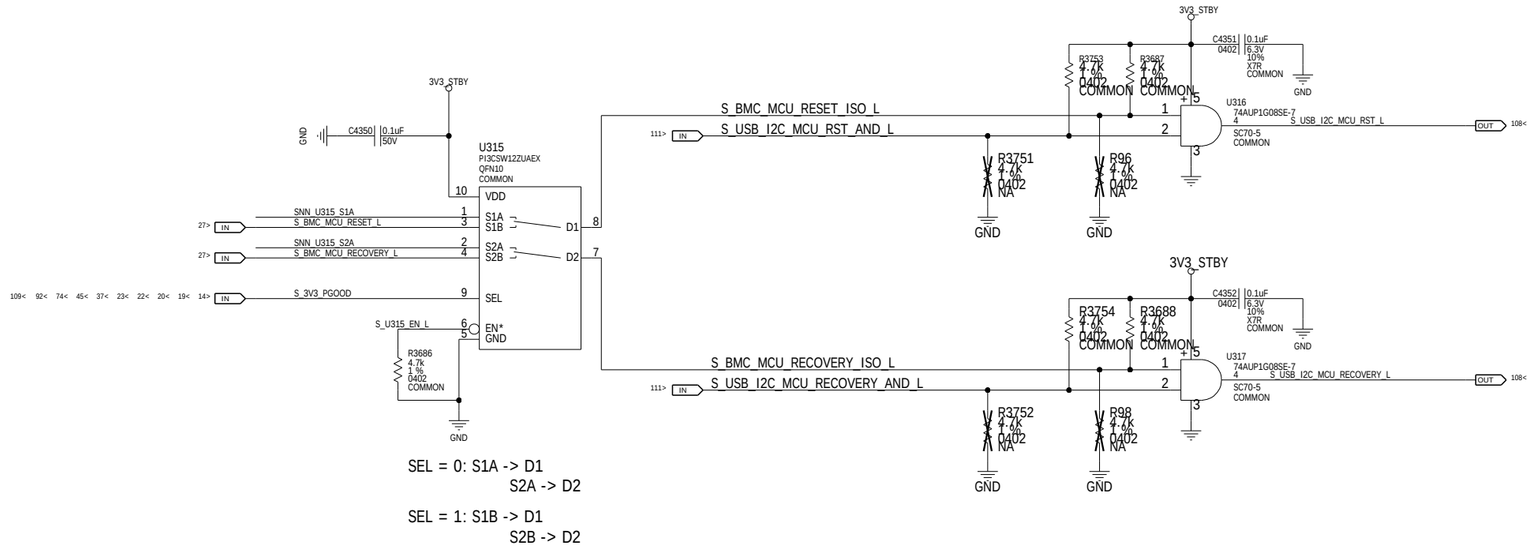
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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Isolators #3

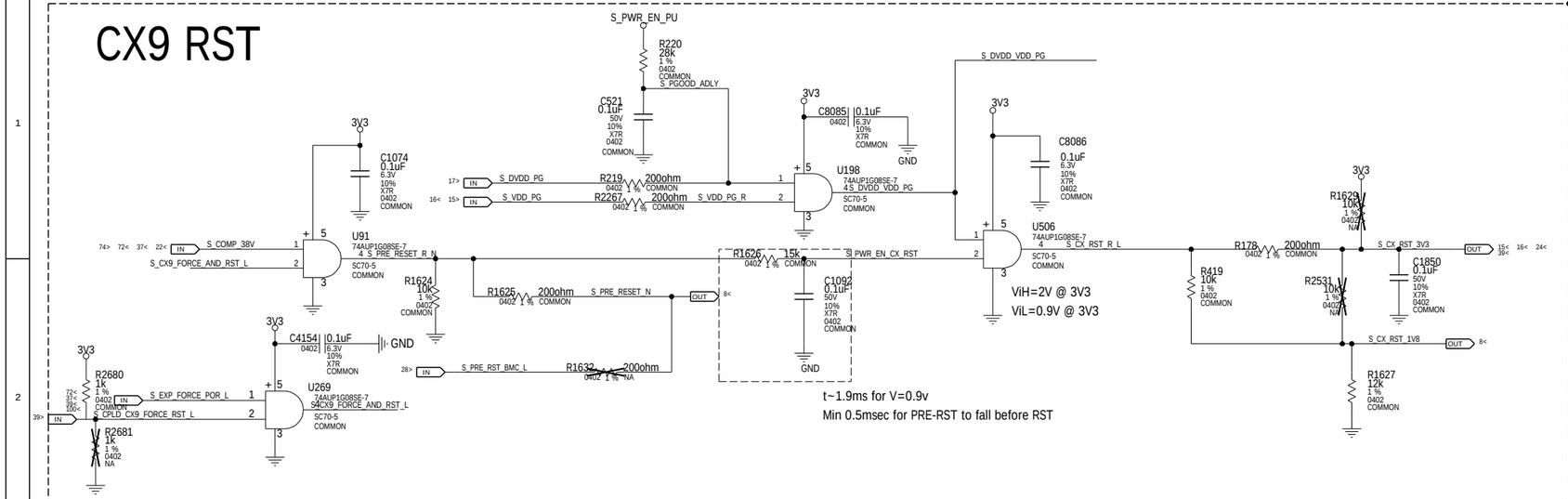
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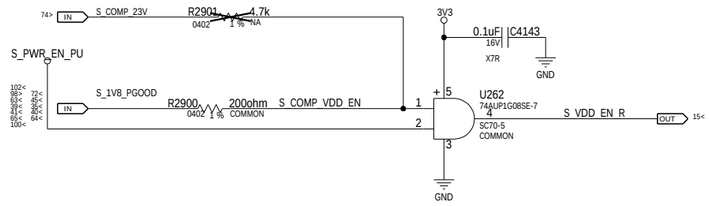
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CX9 RST





Board PG

PWR EN PU

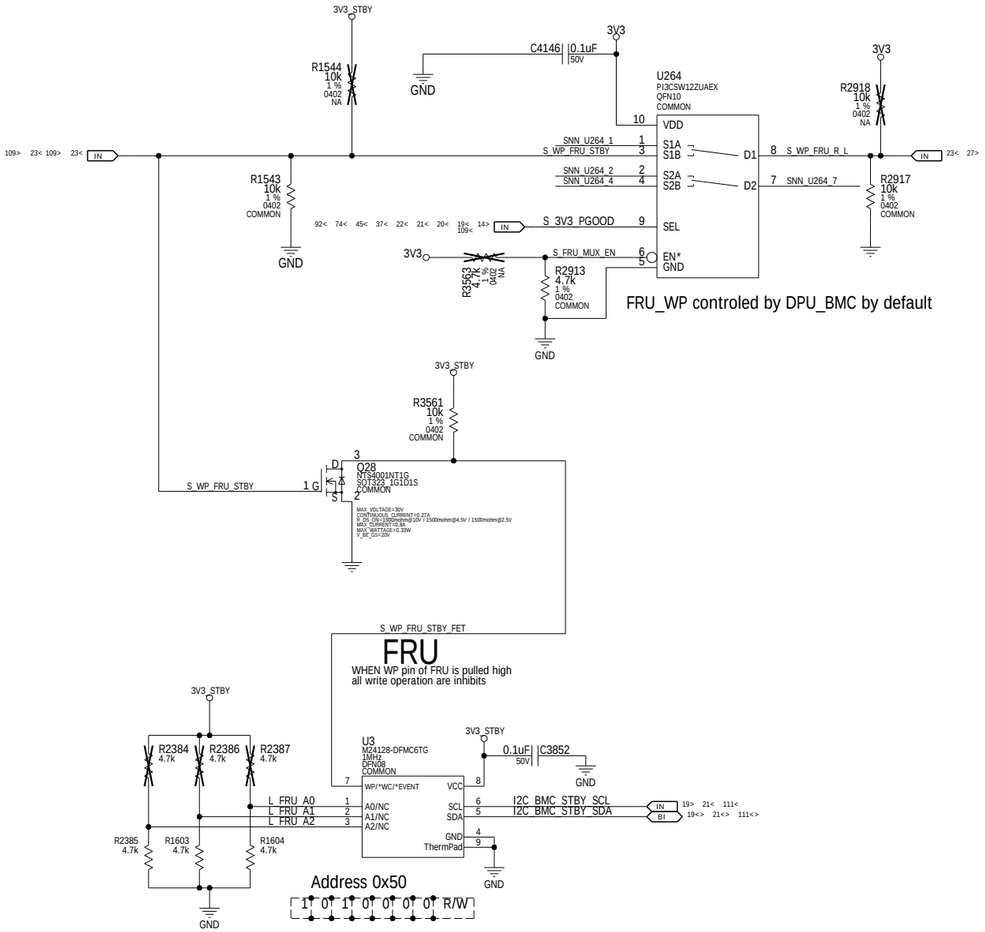
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Resets and PG

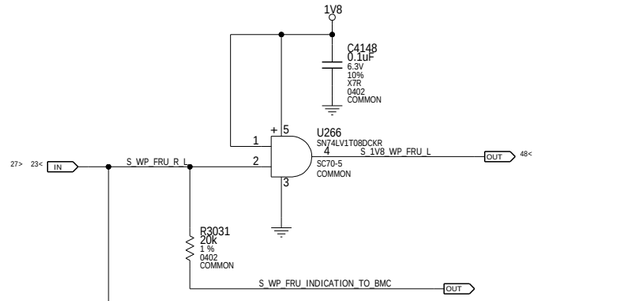
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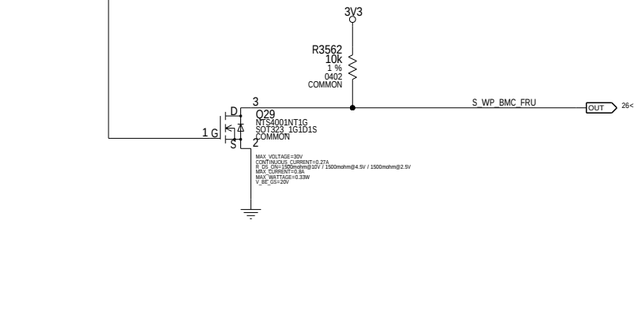
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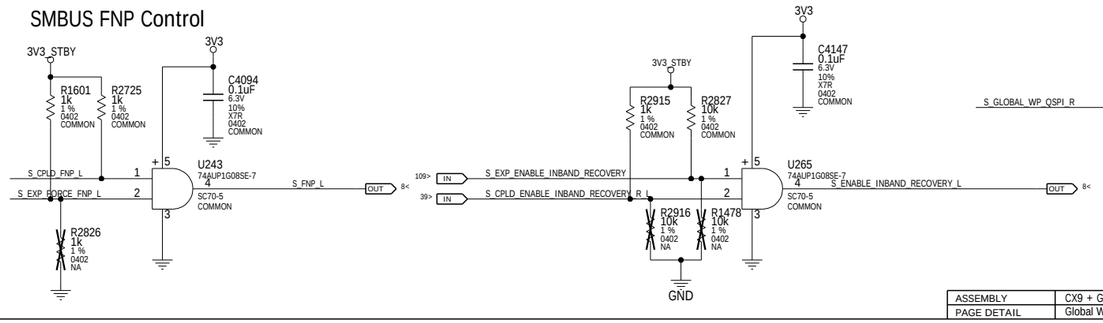
FRU

WHEN WP pin of FRU is pulled high
all write operation are inhibits

GLOBAL WP

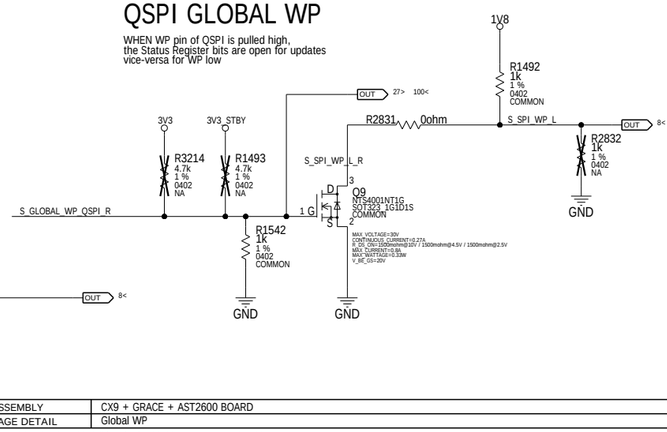
Default Mode defines that QSPI is open for updates
FRU is NOT open for updates

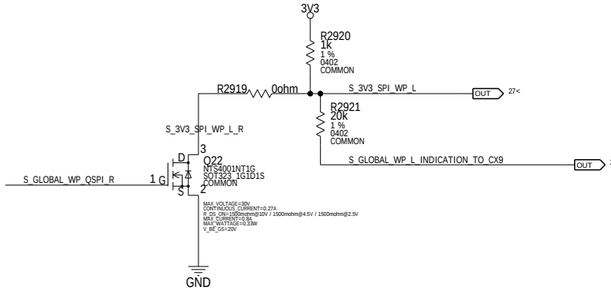
SMBUS INBAND RECOVERY Control



QSPI GLOBAL WP

WHEN WP pin of QSPI is pulled high,
the Status Register bits are open for updates
vice-versa for WP low





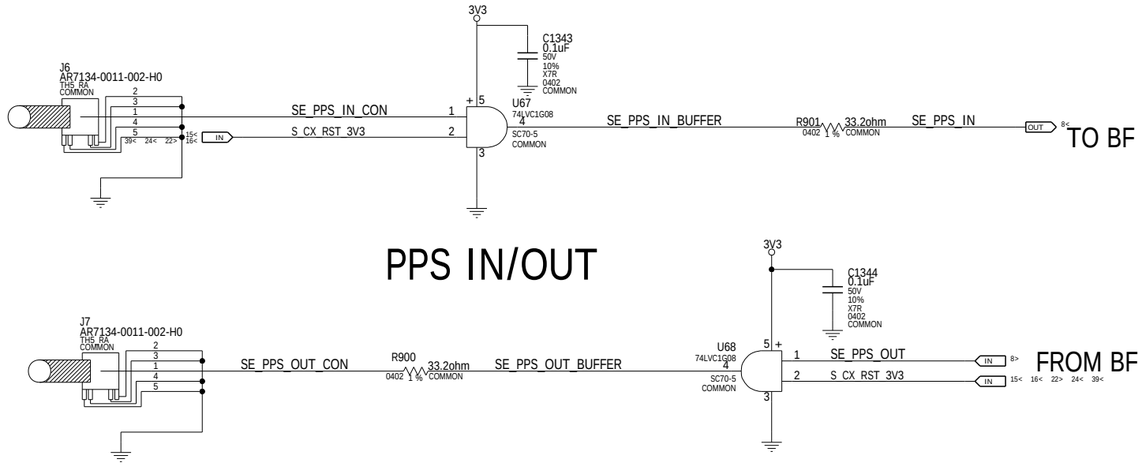
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Global WP

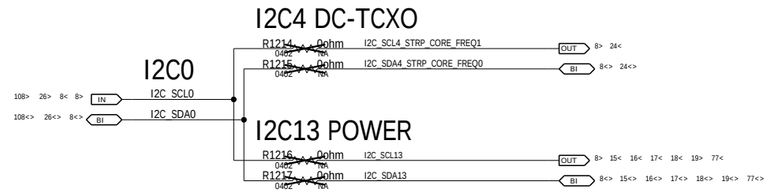
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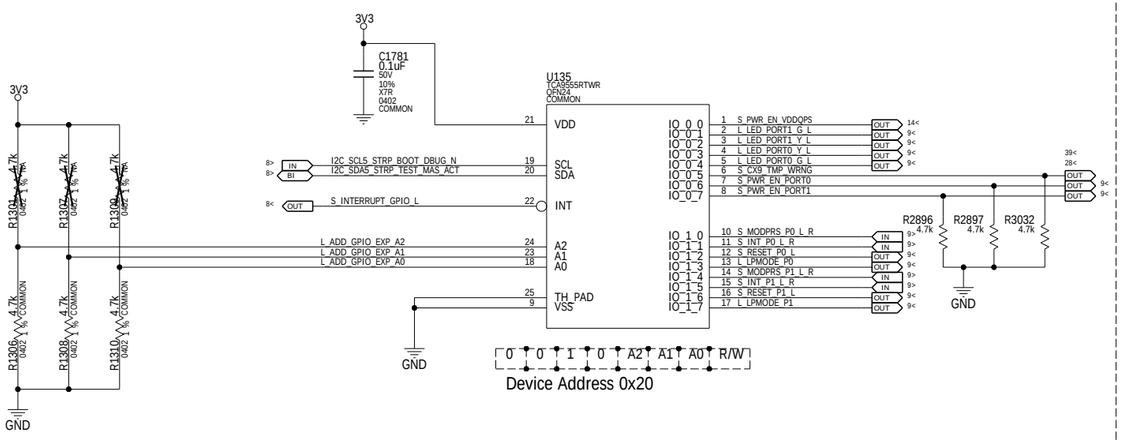
NV_PN	600-9D4A4-0300-000		
PCB REV	PD4A4-D00	PAGE	23 OF 112
BOM REV	M	DATE	16-JUL-2026

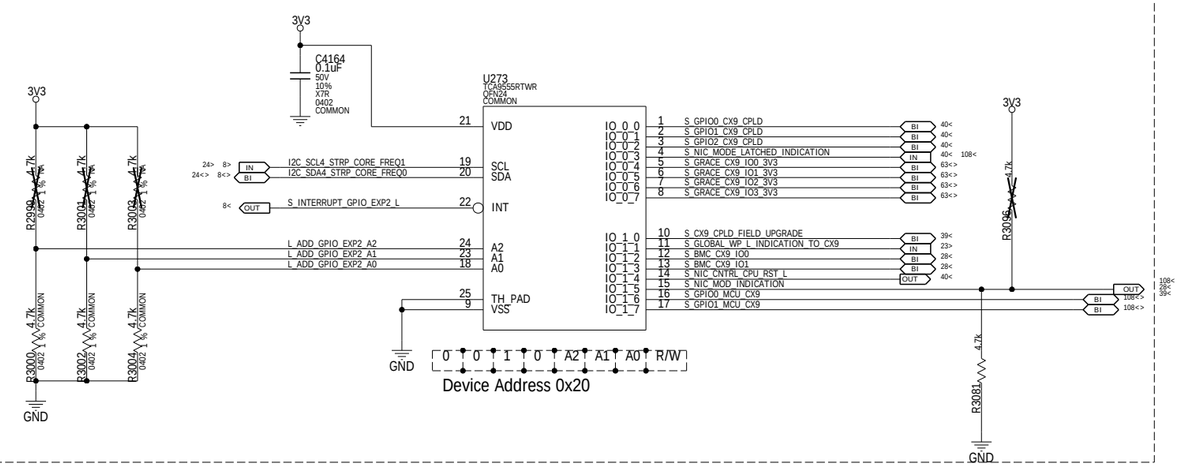
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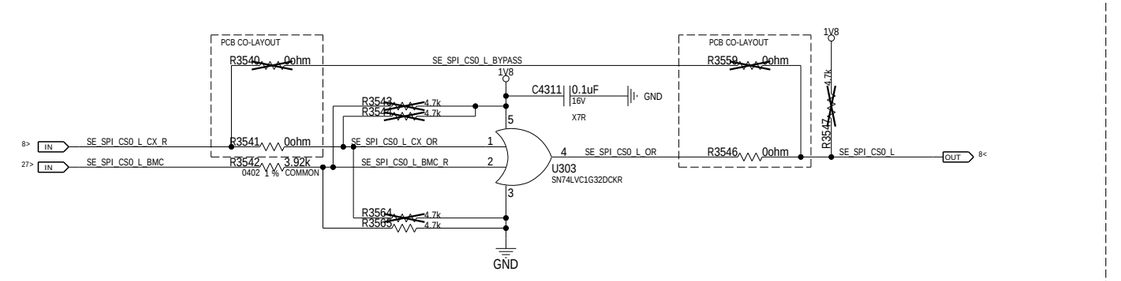




R1216 and R1217 should be located in an accessible location







ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	CX IOXs, PPS IN/OUT

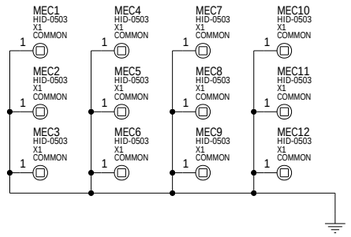
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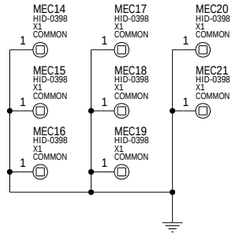
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PCB REV	PD4A4-D00	PAGE	24 OF 112
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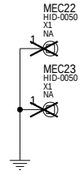
HOLE cold plate



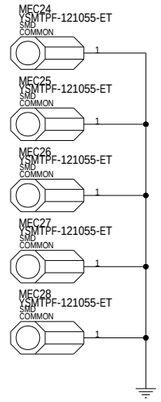
hole for asics cold plate



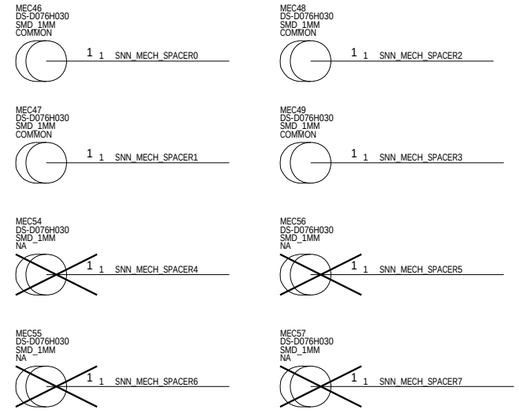
hole for guide pin



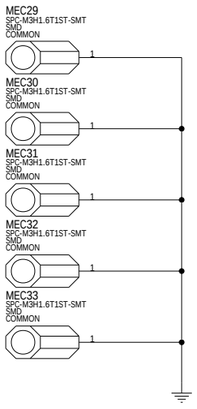
mounting screw to bay



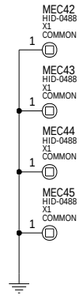
SPACERS



smt nut for M.2



plated holes



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	MECHANIC

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BMC #1

EEPROM OF BMC

I2C ADDR:0x50

U99
AST2600A3-GP
TFBGA624
COMMON

(1/6)

PCI-Express (End Point)	PERST	A7	S_PCIE_BMC_PERST_L_3V3_RGM
	PEREFCLK	AD5	HCSL_BMC_REFCLK_P
	PEREFCLK	AD6	HCSL_BMC_REFCLK_N
	PERX	AF5	PCIE_BMC_RXP
	PERX	AF6	PCIE_BMC_RXN
	PETX	AE4	PCIE_BMC_TXP
	PETX	AE5	PCIE_BMC_TXN
PCI-Express (Root Complex)	RCREFCLK	AE1	SNN_BMC_RCREFCLK
	RCREFCLK	AE2	SNN_BMC_RCREFCLK_L
	RCRX	AD2	SNN_BMC_RCRX
	RCRX	AD3	SNN_BMC_RCRX_L
	RCTX	AF2	SNN_BMC_RCTX
	RCTX	AF3	SNN_BMC_RCTX_L
Display Port	DPAUX	AC3	SNN_BMC_DPAUX
	DPAUX	AB3	SNN_BMC_DPAUX_L
	DPHPD	C7	GND
	DPTX0	AC1	SNN_BMC_DPTX0
	DPTX0	AB1	SNN_BMC_DPTX0_L
	DPTX1	AB2	SNN_BMC_DPTX1
	DPTX1	AA2	SNN_BMC_DPTX1_L
DDR4	MDQ0	T3	DDR4_BMC_DQ0
	MDQ1	R4	DDR4_BMC_DQ1
	MDQ2	U1	DDR4_BMC_DQ2
	MDQ3	R3	DDR4_BMC_DQ3
	MDQ4	R1	DDR4_BMC_DQ4
	MDQ5	P1	DDR4_BMC_DQ5
	MDQ6	P2	DDR4_BMC_DQ6
	MDQ7	P3	DDR4_BMC_DQ7
	MDQ8	W2	DDR4_BMC_DQ8
	MDQ9	V3	DDR4_BMC_DQ9
	MDQ10	W1	DDR4_BMC_DQ10
	MDQ11	W3	DDR4_BMC_DQ11
	MDQ12	W4	DDR4_BMC_DQ12
	MDQ13	U4	DDR4_BMC_DQ13
	MDQ14	V4	DDR4_BMC_DQ14
	MDQ15	U3	DDR4_BMC_DQ15
	MDQS0	T2	DDR4_BMC_DQS0_P
	MDQS1	V2	DDR4_BMC_DQS1_P
	MDQS0	T1	DDR4_BMC_DQS0_N
	MDQS1	V1	DDR4_BMC_DQS1_N
	MRESET	L3	DDR4_BMC_RESET_L
	MALERT	N4	DDR4_BMC_ALERT_L
	MVREF	T5	0.6V_BMC_VREFF
	MVREF	U5	
	MIOZ	P5	L_DDR4_BMC_MIOZ
	MCK	K2	DDR4_BMC_CK_P
	MCK	K1	DDR4_BMC_CK_N
	MCKE	L5	DDR4_BMC_CKE
	MODT	J1	DDR4_BMC_ODT
	MCS	H2	DDR4_BMC_CS_L
	MRAS	J5	DDR4_BMC_RAS_L
	MCAS	J4	DDR4_BMC_CAS_L
	MWE	G5	DDR4_BMC_WE_L
	MBA0	G4	DDR4_BMC_BA0
	MBA1	H5	DDR4_BMC_BA1
	MBA2_MBG0	G3	DDR4_BMC_BG0
	MA0	F3	DDR4_BMC_A0
	MA1	K5	DDR4_BMC_A1
	MA2	F1	DDR4_BMC_A2
	MA3	H3	DDR4_BMC_A3
	MA4	J3	DDR4_BMC_A4
	MA5	L1	DDR4_BMC_A5
	MA6	M5	DDR4_BMC_A6
	MA7	M2	DDR4_BMC_A7
	MA8	M1	DDR4_BMC_A8
	MA9	N1	DDR4_BMC_A9
	MA10	F2	DDR4_BMC_A10
	MA11	G1	DDR4_BMC_A11
	MA12	L4	DDR4_BMC_A12
	MA13	K3	DDR4_BMC_A13
	MA14_MACT*	H1	DDR4_BMC_ACT_L
	MA15_MBG1	M3	SNN_BMC_DDR_MBG1
	MDM0	N3	DDR4_BMC_DBI_B0
	MDM1	R5	DDR4_BMC_DBI_B1
LPC ESPI	GPIOW4_LCLK_ESPICK	AE7	L_BMC_NC_PD3
	GPIOW5_LFRAME*_ESPICS*	AF7	
	GPIOW6_LSIRQ*_ESPIALT*	AD7	
	GPIOW7_LPCRST*_ESPIRST*	AD8	
	GPIOW0_LAD0_ESPID0	AB7	S_BMC_STRAP_D0
	GPIOW1_LAD1_ESPID1	AB8	S_BMC_STRAP_D1
	GPIOW2_LAD2_ESPID2	AC8	S_BMC_STRAP_D2
	GPIOW3_LAD3_ESPID3	AC7	S_BMC_STRAP_D3
SD1/PWM	GPIOF0_SD1CLK_PWM8	D22	L_BMC_NC_PD1
	GPIOF1_SD1CMD_PWM9	E22	
	GPIOF2_SD1DAT0_PWM10	D23	
	GPIOF3_SD1DAT1_PWM11	C23	
	GPIOF4_SD1DAT2_PWM12	C22	
	GPIOF5_SD1DAT3_PWM13	A25	
	GPIOF6_SD1CD*_PWM14	A24	
	GPIOF7_SD1WP*_PWM15	A23	
I2C/I3C	GPIOJ0_SCL1_HVI3C3SCL	B20	I3C_BMC_CX_SCL
	GPIOJ1_SDA1_HVI3C3SDA	A20	I3C_BMC_CX_SDA
	GPIOJ2_SCL2_HVI3C4SCL	E19	I2C_BMC_GRACE_SSIF_SCL
	GPIOJ3_SDA2_HVI3C4SDA	D20	I2C_BMC_GRACE_SSIF_SDA
	GPIOJ4_SCL3_HVI3C5SCL	C19	I2C_BMC_ETH_SW_SCL_ISO
	GPIOJ5_SDA3_HVI3C5SDA	A19	I2C_BMC_ETH_SW_SDA_ISO
	GPIOJ6_SCL4_HVI3C6SCL	C20	I2C_SSD_AUX_SCL
	GPIOJ7_SDA4_HVI3C6SDA	D19	I2C_SSD_AUX_SDA
SD2/UART	GPIOG0_TXD6_SD2CLK_SALT9	E21	
	GPIOG1_RXD6_SD2CMD_SALT10	B22	
	GPIOG2_TXD7_SD2DAT0_SALT11	C21	
	GPIOG3_RXD7_SD2DAT1_SALT12	A22	
	GPIOG4_TXD8_SD2DAT2_SALT13	A21	
	GPIOG5_RXD8_SD2DAT3_SALT14	E20	
	GPIOG6_TXD9_SD2CD*_SALT15	D21	
	GPIOG7_RXD9_SD2WP*_SALT16	B21	
I2C/MII	GPIOK0_SCL5	A11	I2C_BMC_FRU_SCL
	GPIOK1_SDA5	C11	I2C_BMC_FRU_SDA
	GPIOK2_SCL6	D12	I2C_BMC_RTC_SCL
	GPIOK3_SDA6	E13	I2C_BMC_RTC_SDA
	GPIOK4_SCL7	D11	I2C_BMC_GRACE_MNG_SCL
	GPIOK5_SDA7	E11	I2C_BMC_GRACE_MNG_SDA
	GPIOK6_SCL8	F13	I2C_BMC_MCU_SCL
	GPIOK7_SDA8	E12	I2C_BMC_MCU_SDA
	GPIOL0_SCL9	D15	I2C_BMC_GRACE_GW0_SCL
	GPIOL1_SDA9	A14	I2C_BMC_GRACE_GW0_SDA
	GPIOL2_SCL10	E15	I2C_CEC_GRACE_SCL_BMC
	GPIOL3_SDA10	A13	I2C_CEC_GRACE_SDA_BMC
	GPIOA0_MDC3_SCL11	M24	I2C_TERM_SCL
	GPIOA1_MDIO3_SDA11	M25	I2C_TERM_SDA
	GPIOA2_MDC4_SCL12	L26	I2C_INA_GW12_SCL
	GPIOA3_MDIO4_SDA12	K24	I2C_INA_GW12_SDA
I2C/SGPIO	GPIOH0_SGPM1CK	A18	S_BMC_GRACE_RESET
	GPIOH1_SGPM1LD	B18	SE_NIC_EEPROM_SCLK_BMC
	GPIOH2_SGPM1O	C18	SE_NIC_EEPROM_CS_BMC
	GPIOH3_SGPM1I	A17	S_SYS_RST_IN_L
	GPIOH4_SGPS1CK_SCL15	D18	I2C_BMC_SCL0
	GPIOH5_SGPS1LD_SDA15	B17	I2C_BMC_SDA0
	GPIOH6_SGPS1I0_SCL16	C17	I2C_CEC_AP_SCL_R
	GPIOH7_SGPS1I1_SDA16	E18	I2C_CEC_AP_SDA_R
I2C/MACLINK/SGPIO	GPIOA4_MAC1LINK_SCL13_SGPS2CK_SGPM2CK	K26	S_PEXVDD_EN
	GPIOA5_MAC2LINK_SDA13_SGPS2LD_SGPM2LD	L24	SE_NIC_EEPROM_DIDO_BMC
	GPIOA6_MAC3LINK_SCL14_SGPS2I0_SGPM2O	L23	I2C_BMC_GW14_SCL
	GPIOA7_MAC4LINK_SDA14_SGPS2I1_SGPM2I	K25	I2C_BMC_GW14_SDA
GPIO	GPIOS2_PEWAKE*	T26	SNN_BMC_PEWAKE
	GPIOS3_OSCCLK	R24	SNN_BMC_OSCCLK

R2894 10k 1 % 0402 NA
R2893 10k 1 % 0402 COMMON
R577 4.7k 1 % 0402 COMMON
C3860 0.1uF 50V 10% X7R 0402 COMMON
R6 240ohm 1 % 0402 COMMON
R1454 100k 1 % 0402 COMMON
R774 100k 1 % 0402 COMMON
R2927 1.5k
R2928 1.5k
R2435 1.5k
R2436 1.5k
R2435 0ohm
R2436 0ohm
C149 0.1UF 50V 10% X7R 0402 COMMON
U51 M24128-DFMC6TG UMFC DFN8 COMMON
R2339 4.7k
R2341 4.7k
R2343 4.7k
R2340 4.7k
R2342 4.7k
R2344 4.7k
L_FRU_BMC_A0
L_FRU_BMC_A1
L_FRU_BMC_A2
I2C_BMC_FRU_SCL
I2C_BMC_FRU_SDA

R1972 4.7k 1 % 0402 NA
R1974 4.7k 1 % 0402 NA
R1976 4.7k 1 % 0402 NA
R1978 4.7k 1 % 0402 NA
R1973 4.7k 1 % 0402 COMMON
R1975 4.7k 1 % 0402 COMMON
R1977 4.7k 1 % 0402 COMMON
R1979 4.7k 1 % 0402 COMMON
S_BMC_STRAP_D0
S_BMC_STRAP_D1
S_BMC_STRAP_D2
S_BMC_STRAP_D3

R2437 33.2ohm
R2438 33.2ohm
I2C_MCU_BMC_BMCCEC_GREAECEC_SCL
I2C_MCU_BMC_BMCCEC_GREAECEC_SDA

R2539 0ohm
R2540 0ohm
I2C_CEC_GRACE_SCL_BMC_R
I2C_CEC_GRACE_SDA_BMC_R

R1394 1.5k
R1395 1.5k
R244 100k 1 % 0402 NA
R245 100k 1 % 0402 NA

TO ENABLE BSDL MODE

PIN NMAE	DESCRIPTION
ENTEST	force to '1'
GPIOA2	force to '1'
GPIOA3	force to '0'
TXD5	force to '0'
RXD5	force to '1'

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	BMC AST2600 1

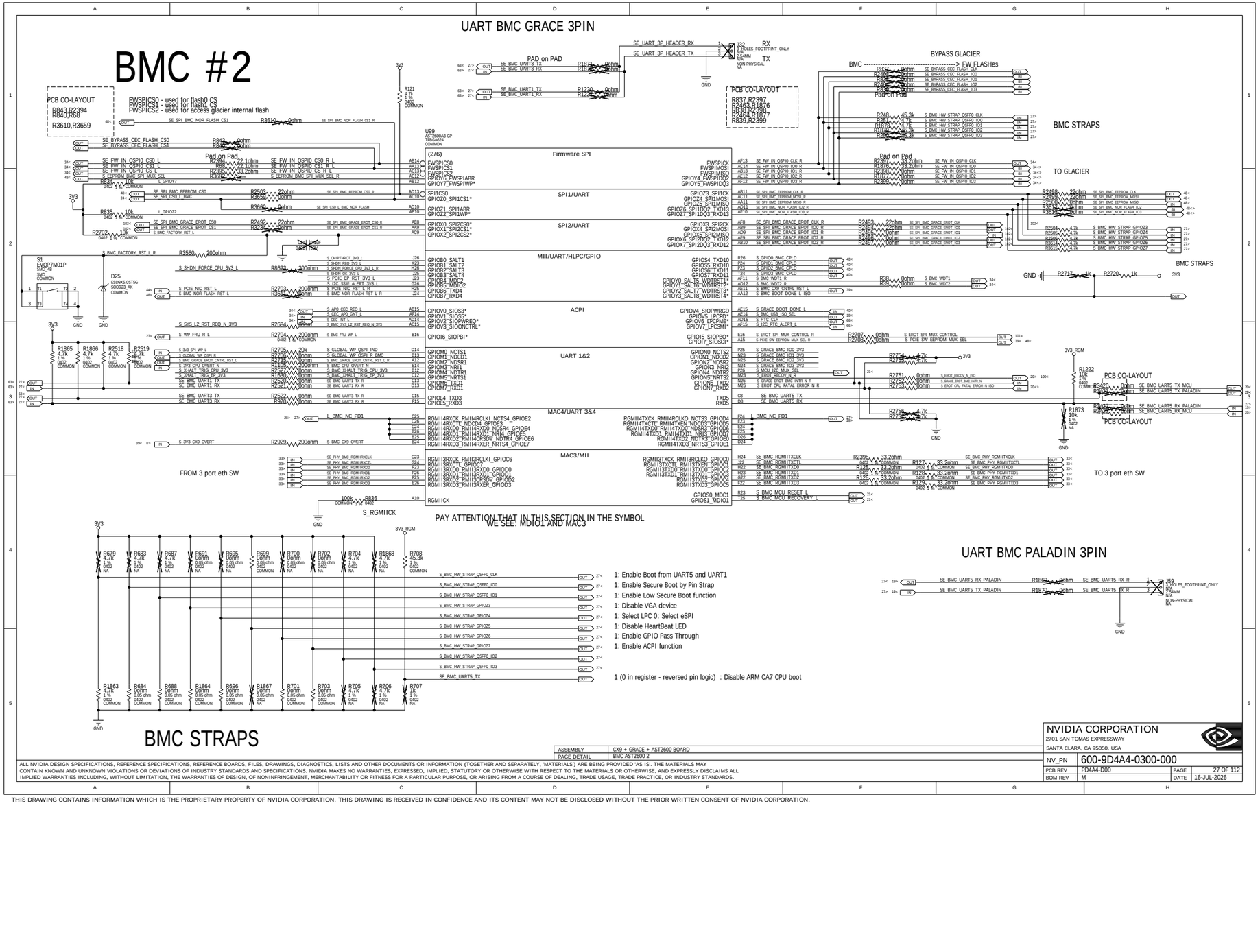
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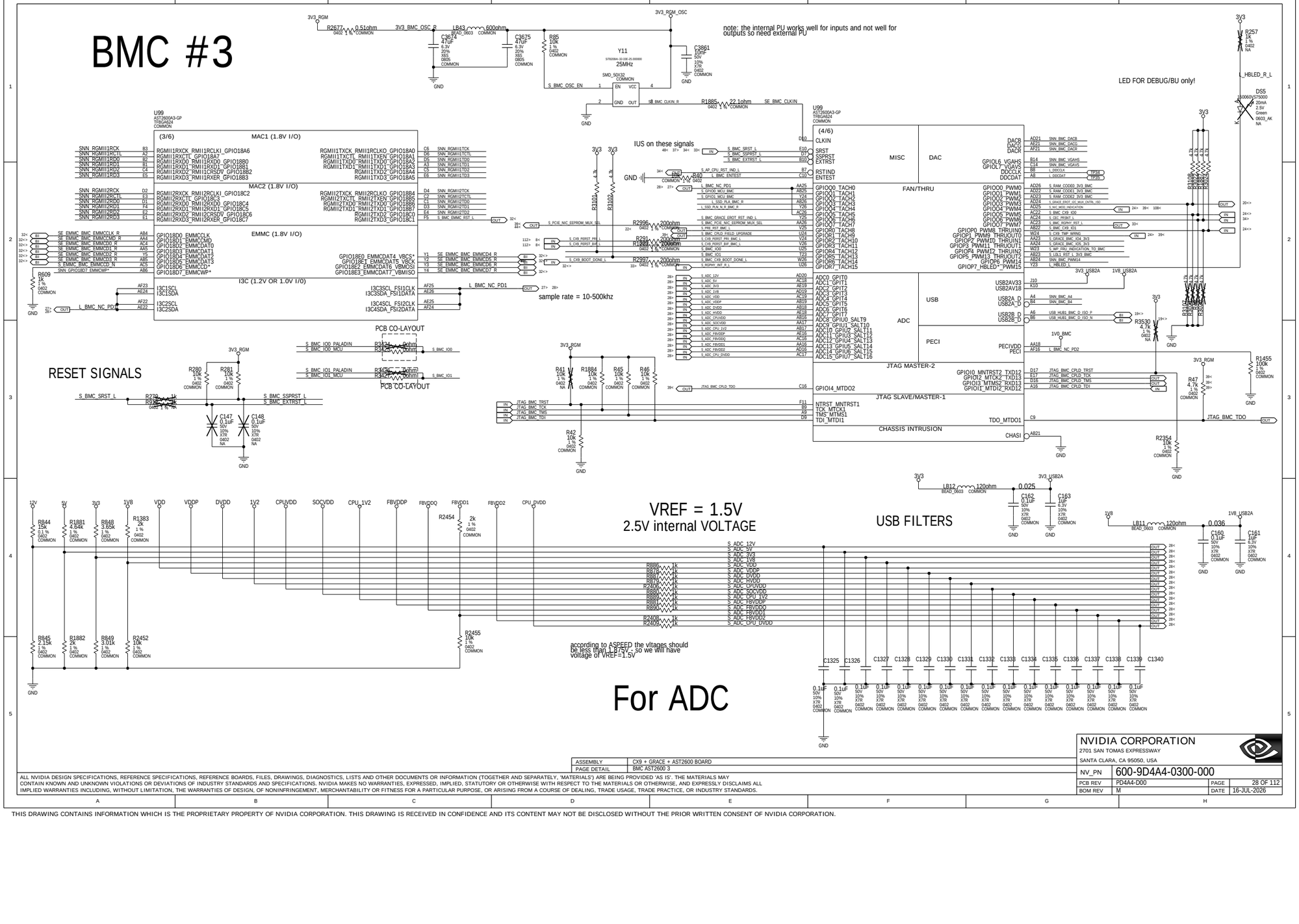
Pin		Description	Value
FWSPICK	0	Disable Boot from UART5 and UART 1	0
	1	Enable Boot from UART5 and UART 1	
FWSPIMOSI	0	Disable Secure boot by pin strap	0
	1	Enable Secure boot by pin strap	
FWSPIMISO	0	Disable Low Secure boot function	0
	1	Enable Low Secure boot function	
GPIOZ3	0	Enable VGA Device	0
	1	Disable VGA device	
GPIOZ4	0	Host is eSPI	0
	1	Host is LPC mode	
GPIOZ5	0	Enable Heart Beat LED	1
	1	Disable Heart Beat LED	
GPIOZ6	0	Disable GPIO Pass Through	0
	1	Enable GPIO Pass Through	
GPIOZ7	0	Disable ACPI FUNCTION	0
	1	Enable ACPI Function	
GPIOY4	0	Reserved	
	1		
GPIOY5	0	Reserved	
	1		
TXD5	0	Strap value = 0 (Enable) just leave pin floating or **pull up**	1
	1	Strap value = 1 (Disable ARM CPU Boot) **pull-down** the pin to ground by a resistor of value 1K ~ 4.7K Ω	



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BMC #3



Heart Beat LED indicator output.(default)
It is open-drain active low output.
The HW blinking modes are shown below:

1 0Hz : LED dark indicates ARM_CA7 is reset or secure boot is running (LED off)
2 80Hz : Secure boot failed (LED 50% on)
3 20Hz : Boot passed with ABR (show about 3 seconds and then exit)
4 4Hz : ARM CA7 is reading address range 0x0000_0000 to 0x0FFF_FFFF (reading flash or SRAM)
5 1Hz : ARM CA7 is reading address range 0x1000_0000 to 0x7FFF_FFFF (reading peripherals)
6 0.5Hz : ARM CA7 is reading address range 0x8000_0000 to 0xFFFF_FFFF (reading DRAM)
7 0.2Hz : No ARM CA7 read access cycles
8 0.1Hz : No ARM CA7 read access cycles over 6-8 seconds
Stay in mode 8 if there is no ARM_CA7 reading access
Exit to mode 4-6 if there is any ARM_CA7 reading access

HW blinking modes can be disabled by HW pin strap.
SW driven mode of Heart Beat needs to disable Multi-funtion pin control 1-3 and use GPIOP7.

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	BMC AST2600 3

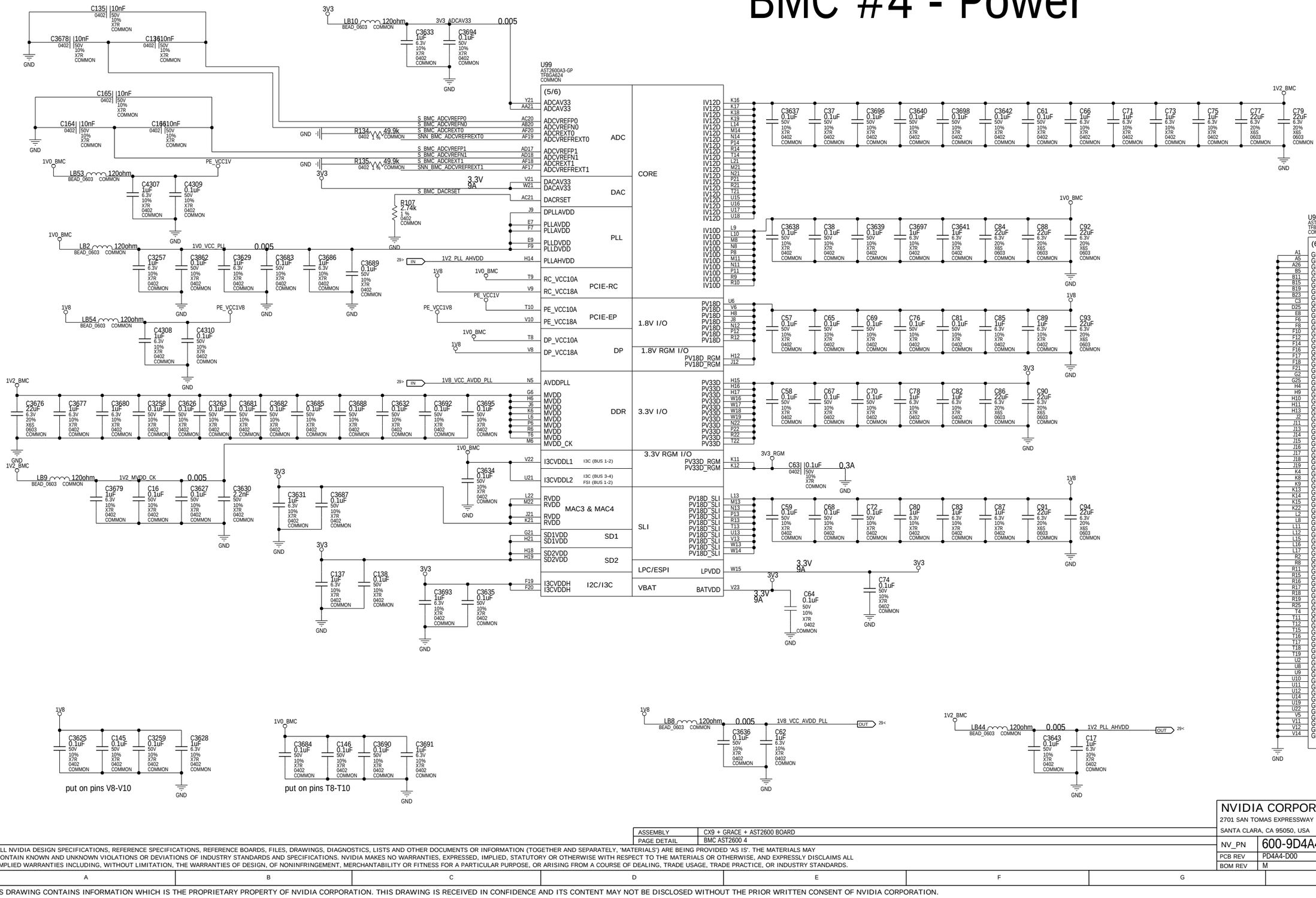
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BMC #4 - Power



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	BMC AST2600 4

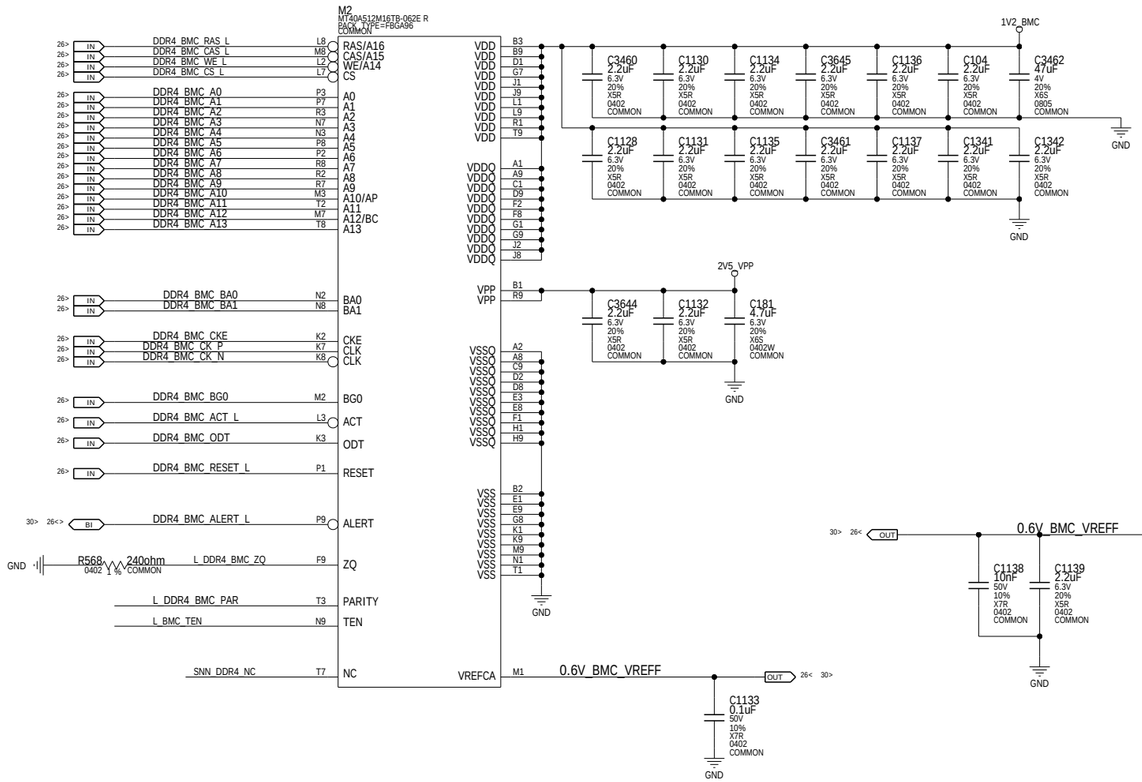
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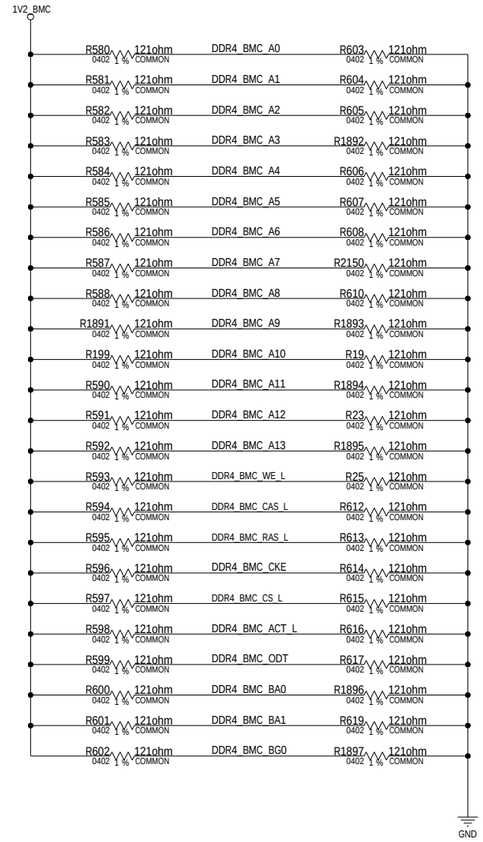
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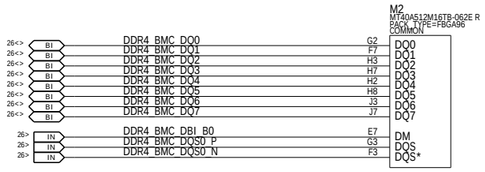
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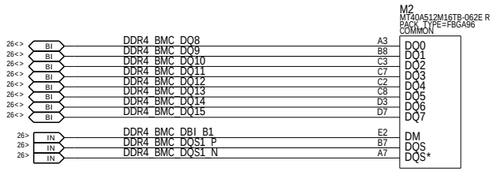
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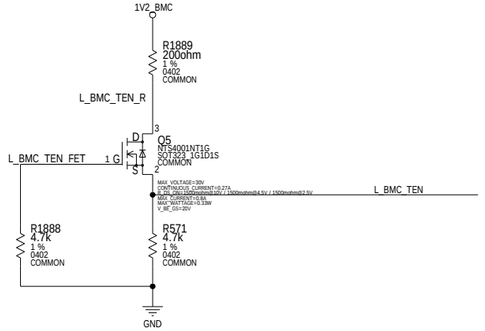
BMC DDR

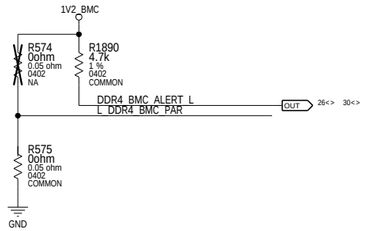


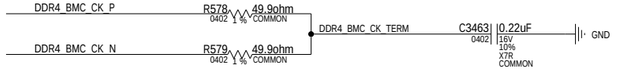












ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	BMC DDR4

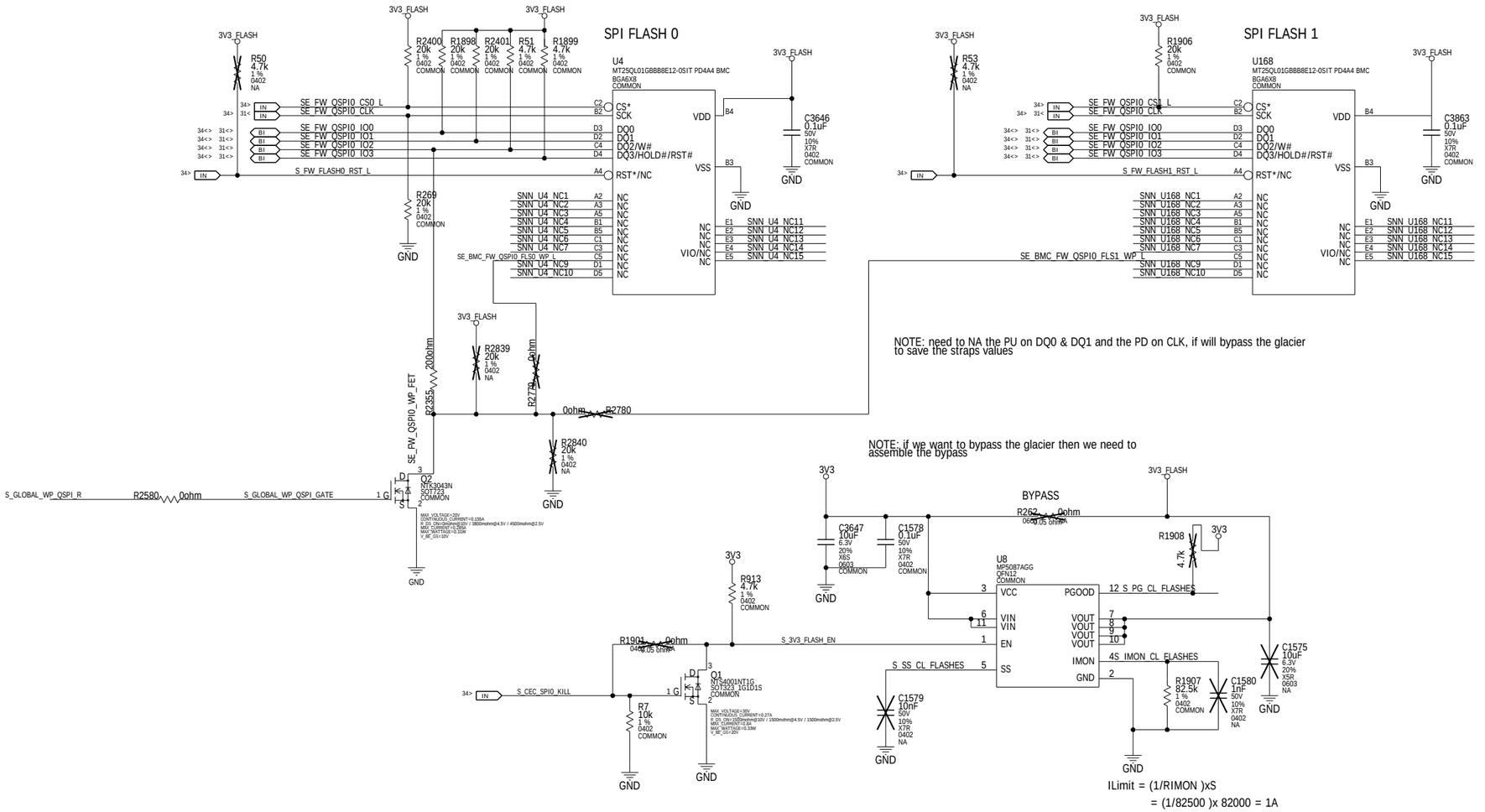
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BMC FLASH



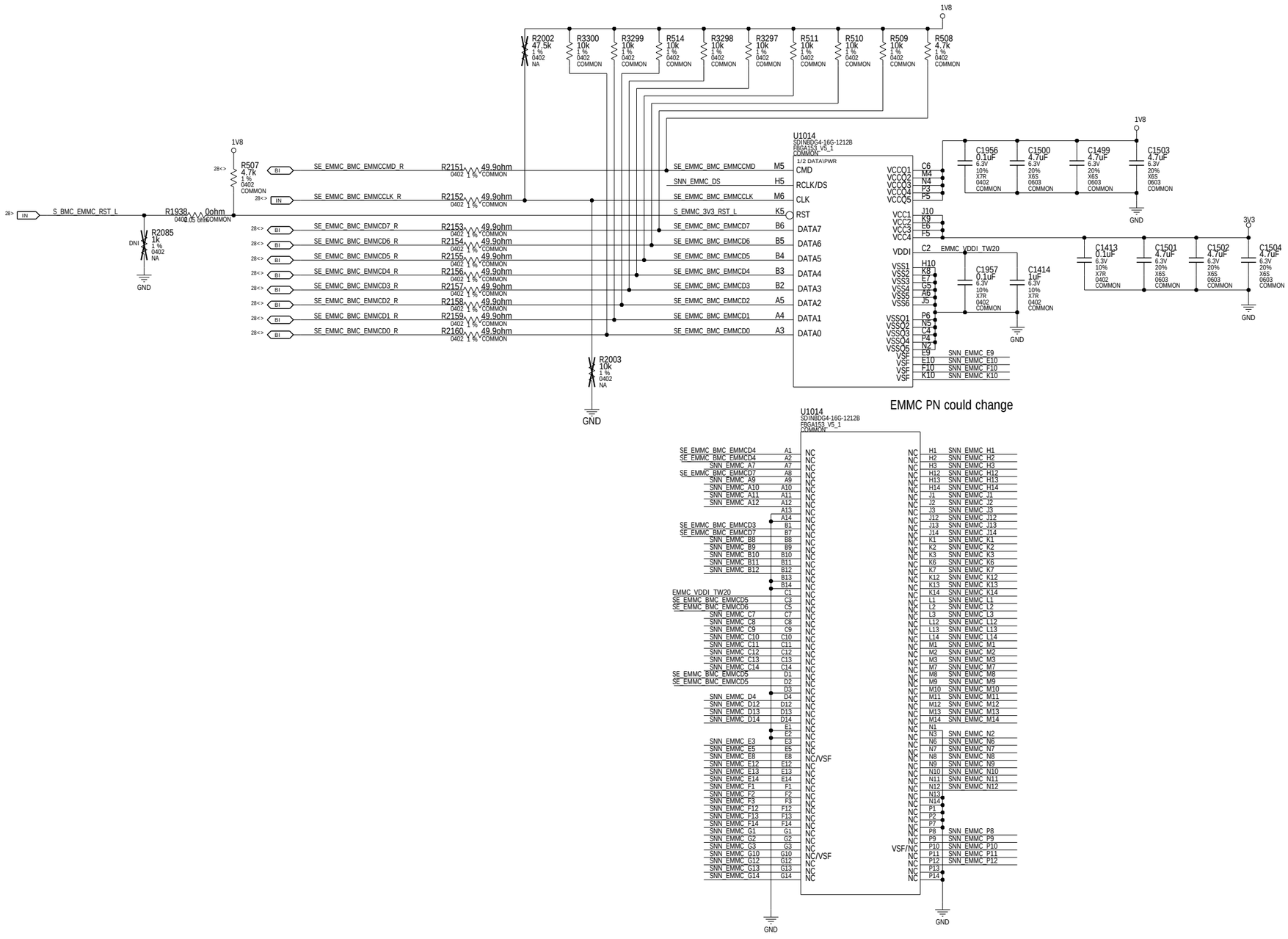
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	BMC Flashes

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	BMC EMMC

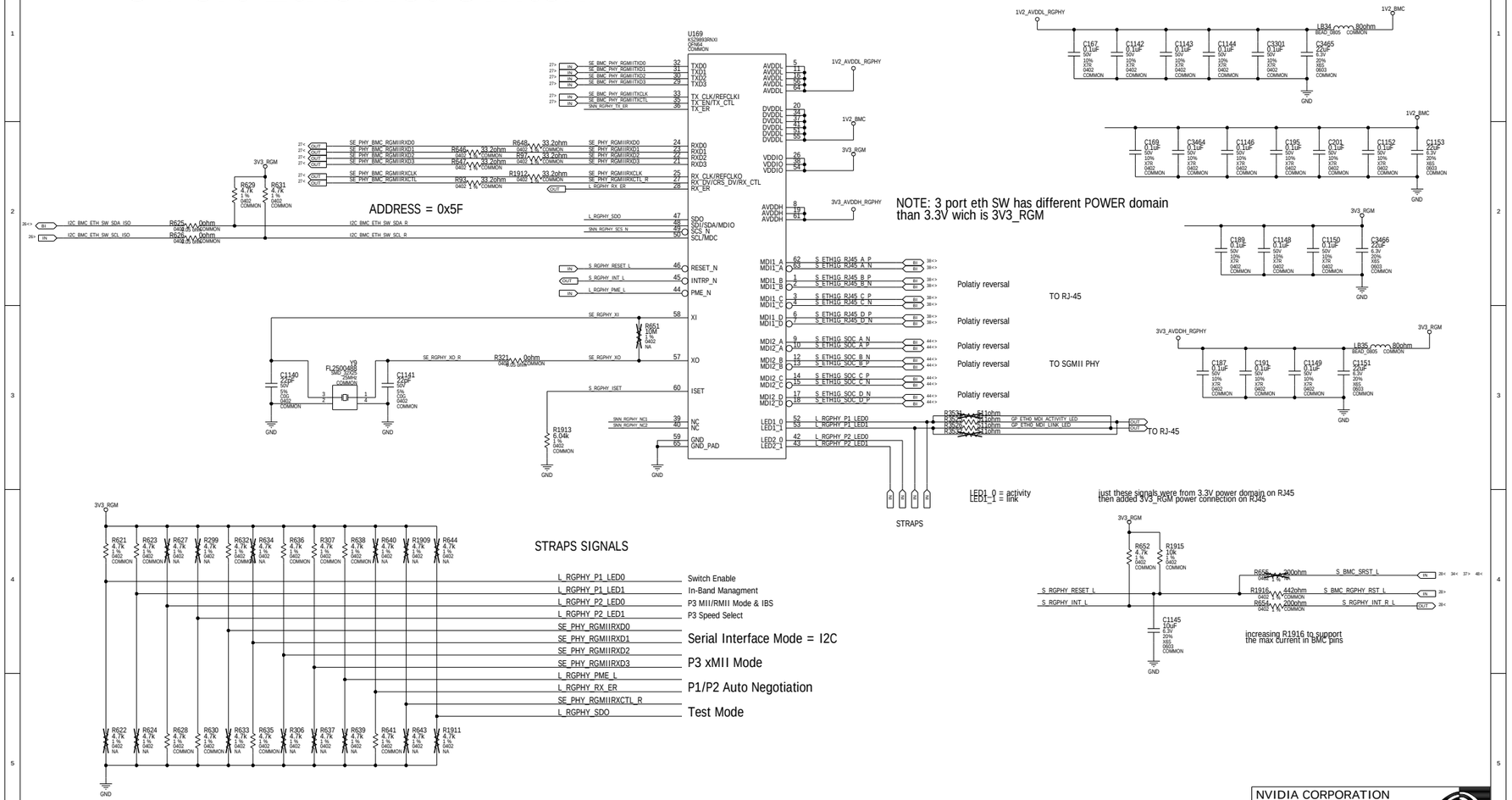
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3 Port Ethernet Switch



Pin Number	Pin Name	Description
52	LED1_0	Switch Enable at startup 0 - Disable 1 - Enable (Default)
53	LED1_1	In-Band Management 0 - Enable 1 - Disable (Default)
42	LED2_0	Port 3 MII/RMII Mode 0 - MII PHY Mode / RMII Clock Mode / RGMII IBS disabled 1 - MII MAC Mode / RMII Normal Mode / RGMII IBS enabled
43	LED2_1	Port 3 Speed Select 0 - 1000Mbps Mode 1 - 100Mbps Mode (Default)
[21, 22]	[RXD3, RXD2]	Port 3 xMII Mode 00 - MII (Default) 01 - RMII 10 - RESRSEVED 11 - RGMII
[23, 24]	[RXD1, RXD0]	Serial Interface Mode 00 - MIIM (Default) 01 - I2C 1x - SPI
[28, 44]	[RX_ER, PME_N]	Ports 1 and 2 (PHY) Configuration 01 - Auto-negotiation enable with EEE (Default) 10 - Auto-negotiation disabled and Auto-MDIX disabled (MDI-X mode) 11 - Auto-negotiation disabled and Auto-MDIX disabled (MDI mode)
27	RX_DV/CRS_DV/RX_CTL	Normal / Factory Test Mode 0 - Normal Mode (Default) 1 - Factory Test Mode
47	SDO	Normal / Factory Test Mode 0 - Normal Mode (Default) 1 - Factory Test Mode

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	ETH SWITCH

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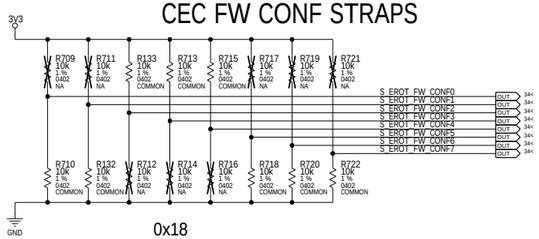
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STRAPPING TABLE		
Pin	**Function**	**Description**
GPIO170	JTAG_STRAP	1=Use the JTAG TAP Controller for Boundary Scan 0=The JTAG TAP Controller is used for debug (normal operation)
GPIO201/32KHZ_OUT	CR_FLASH	Crisis Flash Recovery 1=Don't Boot from Crisis Recovery Flash Component 0=Boot from Crisis Recovery Flash Component
GPIO015/ICT10	BSTRAP	0=Boot Strap (Boot ROM I2C or UART Crisis port) 1=Normal operation
GPIO026/SP0_AP_INTR	I2C_ADDR0	I2C Address Strap Option pin 0. Default Value = 1
GPIO034/SP1_AP_INTR	I2C_ADDR1	I2C Address Strap Option pin 1. Default Value = 0

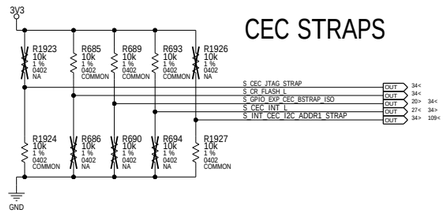
BMC GLACIER

CEC FW CONF STRAPS

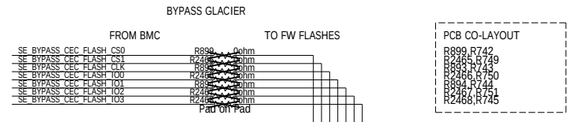


0x18

CEC STRAPS



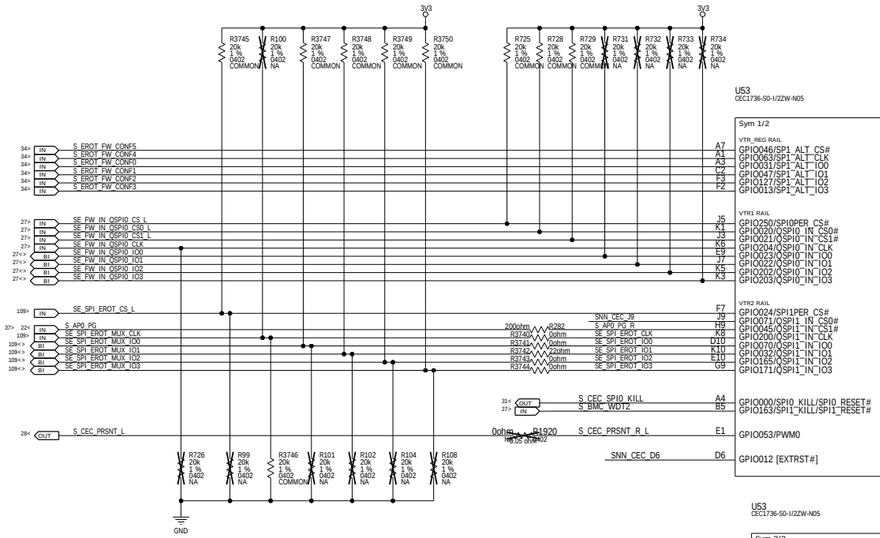
BYPASS GLACIER

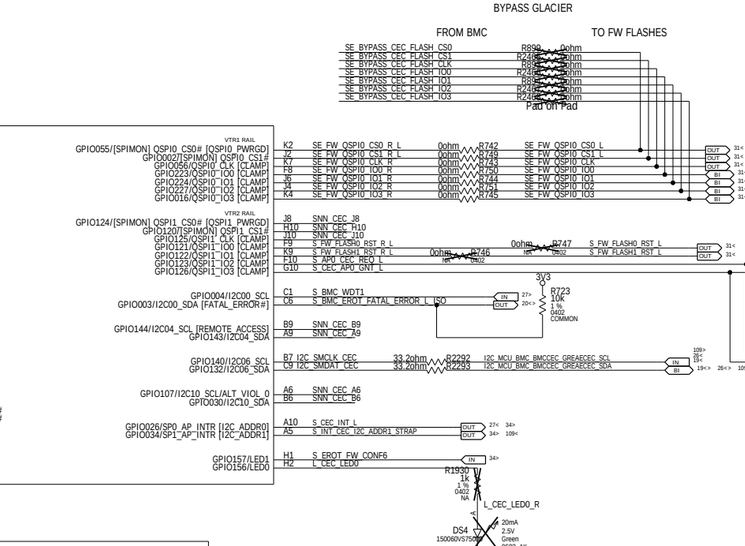


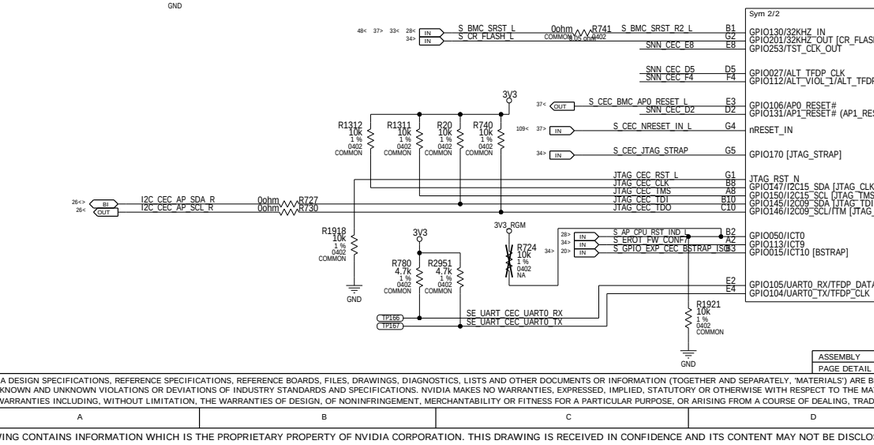
PCB CO-LAYOUT

R899,R742
R2465,R749
R893,R743
R2466,R750
R894,R744
R2467,R751
R2468,R745

TO FW FLASHES







Remove R1388 for silicon Version B

to opne Agile PN for 115-3442-000

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	BMC Glacier

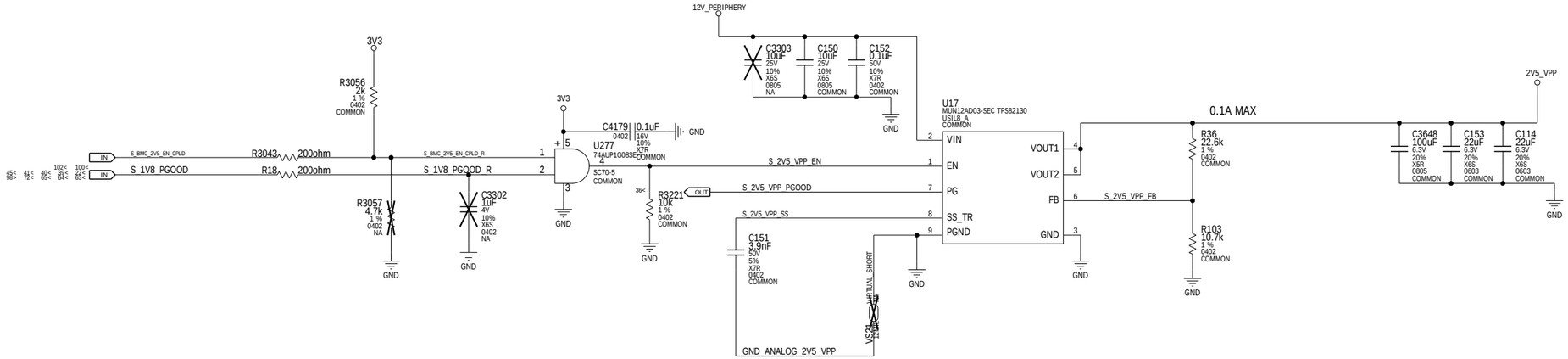
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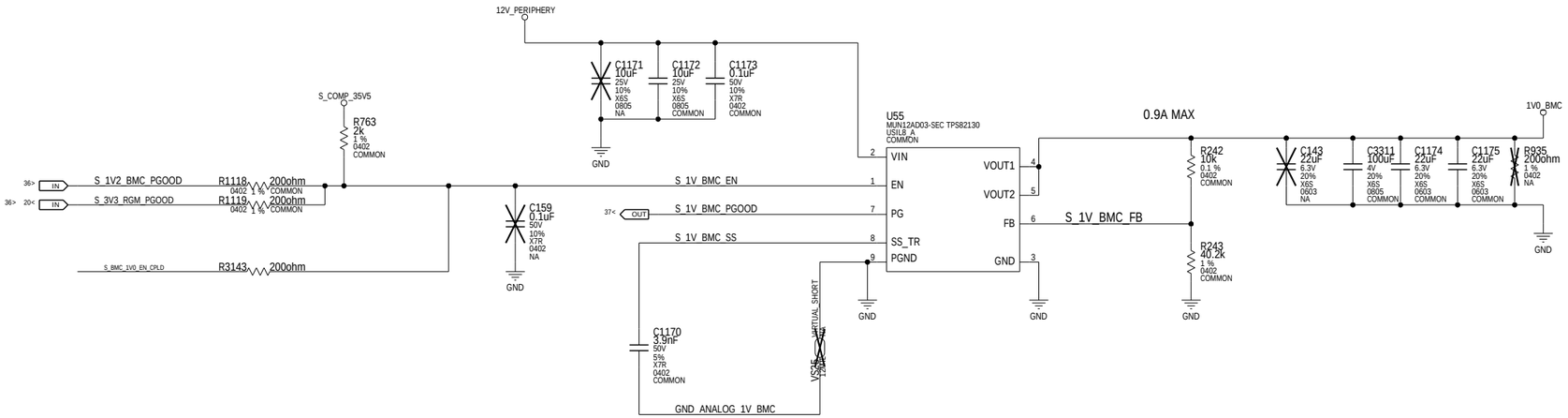
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PS: 2.5V_VPP, 1V_BMC





ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS: 2V5_VPP, 1V_BMC

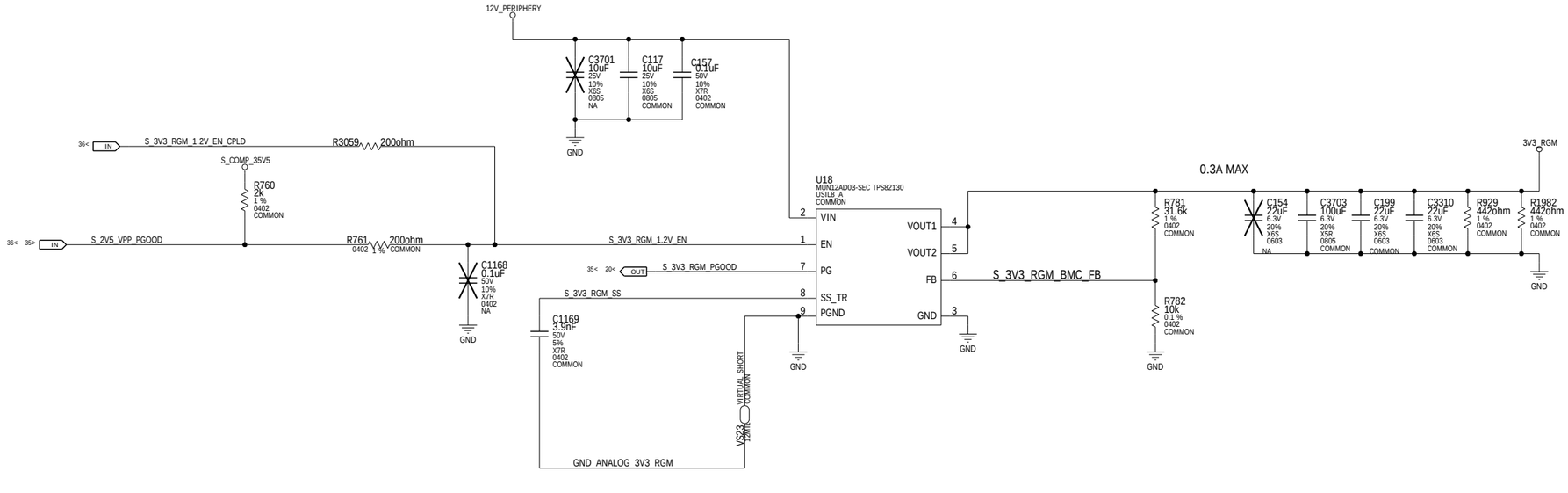
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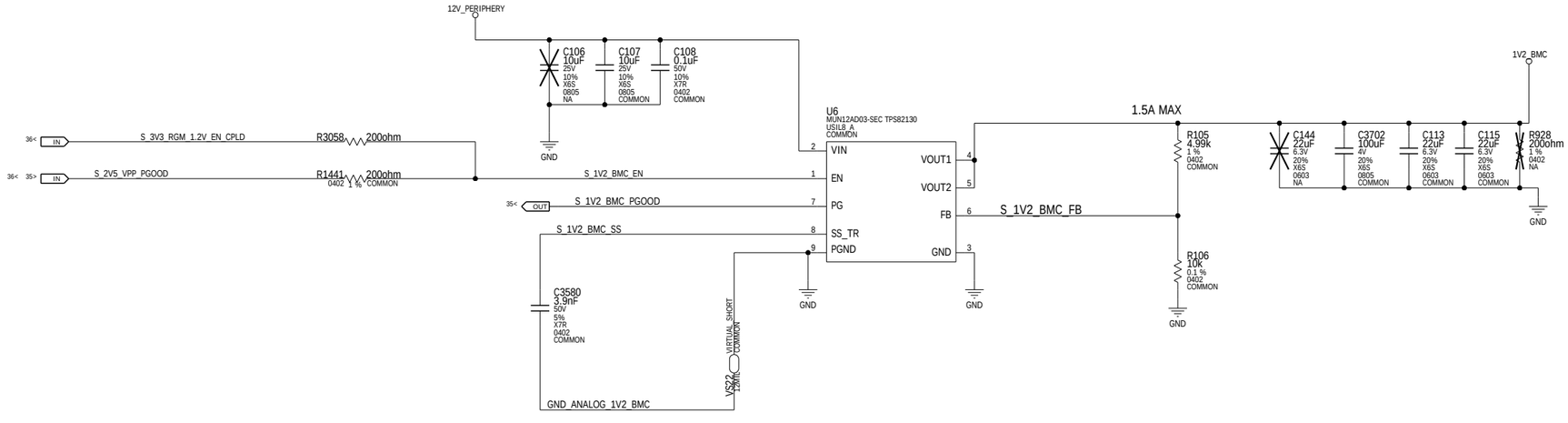
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PS: 3.3V_RGM and 1.2V_BMC





ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS: 3V3_RGM, 1V2_BMC

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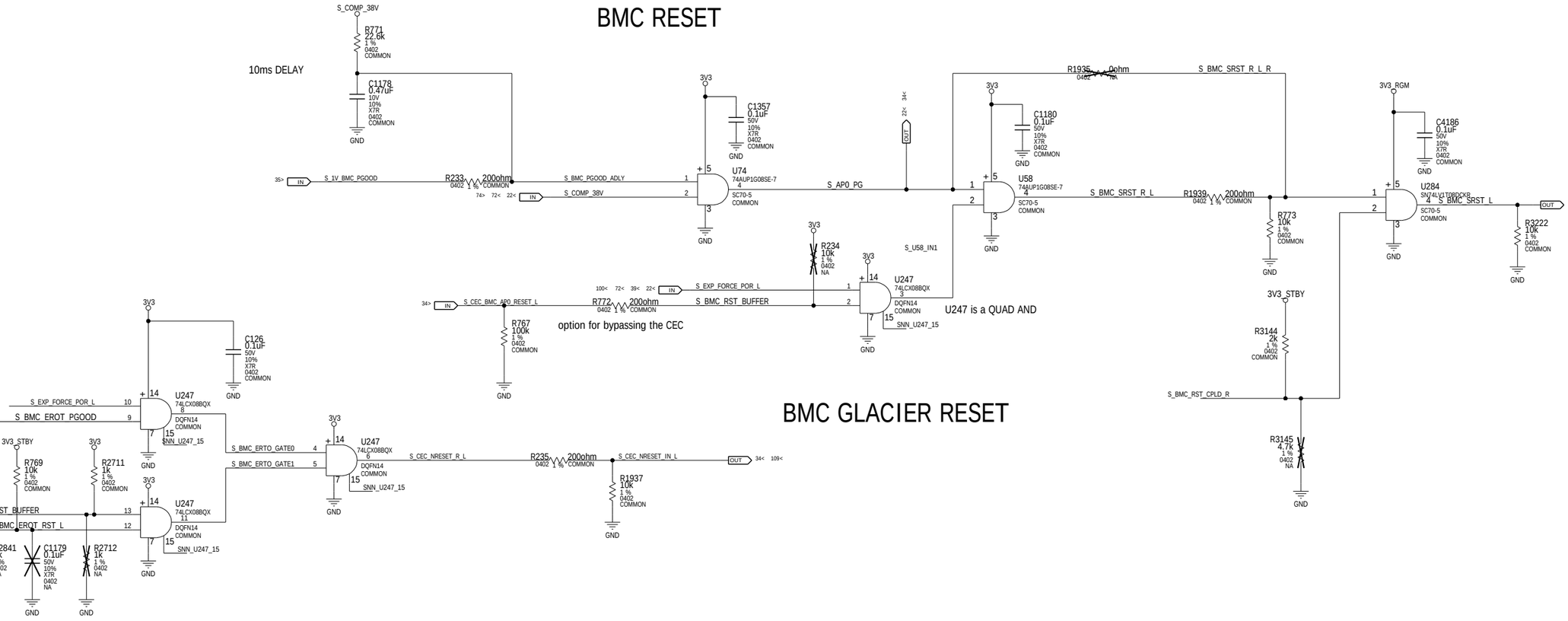
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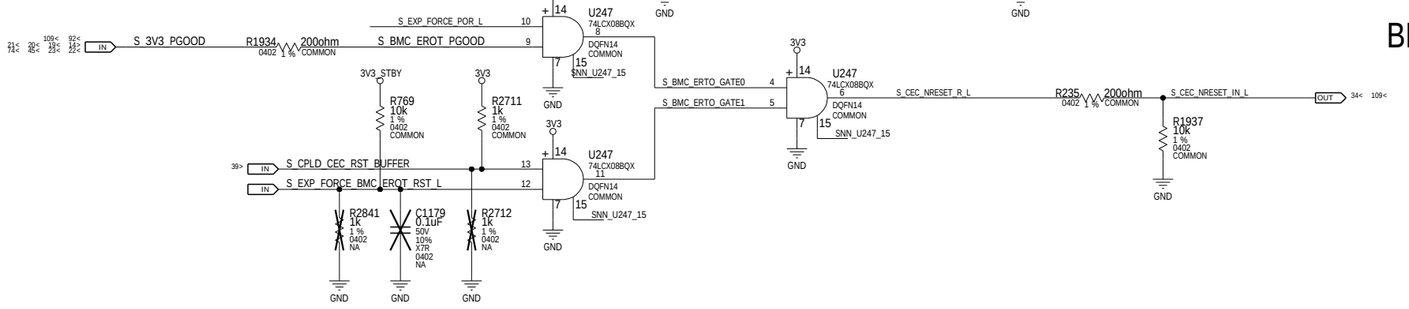
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BMC and GLACIER RESET

BMC RESET



BMC GLACIER RESET



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	BMC Glacier RESET

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RJ-45

P4103_B00(P34)

T-SIDE
CONNECT TO PHY

M-SIDE
CONNECT TO CONN

S_ETH1G_RJ45_A_P 0ohm R1274 S_ETH1G_RJ45_A_P_R
S_ETH1G_RJ45_A_N 0ohm R1275 S_ETH1G_RJ45_A_N_R
S_ETH1G_RJ45_B_P 0ohm R1959 S_ETH1G_RJ45_B_P_R
S_ETH1G_RJ45_B_N 0ohm R1277 S_ETH1G_RJ45_B_N_R
S_ETH1G_RJ45_C_P 0ohm R1278 S_ETH1G_RJ45_C_P_R
S_ETH1G_RJ45_C_N 0ohm R1279 S_ETH1G_RJ45_C_N_R
S_ETH1G_RJ45_D_P 0ohm R1960 S_ETH1G_RJ45_D_P_R
S_ETH1G_RJ45_D_N 0ohm R1281 S_ETH1G_RJ45_D_N_R

T2
M3465NL
SOE214
COMMON
PHY / CONNECTOR
MCT0 MCT1 MCT2 MCT3
295-0039-000

TCT_C
C3744 10nF 25V 10% X7R 0402 COMMON
C3745 1nF 2000V 10% X7R 1208W COMMON
R2245 R2246 R2247 R2248 750ohm 1% 0402 COMMON (R2247 COMMON MCT_C)

D22 D22 D23 D23 TPD4E02B04DQAR DUAL COMMON

J5
AC7400-0011-009-H0
TH_RA_1X1_BI_LED_B
COMMON

S_ETH1G_RJ45_A_P_P / S_ETH1G_RJ45_A_N_P	1 / 2	TRD0+ / TRD0-	
S_ETH1G_RJ45_B_P_P / S_ETH1G_RJ45_B_N_P	3 / 6	TRD1+ / TRD1-	
S_ETH1G_RJ45_C_P_P / S_ETH1G_RJ45_C_N_P	4 / 5	TRD2+ / TRD2-	
S_ETH1G_RJ45_D_P_P / S_ETH1G_RJ45_D_N_P	7 / 8	TRD3+ / TRD3-	
	13	GND	
	14	GND	GND
Link	9	LGLED+/LOLED-	RGLED+/ROLED- 11 Activity
	10	LGLED-/LOLED+	RGLED-/ROLED+ 12

LED:
Vf=(1.7-2.6)V
If=20mA

LED ARE ACTIVE LOW

064-1072-000
J5

GP_ETH0_MDI_LINK_LED_9
GP_ETH0_MDI_LINK_LED_10
GP_ETH0_MDI_ACTIVITY_LED_12
GP_ETH0_MDI_ACTIVITY_LED_11

3V3_RGM
GP_ETH0_MDI_LINK_LED R3533 0ohm
R3527 0ohm 0402 COMMON
GP_ETH0_MDI_LINK_LED R3534 0ohm 0402 COMMON
R3535 0ohm
R3538 0ohm GP_ETH0_MDI_ACTIVITY_LED
R3539 0ohm 0402 COMMON
R3536 0ohm 0402 COMMON GP_ETH0_MDI_ACTIVITY_LED
R3537 0ohm

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	RJ-45

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CPLD

U190
LCMXO3D-4300UHE-5UTG69CTRBES
WLCSP69_PD4A4
COMMON"

1V2_CPLD_VCC → B8 VCC1, E4 VCC2
3V3_STBY → B4 VCCIO0

BANK0
B3 PT38B
A3 PT38A
C4 PT39B
D4 PT39A
E6 PT28B

S_CPU_THERM_WARN_3V3_N → R2212 0ohm → S_CPU_THERM_WARN_3V3_N_R
S_BOARD_THERM_WARN_R
S_1V8_PGOOD → R8694 0ohm → S_PGOOD_1V8_CPLD
S_NIC_MOD_INDICATION
S_CX9_TMP_WRNG

PT11A A8 S_CPLD_ENABLE_INBAND_RECOVERY_L → R1835 22ohm → S_CPLD_ENABLE_INBAND_RECOVERY_R_L
PT28A D6 S_CPLD_FNP_R_L → R1834 0ohm → S_CPLD_FNP_L
PT11B A7 S_NIC_MODE_LATCHED
PT23B/PCLKC0_1 A5 S_CPLD_CEC_RST → R2329 0ohm → S_CPLD_CEC_RST_BUFFER
PT23A/PCLKT0_1 B5 S_DVDD_VDD_PG

I2C INIT
D7 PT27D/SDA/PCLKC0_0
C6 PT27C/SCL/PCLKT0_0

S_BMC_RST_CPLD_R → R3146 0ohm → S_BMC_RST_CPLD
S_BMC_1V0_EN_CPLD

PT44D/DONE A1 S_CPLD_DONE → R2227 1k
PT44C/INITN A2 S_CPLD_INITN → R2228 1k
PT31D/PROGRAMN D5 S_PROGRAMN_L → R2229 1k
3V3_STBY

JTAG
B6 JTAG_CPLD_TCK
C7 JTAG_CPLD_TMS
A6 JTAG_CPLD_TDI
C5 S_JTAG_CPLD_EN
PT15C/TDO B7 JTAG_CPLD_TDO → R2224 1k 1% 0402 NA; R2225 1k 1% 0402 NA; C3735 0.1uF 6.3V 10% X7R 0402 NA; GND
R2721 1k → R2723 1k → 3V3

BANK1
3V3_STBY → D3 VCCIO1
H1 PR28A
E2 PR10C
E1 PR10D
F3 PR17B/PCLKC1_0
F2 PR17A/PCLKT1_0
G2 PR24B
G1 PR24A
PR5A D1 S_CPLD_CX9_RST_R_L → R2226 0ohm → S_CPLD_CX9_FORCE_RST_L
PR3B/R_GPLLC_IN D2 S_GPIO0_CX9_CPLD_OR_GPIO0_MCU_R
PR3A/R_GPLLT_IN C3 S_GPIO1_CX9_CPLD_R
PR2B/R_GPLLC_FB C2 S_GPIO2_CX9_CPLD_R
PR2C B1 S_BMC_CX9_CNTRL_RST_R → R2666 0ohm → S_BMC_CX9_CNTRL_RST_L
PR2A/R_GPLLT_FB B2 S_NIC_CNTRL_CPU_RST_R_L

S_GPIO0_BMC_CPLD_OR_S_GPIO2_MCU_R
S_GPIO1_BMC_CPLD_R
S_GPIO2_BMC_CPLD_R
S_GPIO3_BMC_CPLD_R
S_GRACE_BOOT_DONE_R_L
S_CX9_BOOT_DONE_R_L
S_INA_MUX_S

3V3 → R3104 4.7k → S_GPIO0_BMC_CPLD_OR_S_GPIO2_MCU_R
R3105 4.7k → S_INA_MUX_S → GND

BANK2
3V3_STBY → F4 VCCIO2
PB7B J7
PB18B H6
PB18A J6
PB22A/PCLKT2_0 H5
PB22B/PCLKC2_0 J5
PB29A/PCLKT2_1 H4
PB29B/PCLKC2_1 J4
PB32A G4
PB35A H3

S_PCIE_NIC_EEPROM_MUX_SEL → R2651 0ohm → S_PCIE_NIC_EEPROM_MUX_SEL_CPLD
S_PCIE_SW_EEPROM_MUX_SEL → R2652 0ohm → S_PCIE_SW_EEPROM_MUX_SEL_CPLD
S_3V3_RGM_1.2V_EN_CPLD
S_NCSI_MUX_S → R2665 0ohm → S_NCSI_MUX_S_R
S_BMC_2V5_EN_CPLD
S_SHDN_FORCE_CPU_3V3_L_CPLD → R3051 0ohm → S_SHDN_FORCE_CPU_3V3_L_CPLD_R
S_PCIE_SW_0V95_EN_CPLD
S_FBVDD1_EN_CPLD
S_FBVDD0_EN_CPLD

PB35B J3 S_BMC_GRACE_CNTRL_RST_R_L → R2668 0ohm → S_BMC_GRACE_EROT_CNTRL_RST_L
PB41A G3 S_CPLD_EXP_FORCE_POR_L → R2738 0ohm → S_EXP_FORCE_POR_L
PB41B H2 S_BMC_BOOT_DONE_R_L → R2669 0ohm → S_BMC_BOOT_DONE_L
PB7A/CSSPIN J8 S_FORCE_GRACE_EROT_RST_N_R → R2737 0ohm → S_FORCE_GRACE_EROT_RST_N
PB16A/MCLK/CCLK G5 S_CPLD_CX9_OVERT → R2992 0ohm → S_3V3_CX9_OVERT
PB16B/SO/SPISO G6 S_FBVDD2_FBVDDP_EN_CPLD
PB46A/SN J2 S_SHDN_OK_3V3_L_CPLD
PB46B/SI/SISPI J1 S_SHDN_REQ_3V3_L_CPLD

SPI

BANK3
3V3_STBY → D8 VCCIO3
PL25A/I3C
PL25B/I3C
PL27A/PCLKT3_0/I3C
PL27B/PCLKC3_0/I3C
PL28A

PL28B F6 S_SOCVDD_EN_CPLD
PL27C F7 S_CPUVDD_CPU_DVDD_EN_CPLD
PL28D H7 S_BOARD_THERM_SHTDN_R
PL28C G7 S_3V3_CPU_OVERT_N

GND E5, E3, C8, A4

S_DPU_BMC_PWR_OFF_R → R3060 200ohm
S_BOARD_PG → R3061 0ohm → S_DPU_BMC_PWR_OFF_OR_BOARD_PG
S_P_ON_COUNTER_EXPIRED_R
S_GRACE_CPLD_IO1_3V3_R
S_GRACE_CPLD_IO2_3V3_R
S_GRACE_CPLD_IO3_3V3_R
S_JTAG_CPLD_EN → R3629 0ohm

Place together
3V3_STBY
J49 J33 J29 J31 J30 J50
GND

CPLD POWER

3V3_STBY
C1942 0.1uF 6.3V 10% X7R 0402 COMMON
C3736 0.1uF 6.3V 10% X7R 0402 COMMON
C3738 0.1uF 6.3V 10% X7R 0402 COMMON
C1965 10uF 6.3V 20% X6S 0603 COMMON
1V2_CPLD_VCC → R2234 0.25ohm 0402 1% COMMON → 1V2_CPLD
C3737 0.1uF 6.3V 10% X7R 0402 COMMON
C1909 0.1uF 6.3V 10% X7R 0402 COMMON
C1963 10uF 6.3V 20% X6S 0603 COMMON
C1915 10uF 6.3V 20% X6S 0603 COMMON
GND

Locate close to CPLD power pins

Power: 1.2V CPLD

3V3_STBY
C1945 2.2uF 6.3V 20% X6S 0402 NA
C1939 2.2uF 6.3V 20% X6S 0402 COMMON
GND

U192
TLV75901PDRVR
0.95V
WSON6
COMMON

3V3_STBY → R1718 10k 1% 0402 COMMON → VIN
R1719 11k 1% 0402 NA
C1949 1uF 25V
GND
VEN HIGH >1V
VEN LOW <0.3V
S_1V2_CPLD_EN → EN
VIN 6, VOUT 1, GND 3, EPAD, EN 4, NC(ADJ)/NC/PG 2, SNS_LDO_5 5
TP103
R1715 13k 1% 0402 COMMON
R1716 11k 1% 0402 COMMON
GND
VOUT=VFBx(1+R1/R2)
VFB=0.55V
C1940 2.2uF 6.3V 20% X6S 0402 NA
C1941 2.2uF 6.3V 20% X6S 0402 COMMON
R1720 1k 1% 0402 COMMON
1V2_CPLD
GND

3V3_STBY
C1967 0.1uF 16V 10% X7R 0402 COMMON
GND

U194
TMUX1574
UQFN16
COMMON

JTAG_BMC_CPLD_TDO	R2361 0ohm	PCB CO-LAYOUT	SE_BMC_CX9_DO
SE_SER_DO_STRP_FNP	R2362 0ohm		JTAG_HARNESS_TDO
JTAG_BMC_CPLD_TCK	R2363 0ohm	PCB CO-LAYOUT	SE_BMC_CX9_CLK
SE_SER_CLK_STRP_PLL_BYPASS	R2364 0ohm		JTAG_HARNESS_TCK
JTAG_BMC_CPLD_TMS	R2365 0ohm	PCB CO-LAYOUT	SE_BMC_CX9_TMS
SE_SER_CAPTURE	R2366 0ohm		JTAG_HARNESS_TMS
JTAG_BMC_CPLD_TDI	R2367 0ohm	PCB CO-LAYOUT	SE_BMC_CX9_DI
SE_SER_DI	R2368 0ohm		JTAG_HARNESS_TDI

VDD 14
S1A/S1B 16/1
S2A/S2B 3/4
S3A/S3B 9/8
S4A/S4B 12/11
SEL 15 ← S_JTAG_HARNESS_ENABLE
EN* 13 ← S_EN_CPLD_IN
GND 6
D1 2 → JTAG_CPLD_TDI
D2 5 → JTAG_CPLD_TCK
D3 7 → JTAG_CPLD_TMS
D4 10 → JTAG_CPLD_TDO

3V3_STBY → R2233 1k, R1723 1k, R1725 1k
R1724 4.7k → GND

EN (Active low)	SEL	OUT	
0	0	A->D	By Default, No connector CX->CPLD
0	1	B->D	Connector present,Harness->CPLD
1	X	HI-Z	

JTAG ENABLE

3V3_STBY
J27 22035025 male 2.5mm N/A NORM NA
28> S_BMC_CPLD_FIELD_UPGRADE
24<> S_CX9_CPLD_FIELD_UPGRADE
S_JTAG_HARNESS_ENABLE_IN
R1658 200ohm 0402 1% COMMON → S_JTAG_HARNESS_ENABLE
PCB CO-LAYOUT
R2359 0ohm
R2360 0ohm
S_CPLD_FU
C1966 0.1uF 16V 10% X7R 0402 COMMON
3V3
R2963 4.7k
R2220 4.7k
R2964 4.7k
U191 74LVC1G32 SC70_5 COMMON
S_JTAG_CPLD_EN_R
R2222 200ohm 0402 1% COMMON → S_JTAG_CPLD_EN
R2223 4.7k
GND

3V3_STBY → 1k → S_EN_CPLD → R1722 1k 0402 1% COMMON
R1721
24< 22> 16< 15< S_CX_RST_3V3 → Q15 SOT723 NTK3043N
C1948 1uF 25V
GND

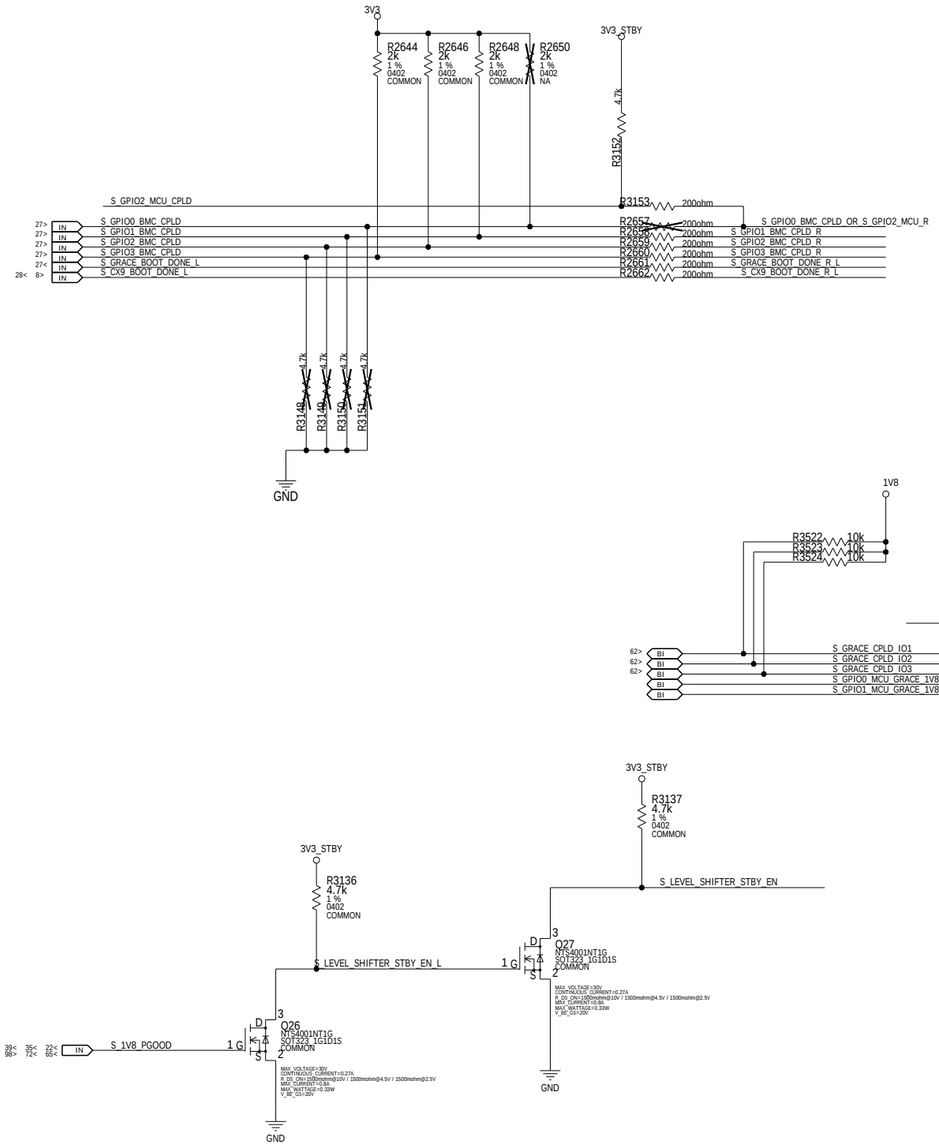
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	CPLD

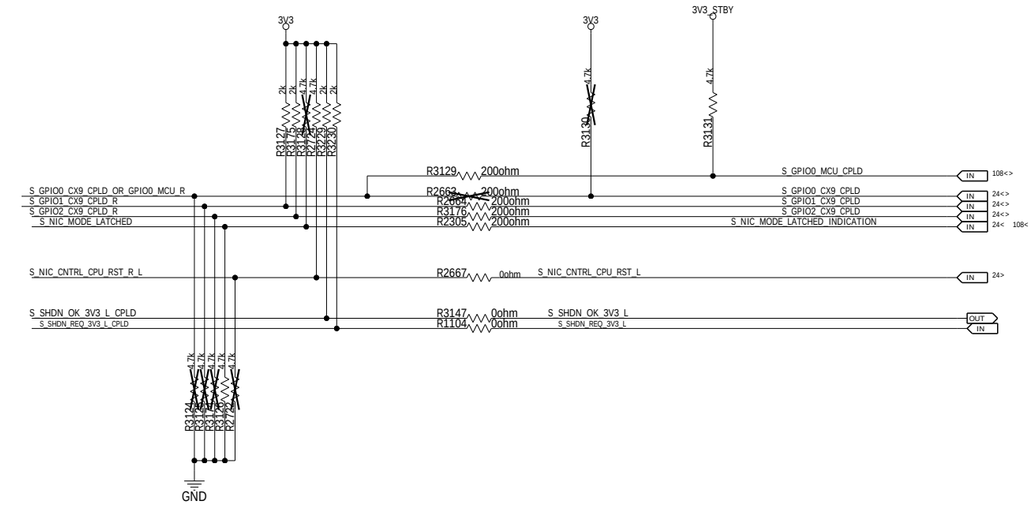
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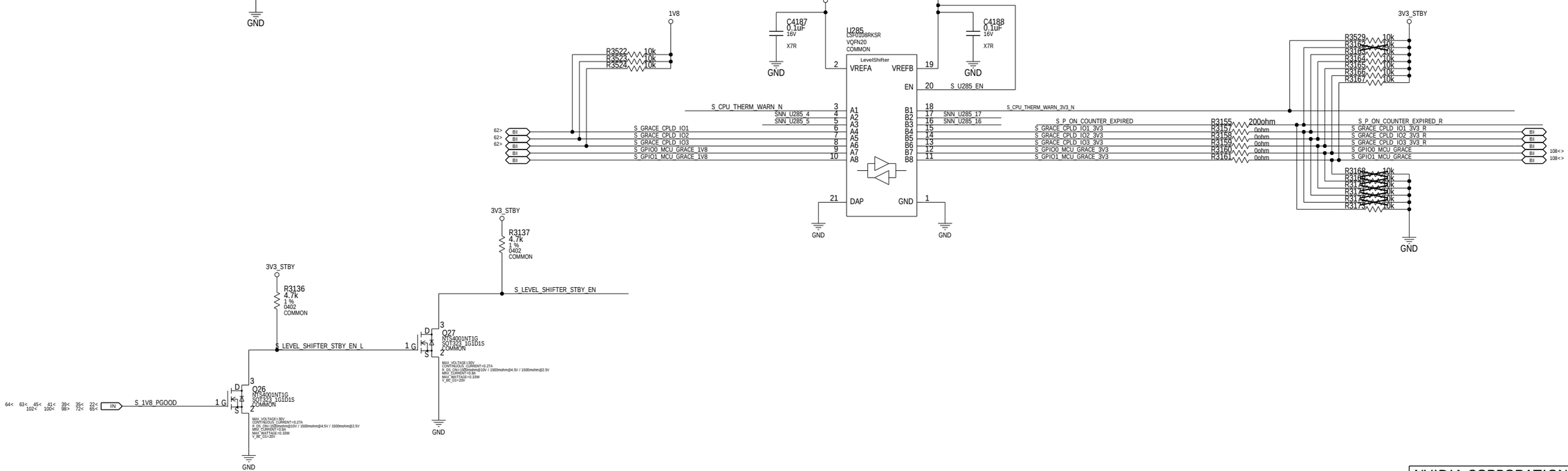
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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	CPLD GPIO

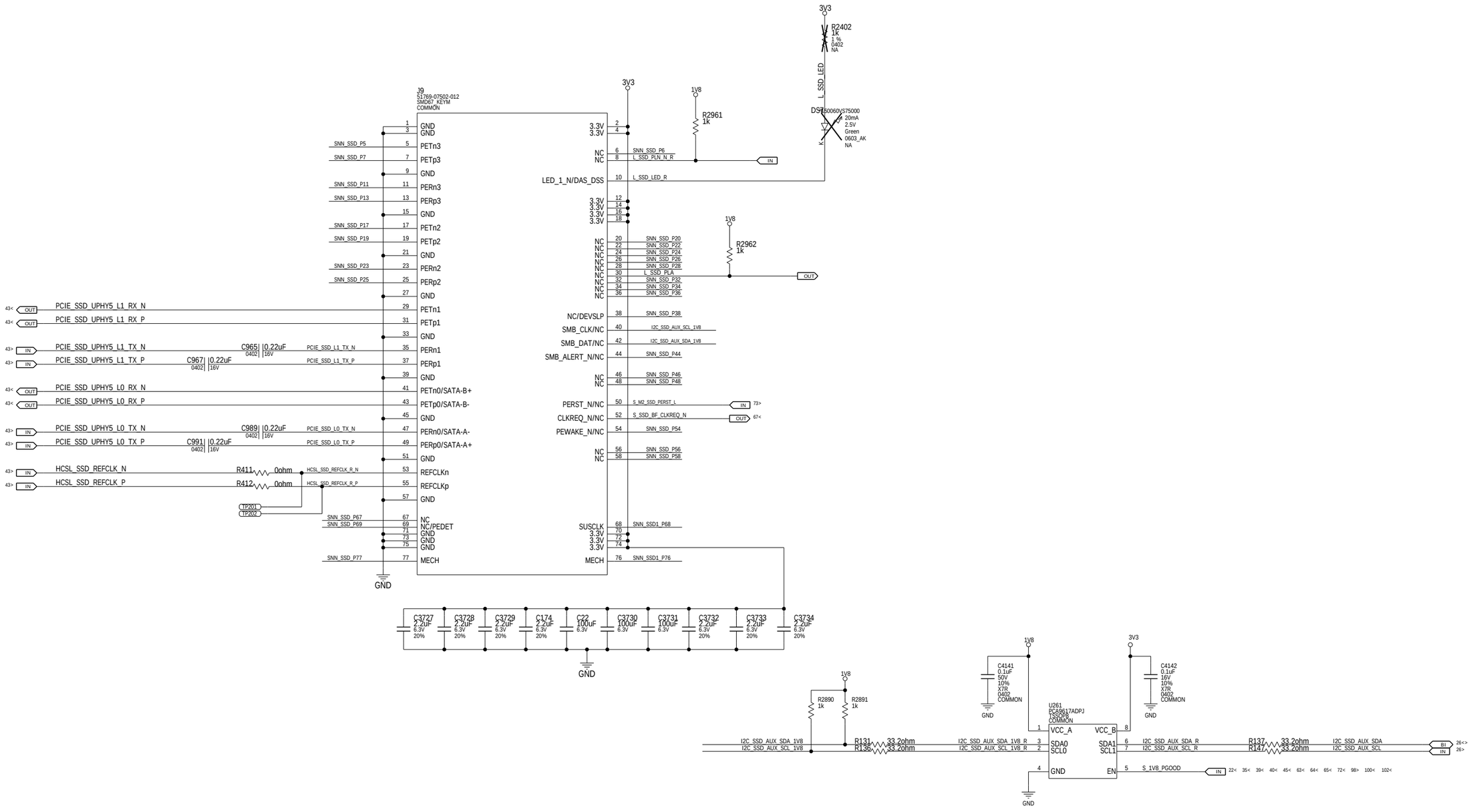
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M.2 CONNECTOR



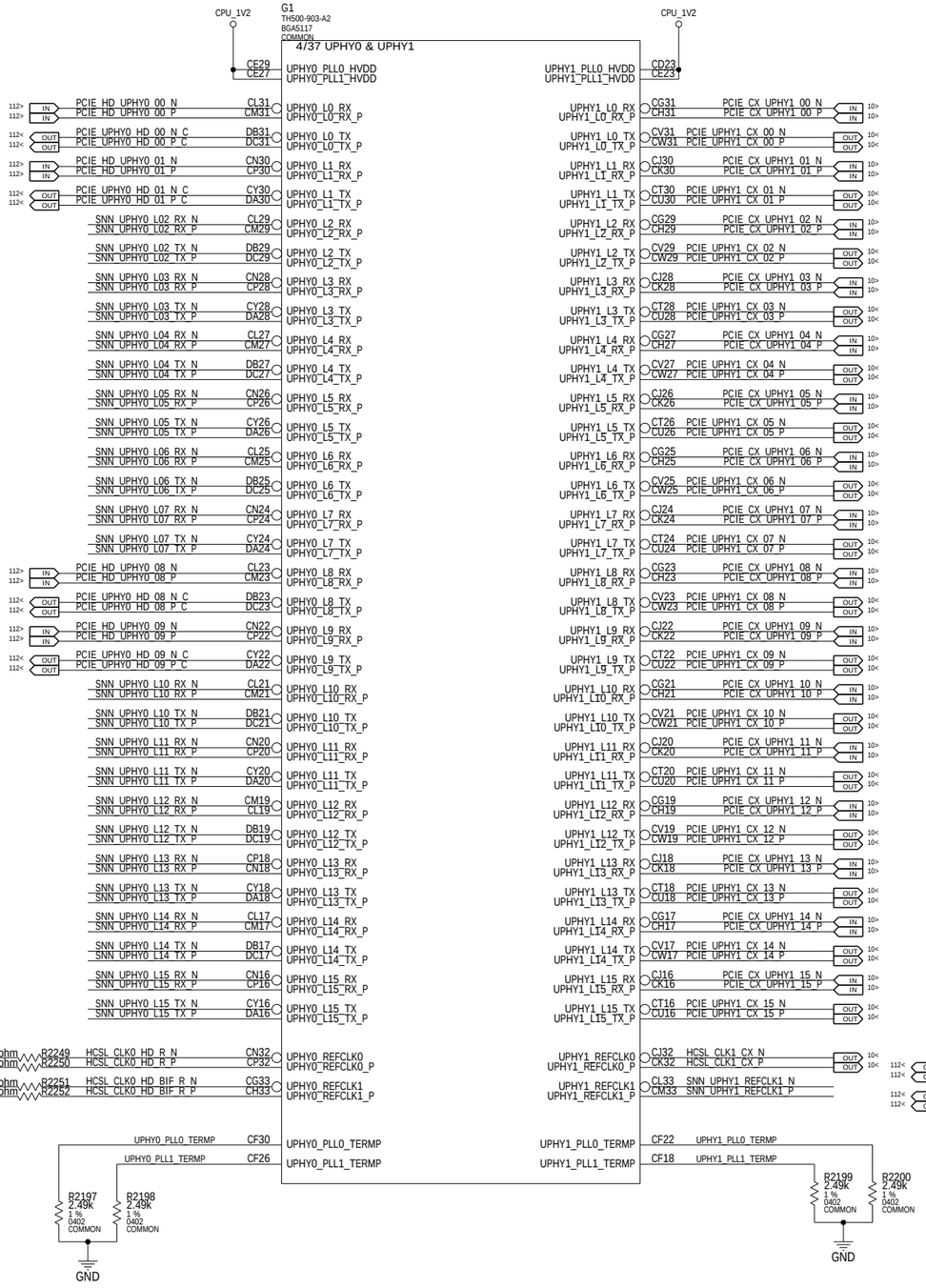
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	M.2 CONNECTOR

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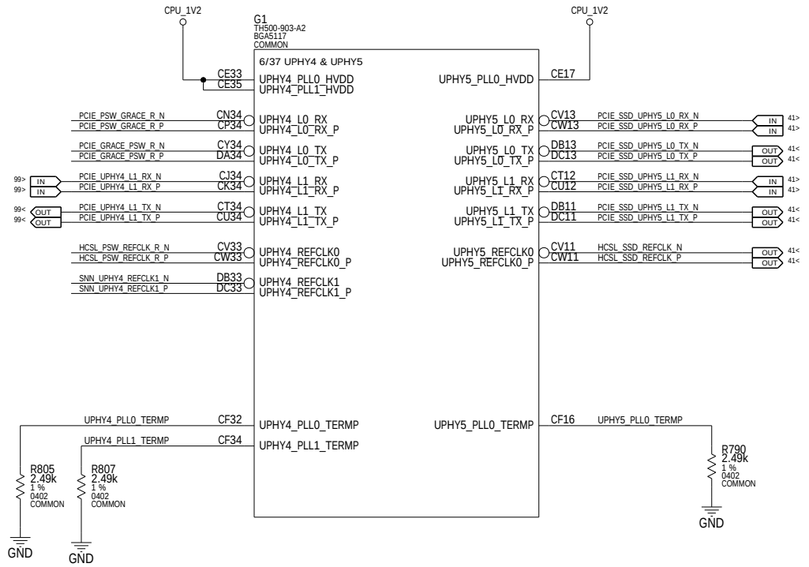
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - UPHY0,UPHY1,UPHY2,UPHY3

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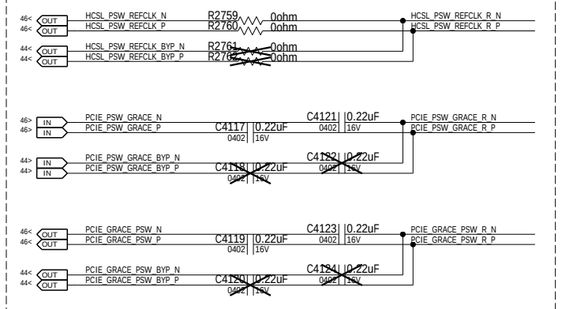
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PCB CO-LAYOUT



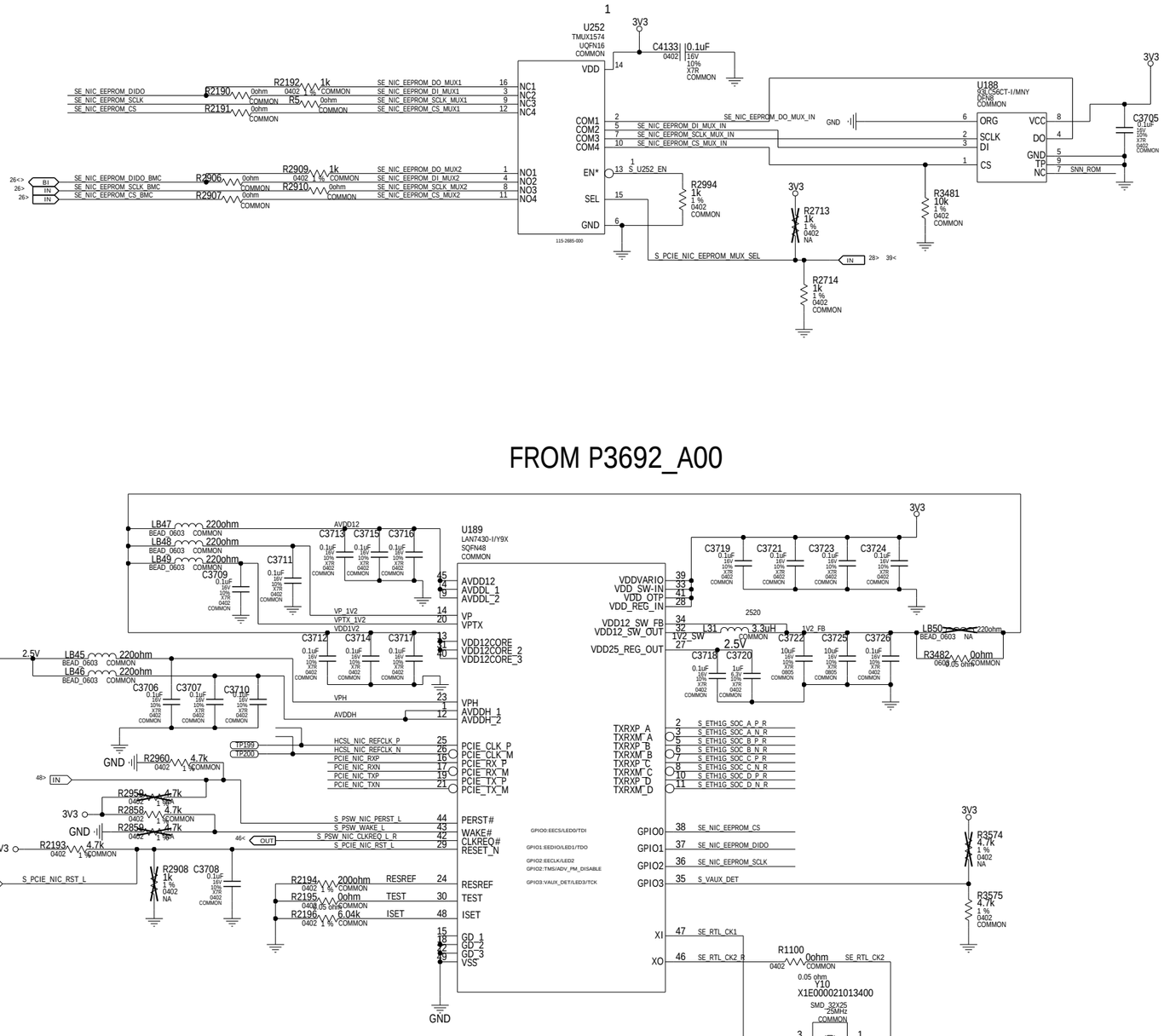
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - UPHY4, UPHY5

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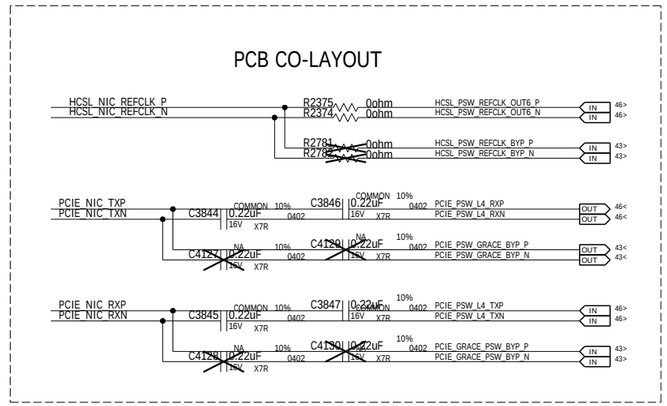
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NV_PN	600-9D4A4-0300-000		
PCB REV	PD4A4-D00	PAGE	43 OF 112
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PCB CO-LAYOUT



FROM P3692_A00

S_ETH1G_SOC_A_P_R	0.1uF	C1557	S_ETH1G_SOC_A_P
S_ETH1G_SOC_A_N_R	0.1uF	C1558	S_ETH1G_SOC_A_N
S_ETH1G_SOC_B_P_R	0.1uF	C1559	S_ETH1G_SOC_B_P
S_ETH1G_SOC_B_N_R	0.1uF	C1560	S_ETH1G_SOC_B_N
S_ETH1G_SOC_C_P_R	0.1uF	C1561	S_ETH1G_SOC_C_P
S_ETH1G_SOC_C_N_R	0.1uF	C1562	S_ETH1G_SOC_C_N
S_ETH1G_SOC_D_P_R	0.1uF	C1563	S_ETH1G_SOC_D_P
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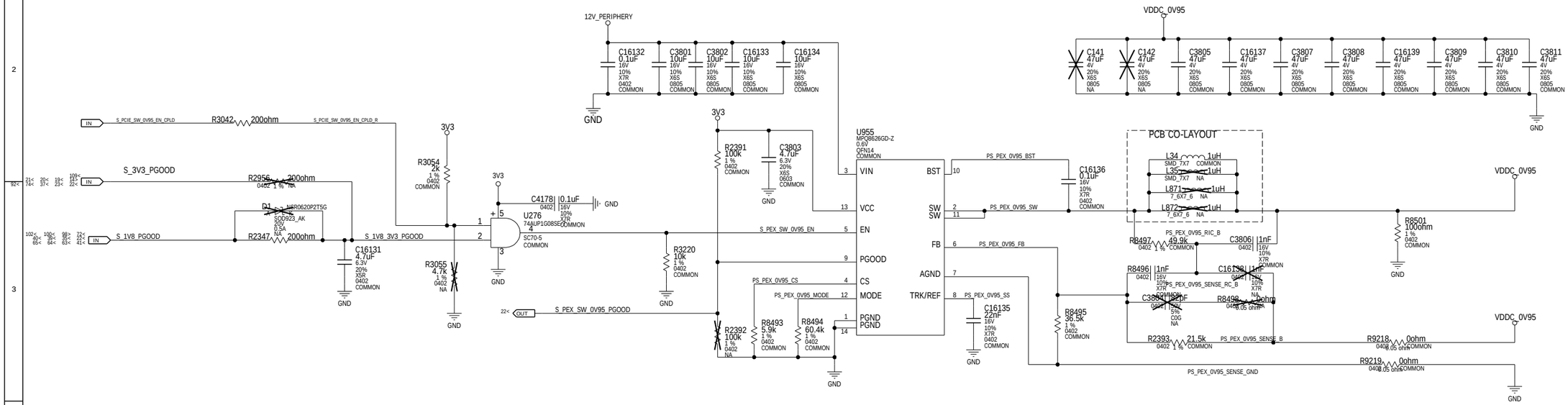
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PCIe NIC

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PCB REV	PD4A4-D00	PAGE	44 OF 112
BOM REV	M	DATE	16-JUL-2026

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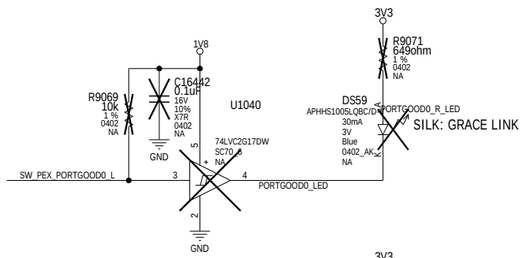
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS: PEX 0V95

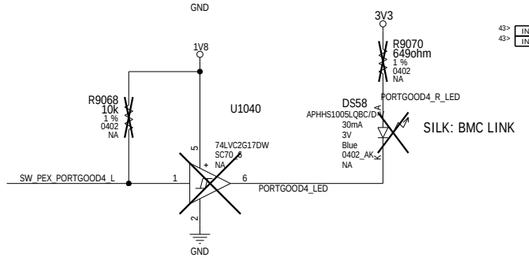
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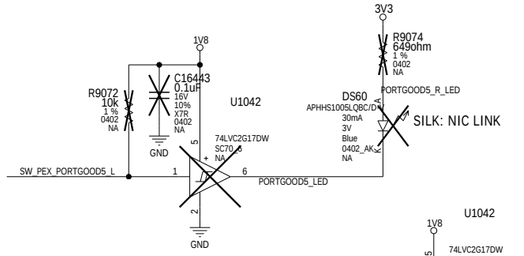
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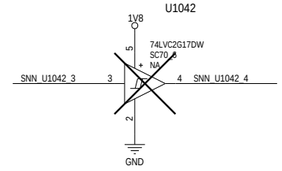
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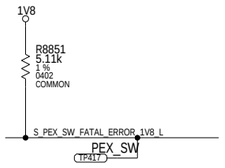
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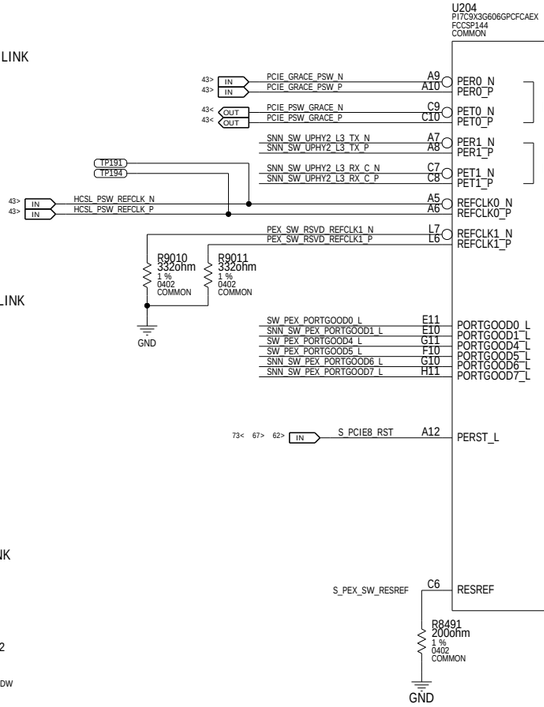


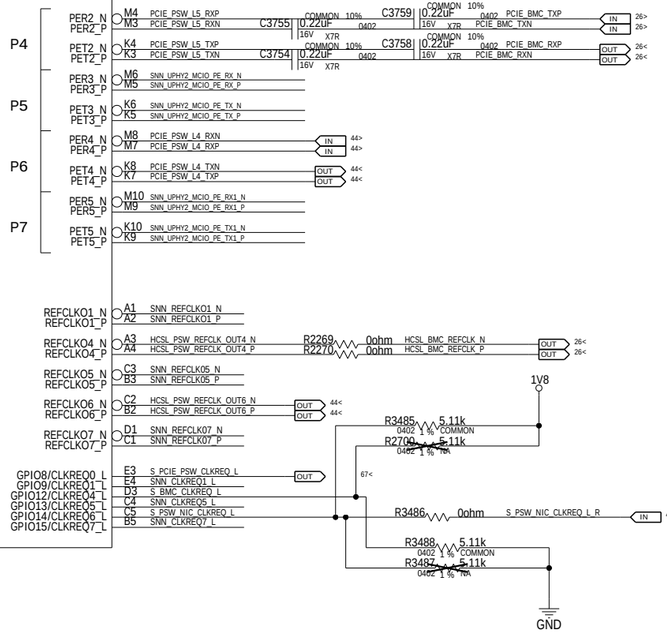


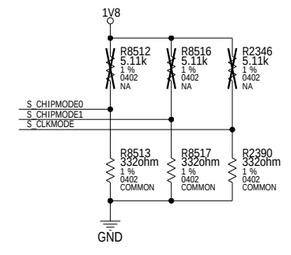


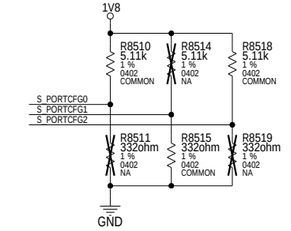


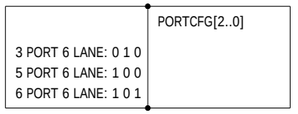


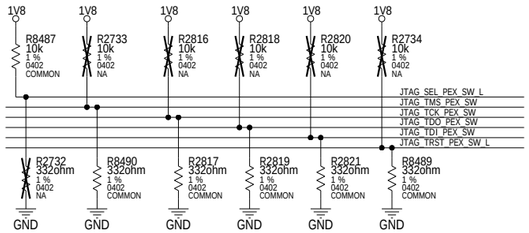


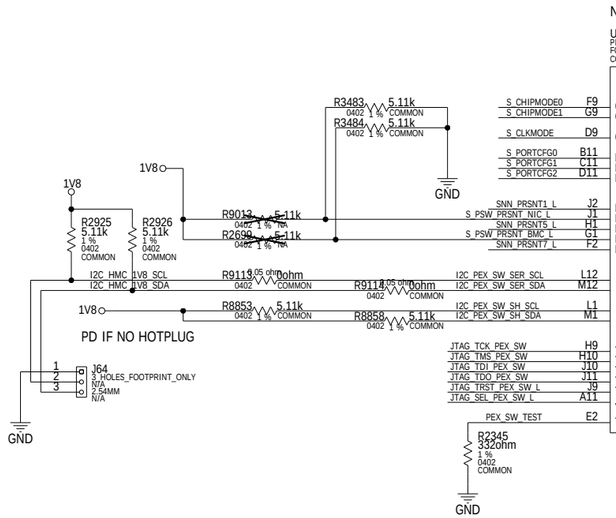


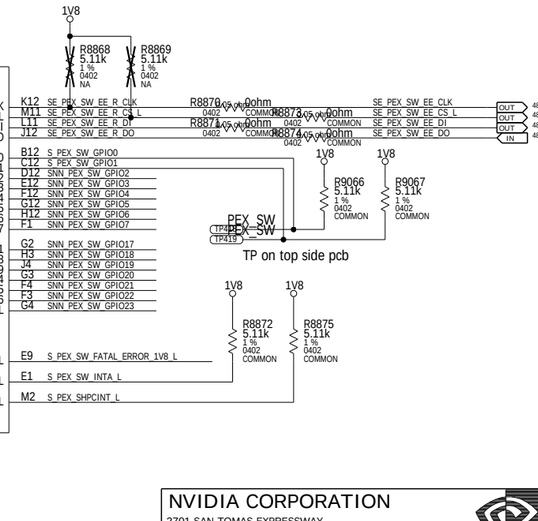












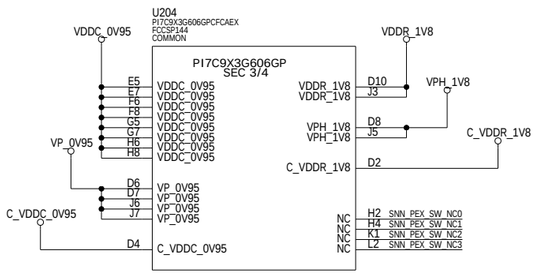
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PCIE GEN 3 SWITCH IO

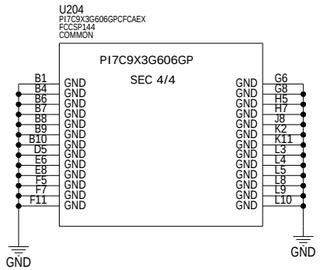
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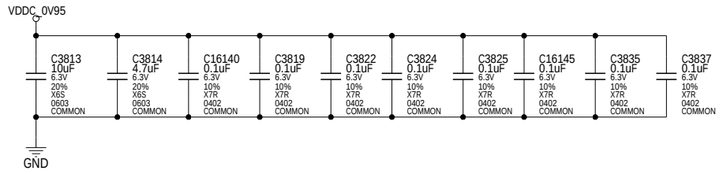
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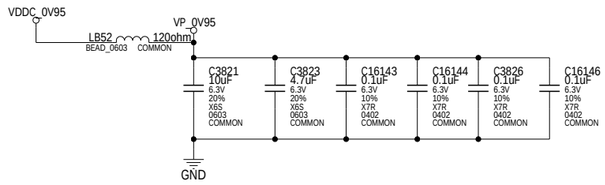
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PCB REV	PD4A4-D00	PAGE	46 OF 112
BOM REV	M	DATE	16-JUL-2026

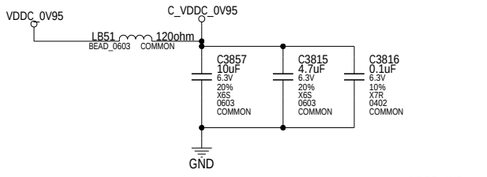
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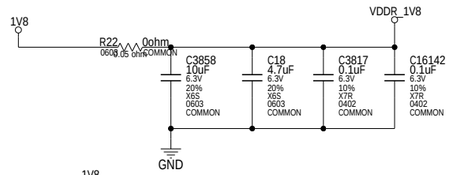


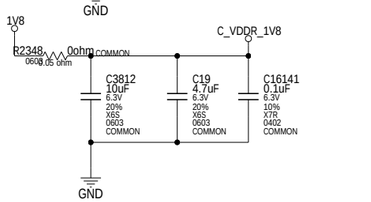


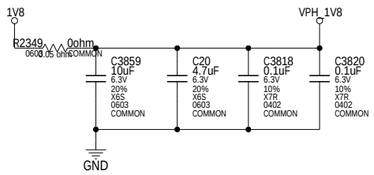












ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PCIE GEN 3 SWITCH POWER & GND

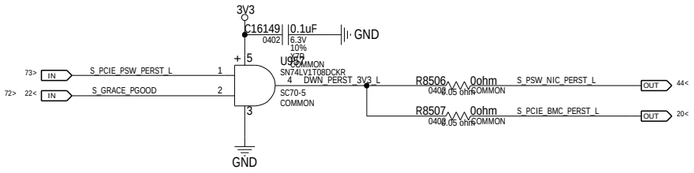
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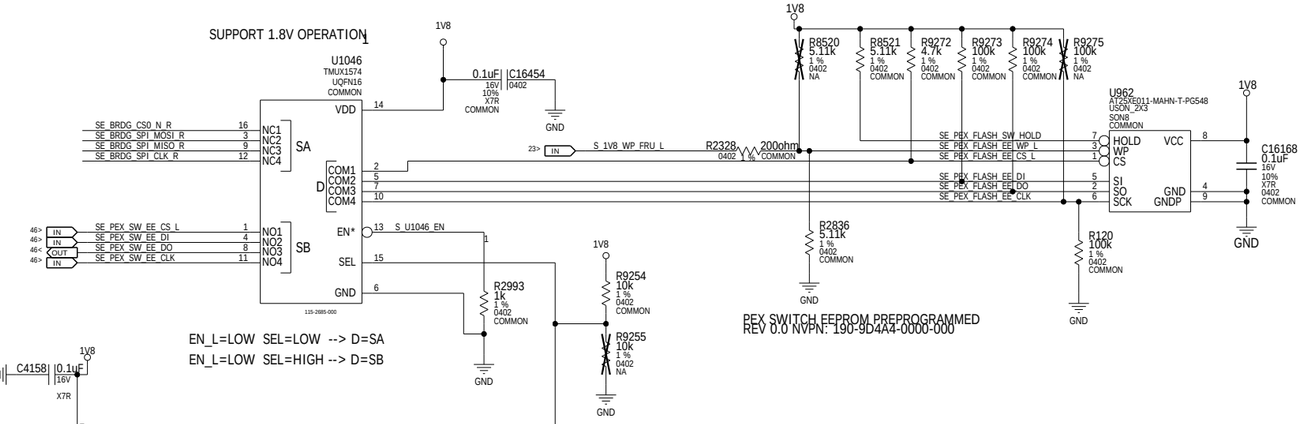
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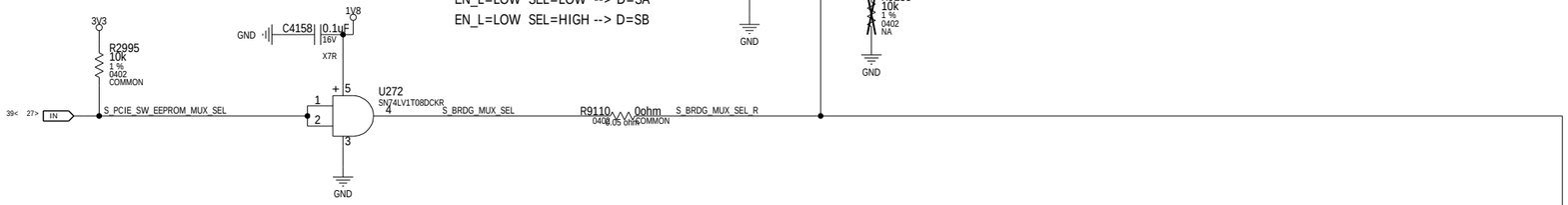
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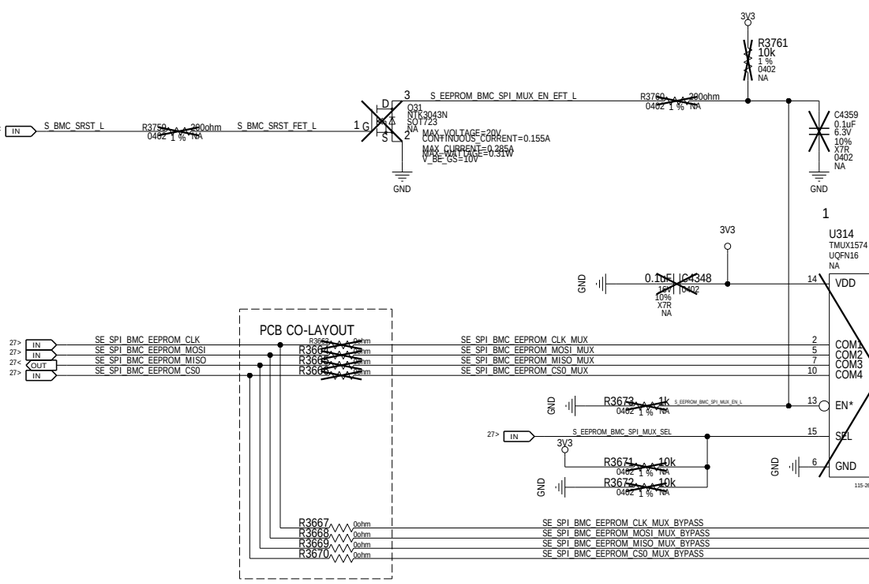
PEX SWITCH PERST FANOUT

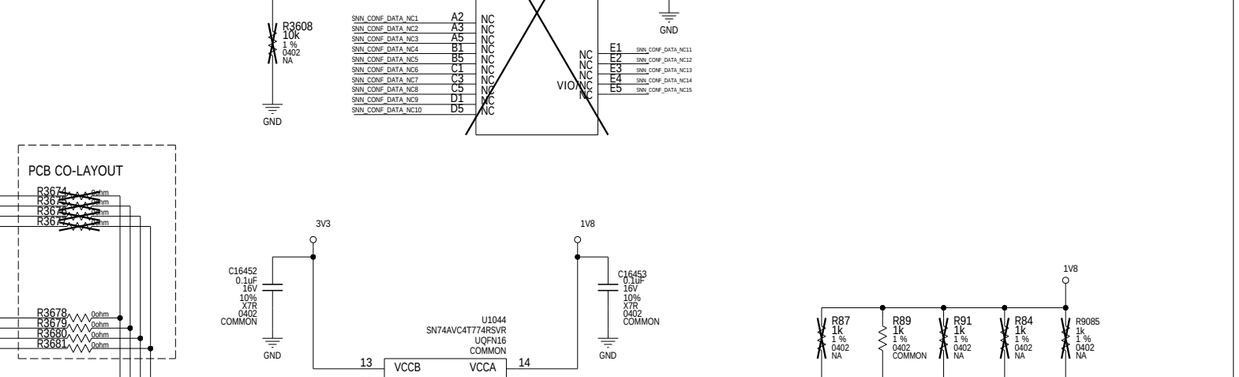


SUPPORT 1.8V OPERATION

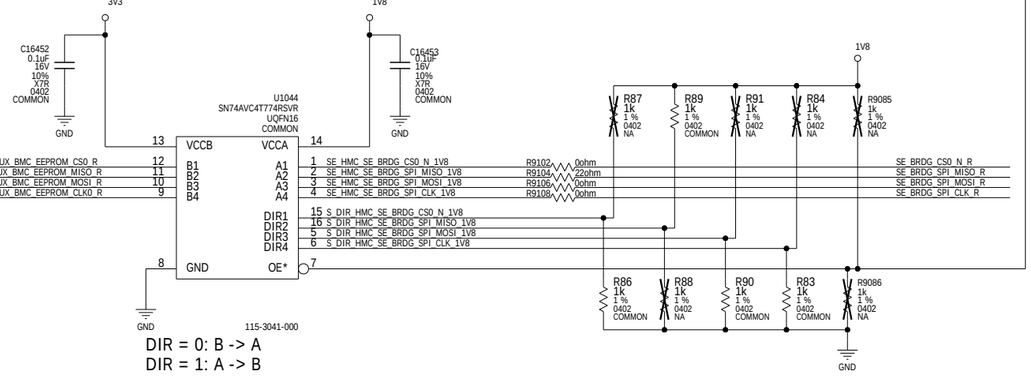








BMC SPI Config Flash backup for EMMC shortage



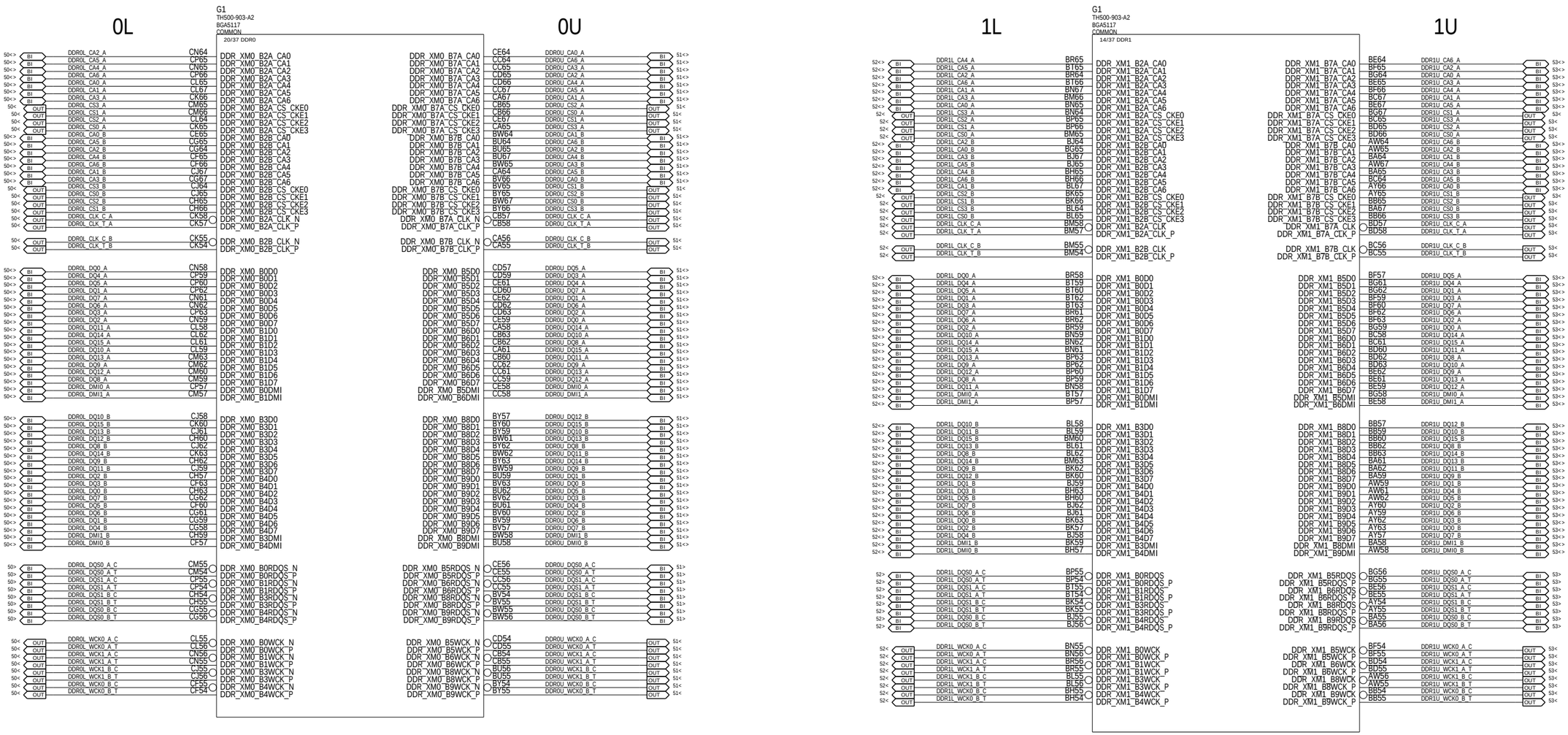
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PCIE GEN 3 FLASH AND BMC SPI FLASH

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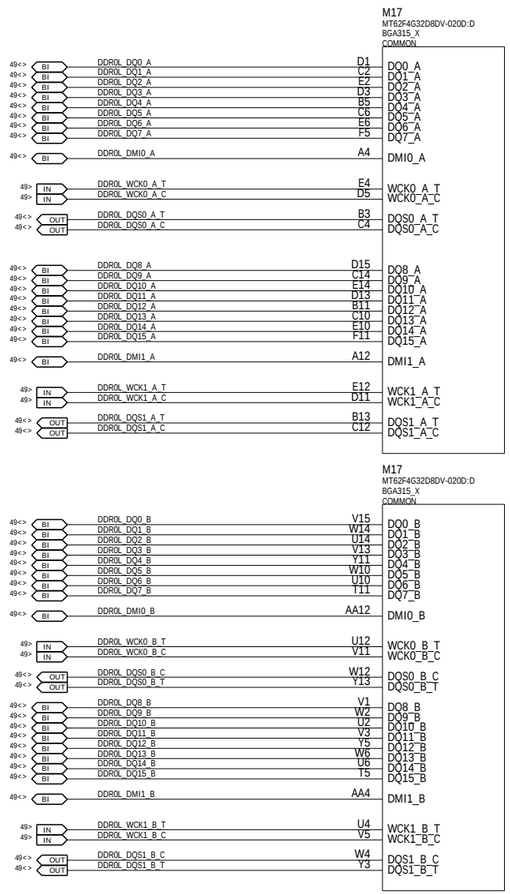
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PAGE DETAIL	GRACE - DDR0(U,L) + DDR1(U,L)

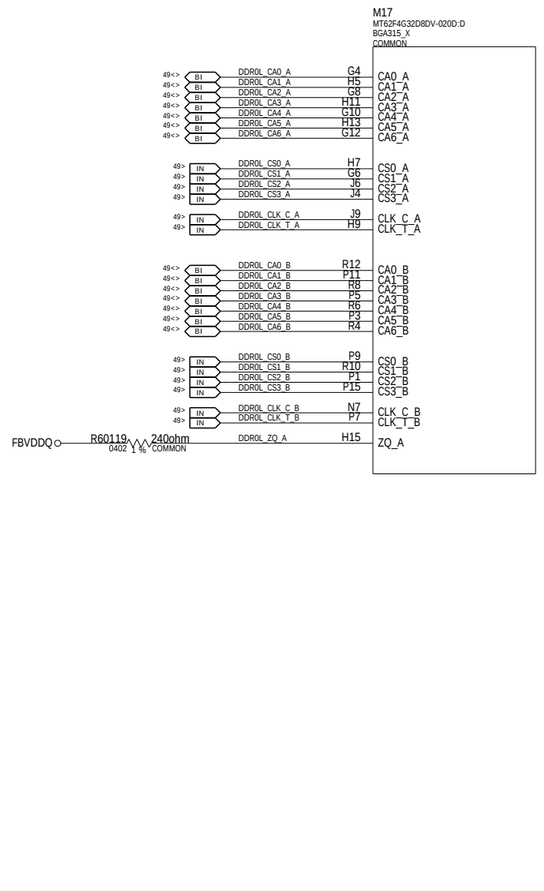
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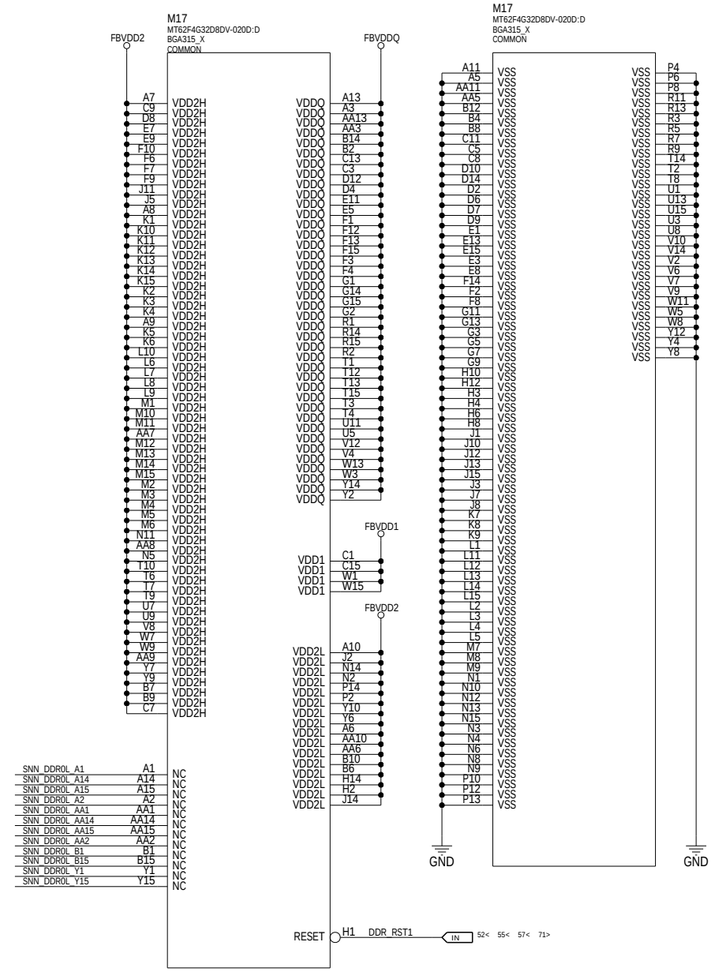
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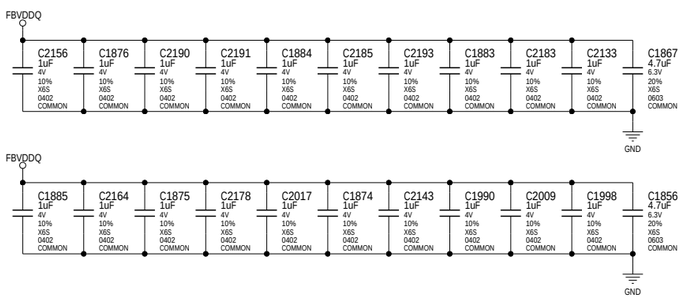
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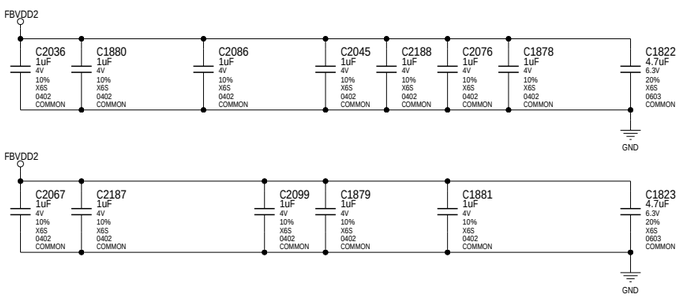
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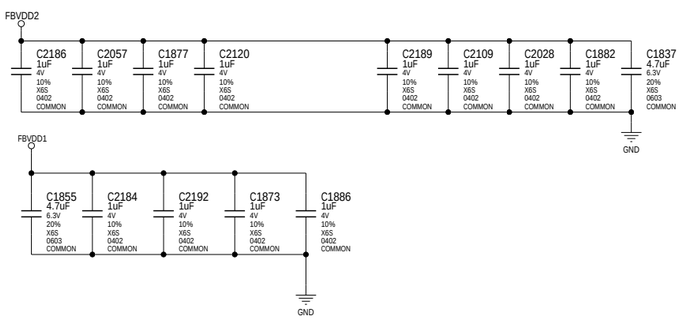












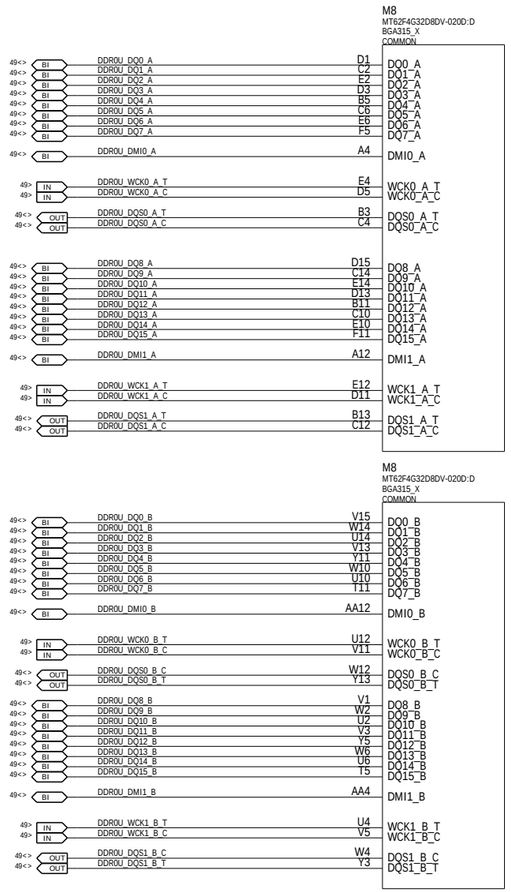
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - DDR0L

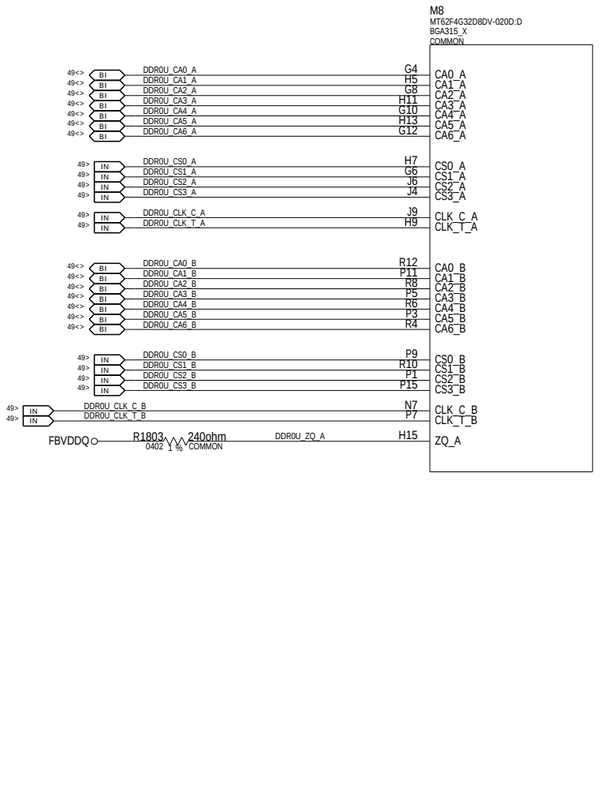
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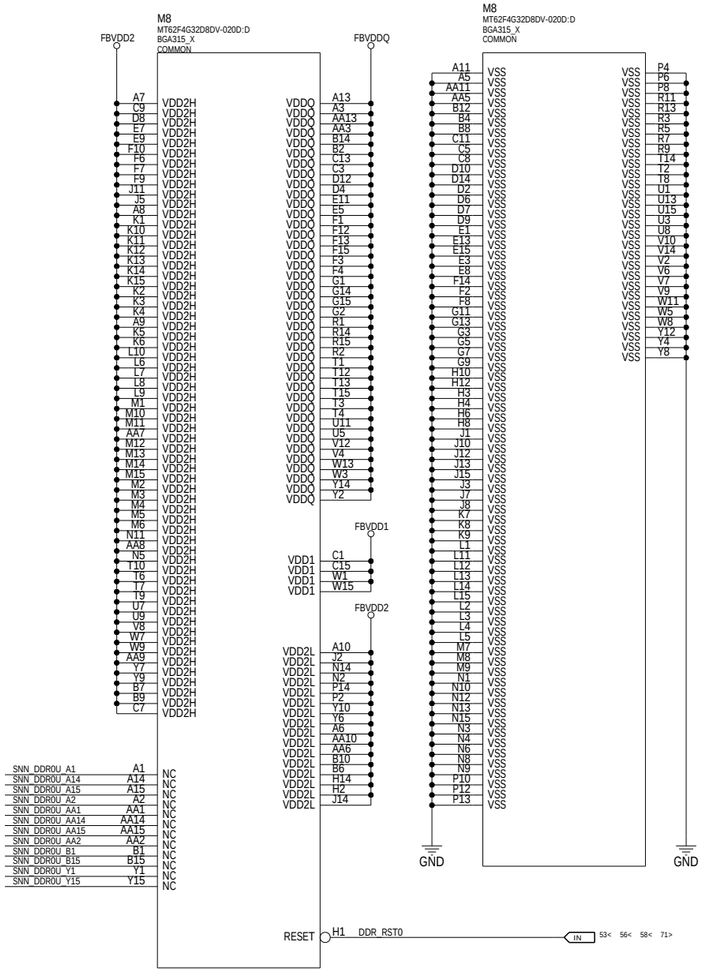
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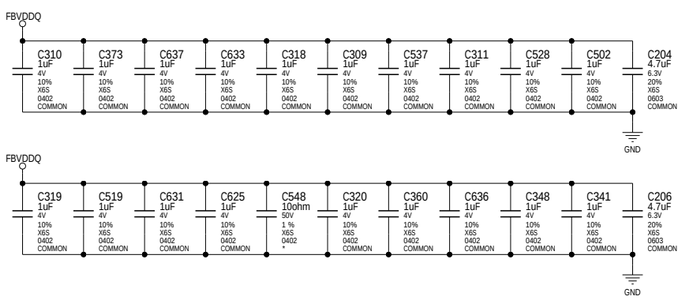
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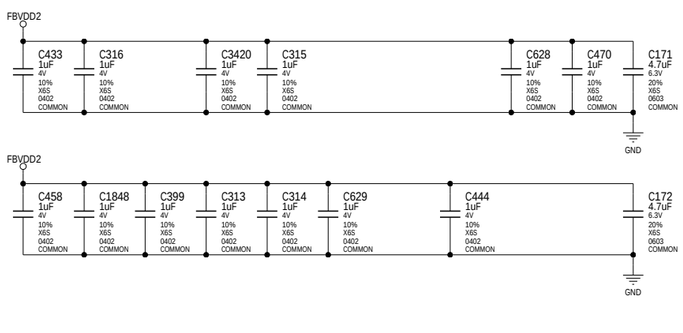
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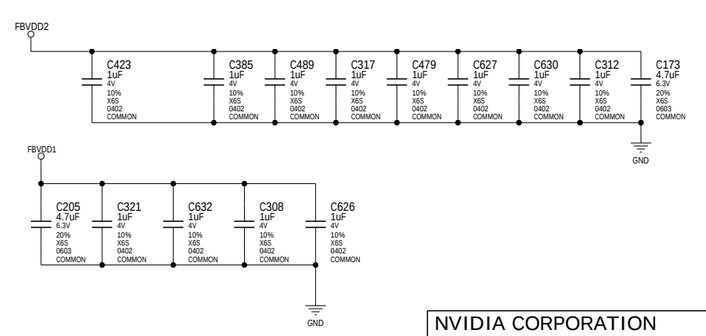












ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - DDR0U

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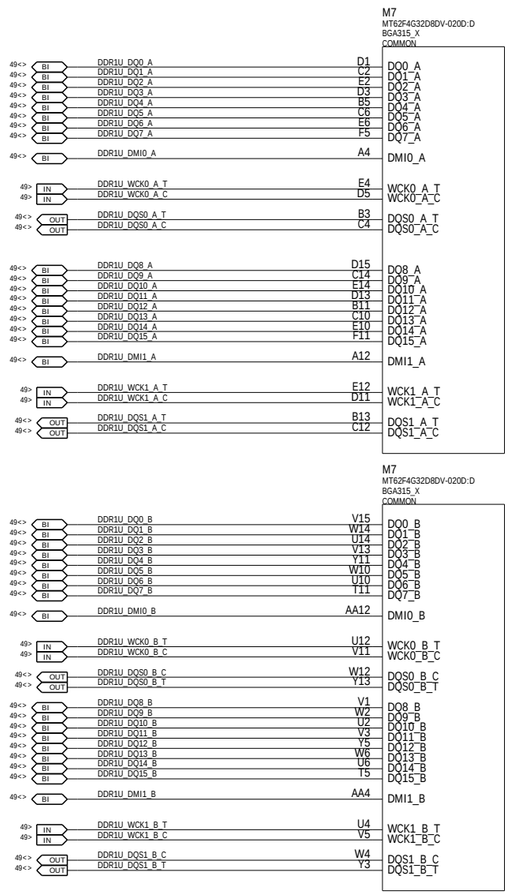
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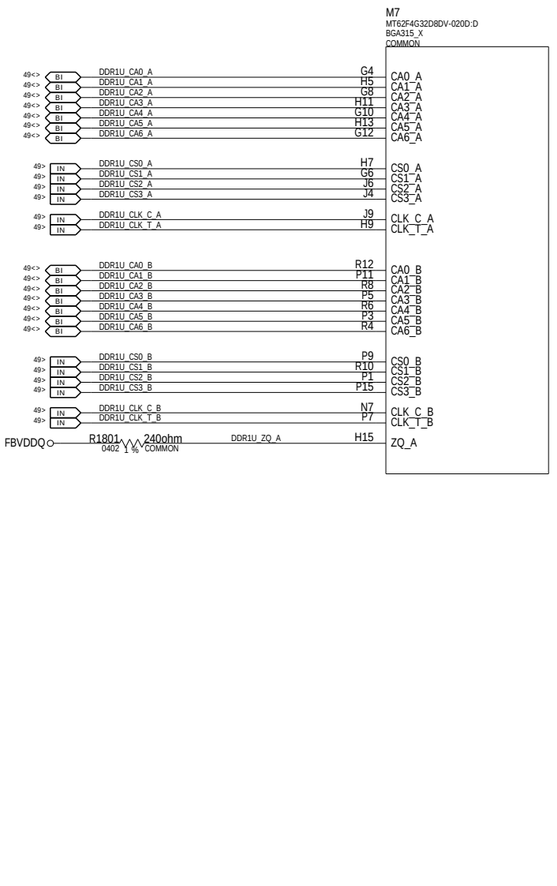
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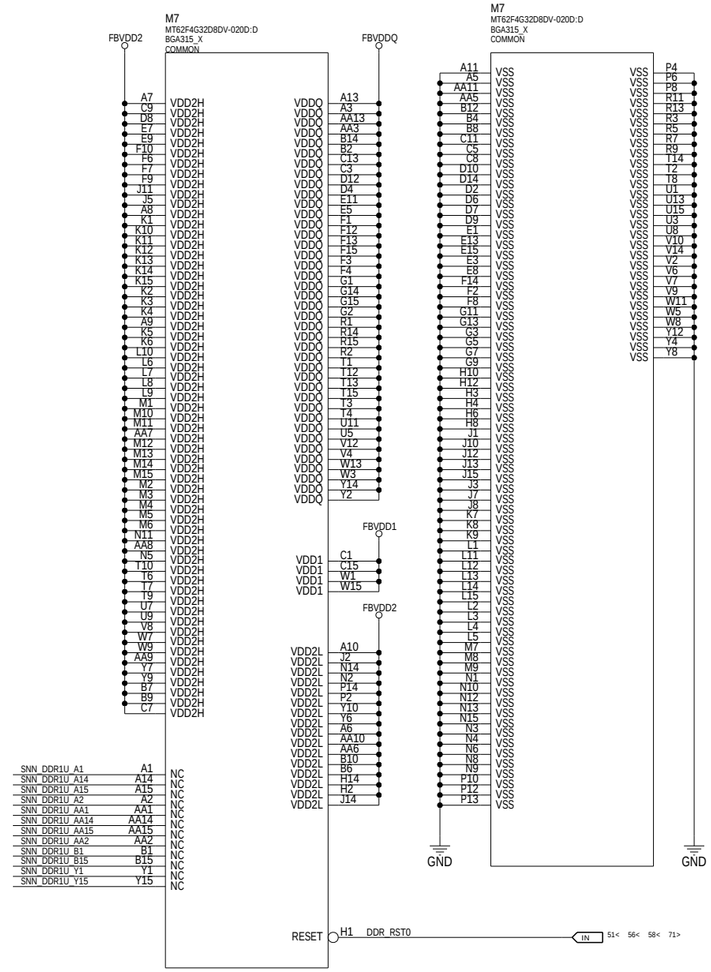
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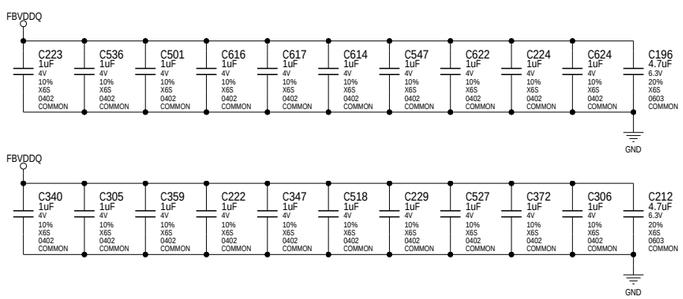
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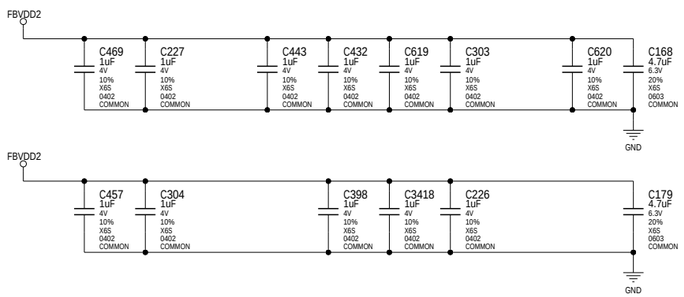
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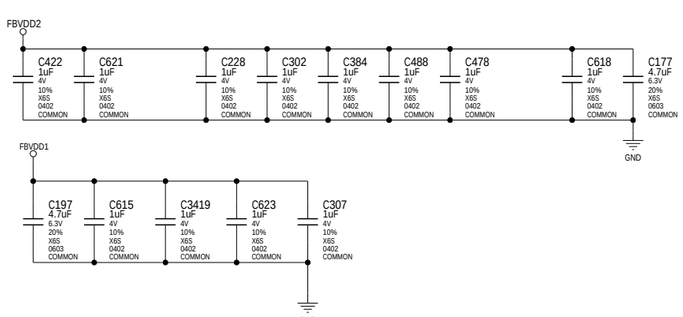












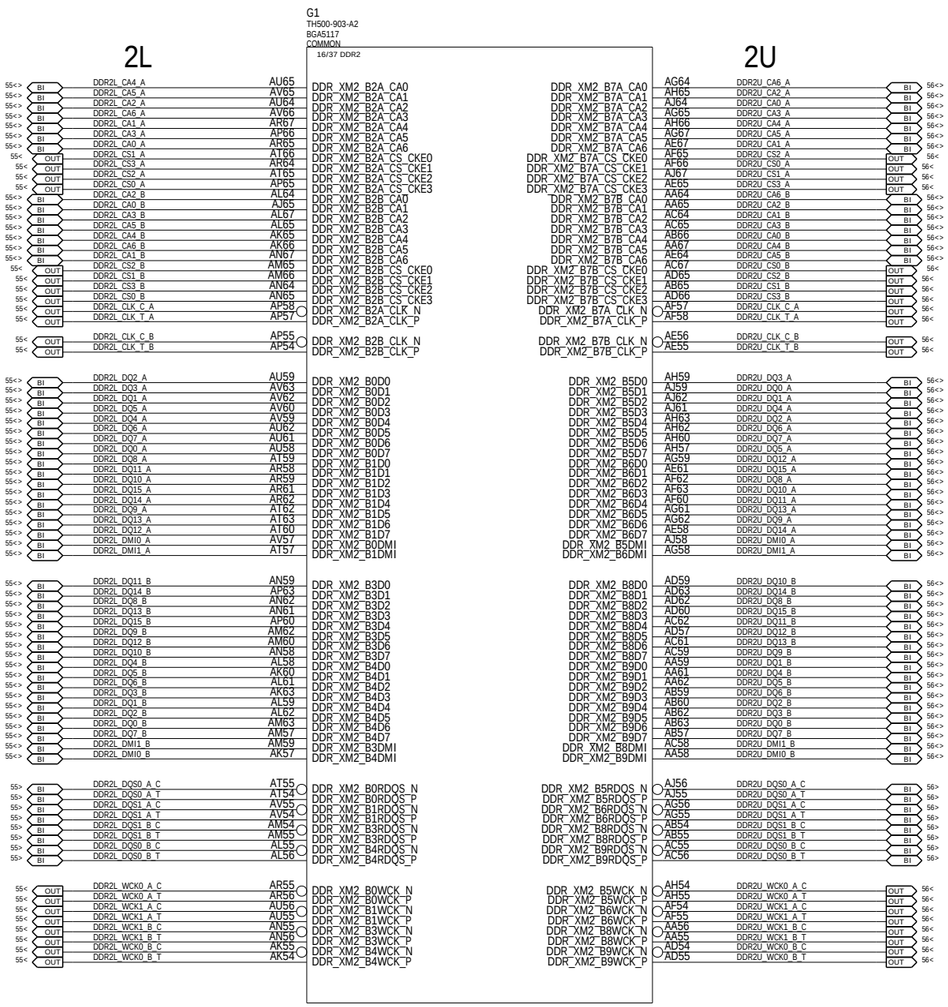
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PAGE DETAIL	GRACE - DDR1U

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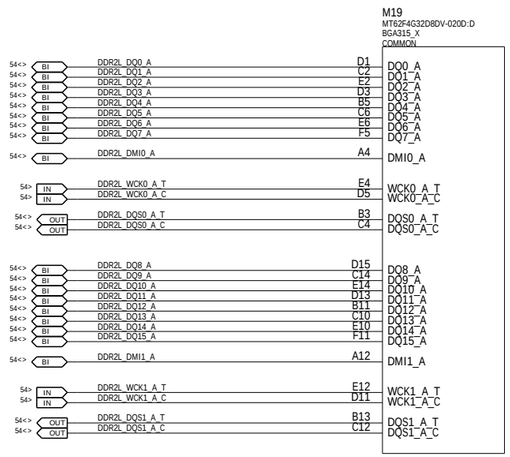
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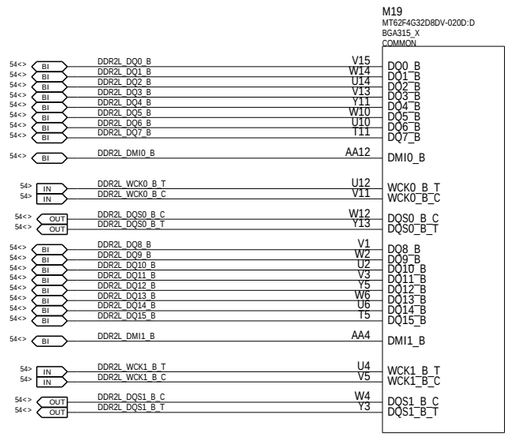
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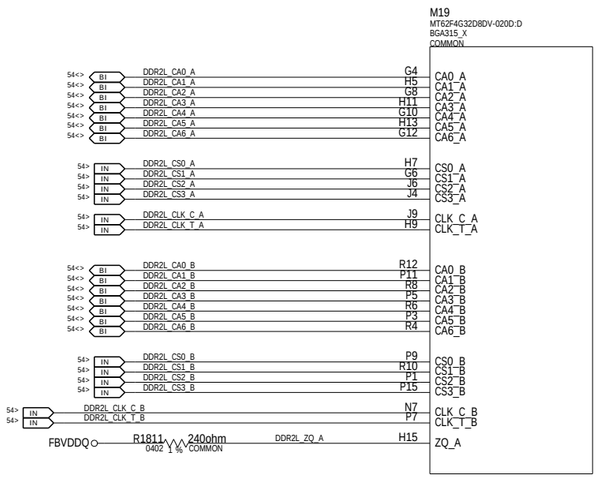
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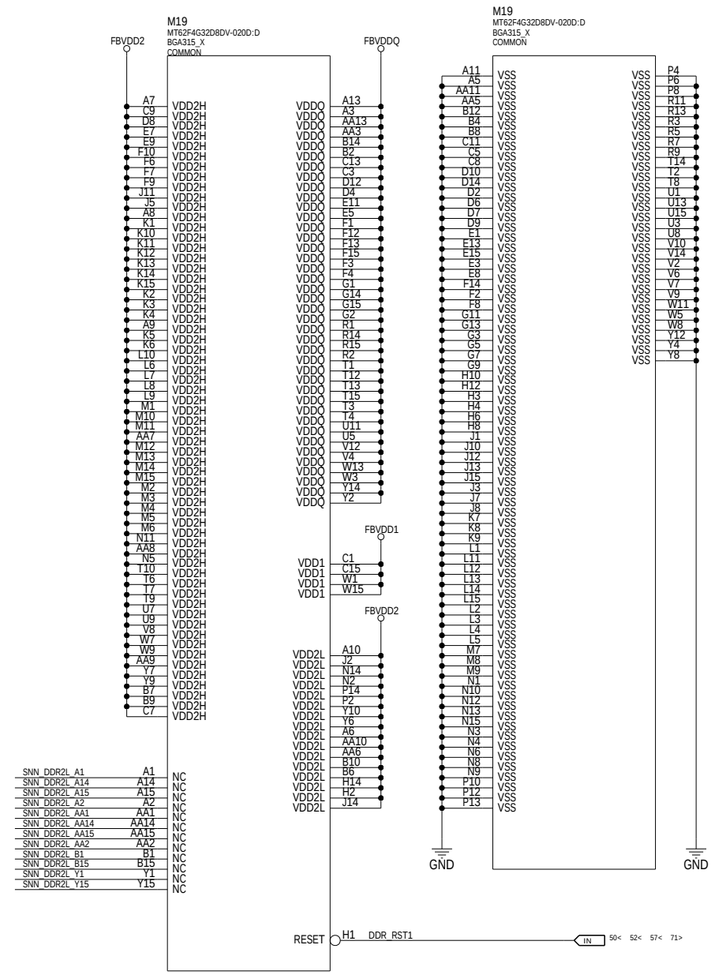
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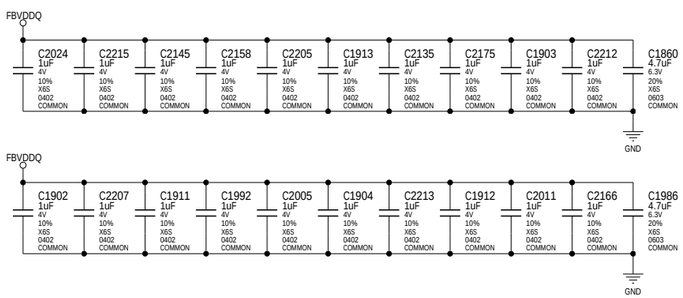
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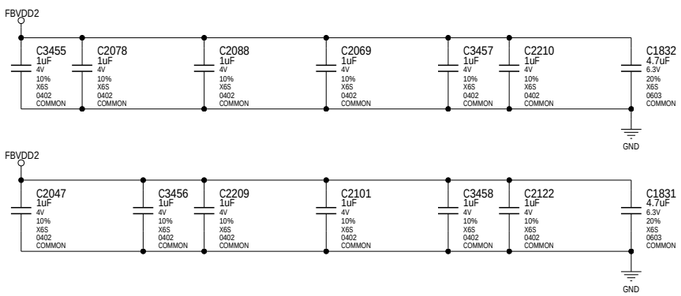


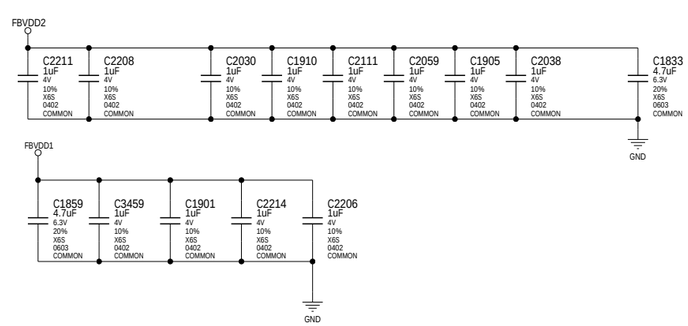












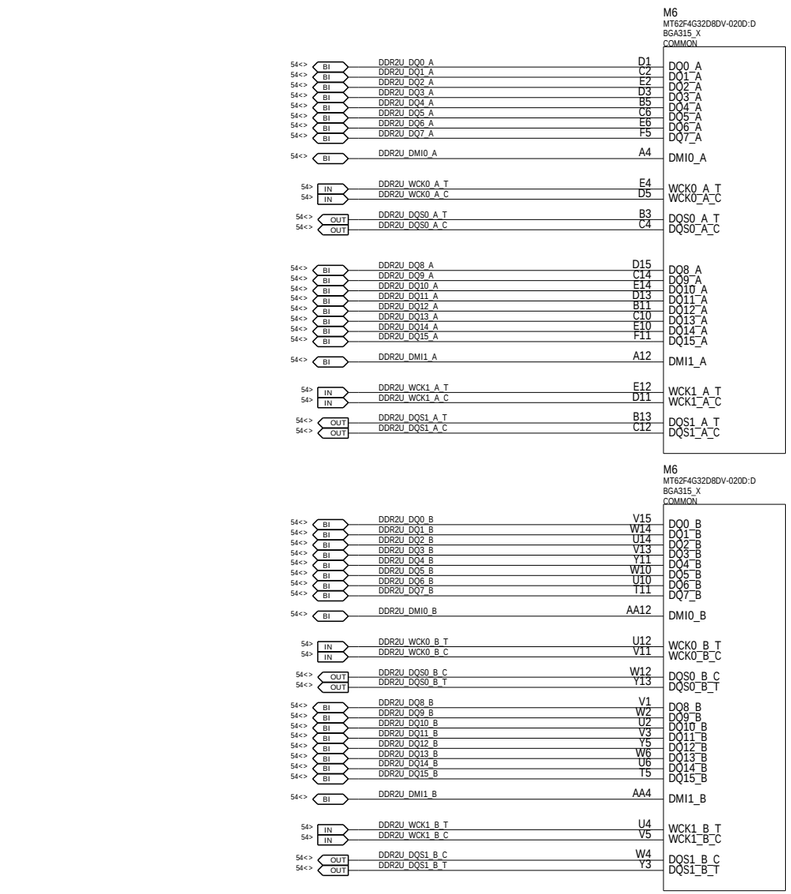
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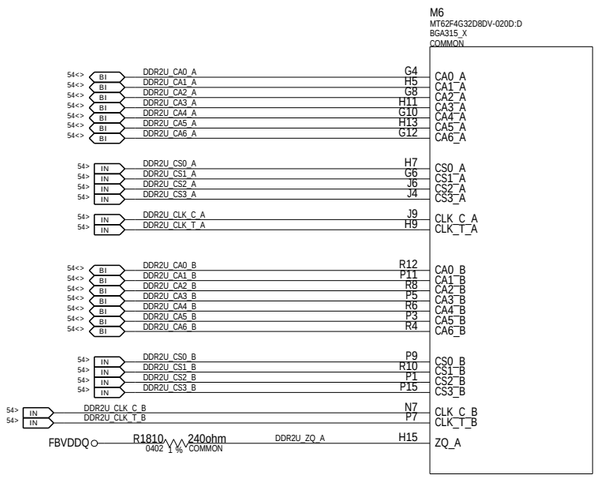
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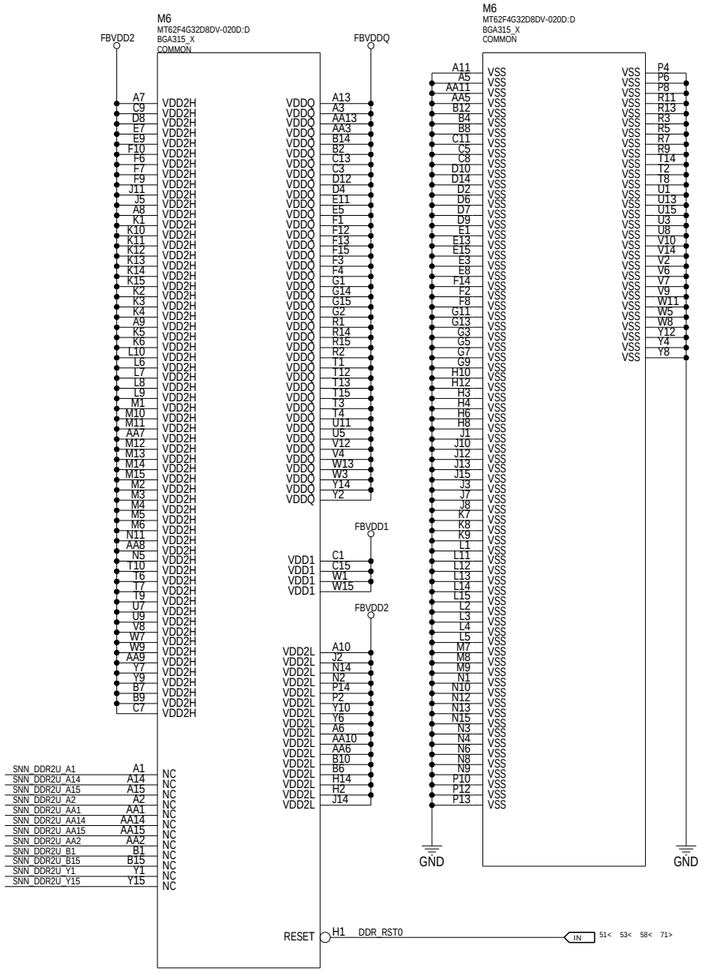
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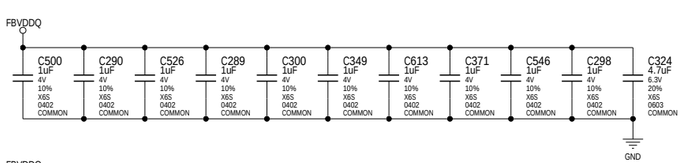
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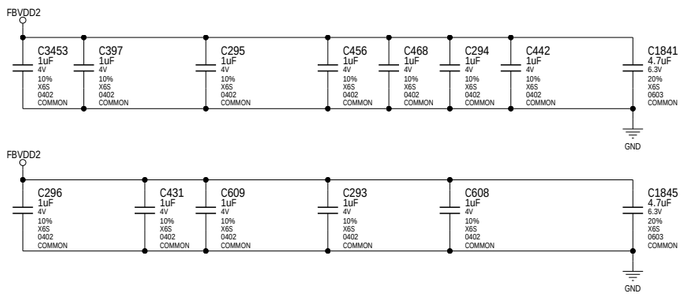
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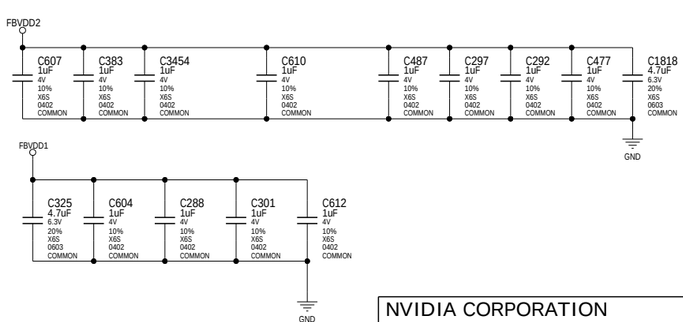












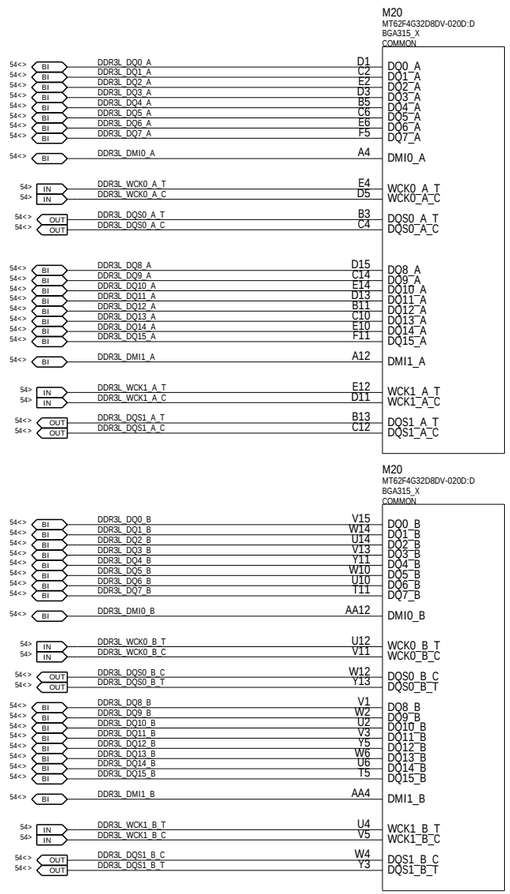
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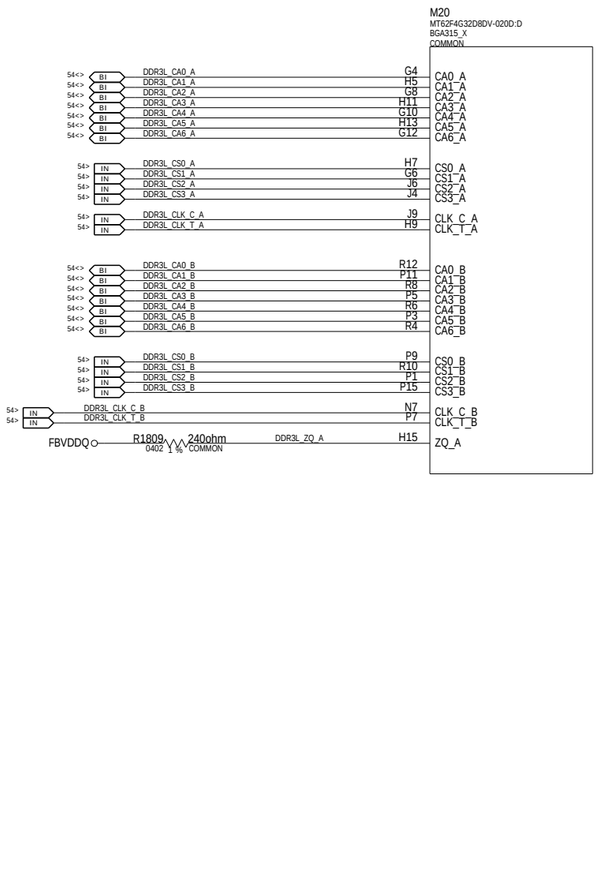
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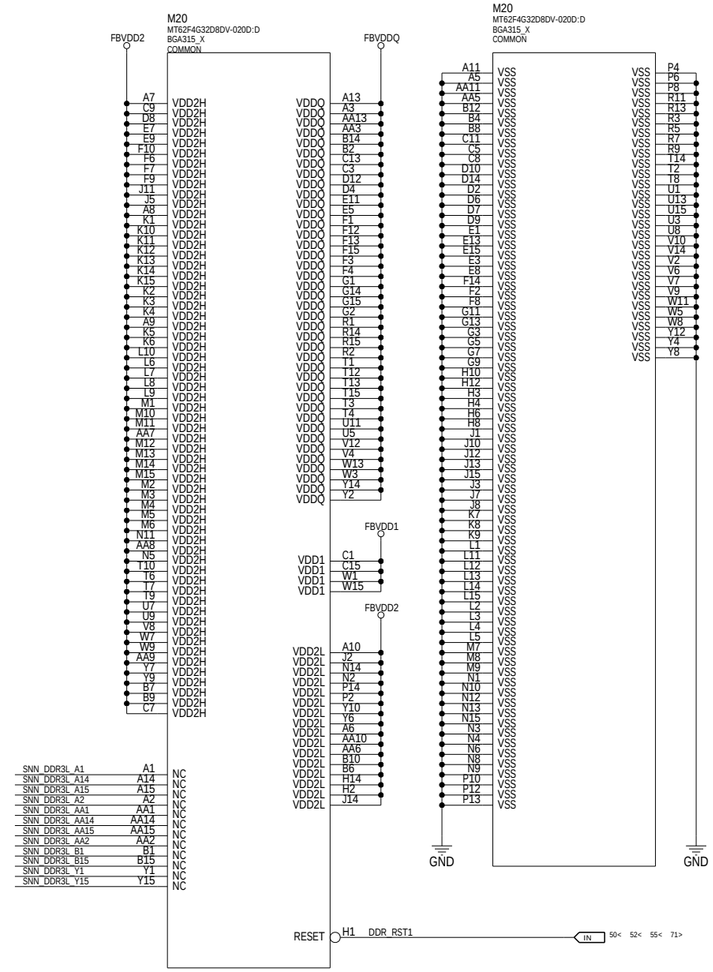
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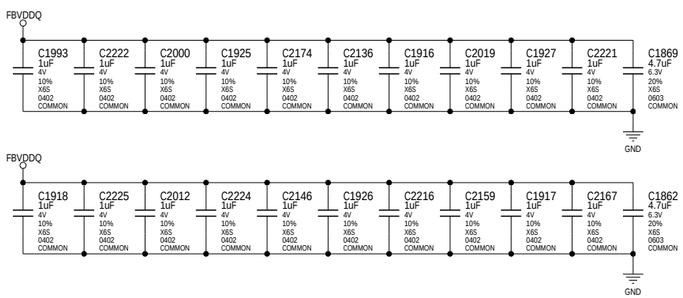
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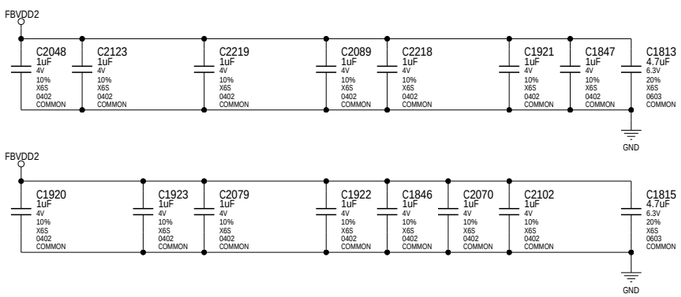
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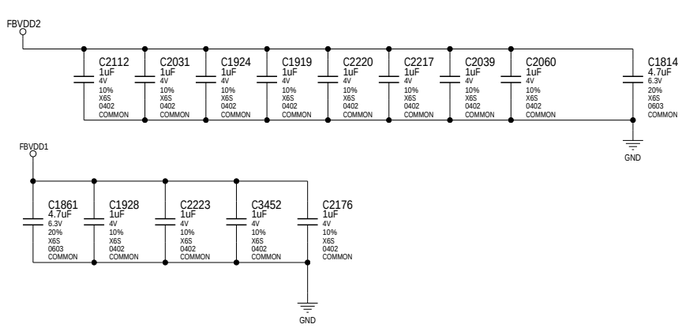












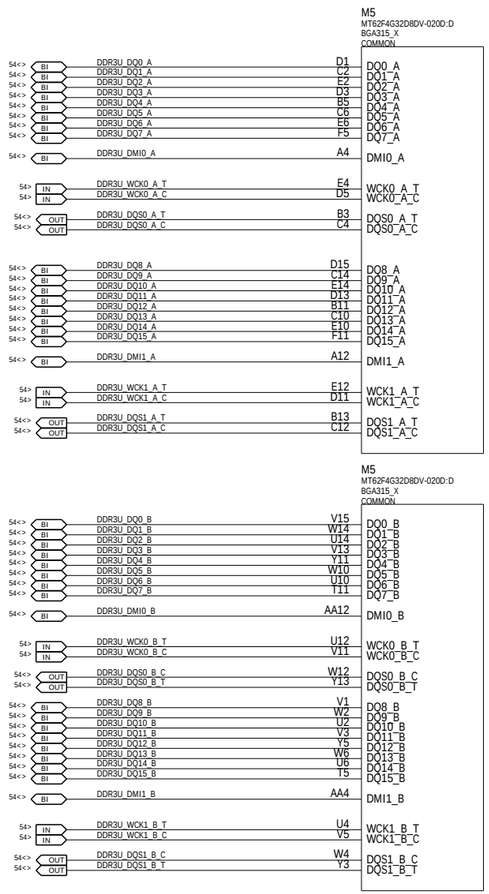
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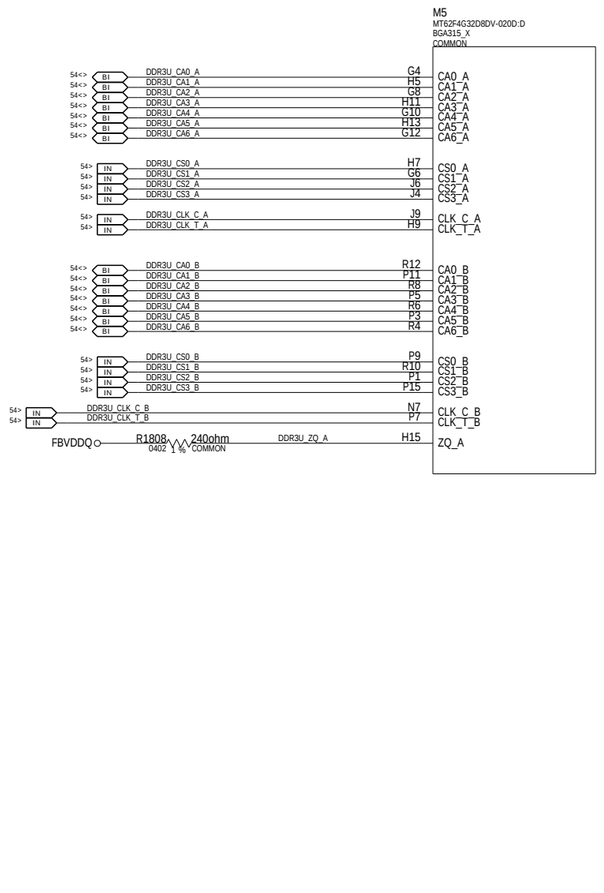
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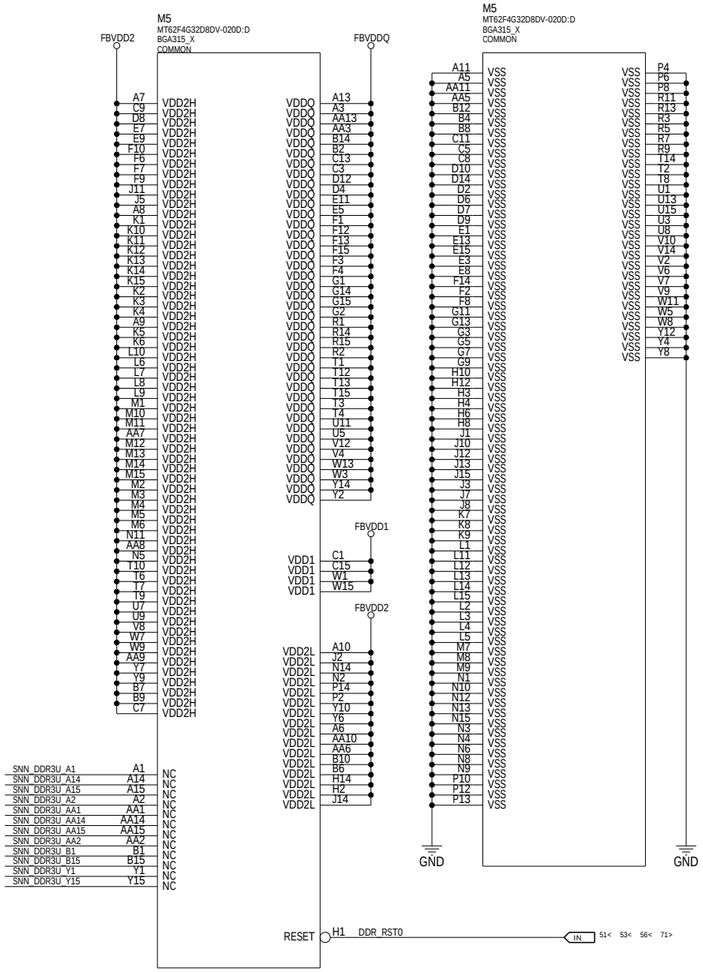
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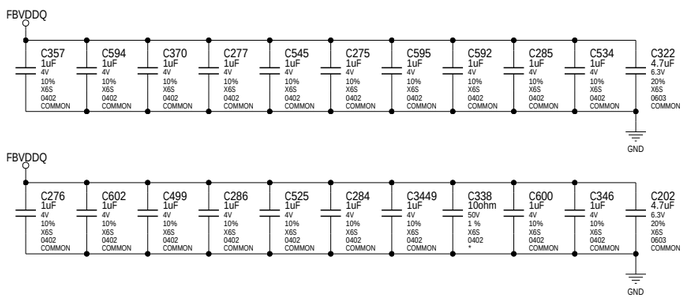
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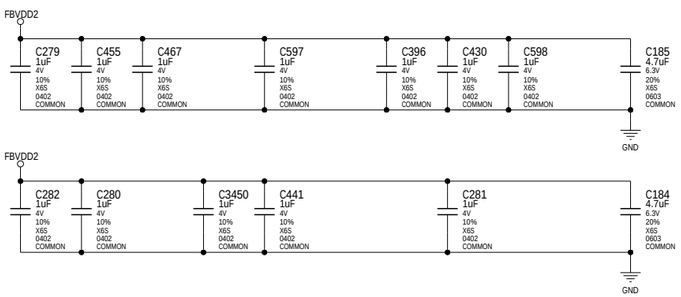
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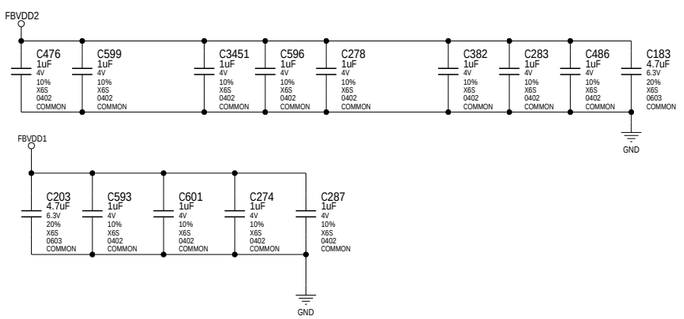












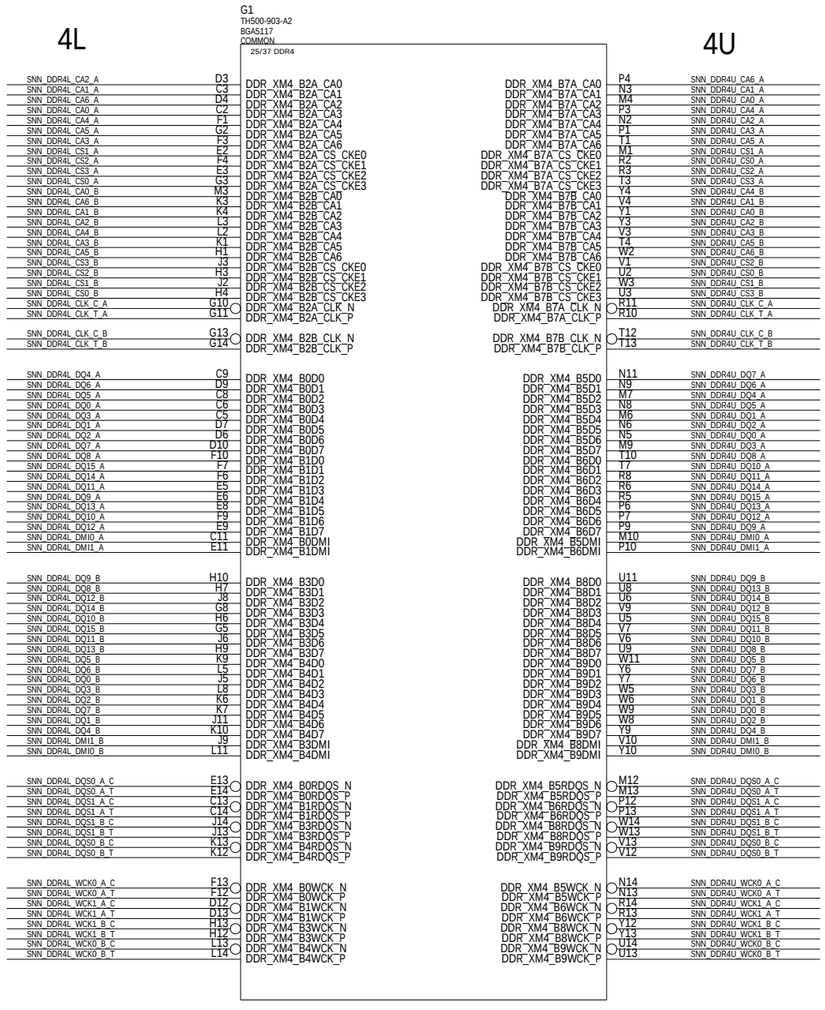
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - DDR3U

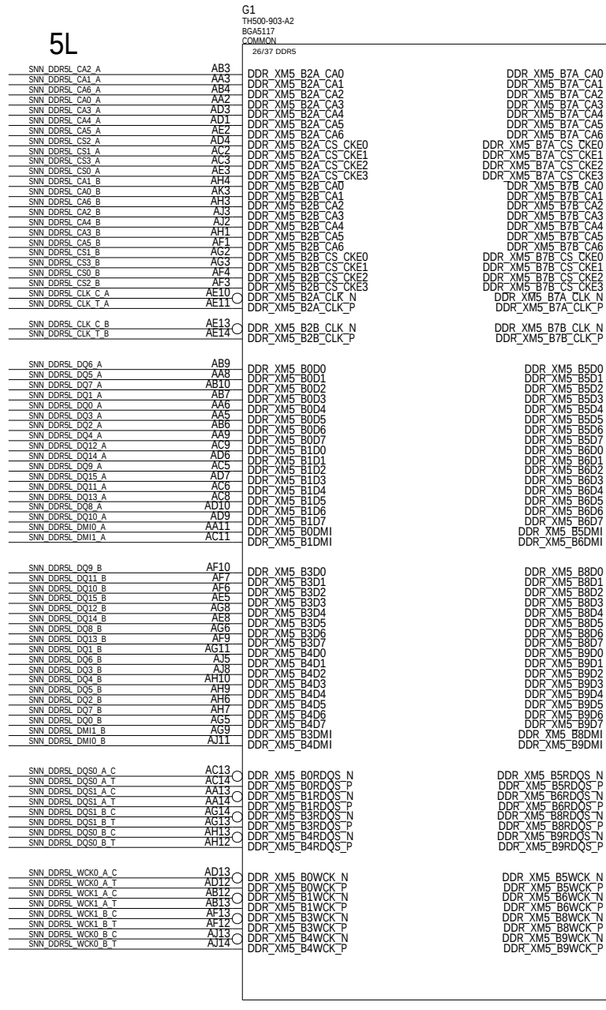
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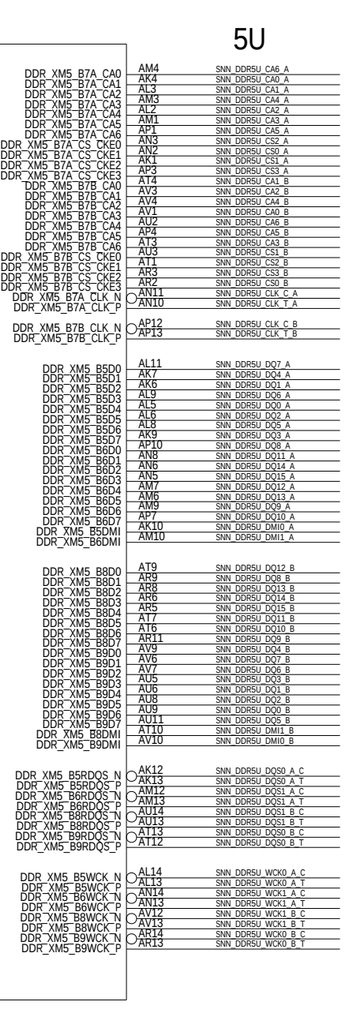
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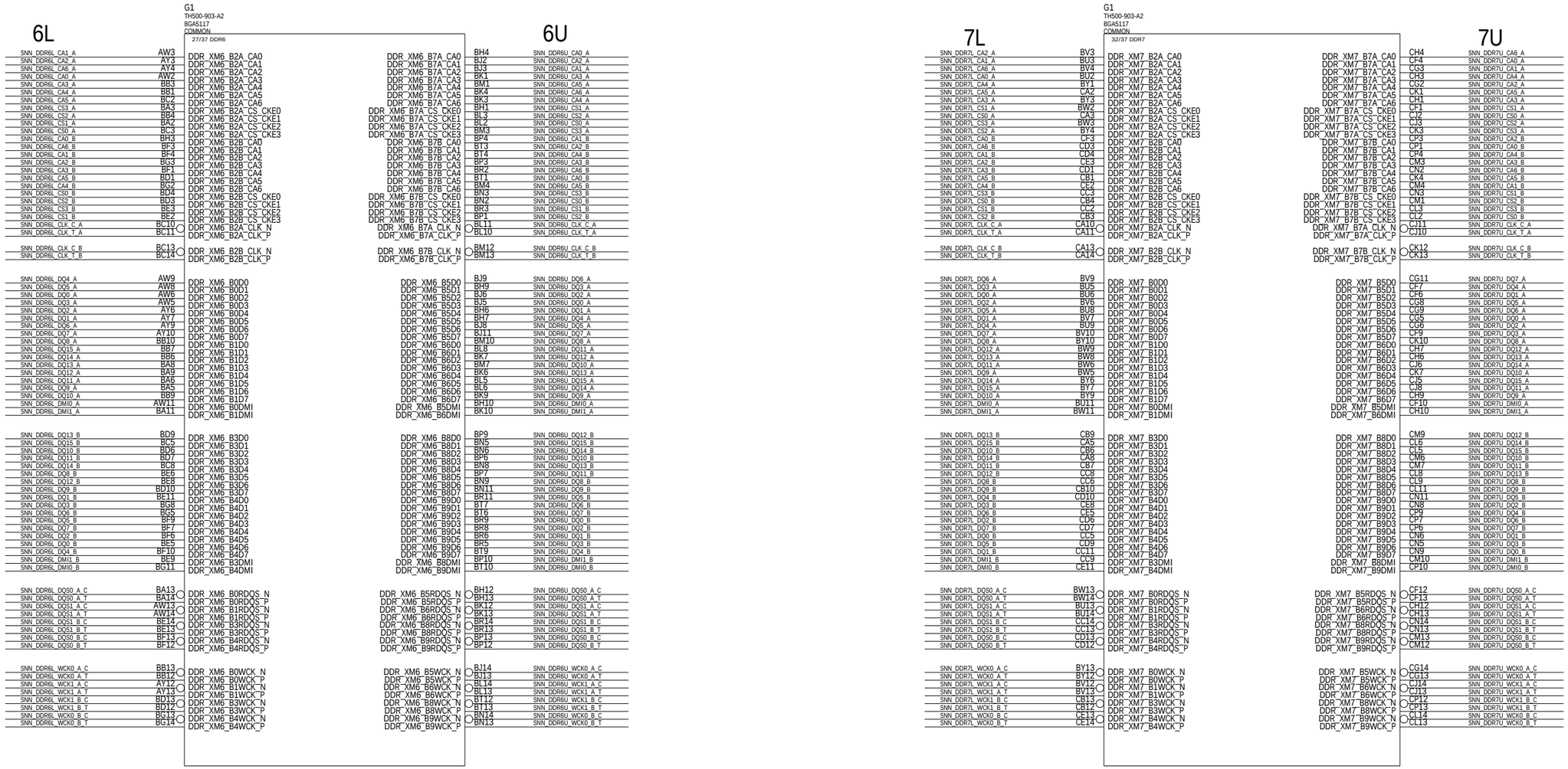
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - DDR4(U,L) + DDR5(U,L)

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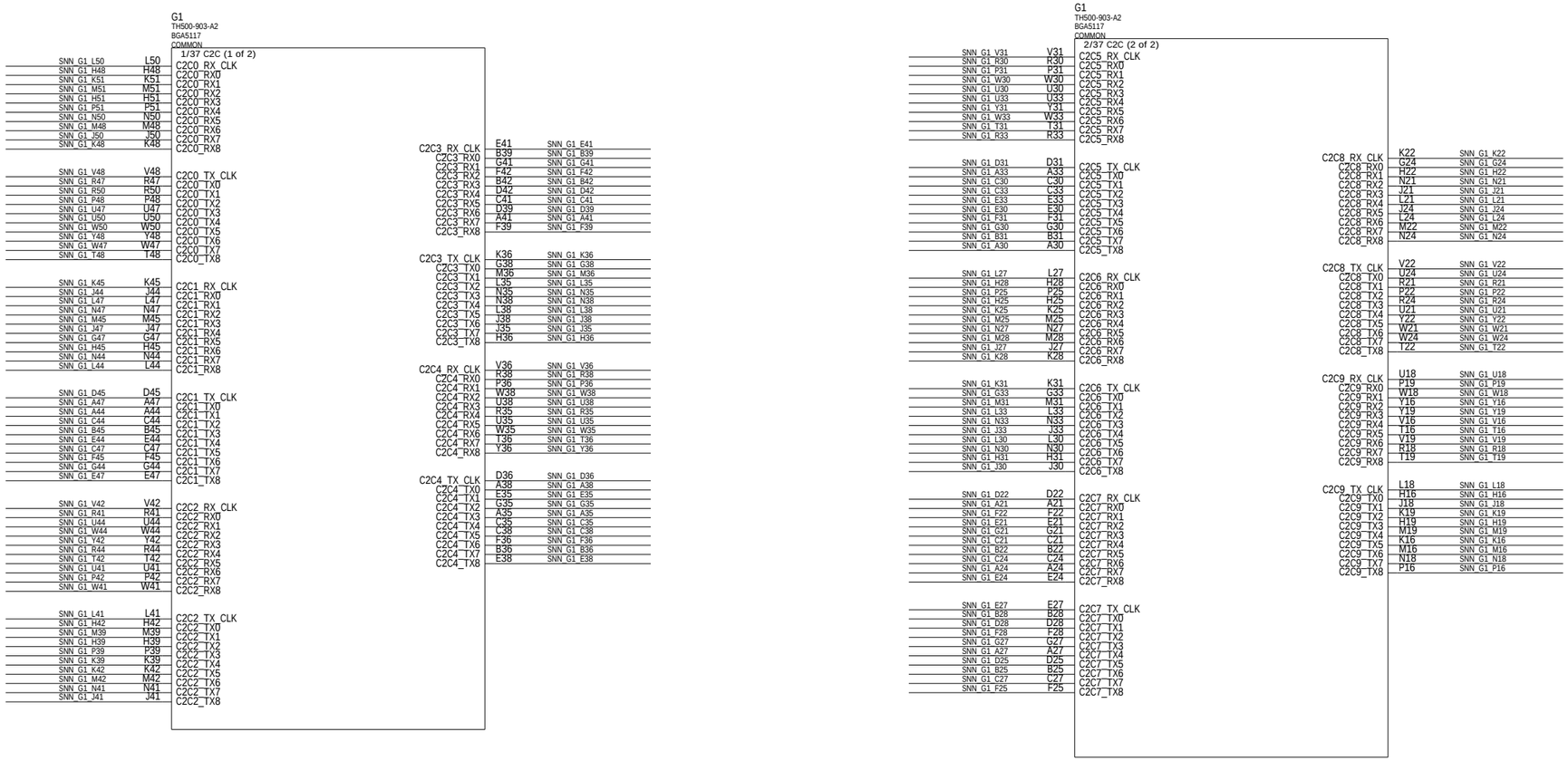
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - DDR6(U,L) + DDR7(U,L)

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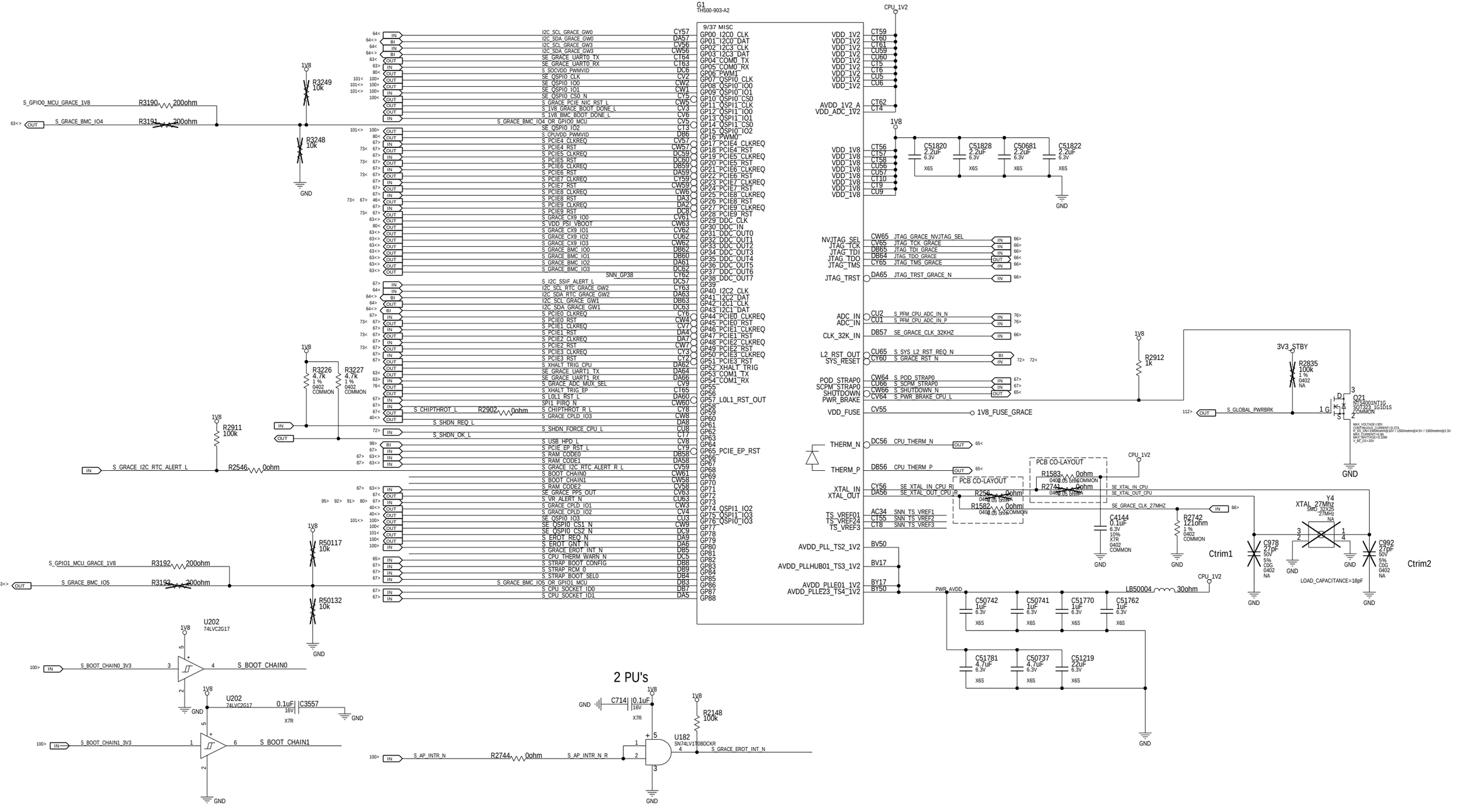
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - C2C (UNUSED)

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - GPIOS

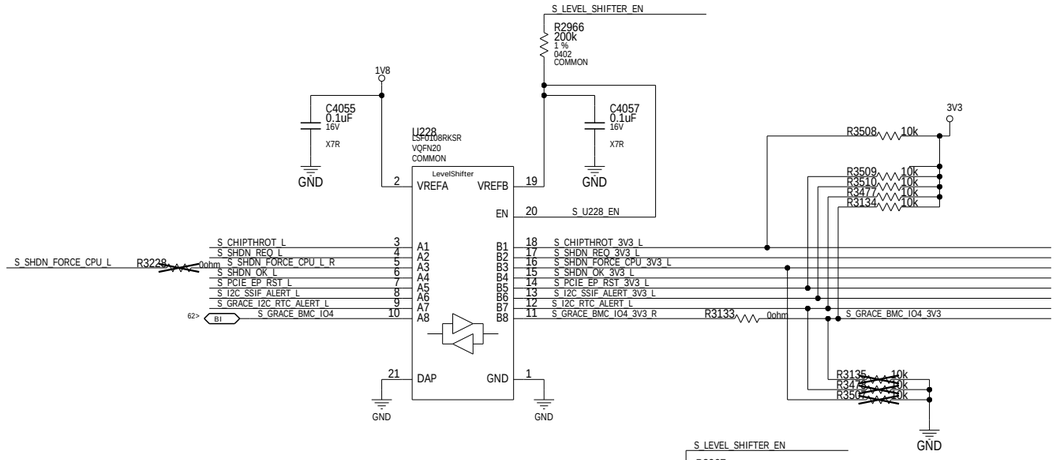
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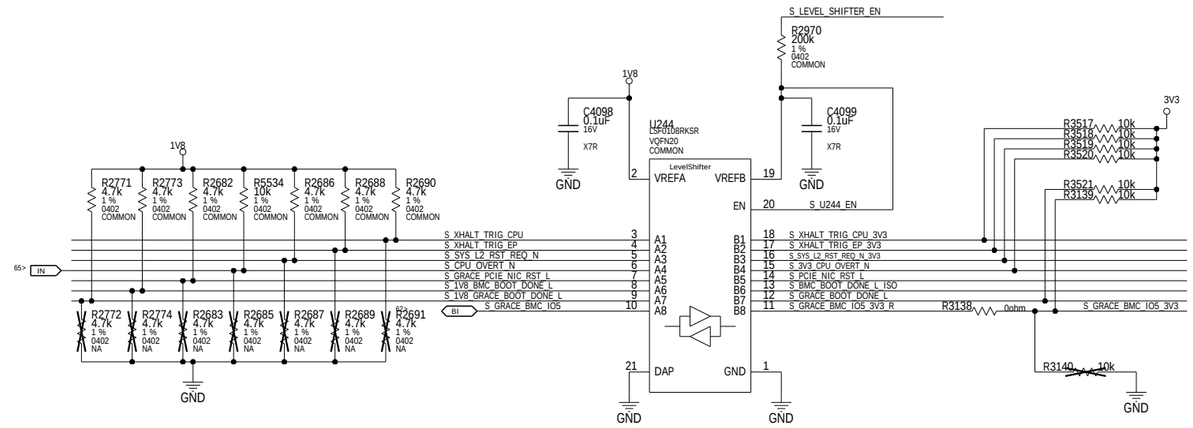
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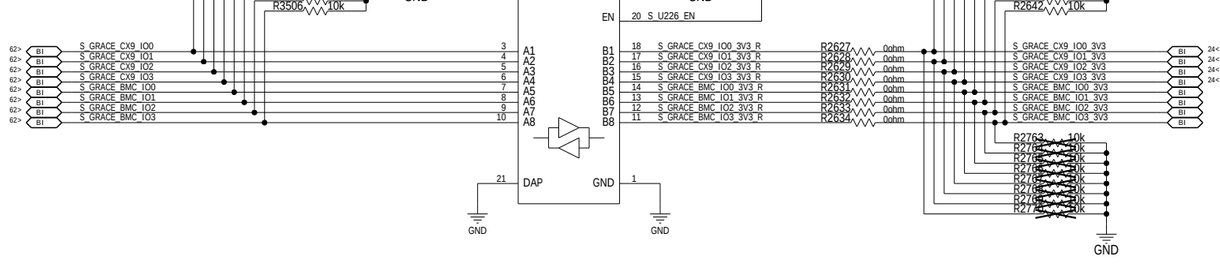
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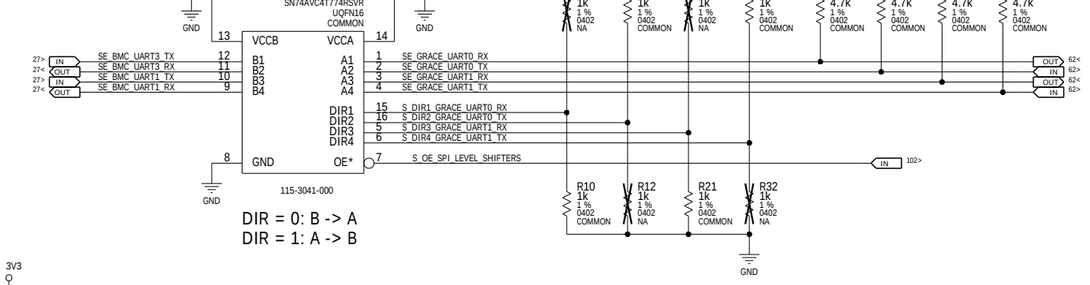
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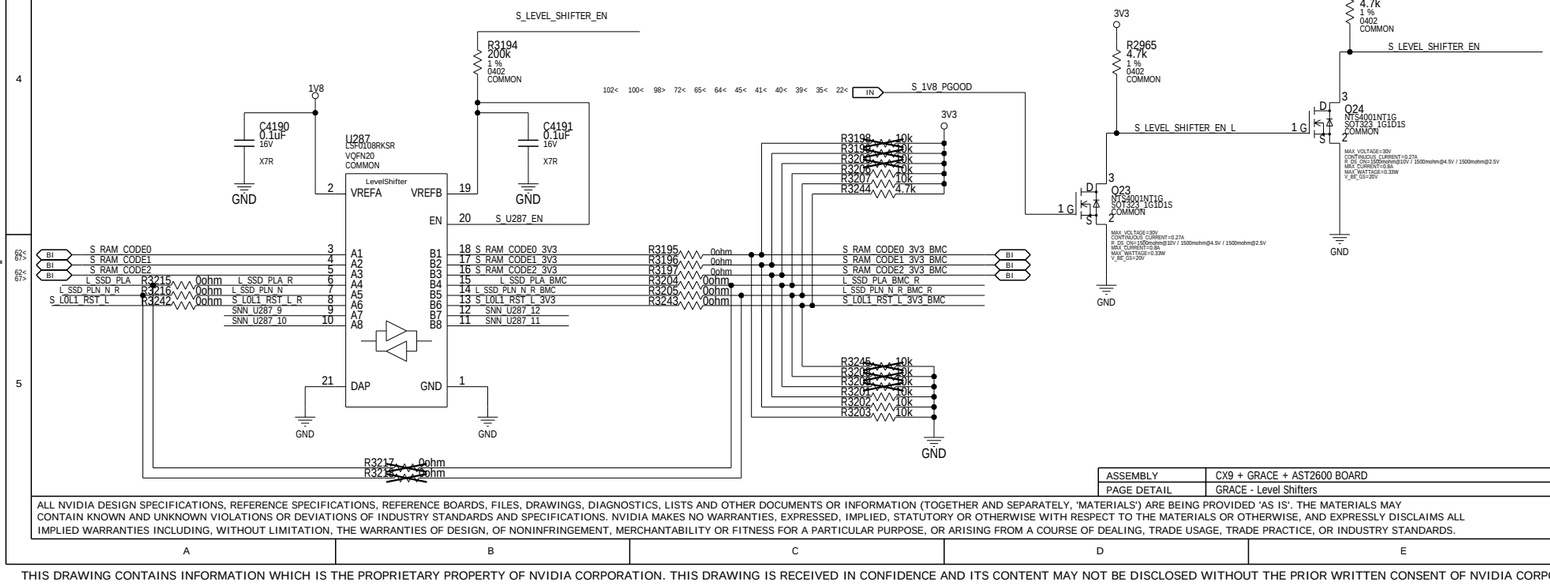
add serial resistors











ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - Level Shifters

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - I2C Level Shifters

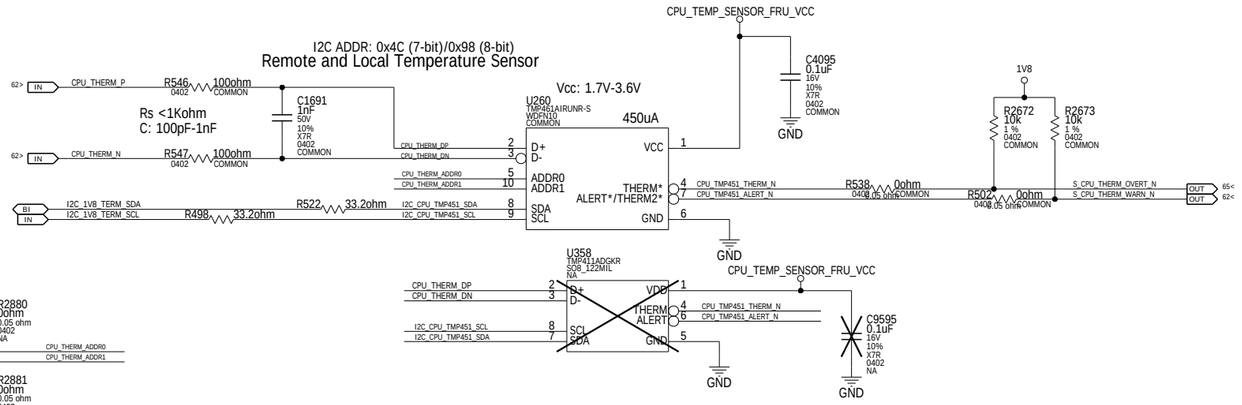
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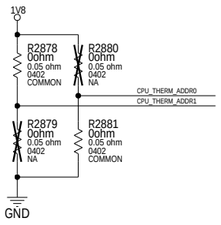
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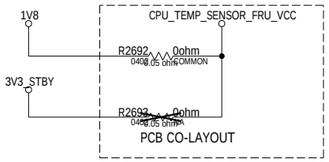
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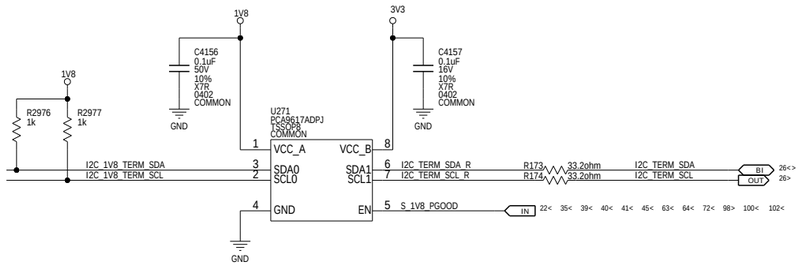
CPU TEMP SENSOR (EMULATED)

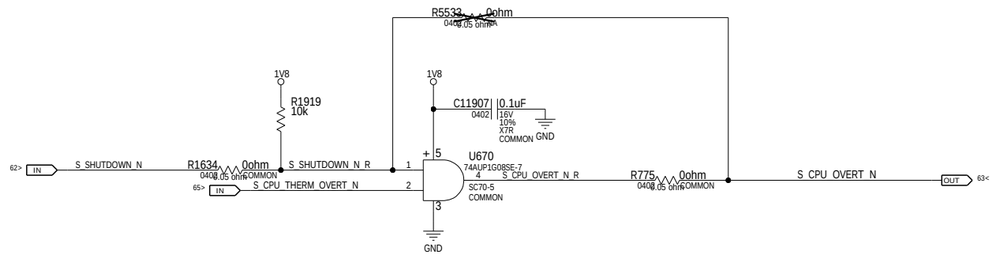






fix naming + connectivity





ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - Temp sensor

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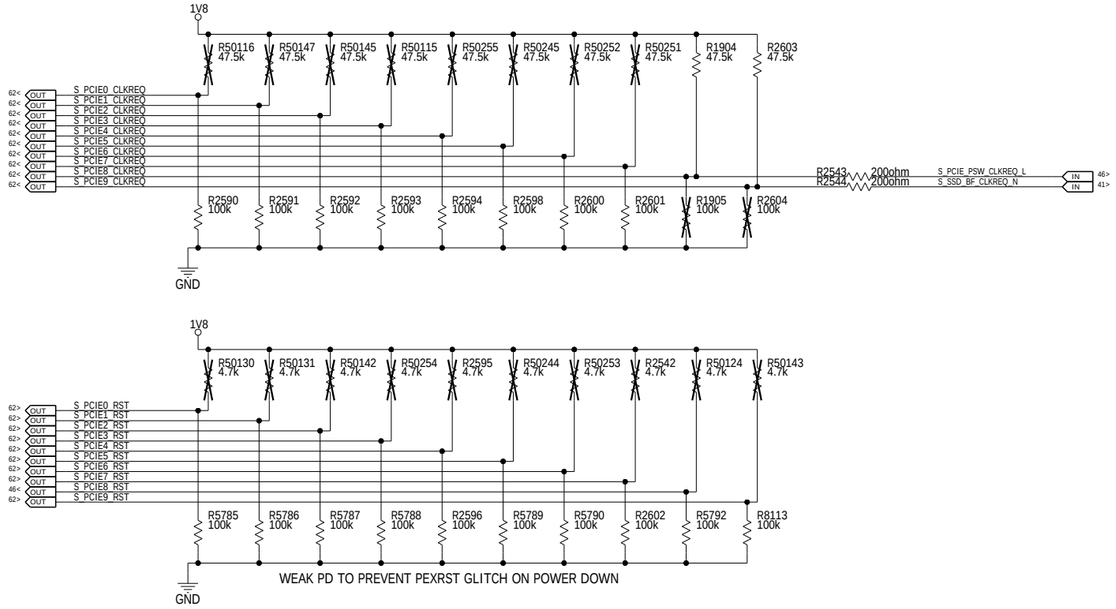
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - MISC

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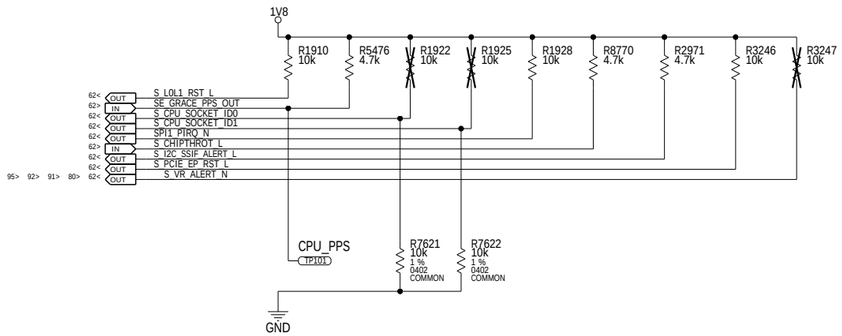
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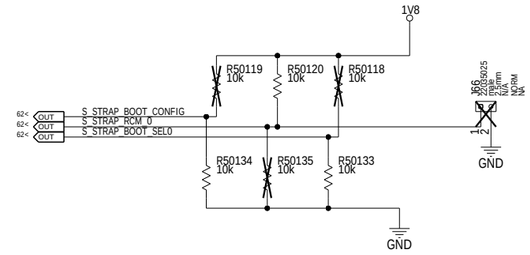
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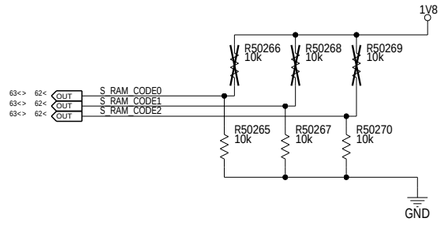
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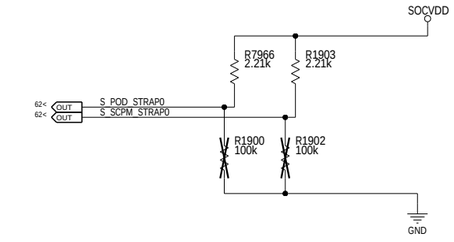


WEAK PD TO PREVENT PEXRST GLITCH ON POWER DOWN









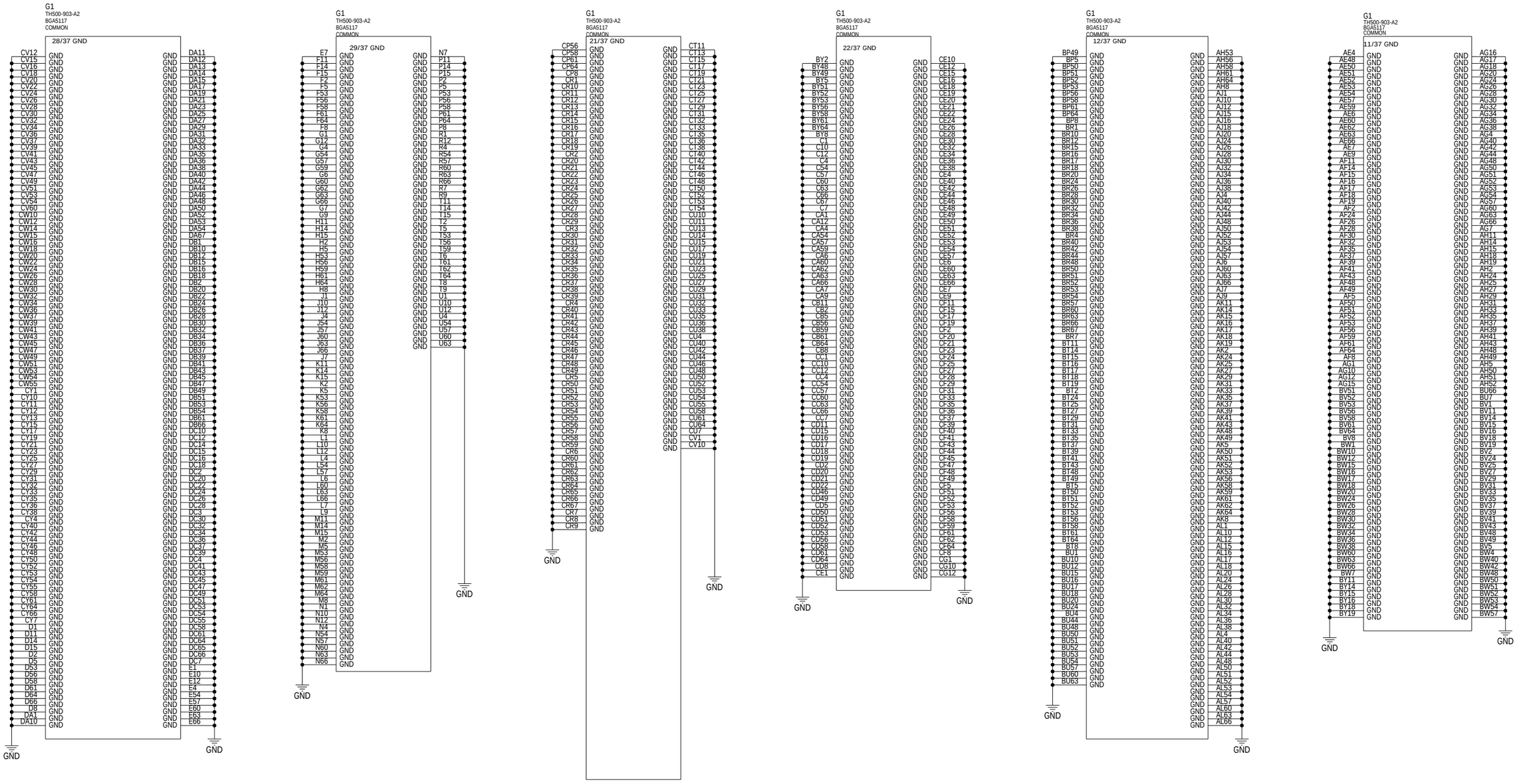
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - STRAPS\PU\PD

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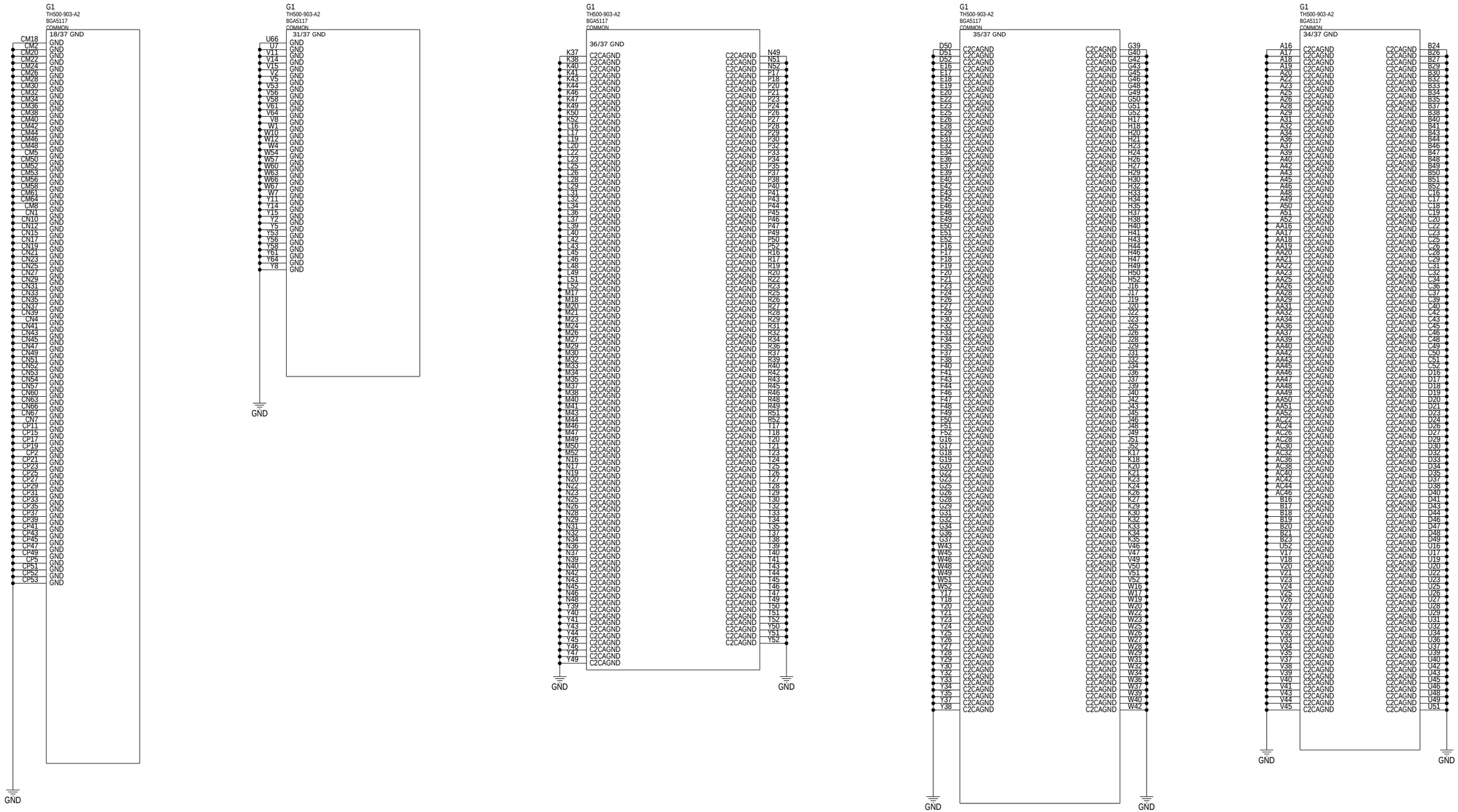
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - GND1

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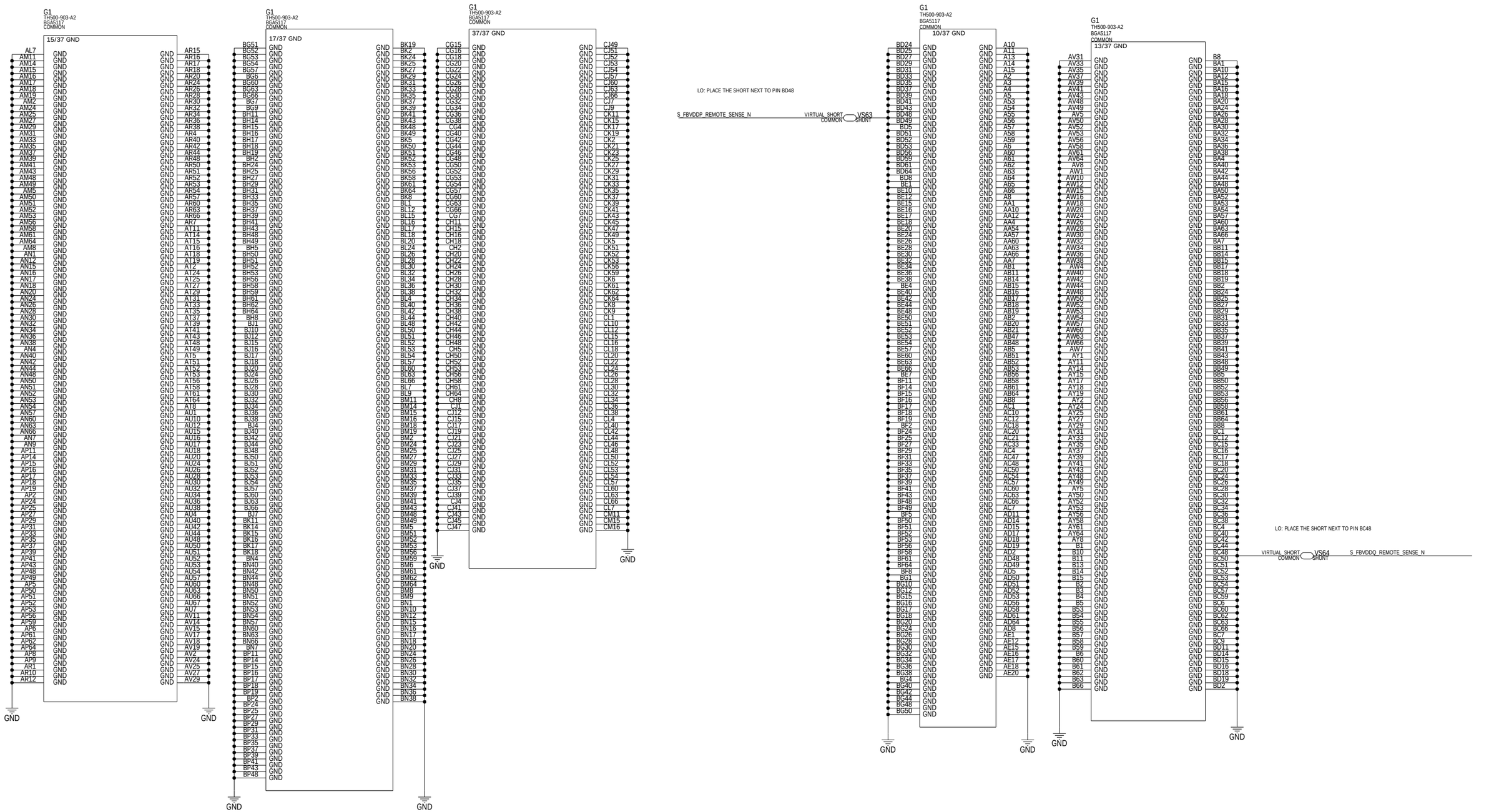
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - GND2

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NV_PN	600-9D4A4-0300-000		
PCB REV	PD4A4-D00	PAGE	69 OF 112
BOM REV	M	DATE	16-JUL-2026

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - GND3

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE - POWER

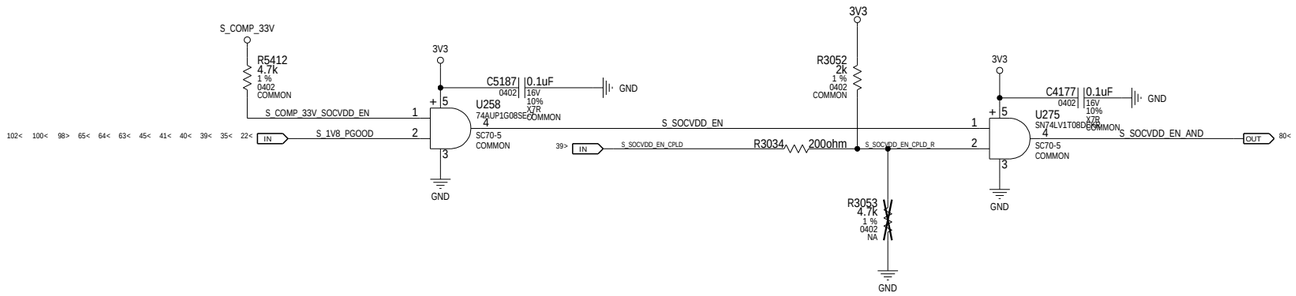
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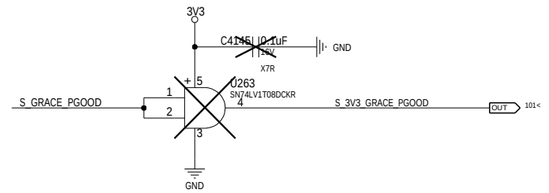
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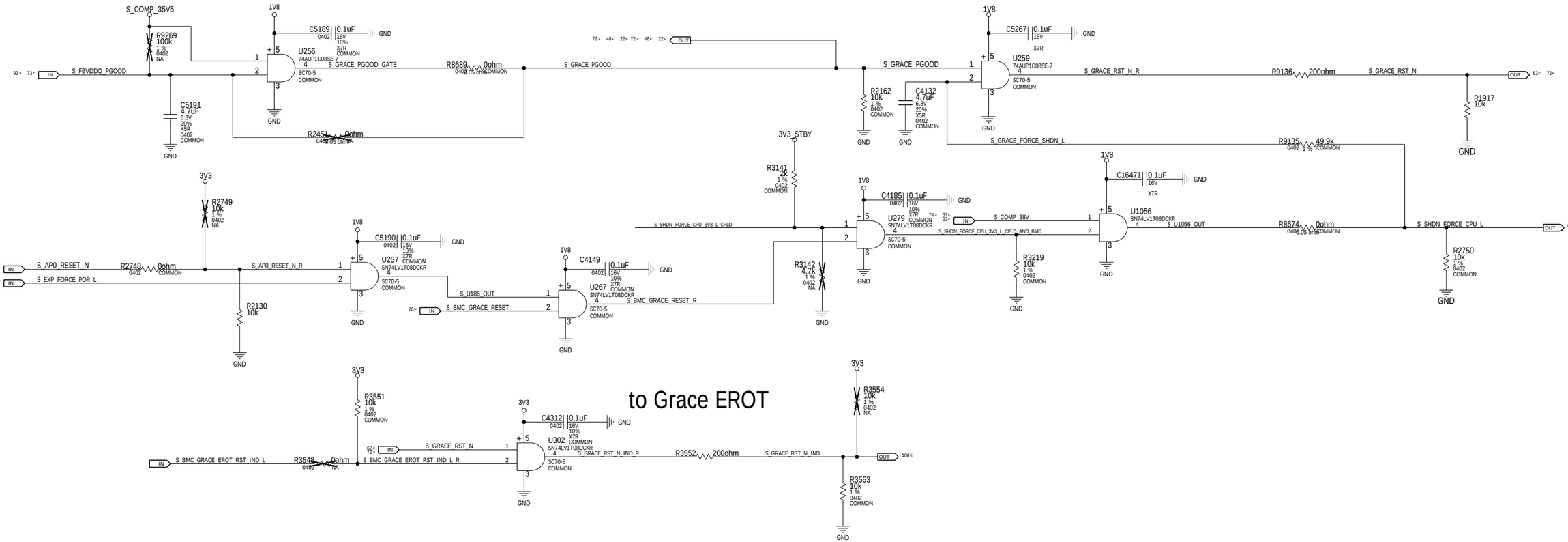
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POWER SEQUENCE







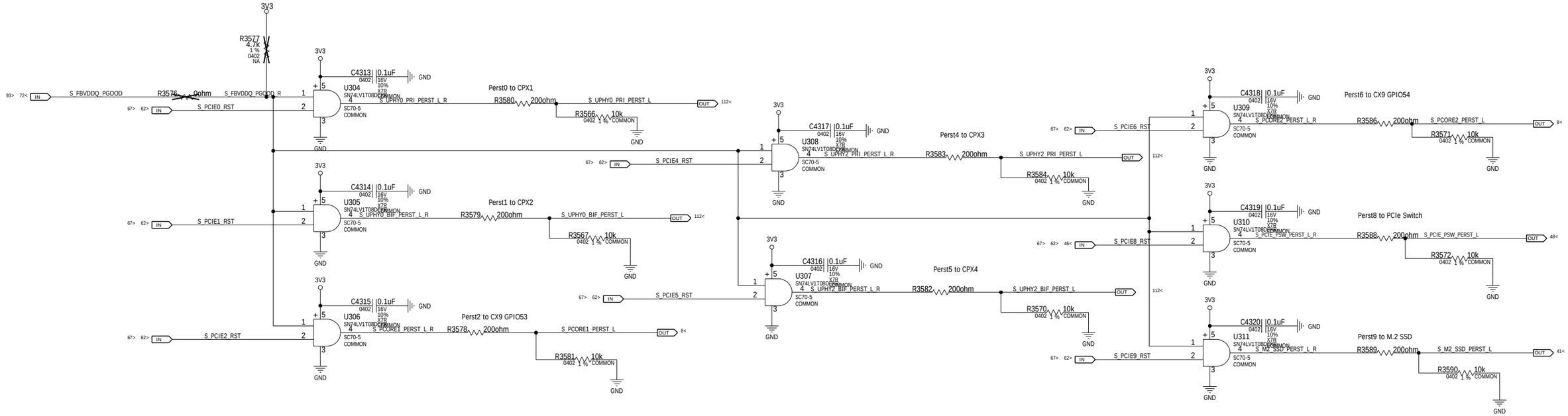
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS: POWER SEQUENCE - CPU1

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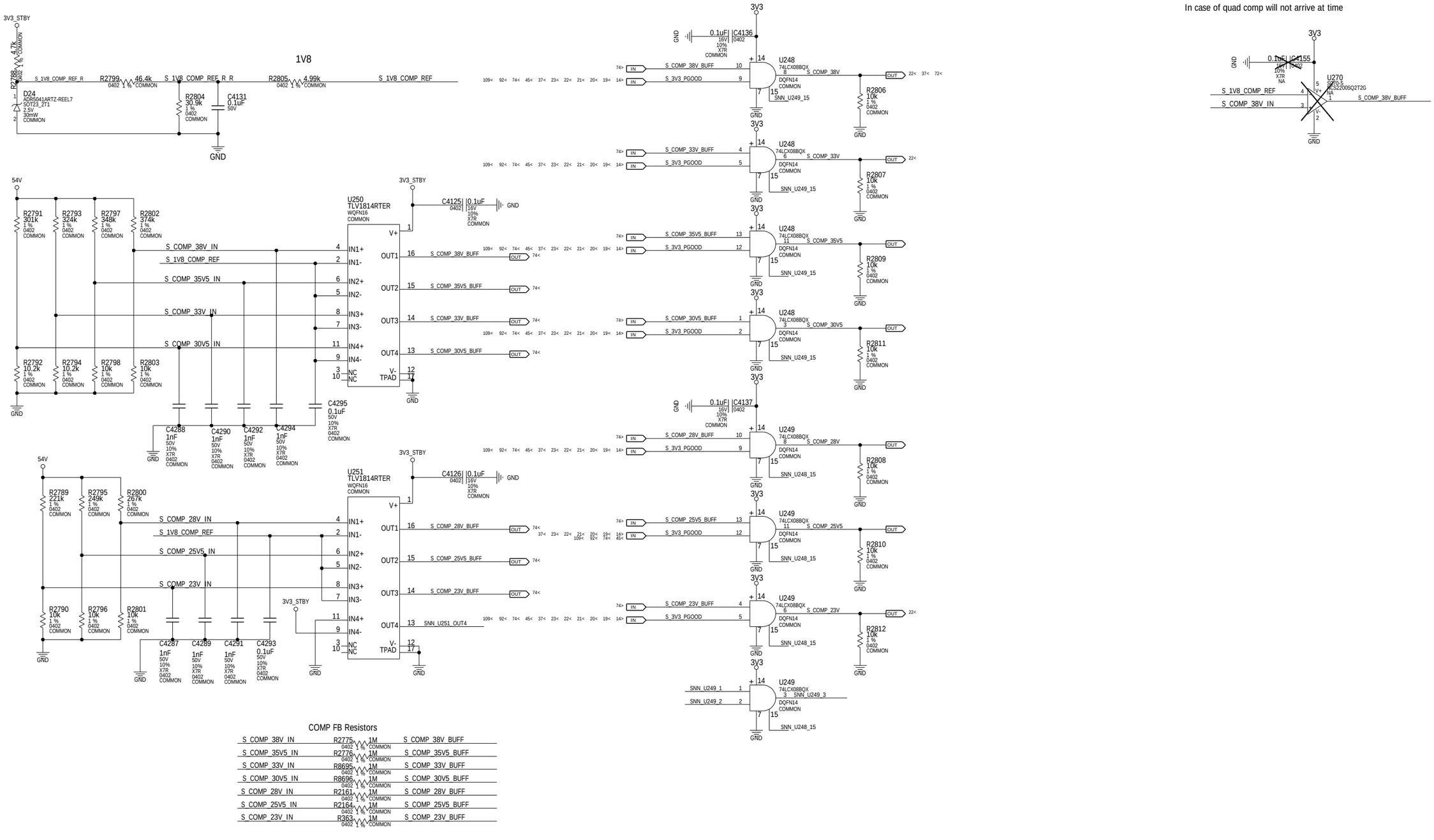
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS: POWER SEQUENCE - CPU2

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COMP FB Resistors

S_COMP_38V_IN	R2775 0402 1 % 1M COMMON	S_COMP_38V_BUFF
S_COMP_35V5_IN	R2776 0402 1 % 1M COMMON	S_COMP_35V5_BUFF
S_COMP_33V_IN	R8695 0402 1 % 1M COMMON	S_COMP_33V_BUFF
S_COMP_30V5_IN	R8696 0402 1 % 1M COMMON	S_COMP_30V5_BUFF
S_COMP_28V_IN	R2161 0402 1 % 1M COMMON	S_COMP_28V_BUFF
S_COMP_25V5_IN	R2164 0402 1 % 1M COMMON	S_COMP_25V5_BUFF
S_COMP_23V_IN	R363 0402 1 % 1M COMMON	S_COMP_23V_BUFF

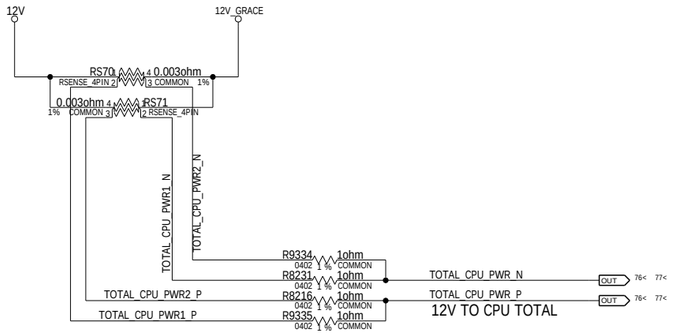
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	54V COMPARATOR

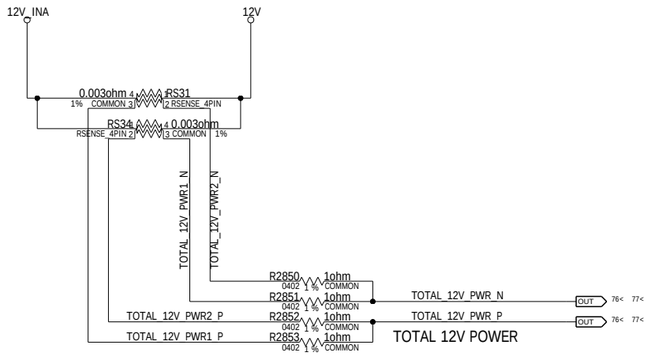
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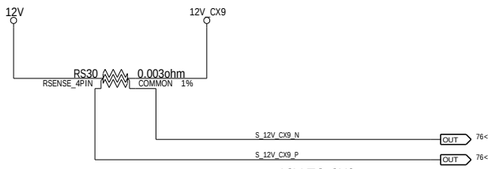
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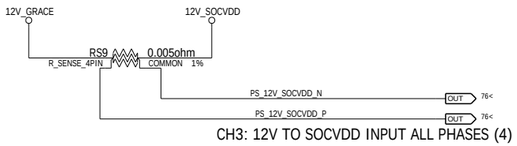
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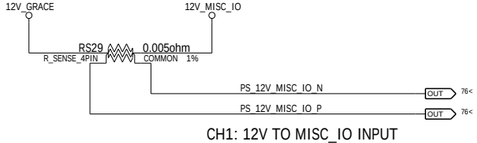
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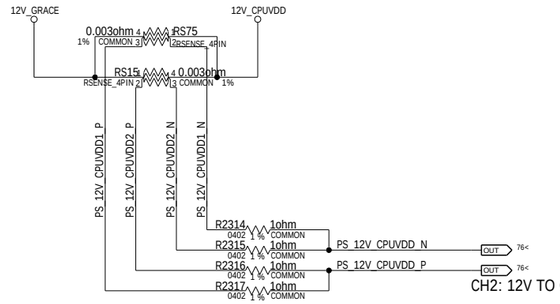


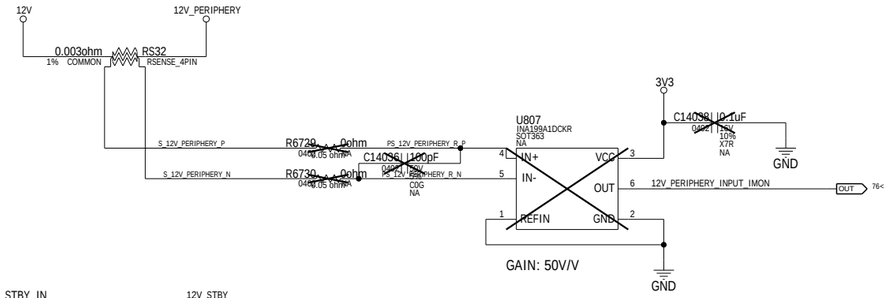


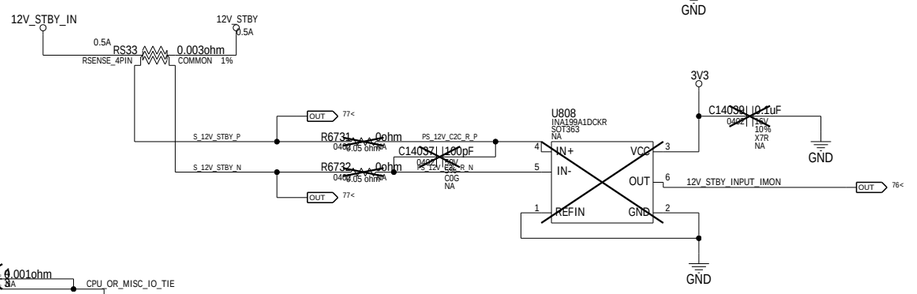


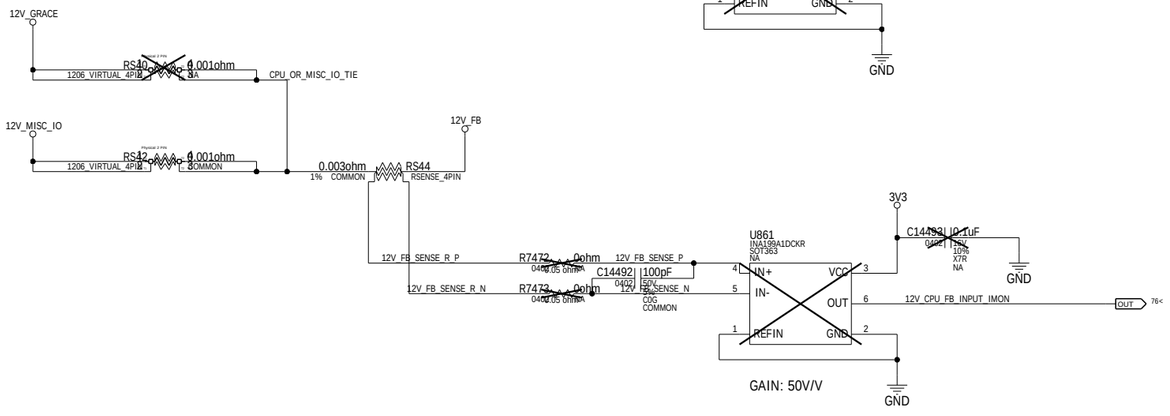












ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	INPUT 12V SENSE TOTAL

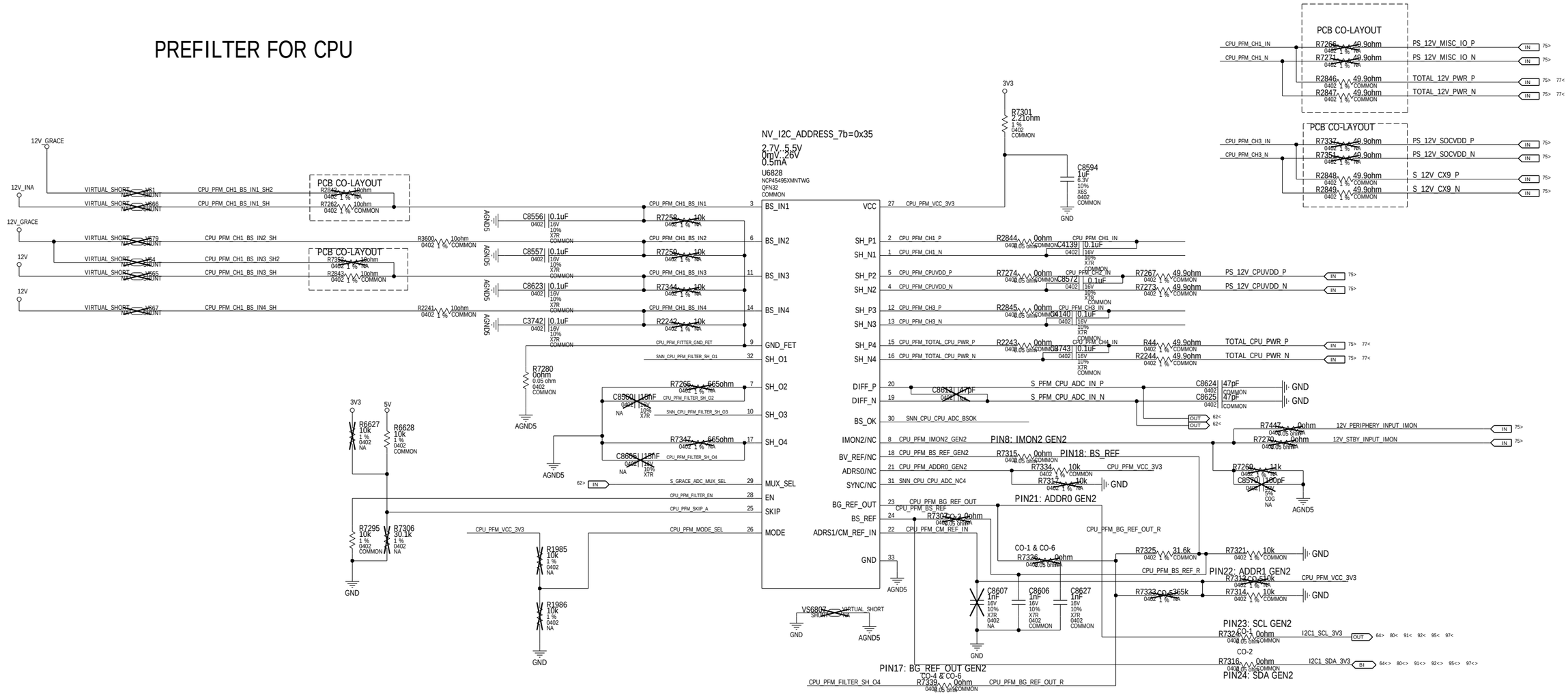
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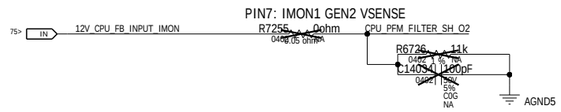
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PREFILTER FOR CPU



OVR-M can support 1.8V signaling without issues



7-BIT I2C ADDR OF OVR-M GEN2:

ADDR1	ADDR0	8-BIT I2C ADDR	
0	0	0x34	
0	1	0x35	<--
1	0	0x36	
1	1	0x37	

OVR-M GEN2 SUPPORT

OVR-M GEN2:
PIN 7, 8, 9, 10, 17, 18, 21, 22, 23, 24, 26, 32
are different from GEN1 OVRM,

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS: CPU Input Pre_Filter OVR-M

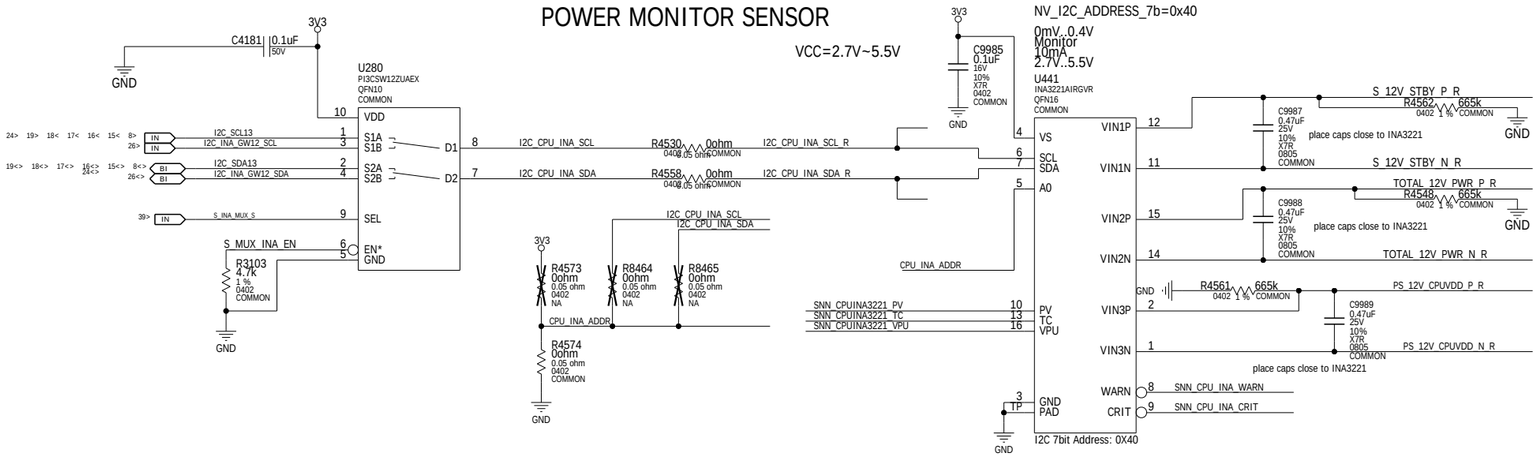
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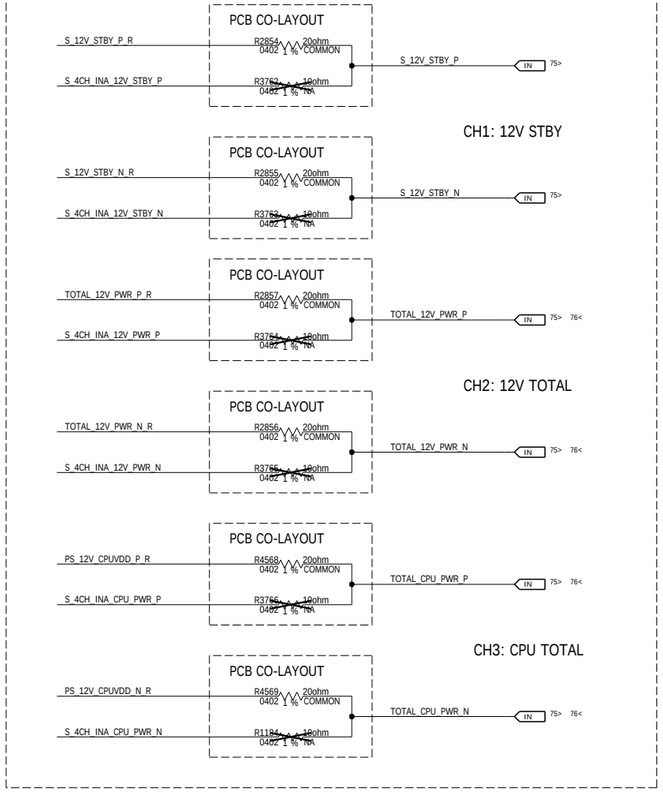
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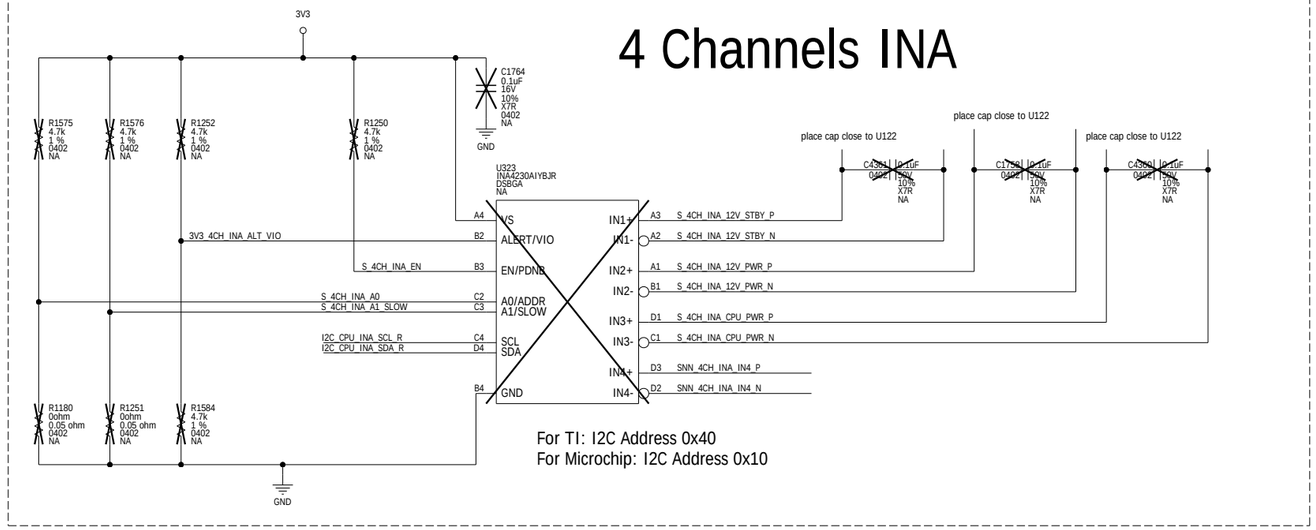
POWER MONITOR SENSOR



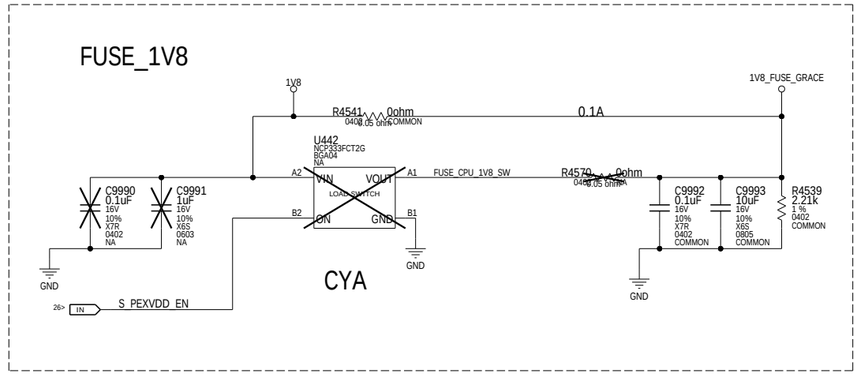
FOR 48V CH1 AND CH2, ACTUAL VOLTAGE(V) = 7.04 X VIN_N VOLTAGE(V)
FOR 48V CH1 AND CH2, ACTUAL CURRENT(A) = (VIN_P - VIN_N) VOLTAGE(mV)*1.33 / 9.96



4 Channels INA



For TI: I2C Address 0x40
For Microchip: I2C Address 0x10



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	CPU INA, FUSE_1V8

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Page: TH500 CPUVDD, FBVDDQ, FBVDDP DECOUPLING

CPUVDD (PLACE UNDER CPU)

CPUVDD

C2908 2.2uF 2.5V 20% X6S 0402 COMMON
C2625 2.2uF 2.5V 20% X6S 0402 COMMON
C2938 2.2uF 2.5V 20% X6S 0402 COMMON
C3100 2.2uF 2.5V 20% X6S 0402 COMMON
C3019 2.2uF 2.5V 20% X6S 0402 COMMON
C2941 2.2uF 2.5V 20% X6S 0402 COMMON
C3023 2.2uF 2.5V 20% X6S 0402 COMMON
C3024 2.2uF 2.5V 20% X6S 0402 COMMON
C3154 2.2uF 2.5V 20% X6S 0402 COMMON
C3104 2.2uF 2.5V 20% X6S 0402 COMMON
C2663 2.2uF 2.5V 20% X6S 0402 COMMON
C2937 2.2uF 2.5V 20% X6S 0402 COMMON
C2660 2.2uF 2.5V 20% X6S 0402 COMMON
C2672 2.2uF 2.5V 20% X6S 0402 COMMON
C3145 2.2uF 2.5V 20% X6S 0402 COMMON

C2671 2.2uF 2.5V 20% X6S 0402 COMMON
C3103 2.2uF 2.5V 20% X6S 0402 COMMON
C3020 2.2uF 2.5V 20% X6S 0402 COMMON
C2781 2.2uF 2.5V 20% X6S 0402 COMMON
C3101 2.2uF 2.5V 20% X6S 0402 COMMON
C2814 2.2uF 2.5V 20% X6S 0402 COMMON
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C2748 2.2uF 2.5V 20% X6S 0402 COMMON
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C2592 2.2uF 2.5V 20% X6S 0402 COMMON
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C3015 2.2uF 2.5V 20% X6S 0402 COMMON
C2815 2.2uF 2.5V 20% X6S 0402 COMMON
C3140 2.2uF 2.5V 20% X6S 0402 COMMON
C2749 2.2uF 2.5V 20% X6S 0402 COMMON
C2739 2.2uF 2.5V 20% X6S 0402 COMMON
C2664 2.2uF 2.5V 20% X6S 0402 COMMON
C2782 2.2uF 2.5V 20% X6S 0402 COMMON
C2936 2.2uF 2.5V 20% X6S 0402 COMMON
C2741 2.2uF 2.5V 20% X6S 0402 COMMON
C3038 2.2uF 2.5V 20% X6S 0402 COMMON

C2896 4.7uF 4V 10% X6S 0603 COMMON
C3066 4.7uF 4V 10% X6S 0603 COMMON
C2708 4.7uF 4V 10% X6S 0603 COMMON
C2670 4.7uF 4V 10% X6S 0603 COMMON
C2825 4.7uF 4V 10% X6S 0603 COMMON
C2702 4.7uF 4V 10% X6S 0603 COMMON
C3069 4.7uF 4V 10% X6S 0603 COMMON
C2775 4.7uF 4V 10% X6S 0603 COMMON
C2701 4.7uF 4V 10% X6S 0603 COMMON
C2897 4.7uF 4V 10% X6S 0603 COMMON
C2976 4.7uF 4V 10% X6S 0603 COMMON
C2974 4.7uF 4V 10% X6S 0603 COMMON
C3067 4.7uF 4V 10% X6S 0603 COMMON
C2904 4.7uF 4V 10% X6S 0603 COMMON
C2584 4.7uF 4V 10% X6S 0603 COMMON

C2776 4.7uF 4V 10% X6S 0603 COMMON
C2874 4.7uF 4V 10% X6S 0603 COMMON
C2623 4.7uF 4V 10% X6S 0603 COMMON
C2786 4.7uF 4V 10% X6S 0603 COMMON
C2583 4.7uF 4V 10% X6S 0603 COMMON
C2745 4.7uF 4V 10% X6S 0603 COMMON
C2973 4.7uF 4V 10% X6S 0603 COMMON

GND

28x 1uF 0402
74x 2.2uF 0402
22x 4.7uF 0603

21x 22uF 0805

CPUVDD

C2621 1uF 4V 10% X6S 0402 COMMON
C3022 1uF 4V 10% X6S 0402 COMMON
C2615 1uF 4V 10% X6S 0402 COMMON
C2866 1uF 4V 10% X6S 0402 COMMON
C2620 1uF 4V 10% X6S 0402 COMMON
C2864 1uF 4V 10% X6S 0402 COMMON
C2863 1uF 4V 10% X6S 0402 COMMON
C2869 1uF 4V 10% X6S 0402 COMMON
C3076 1uF 4V 10% X6S 0402 COMMON
C3109 1uF 4V 10% X6S 0402 COMMON
C2991 1uF 4V 10% X6S 0402 COMMON
C2614 1uF 4V 10% X6S 0402 COMMON

C2747 1uF 4V 10% X6S 0402 COMMON
C2812 1uF 4V 10% X6S 0402 COMMON
C2811 1uF 4V 10% X6S 0402 COMMON
C3111 1uF 4V 10% X6S 0402 COMMON
C3124 1uF 4V 10% X6S 0402 COMMON
C2907 1uF 4V 10% X6S 0402 COMMON
C2813 1uF 4V 10% X6S 0402 COMMON
C2736 1uF 4V 10% X6S 0402 COMMON
C2862 1uF 4V 10% X6S 0402 COMMON
C2536 1uF 4V 10% X6S 0402 COMMON
C2616 1uF 4V 10% X6S 0402 COMMON
C2611 1uF 4V 10% X6S 0402 COMMON

C2626 1uF 4V 10% X6S 0402 COMMON
C2622 1uF 4V 10% X6S 0402 COMMON
C2859 1uF 4V 10% X6S 0402 COMMON
C3110 1uF 4V 10% X6S 0402 COMMON

C3021 22uF 4V 20% X6S 0805 COMMON
C3096 22uF 4V 20% X6S 0805 COMMON
C2697 22uF 4V 20% X6S 0805 COMMON
C3095 22uF 4V 20% X6S 0805 COMMON
C2580 22uF 4V 20% X6S 0805 COMMON
C2698 22uF 4V 20% X6S 0805 COMMON
C2826 22uF 4V 20% X6S 0805 COMMON
C2699 22uF 4V 20% X6S 0805 COMMON
C2585 22uF 4V 20% X6S 0805 COMMON
C2949 22uF 4V 20% X6S 0805 COMMON

C2661 22uF 4V 20% X6S 0805 COMMON
C3028 22uF 4V 20% X6S 0805 COMMON
C2582 22uF 4V 20% X6S 0805 COMMON
C2579 22uF 4V 20% X6S 0805 COMMON
C2581 22uF 4V 20% X6S 0805 COMMON
C2744 22uF 4V 20% X6S 0805 COMMON
C2912 22uF 4V 20% X6S 0805 COMMON
C2989 22uF 4V 20% X6S 0805 COMMON
C3080 22uF 4V 20% X6S 0805 COMMON
C3027 22uF 4V 20% X6S 0805 COMMON
C3029 22uF 4V 20% X6S 0805 COMMON

GND

FBVDDP (PLACE UNDER CPU)

FBVDDP

C2376 1uF 4V 10% X6S 0402 COMMON
C2398 1uF 4V 10% X6S 0402 COMMON
C2387 1uF 4V 10% X6S 0402 COMMON
C2384 1uF 4V 10% X6S 0402 COMMON
C2378 1uF 4V 10% X6S 0402 COMMON
C2375 1uF 4V 10% X6S 0402 COMMON
C2370 1uF 4V 10% X6S 0402 COMMON
C2391 1uF 4V 10% X6S 0402 COMMON
C2390 1uF 4V 10% X6S 0402 COMMON
C2397 1uF 4V 10% X6S 0402 COMMON
C2377 1uF 4V 10% X6S 0402 COMMON
C2388 1uF 4V 10% X6S 0402 COMMON
C2411 1uF 4V 10% X6S 0402 COMMON
C2389 1uF 4V 10% X6S 0402 COMMON

C2337 4.7uF 4V 10% X6S 0603 COMMON
C2341 4.7uF 4V 10% X6S 0603 COMMON
C2331 4.7uF 4V 10% X6S 0603 COMMON
C2406 4.7uF 4V 10% X6S 0603 COMMON
C2342 4.7uF 4V 10% X6S 0603 COMMON
C2347 4.7uF 4V 10% X6S 0603 COMMON
C2335 47uF 4V 20% X6S 0805 COMMON
C2351 47uF 4V 20% X6S 0805 COMMON
C2330 47uF 4V 20% X6S 0805 COMMON
C2322 47uF 4V 20% X6S 0805 COMMON
C2367 4.7uF 4V 10% X6S 0603 COMMON
C2374 4.7uF 4V 10% X6S 0603 COMMON
C2369 4.7uF 4V 10% X6S 0603 COMMON

C2338 10uF 4V 20% X6S 0603 COMMON
C2385 10uF 4V 20% X6S 0603 COMMON
C2348 10uF 4V 20% X6S 0603 COMMON

GND

FBVDDQ (PLACE UNDER CPU)

20x 1uF 0402
10x 4.7uF 0603
5x 10uF 0603
9x 47uF 0805

FBVDDQ

C2426 4.7uF 4V 10% X6S 0603 COMMON
C2451 4.7uF 4V 10% X6S 0603 COMMON
C2457 4.7uF 4V 10% X6S 0603 COMMON
C2455 4.7uF 4V 10% X6S 0603 COMMON
C2450 4.7uF 4V 10% X6S 0603 COMMON
C2453 1uF 4V 10% X6S 0402 COMMON
C2459 1uF 4V 10% X6S 0402 COMMON
C2437 1uF 4V 10% X6S 0402 COMMON
C2454 1uF 4V 10% X6S 0402 COMMON
C2436 1uF 4V 10% X6S 0402 COMMON
C2452 1uF 4V 10% X6S 0402 COMMON
C2416 1uF 4V 10% X6S 0402 COMMON
C2435 1uF 4V 10% X6S 0402 COMMON
C2415 1uF 4V 10% X6S 0402 COMMON
R20659 1K 1 % 0402 COMMON

C2448 10uF 4V 20% X6S 0603 COMMON
C2456 10uF 4V 20% X6S 0603 COMMON
C2431 10uF 4V 20% X6S 0603 COMMON
C2427 100uF 4V 20% X6S 0805 COMMON
C2462 100uF 4V 20% X6S 0805 COMMON
C2410 100uF 4V 20% X6S 0805 COMMON
C2425 100uF 4V 20% X6S 0805 COMMON
C2428 100uF 4V 20% X6S 0805 COMMON

GND

30x 1uF 0402
15x 4.7uF 0603
8x 10uF 0603
8x 47uF 0805

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	CPU DECAPS-1

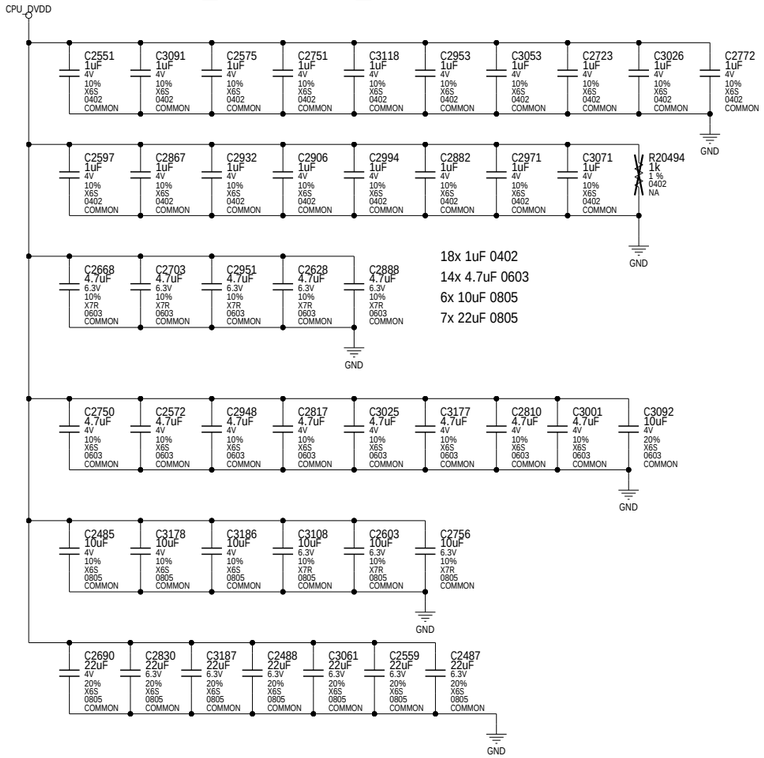
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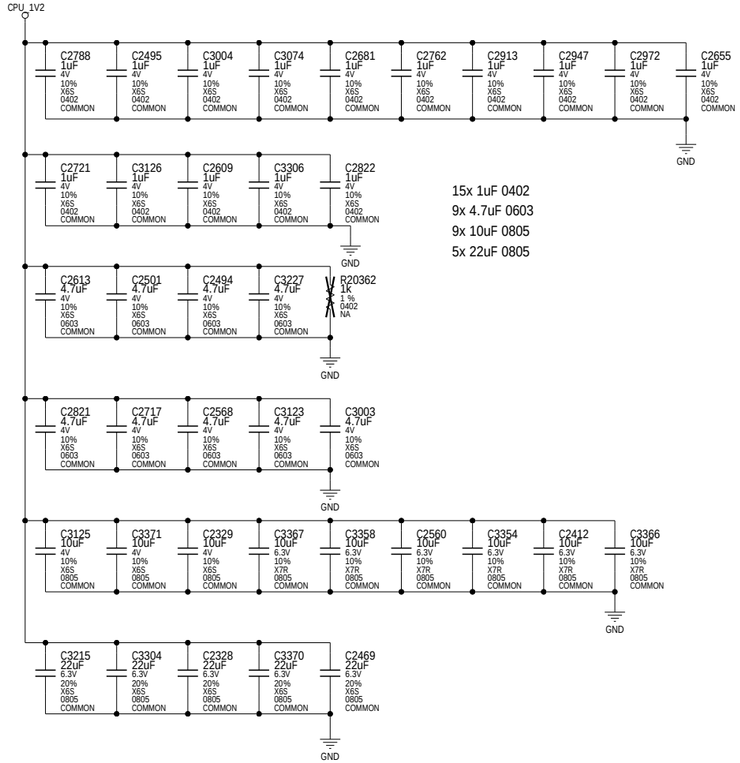
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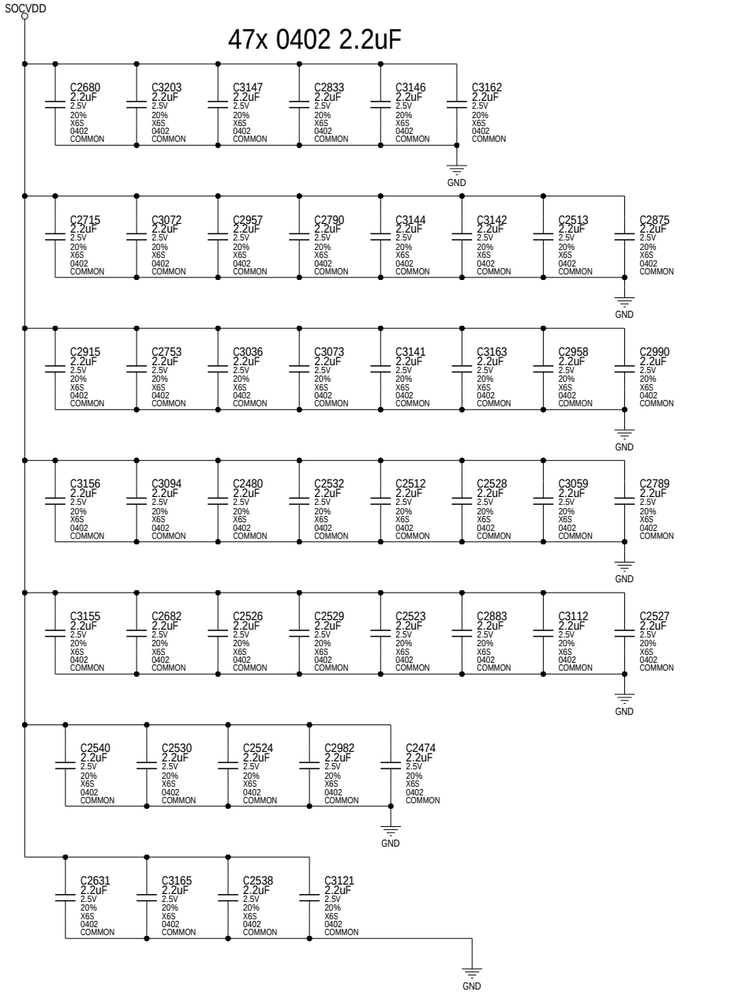
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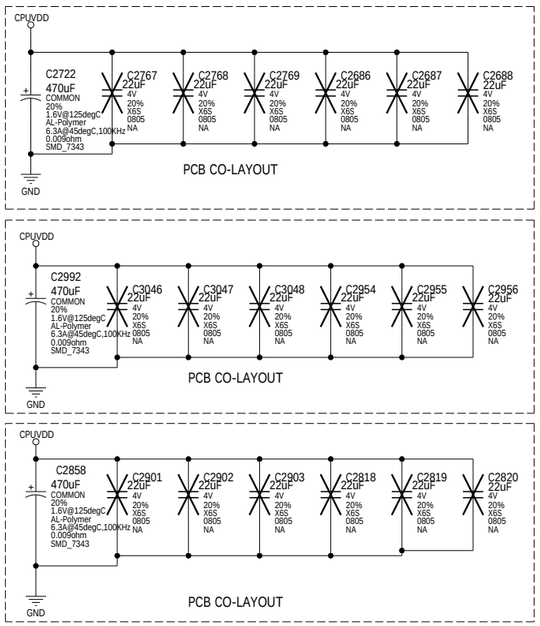
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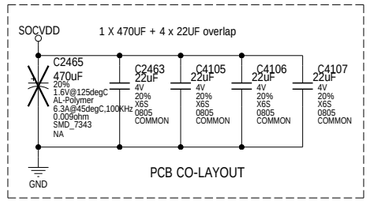
Page: TH500 CPU_DVDD, CPU_1V2, SOCVDD DECOUPLING

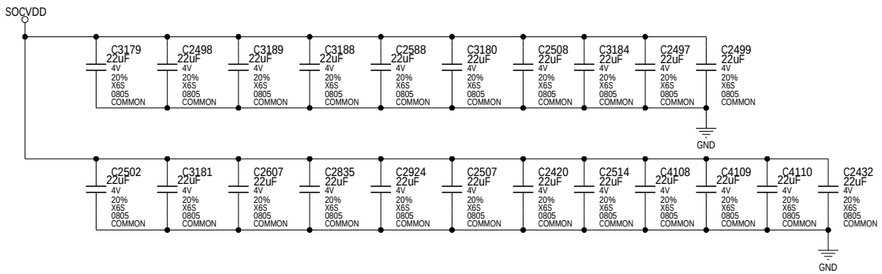


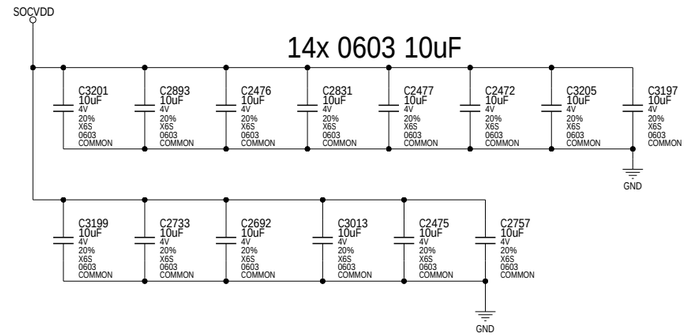












ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	CPU DECAPS-2

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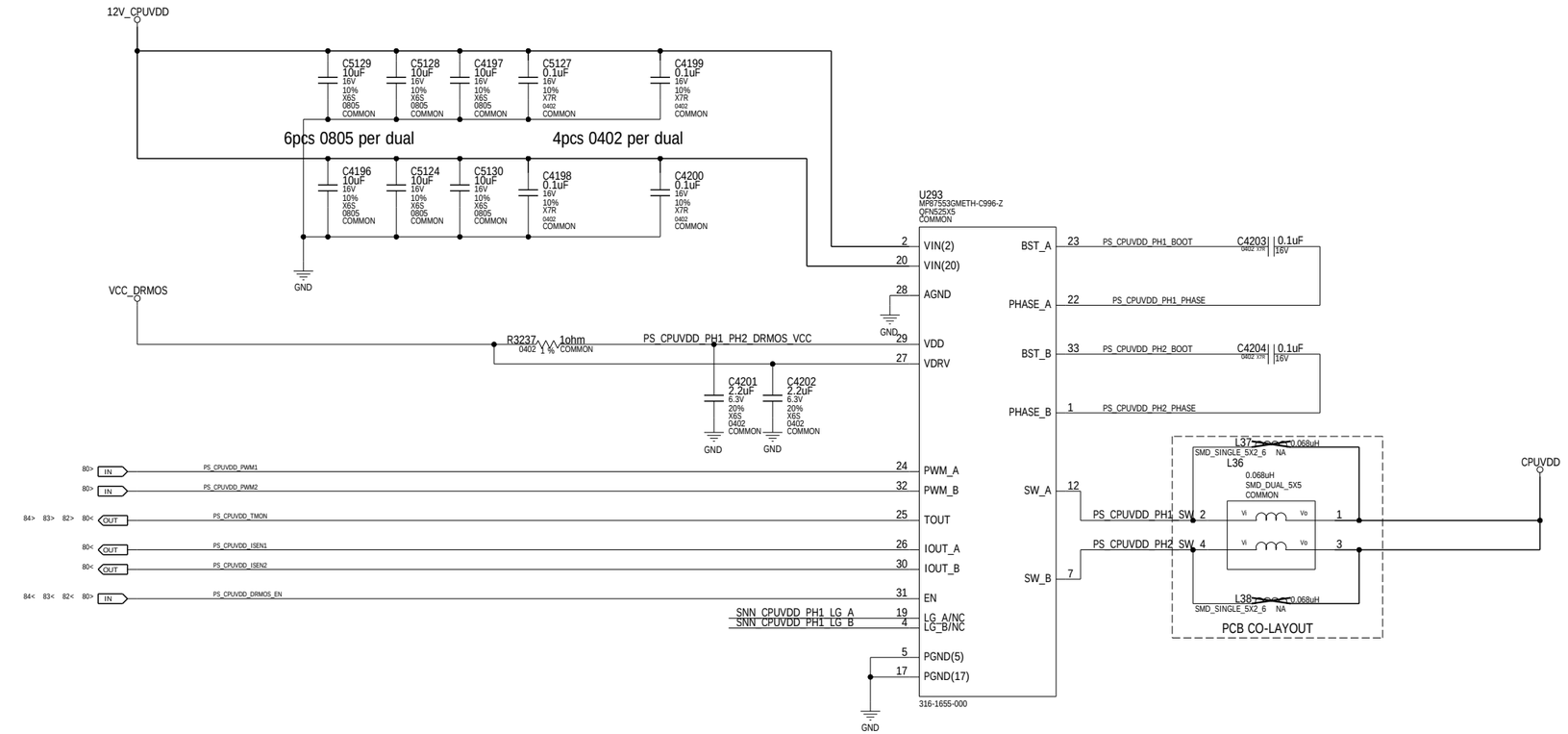
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS CPUVDD_SOCVDD CTO

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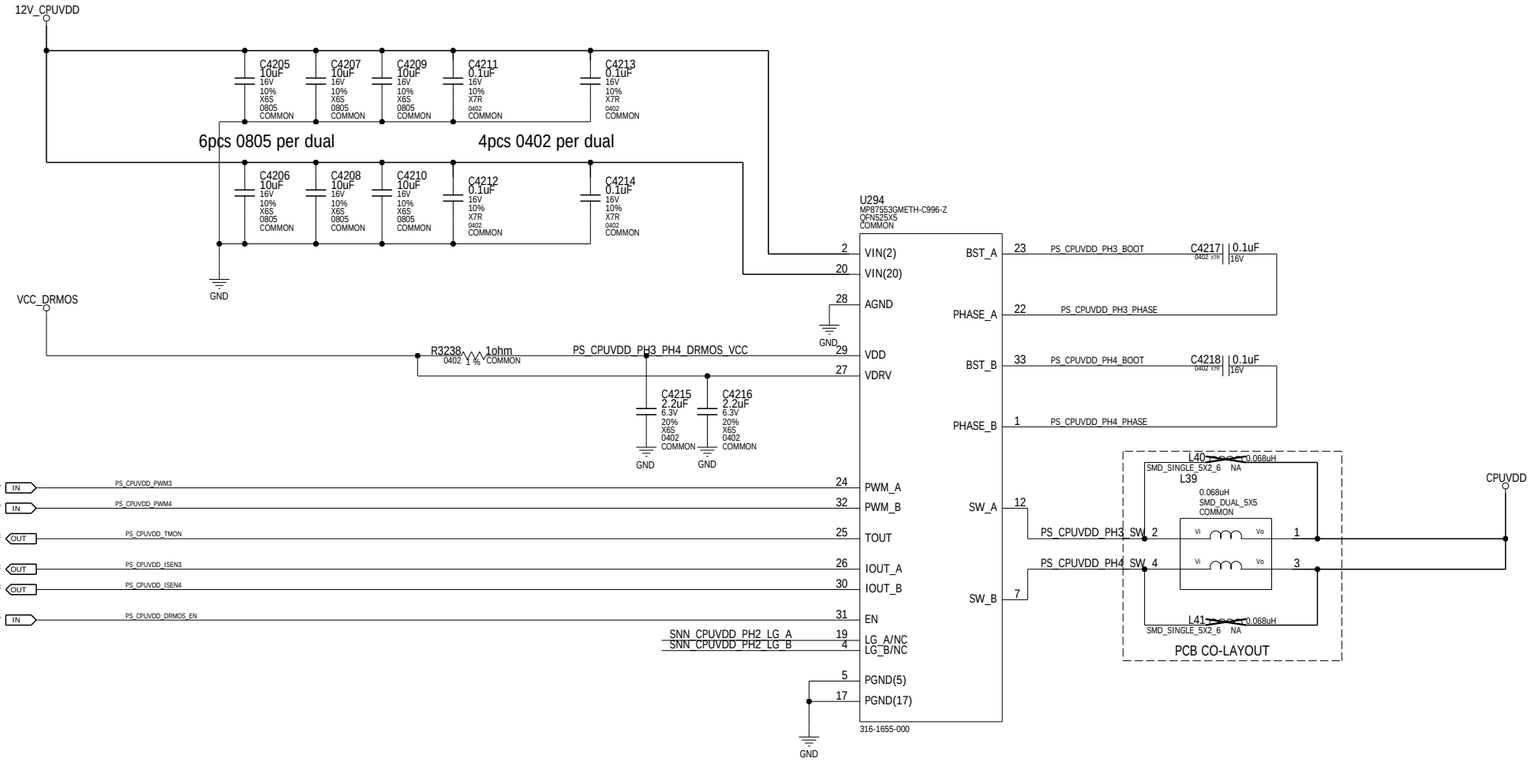
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS CPUVDD PH1,2

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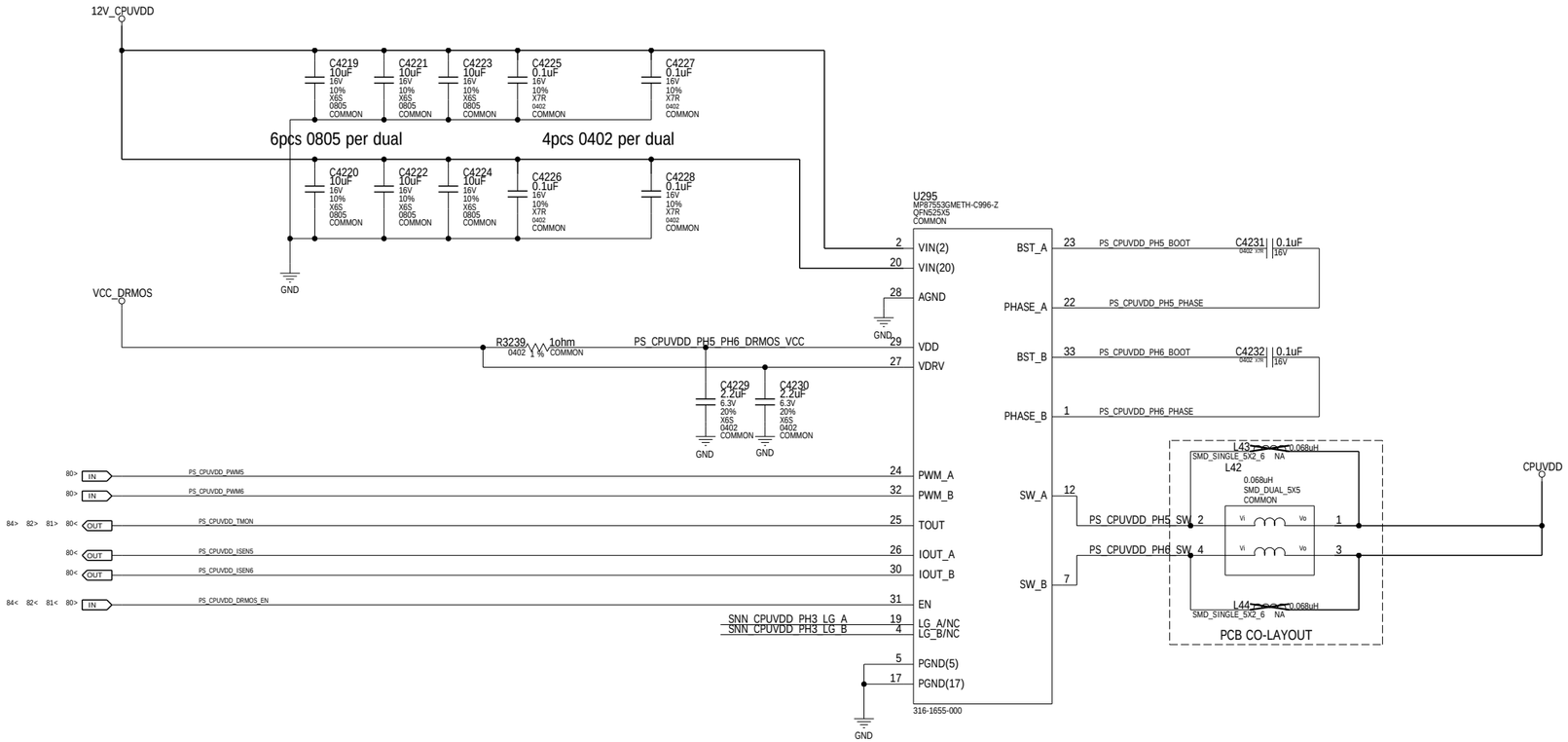
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS CPUVDD PH3,4

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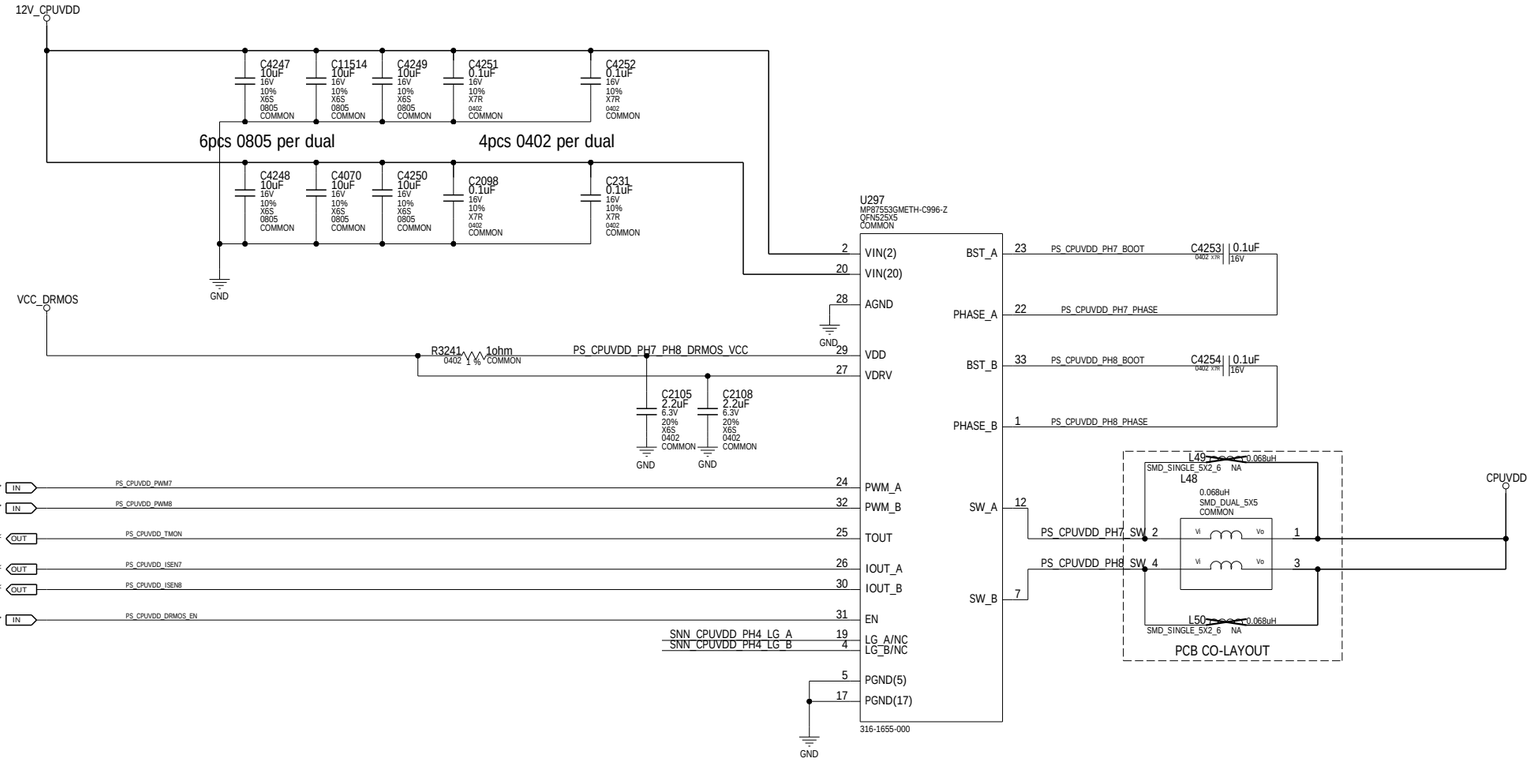
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS CPUVDD PH5,6

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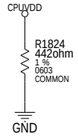
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PAGE DETAIL	PS CPUVDD PH7,8

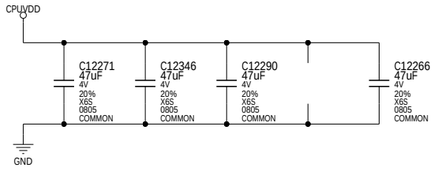
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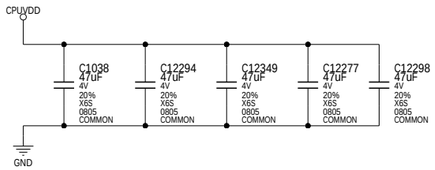
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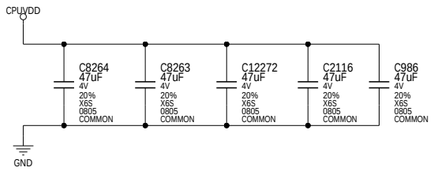
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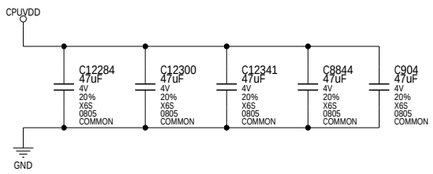
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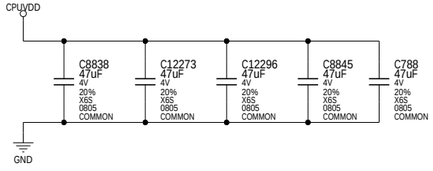


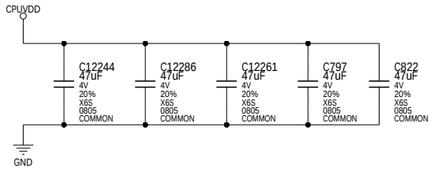


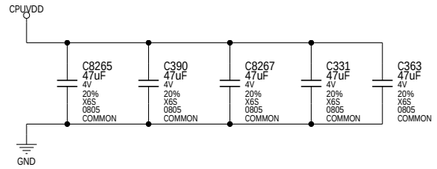


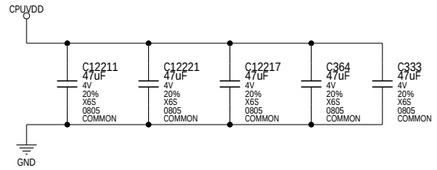


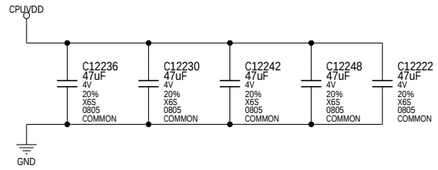


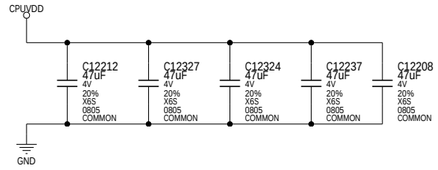


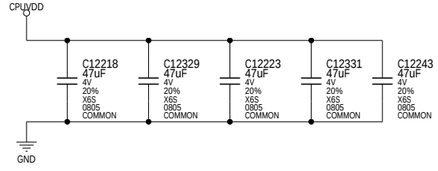


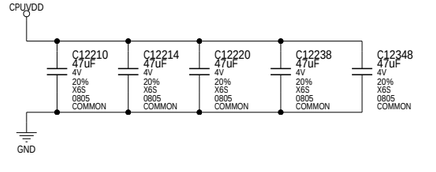












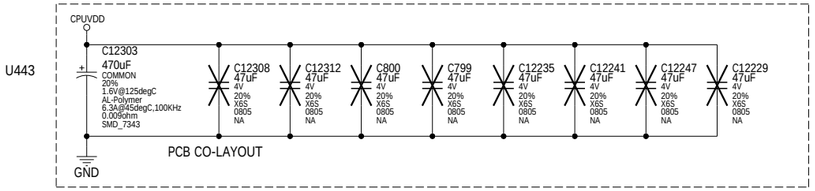
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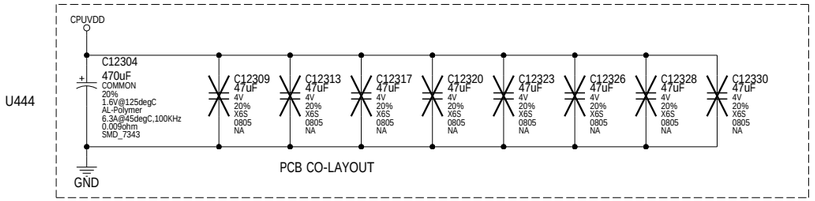
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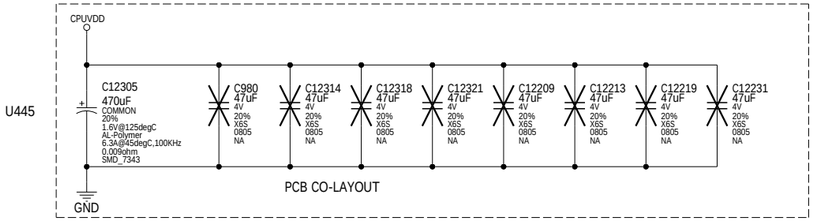
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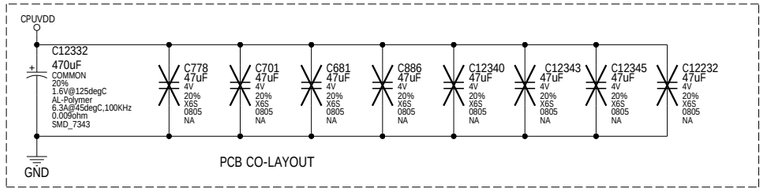
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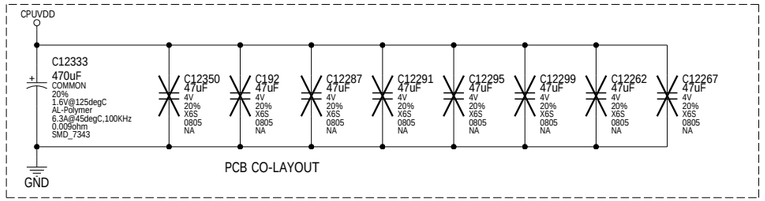
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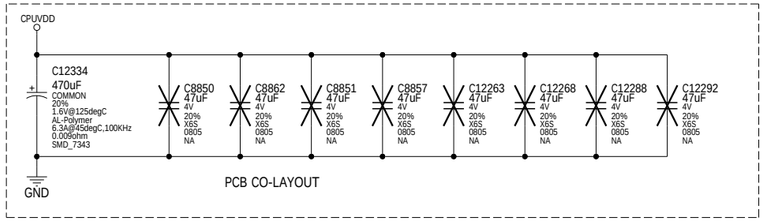


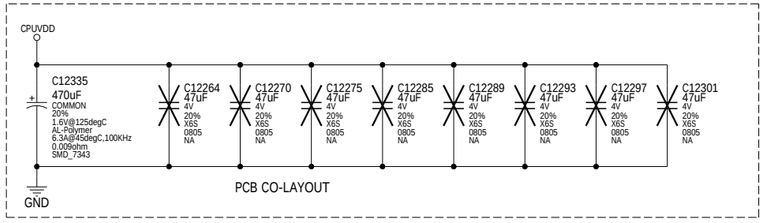












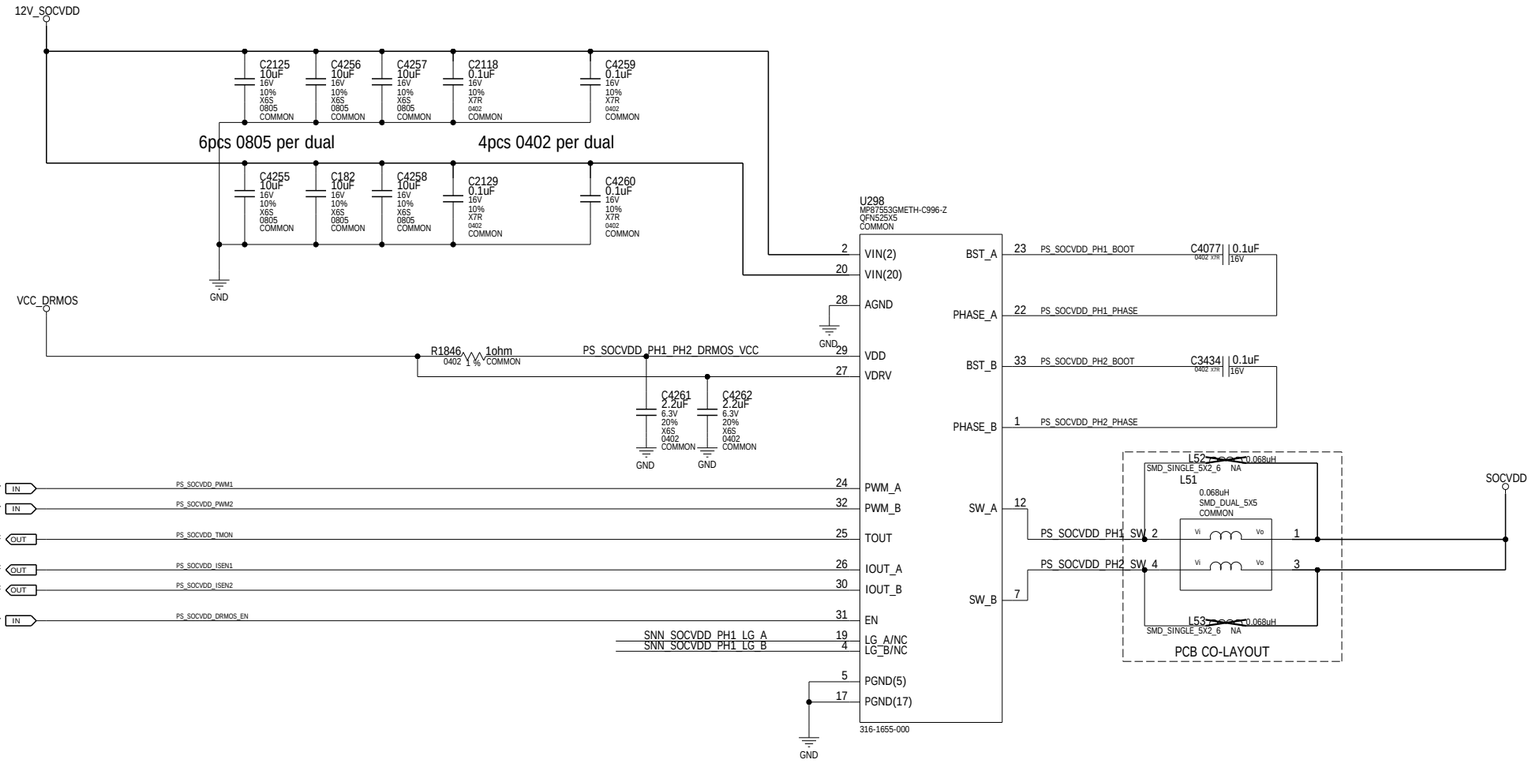
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PAGE DETAIL	PS CPUVDD CAPS 2/2

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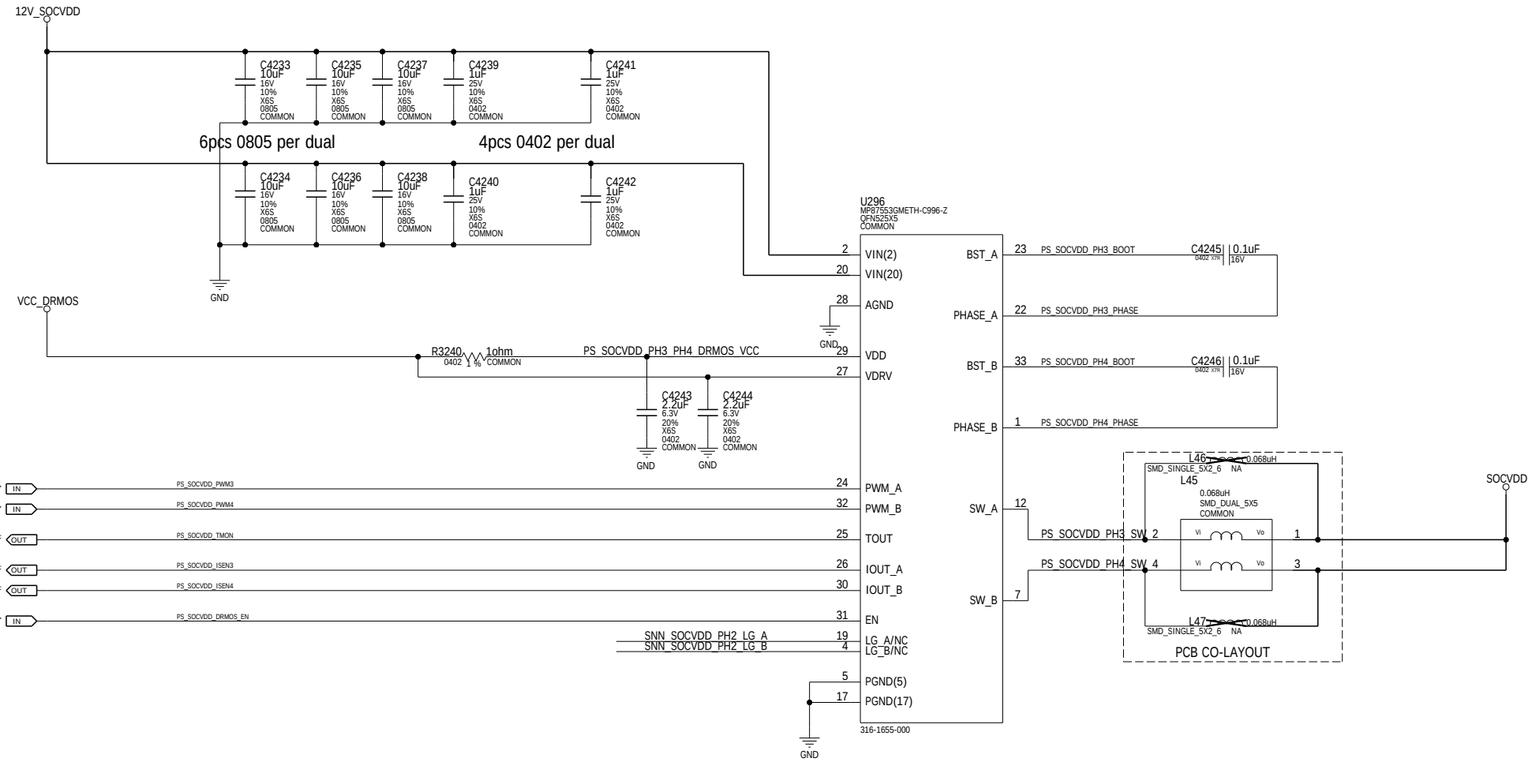
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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS SOCVDD PH3,4

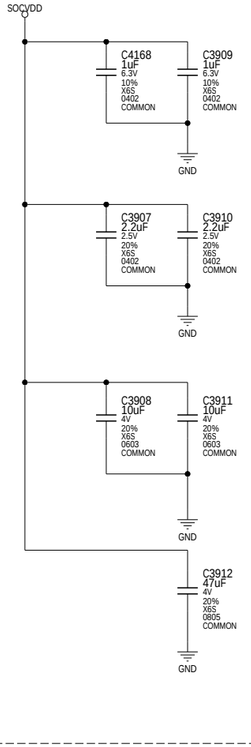
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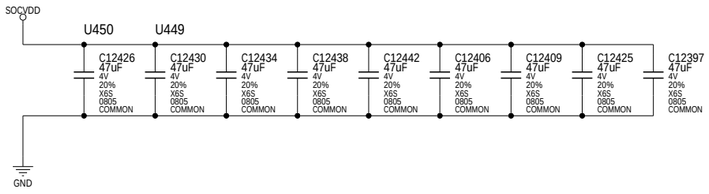
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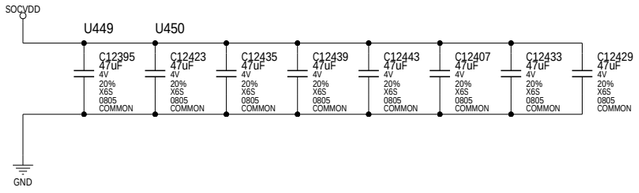
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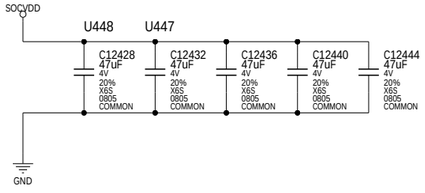
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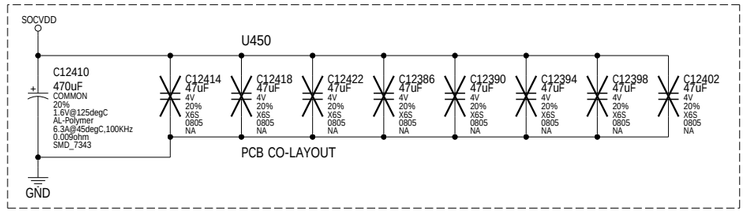
Place caps neer C2C BGA pads

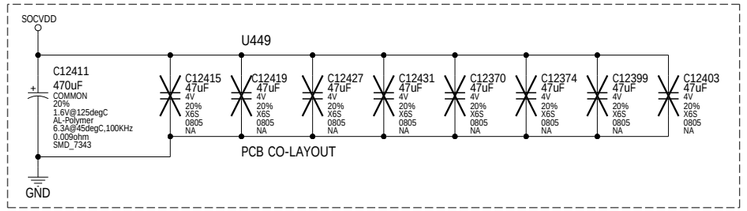


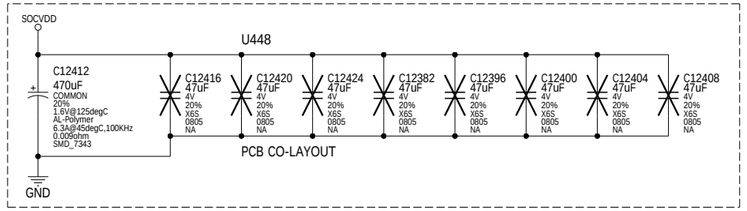


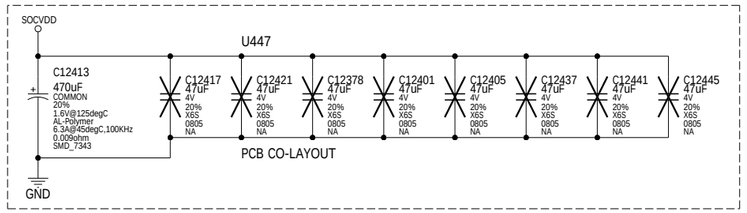


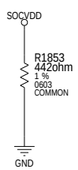












ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS SOCVDD CAPS

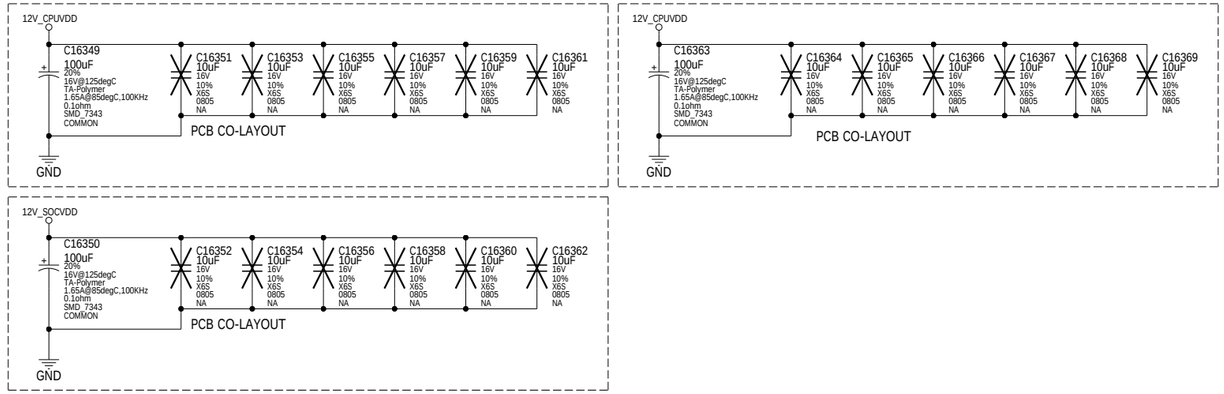
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PS: CPUVDD SCOVDD INPUT BULK CAP



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS CPUVDD SOCVDD INPUT BULK CAP

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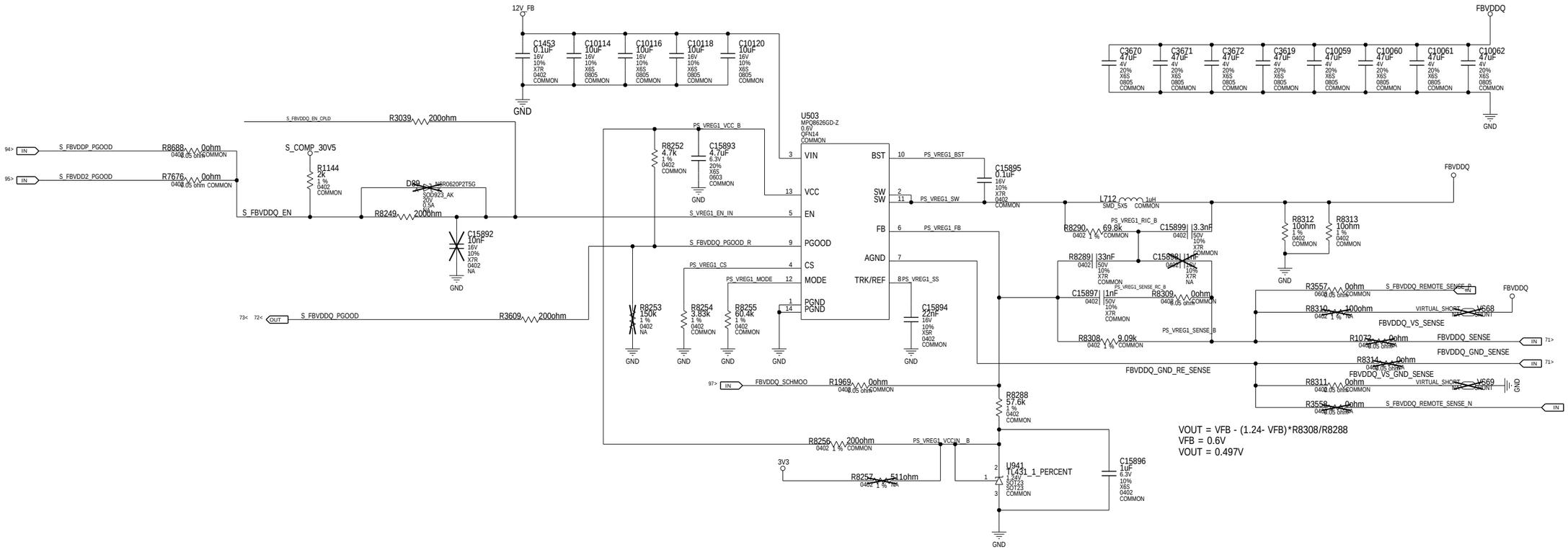
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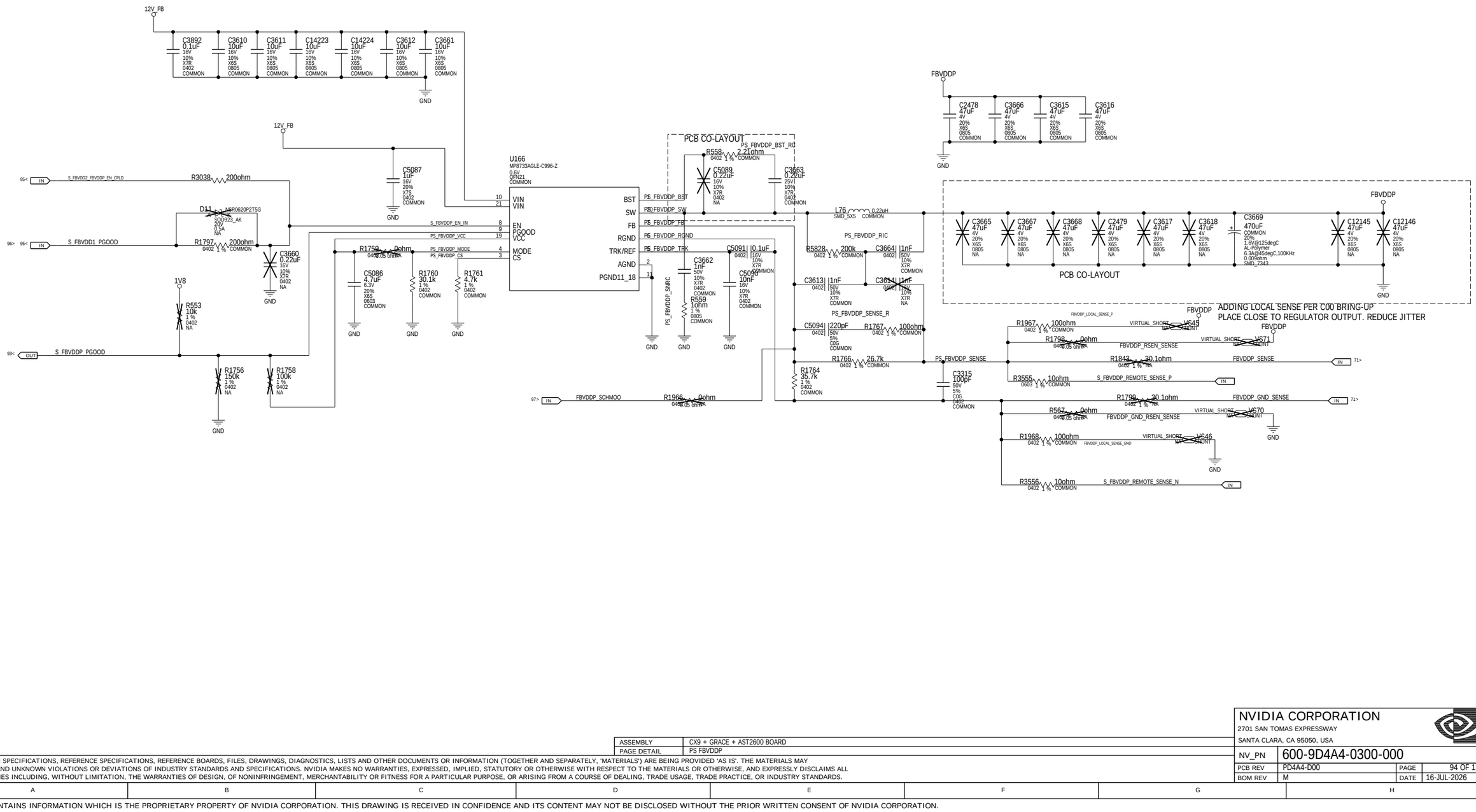
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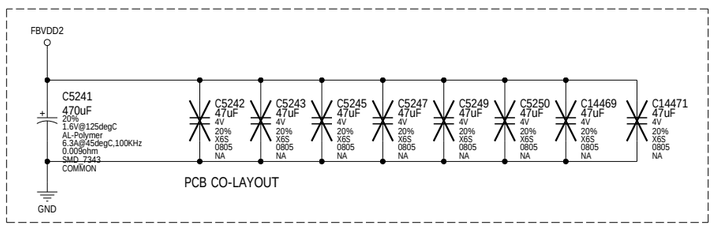
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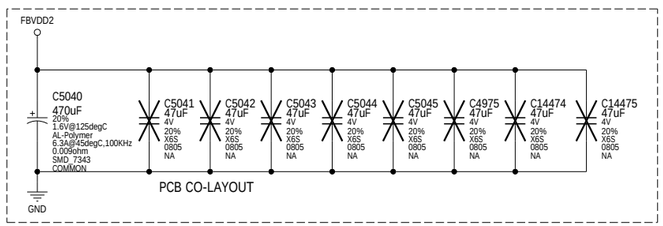
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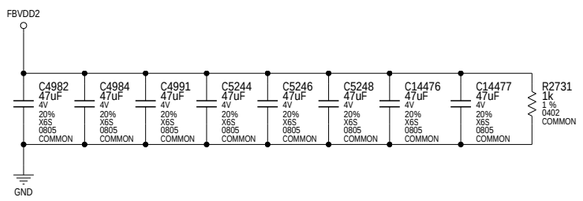
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PCB REV	PD4A4-D00	PAGE	94 OF 112
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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS FBVDD2

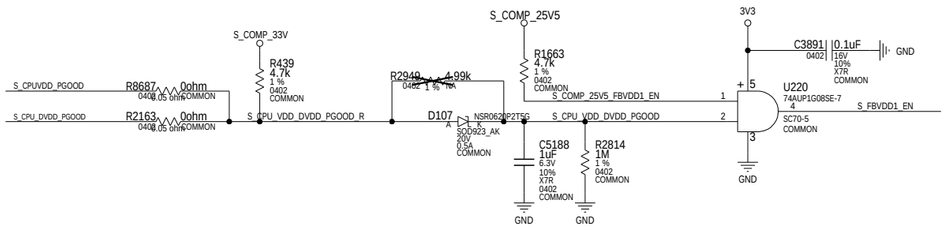
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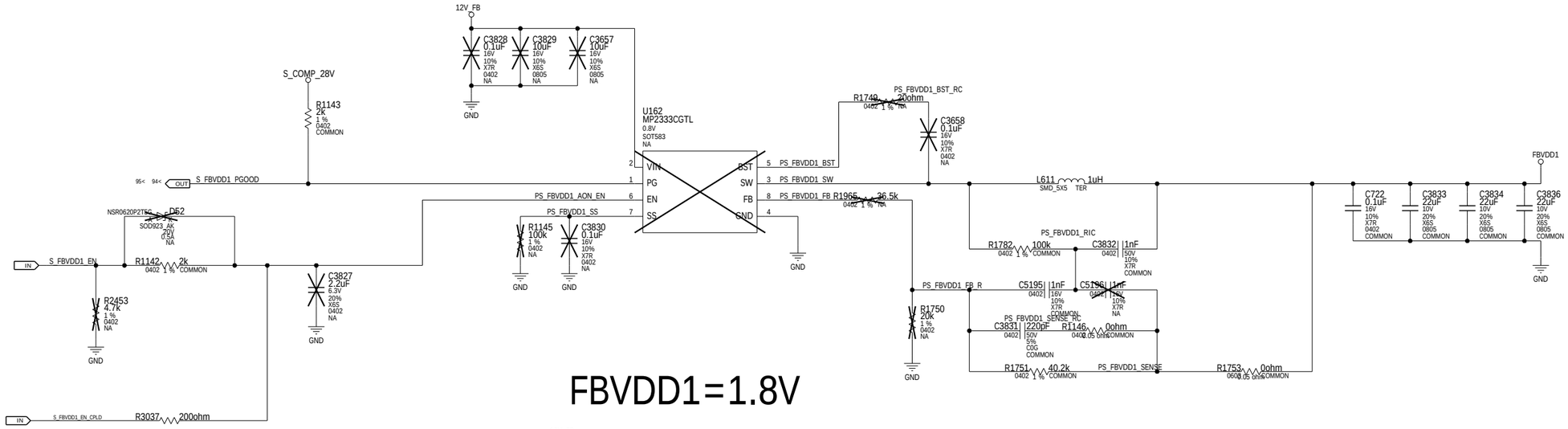
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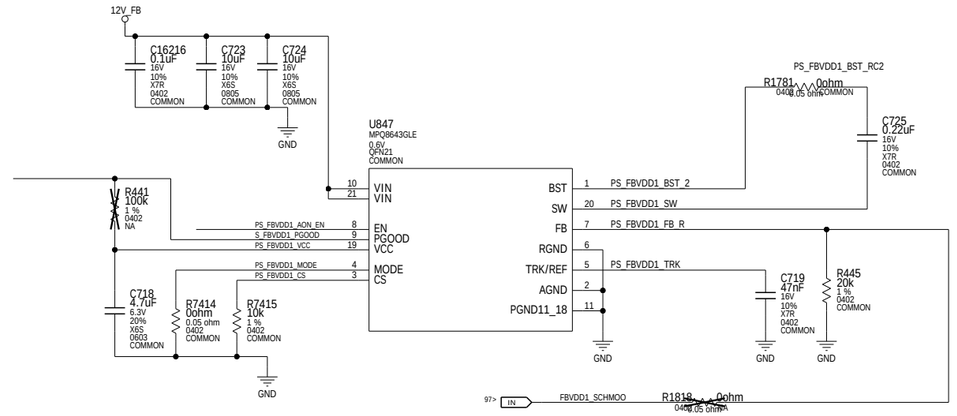
FBVDD1 (1.8V)





FBVDD1=1.8V

PG NEEDS TO BE TIES TO VCC
SEE NVBUG: 4033587



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS FBVDD1

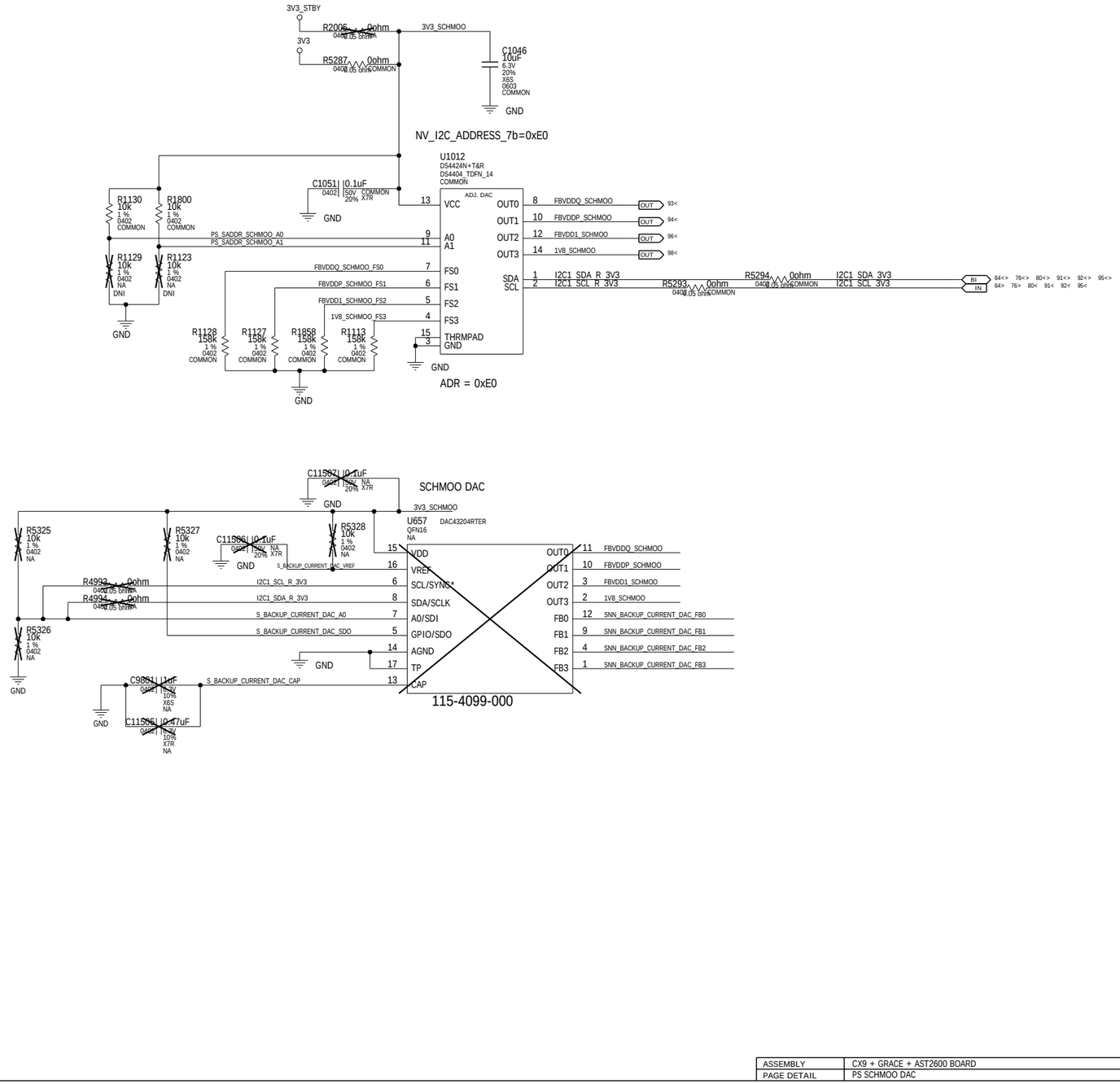
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Schmoo DAC



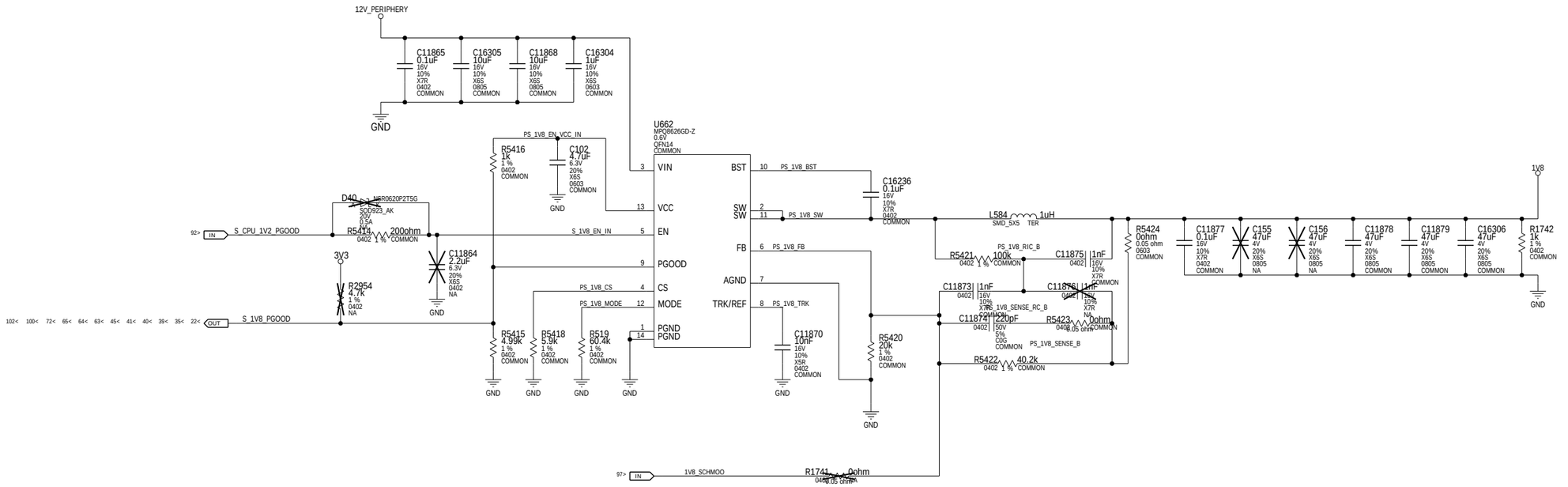
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS SCHMOO DAC

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	PS: 1V8

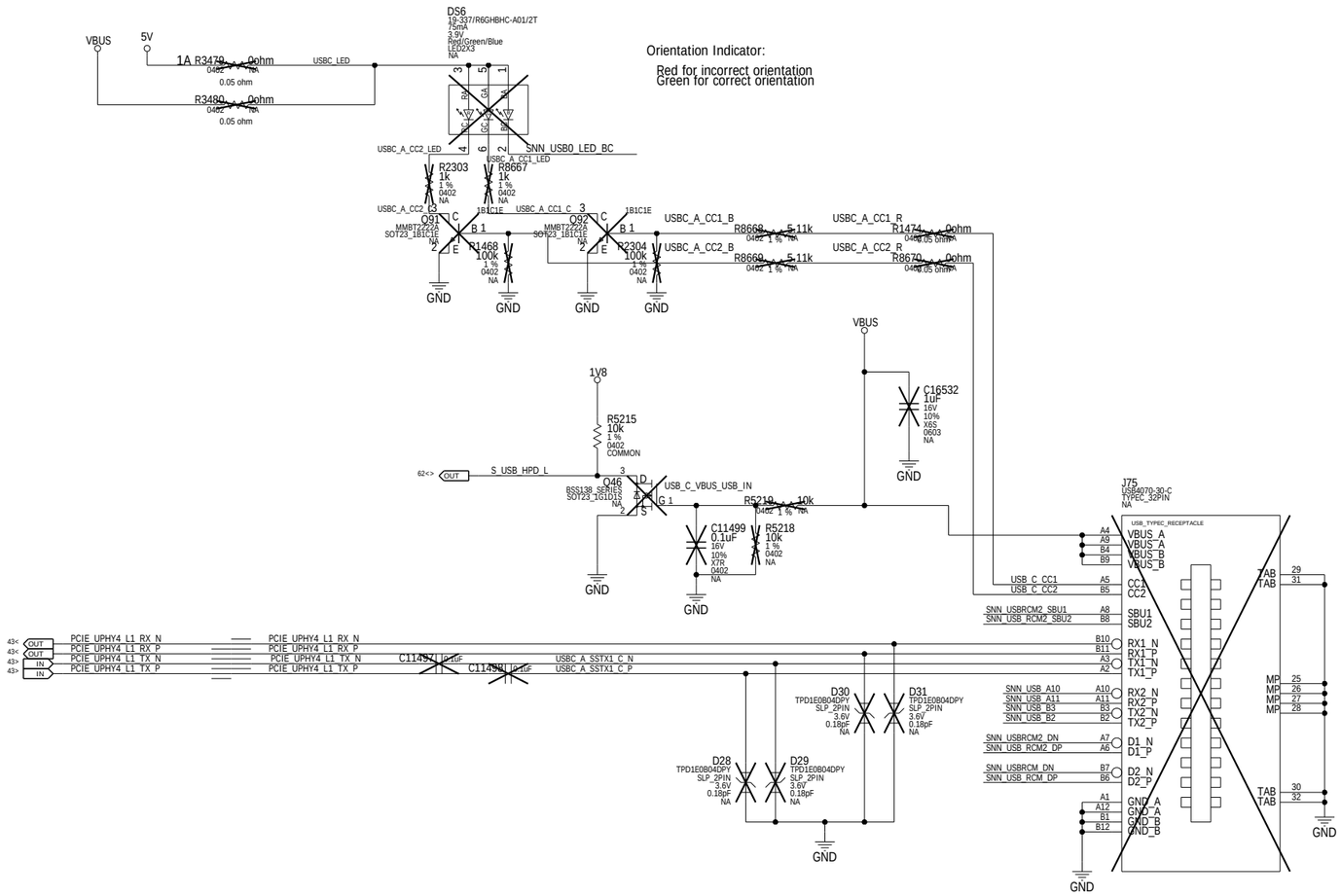
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CPU UPHY4 - USB3.2 USBRCM



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	DEBUG: USB RCM CONN

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CEC CONTROLLER

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE: ERoT

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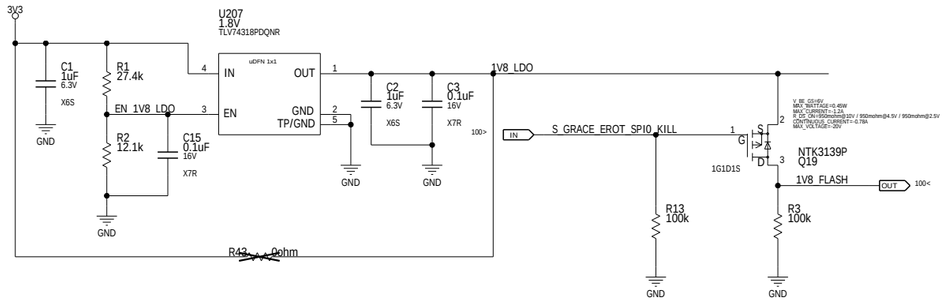
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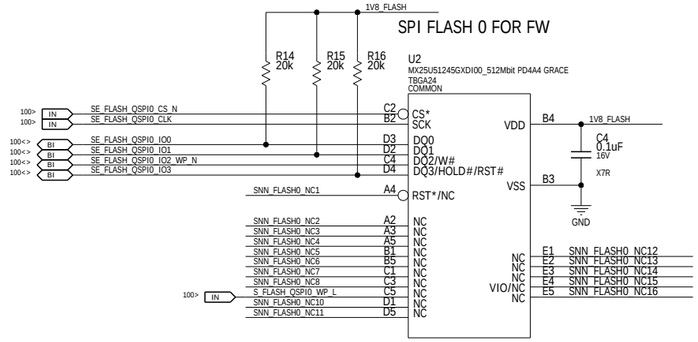
POWER, SPI FLASH

1.8V LDO



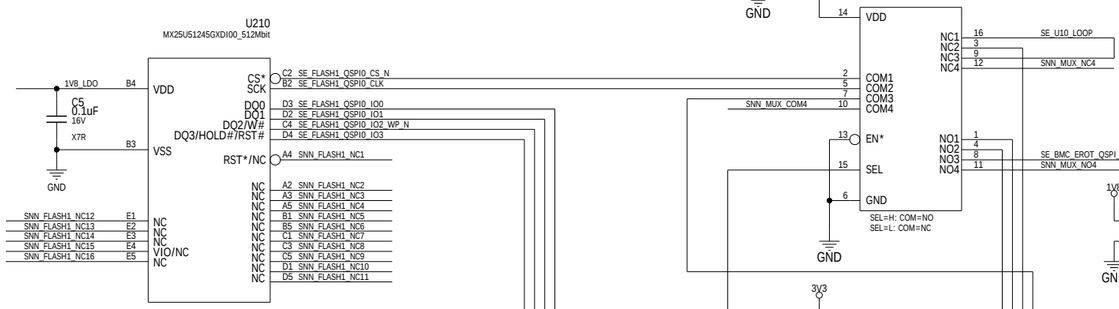
FLASH PWR DISABLE

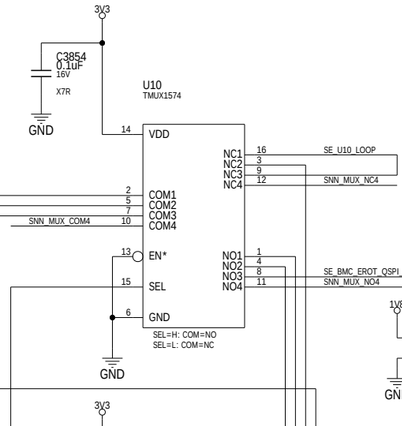
SPI FLASH 0 FOR FW

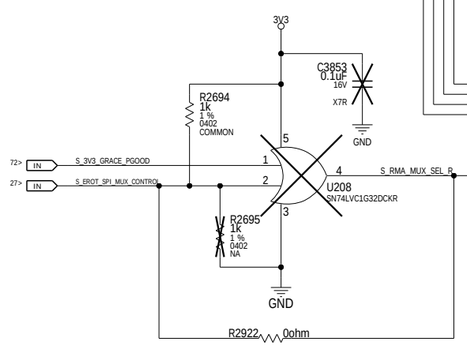


FPGA TO ACCESS R/W SPI

SPI FLASH 1 FOR R/W







TRUETH TABLE

AP_PGOOD	RMA_MUX_SEL_N	OR_OUT	SPI_OWNED_BY
HIGH	HIGH	HIGH	GRACE
HIGH	LOW	HIGH	GRACE
LOW	HIGH	HIGH	GRACE
LOW	LOW	LOW	BMC

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	GRACE: ERoT flash

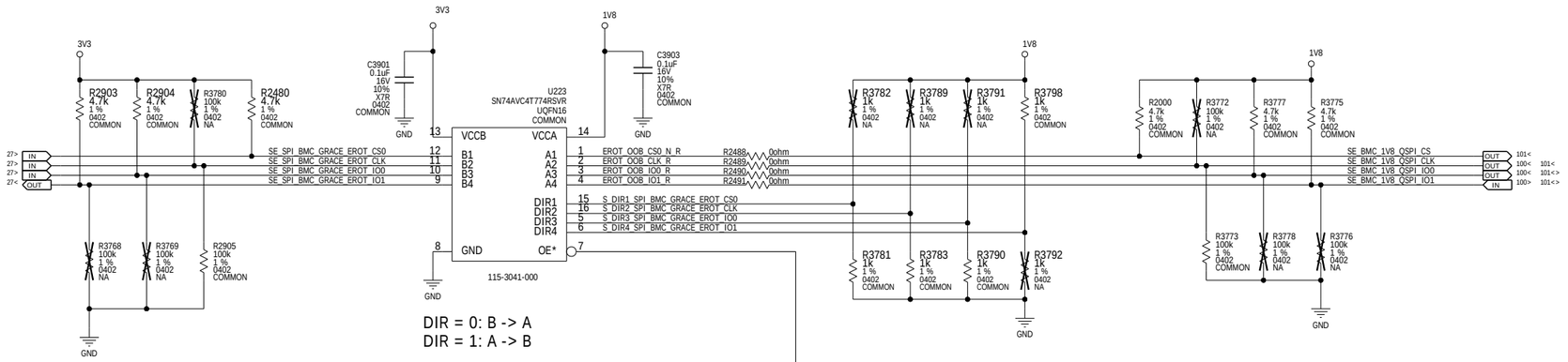
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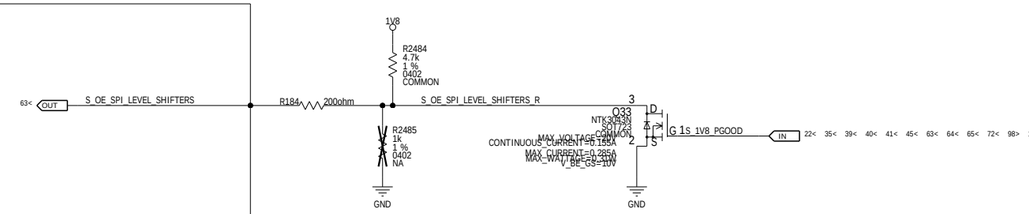
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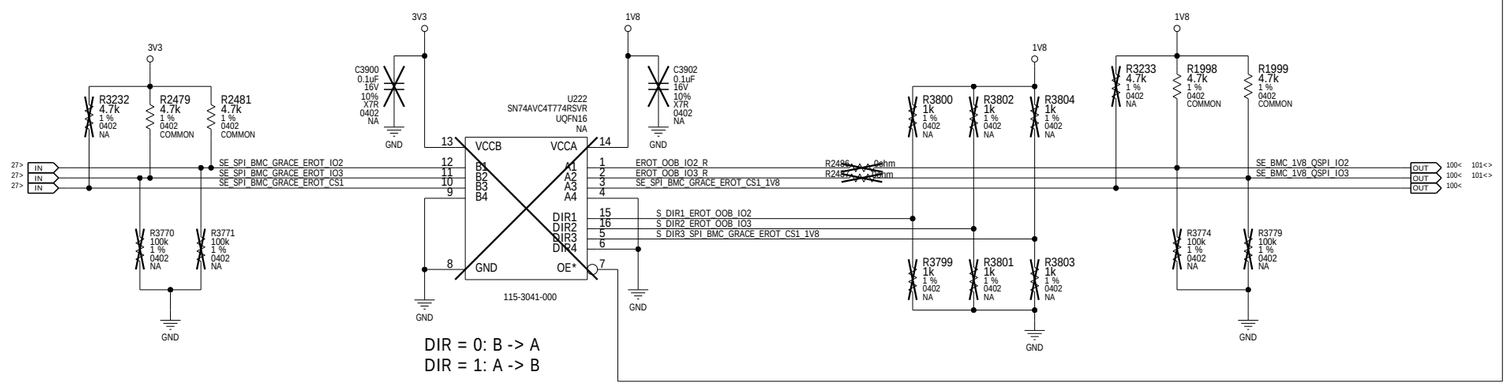
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EROT Level Shifters







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PAGE DETAIL	GRACE: EROT Level Shifters

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PAGE DETAIL	PS: 54V-12V

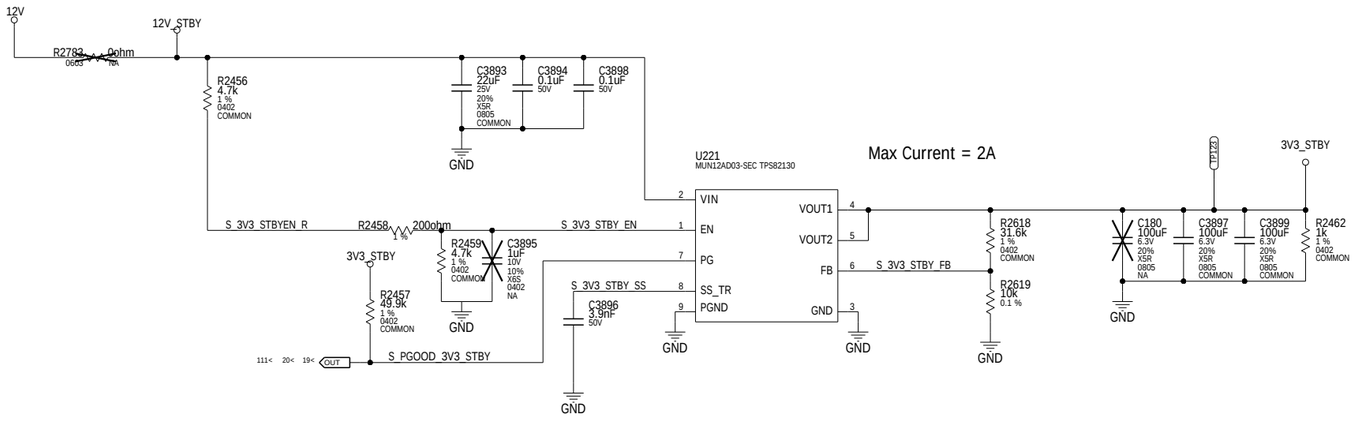
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Power: 3V3_STBY



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PAGE DETAIL	PS: 3V3_STBY

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Management

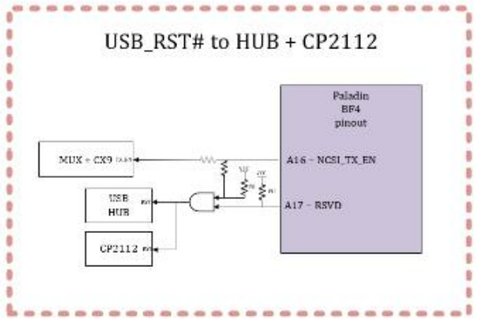
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	------- Management -------

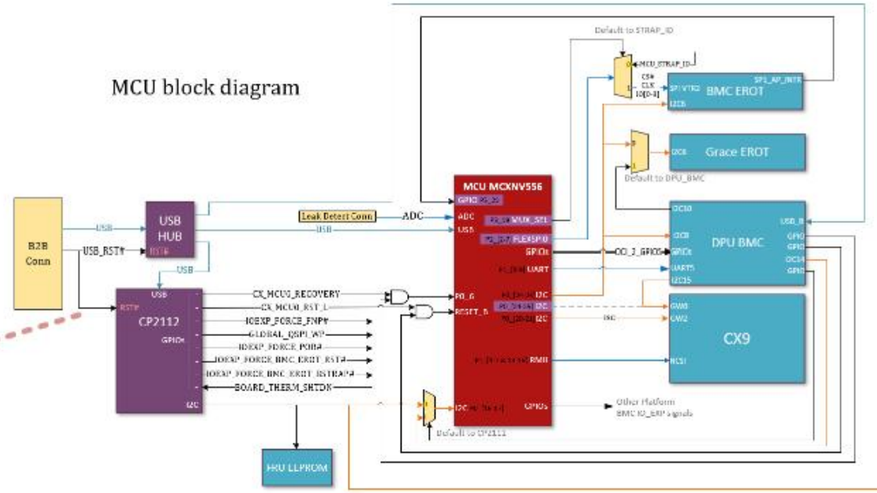
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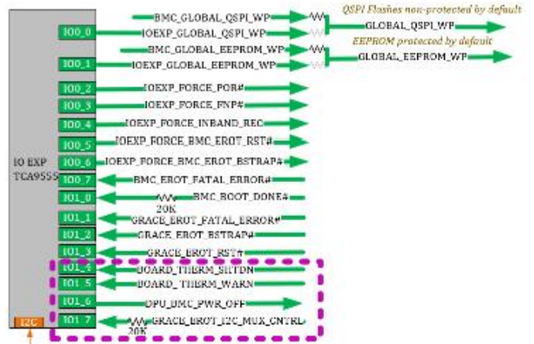
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PAGE DETAIL	MCU BLOCK DIAGRAM

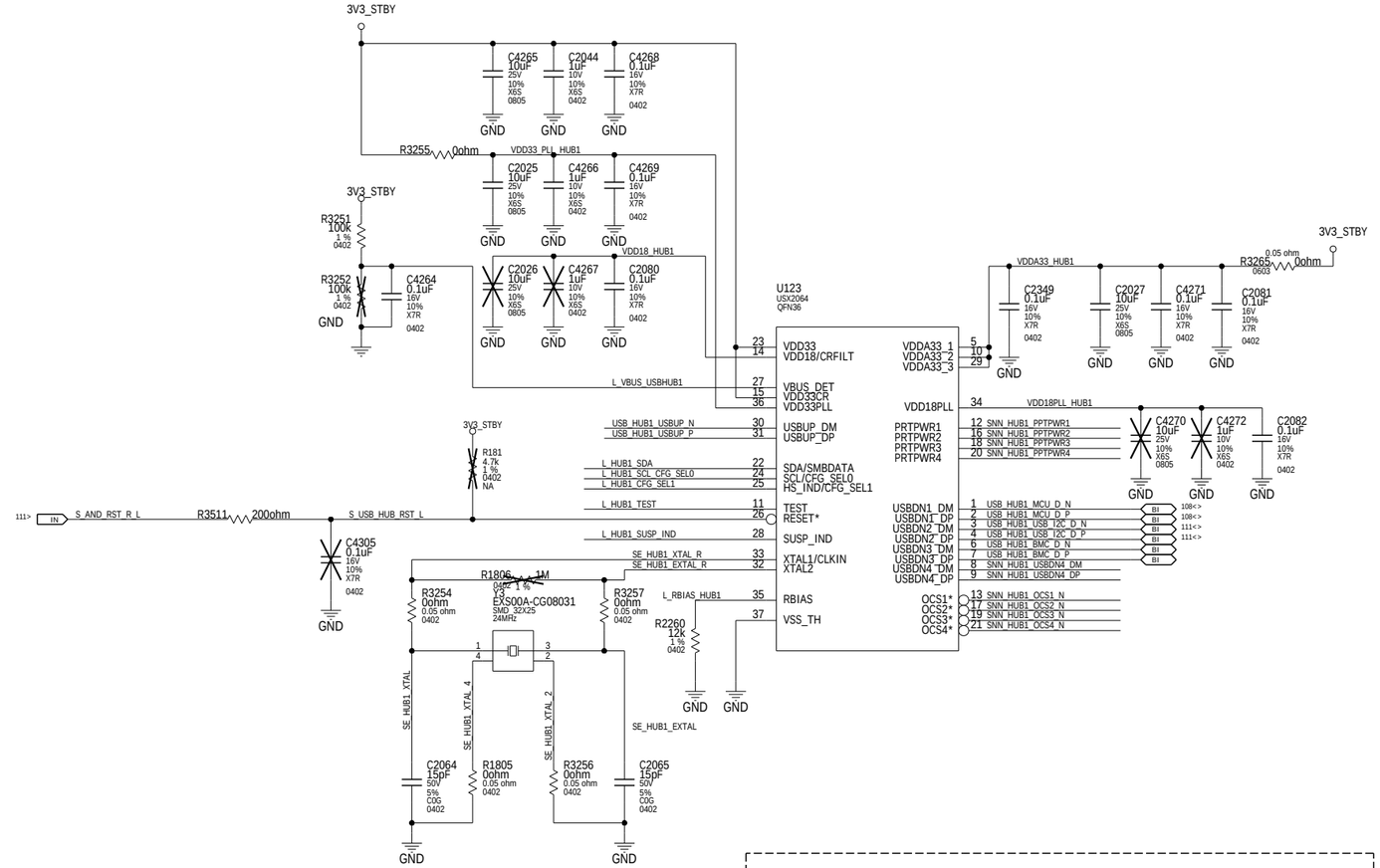
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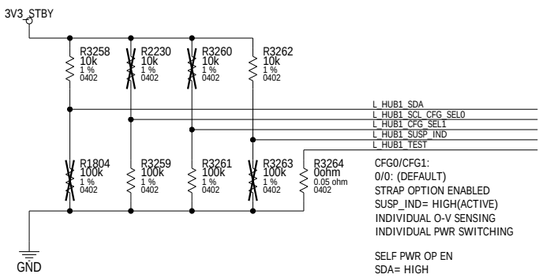
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PCB REV	PD4A4-D00	PAGE	106 OF 112
BOM REV	M	DATE	16-JUL-2026

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USB HUB



USB HUB Straps



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	USB HUB

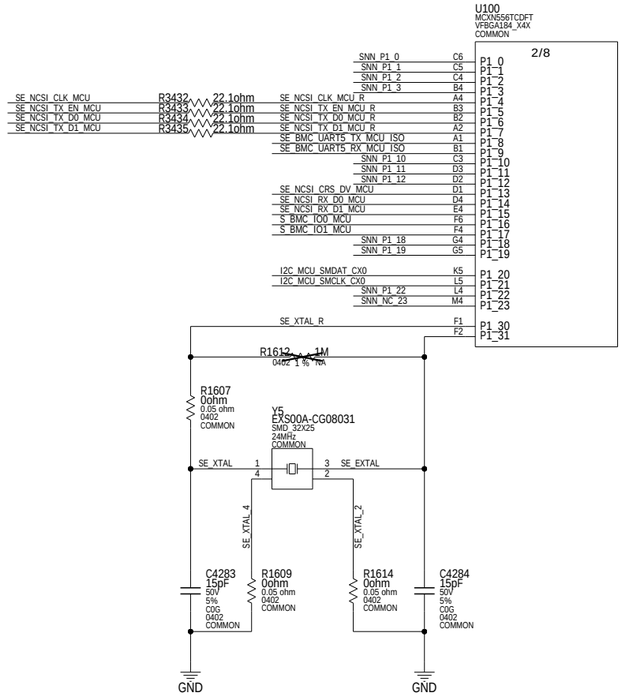
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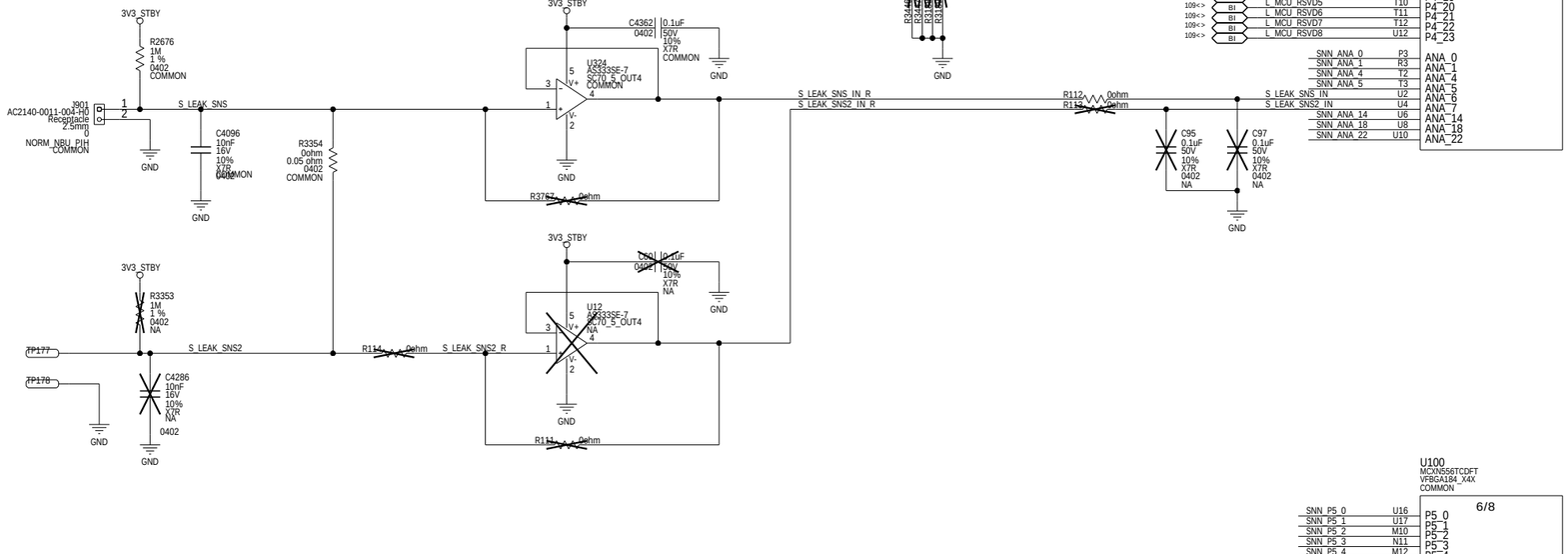
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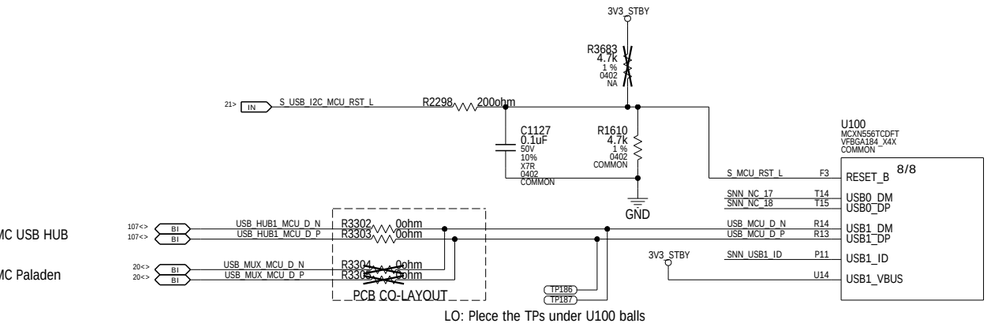
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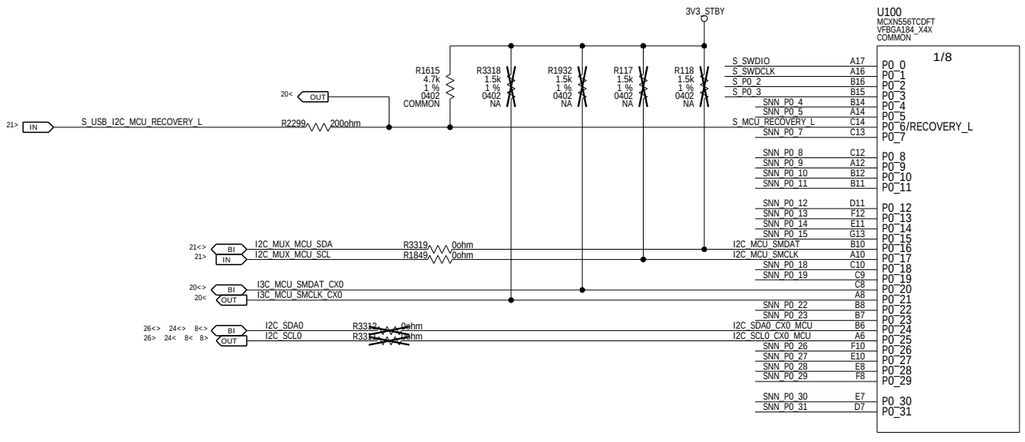
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MCU:GPIO I









Serial Wire Interface

TP156 S_SWDIO
TP157 S_SWDCLK
TP179 S_P0_2
TP180 S_P0_3
TP176 GND

ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	MCU GPIOs Part1

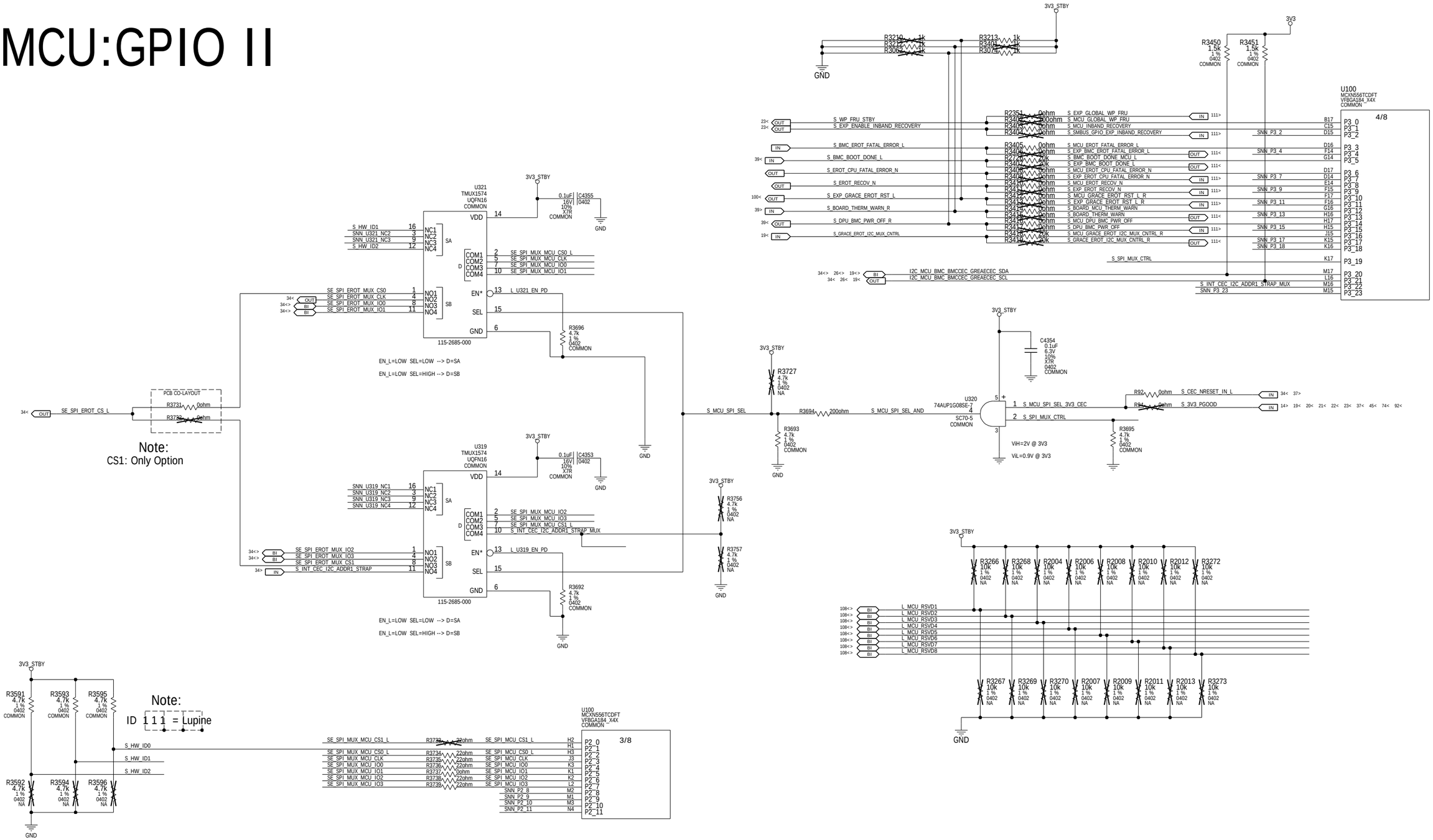
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MCU:GPIO II



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	MCU GPIOs Part2

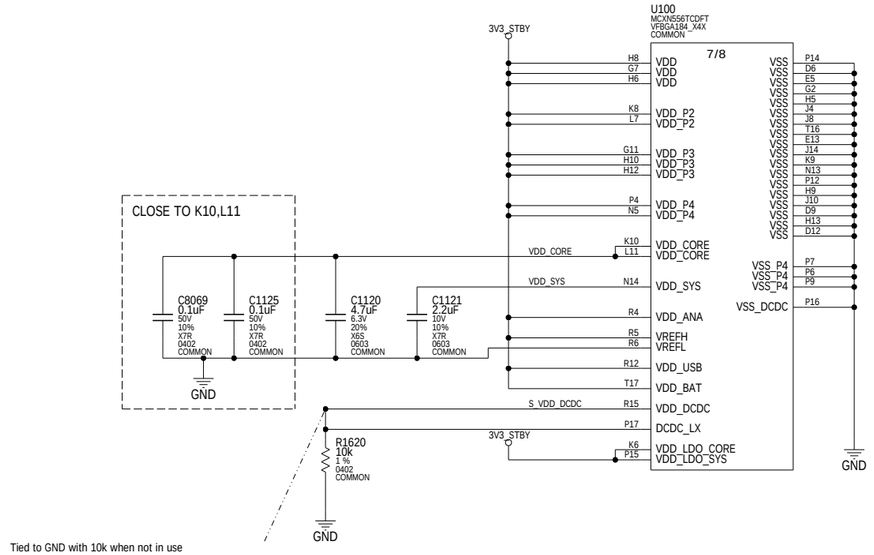
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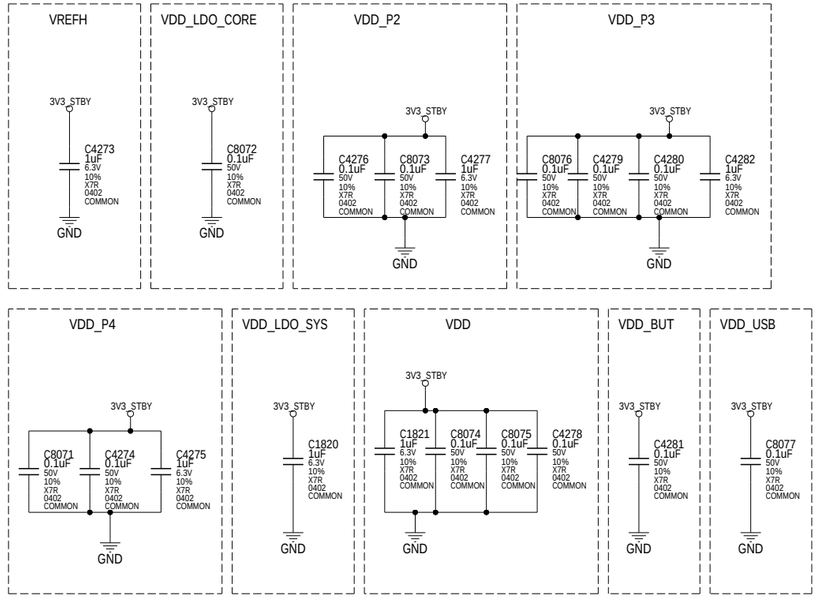
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PCB REV	PD4A4-D00	PAGE	109 OF 112
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MCU:Power



PLACE THE CAPACITORS CLOSE TO THE PINS ON U104
EACH PIN NAME IS WRITTEN BELOW



ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	MCU Power

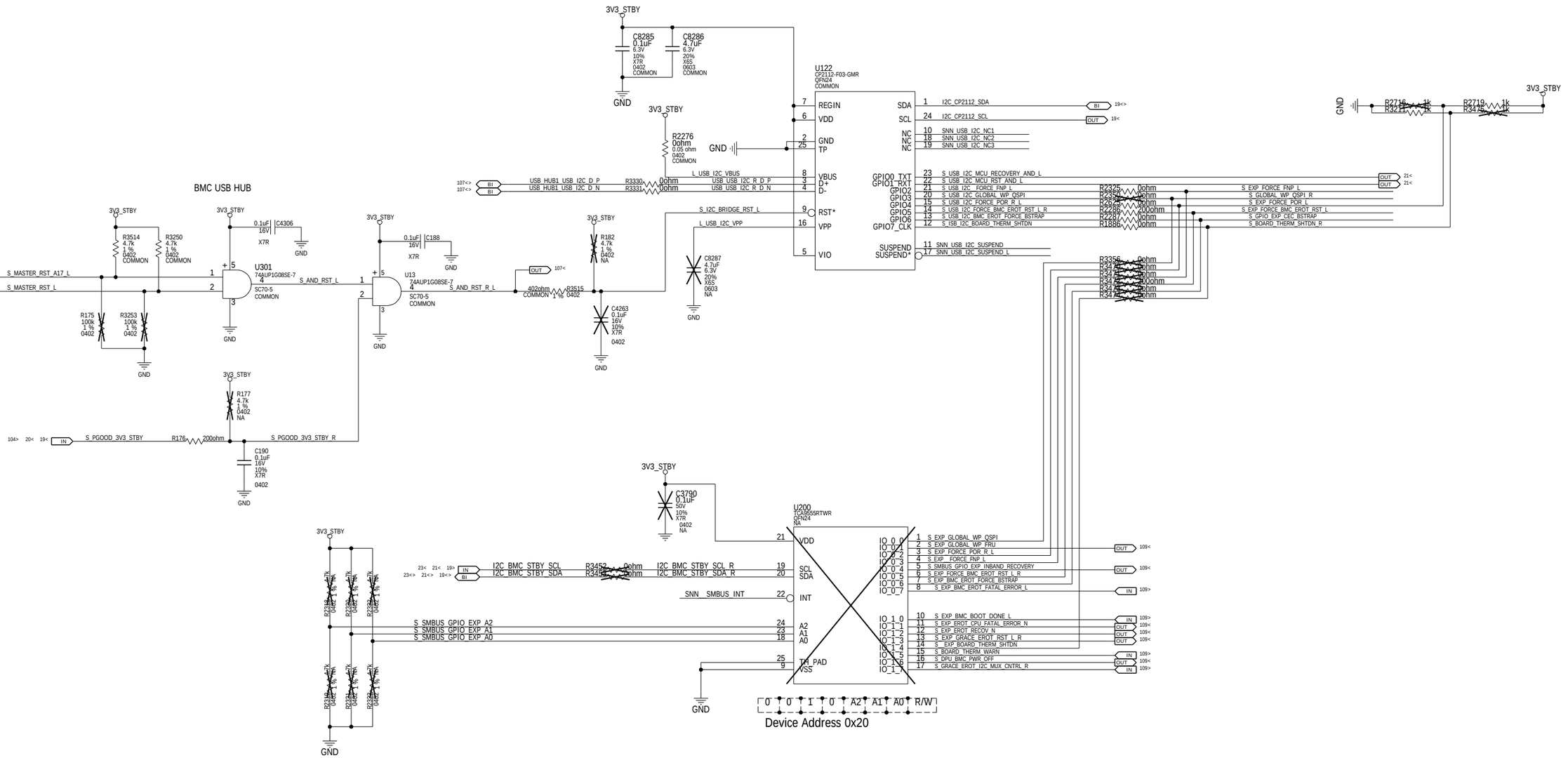
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USB to I2C Bridge



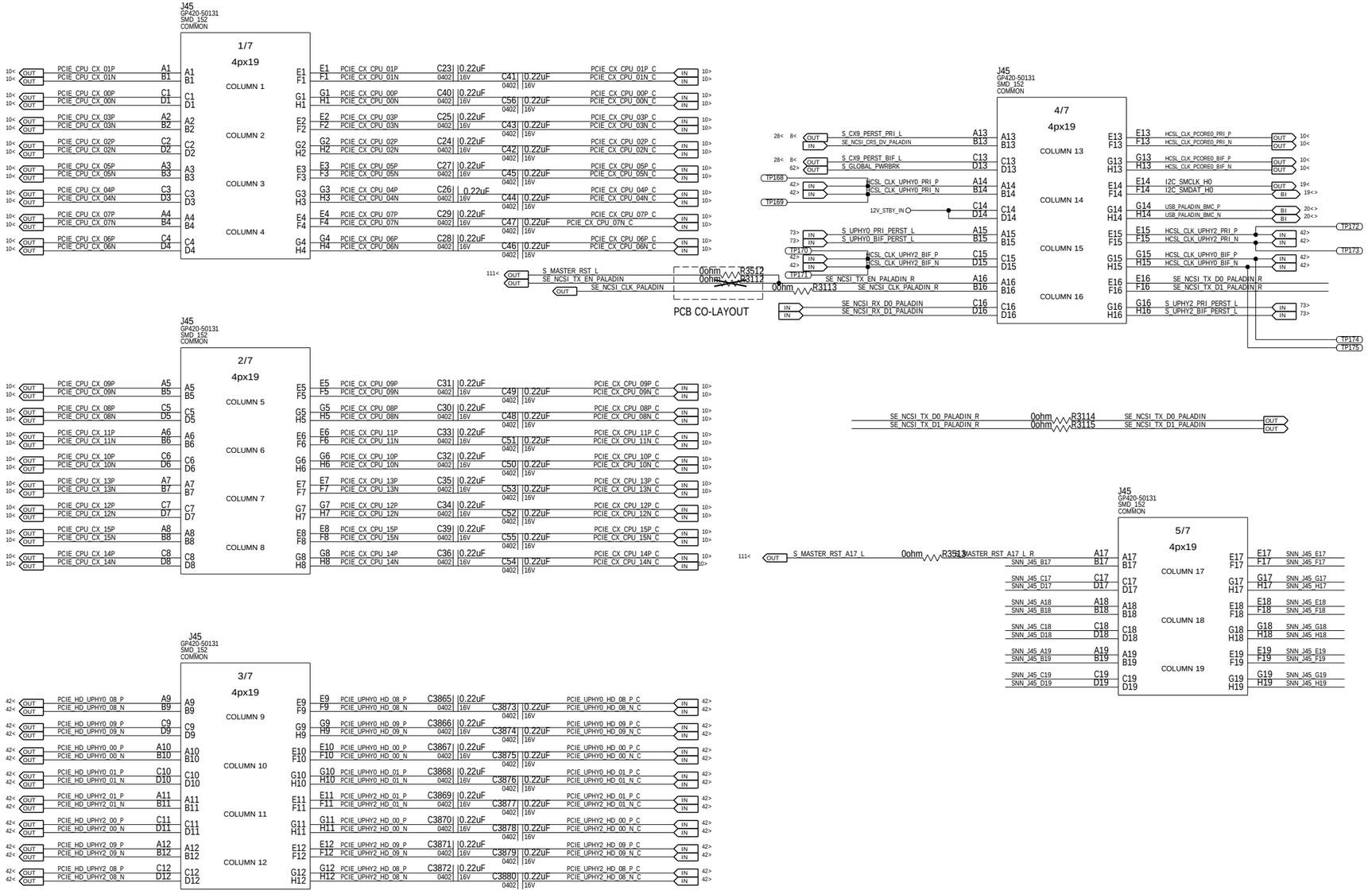
ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	USB to I2C (CP2112)

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ASSEMBLY	CX9 + GRACE + AST2600 BOARD
PAGE DETAIL	Paladin connector

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